

2026

FACULTY of
NATURAL SCIENCES



UNIVERSITY of the WESTERN CAPE
IYUNIVESITHI yaseNTSHONA KOLONI
UNIVERSITEIT van WES-KAAPLAND



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2026

Faculty of Natural Sciences

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Please check the University website (www.uwc.ac.za) for the latest version of this Calendar.

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HOW TO USE THIS CALENDAR

The following provides an overview of the structure of this Calendar to guide users.

General Information

This section provides the contact information for the Faculty and University.

Degrees and Diplomas conferred in the Faculty

This section provides information on each of the qualifications conferred in the Faculty.

Explanation of the National Qualifications Framework (NQF)

This is a brief section on the NQF levels and qualification types.

Faculty Board and Faculty Office Staff

This section lists staff members who constitute the Faculty Board and Faculty Office staff members.

Lecturing and Technical Staff

A comprehensive Faculty staff listing per Department, Centre, Institute, School or Unit is provided.

Rules for Programmes

This section provides information on the rules for each academic programme at undergraduate and postgraduate level offered by the Faculty. In each year level, and depending on the programme for which a student is registered, they are required to complete and pass a certain number of credits in order to promote to the next level of study. Promotion requirements per programme can be found in this section. These rules should be read in conjunction with the academic rules (Section 3) of the General Calendar (Part 1). Students should acquaint themselves with the rules in both Calendars and annually check for rule and curriculum changes.

Module Descriptors

This section provides information on all the modules offered by the faculty at undergraduate and postgraduate level. Module descriptors contain information which relate to the main outcomes and content for each module, including the credit value and NQF level of the module and pre-requisite and co-requisite modules. It is set in alphanumeric order as per undergraduate and postgraduate offering.

Pre-requisite and Co-requisite Modules

A pre-requisite module is a module that must be passed prior to a student being admitted to a higher module or the following year of study as determined in the faculty yearbook.

A co-requisite module is a module that must be passed prior to or simultaneously with another associated module before credit can be granted for the latter module. Requirements are provided in each module descriptor.

Explanation of Symbols and Remarks on Academic Transcript

This section provides an explanation of the symbols used and the remarks on the academic transcript.

GENERAL INFORMATION

CORRESPONDENCE WITH THE UNIVERSITY

All postal correspondence should be addressed to the relevant person or department at:

The University of the Western Cape
Private Bag X17
Bellville
7535

Should you not know the person or department, please direct all correspondence to the Registrar.

Faculty related enquiries can be directly forwarded to:

The Faculty Manager
Faculty of Natural Sciences
The University of the Western Cape
Private Bag X17
Bellville
7535
Tel: +27 (0)21 959 3426
Email: science-undergrad@uwc.ac.za
science-postgrad@uwc.ac.za

CONTACT NUMBERS

UWC Contact Centre: +27 (0)21 959 3900/1/2/3
Email: info@uwc.ac.za

THE UNIVERSITY'S WEBSITE

www.uwc.ac.za

CALENDAR

The Calendar is obtainable in the following separate parts:

- Part 1 General Information
- Part 2 Faculty of Natural Sciences
- Part 3 Faculty of Arts and Humanities
- Part 4 Faculty of Economic and Management Sciences
- Part 5 Faculty of Education
- Part 6 Faculty of Dentistry
- Part 7 Faculty of Law
- Part 8 Faculty of Community and Health Sciences
- Part 9 Schedule of Fees (Refer to the Schedule of Fees available on the University website)

DEGREES AND DIPLOMAS CONFERRED BY THE FACULTY

DEGREES

Qualification	Abbreviation	Minimum period of Study
Bachelor of Science in Applied Geology	BSc (Applied Geology)	3
Bachelor of Science in Biodiversity and Conservation Biology	BSc (Biodiversity and Conservation Biology)	3
Bachelor of Science in Biotechnology	BSc (Biotechnology)	3
Bachelor of Science in Chemical Sciences	BSc (Chemical Sciences)	3
Bachelor of Science in Computer Science	BSc (Computer Science)	3
Bachelor of Science in Environment and Water Science	BSc (Environment and Water Science)	3
Bachelor of Science in Mathematics and Statistical Sciences	BSc (Mathematics and Statistical Sciences)	3
Bachelor of Science in Medical Bioscience	BSc (Medical Bioscience)	3
Bachelor of Science in Physical Science	BSc (Physical Science)	3
Bachelor of Pharmacy	BPharm	4
Bachelor of Science Honours in Applied Geology	BScHons (Applied Geology)	1
Bachelor of Science Honours in Astrophysics	BScHons (Astrophysics)	1
Bachelor of Science Honours in Biodiversity and Conservation Biology	BScHons (Biodiversity and Conservation Biology)	1
Bachelor of Science Honours in Biotechnology	BScHons (Biotechnology)	1
Bachelor of Science Honours in Chemistry	BScHons (Chemistry)	1
Bachelor of Science Honours in Computational Finance	BScHons (Computational Finance)	1
Bachelor of Science Honours in Computer Science	BScHons (Computer Science)	1
Bachelor of Science Honours in Environment and Water Science	BScHons (Environment and Water Science)	1
Bachelor of Science Honours in Mathematics	BScHons (Mathematics)	1
Bachelor of Science Honours in Medical Bioscience	BScHons (Medical Bioscience)	1
Bachelor of Science Honours in Physical Science	BScHons (Physical Science)	1
Bachelor of Science Honours in Population Studies	BScHons (Population Studies)	1
Bachelor of Science Honours in Statistical Science	BScHons (Statistical Science)	1
Master of Pharmacy	MPharm	1
Master of Philosophy in Population Studies	MPhil (Population Studies)	1
Master of Science in Applied Geology	MSc (Applied Geology)	1
Master of Science in Biodiversity and Conservation Biology	MSc (Biodiversity and Conservation Biology)	1
Master of Science in Bioinformatics	MSc (Bioinformatics)	1
Master of Science in Biotechnology	MSc (Biotechnology)	1
Master of Science in Chemical Sciences	MSc (Chemical Sciences)	1
Master of Science in Computational Finance	MSc (Computational Finance)	1
Master of Science in Computer Science	MSc (Computer Science)	1
Master of Science in Environment and Water Science	MSc (Environment and Water Science)	1
Master of Science in Mathematical Science	MSc (Mathematical Science)	1
Master of Science in Medical Bioscience	MSc (Medical Bioscience)	1

Master of Science in Nanoscience	MSc (Nanoscience)	2
Master of Science in Pharmaceutical Sciences	MSc (Pharmaceutical Sciences)	1
Master of Science in Pharmacy Administration and Policy Regulation	MSc (Pharmacy Administration and Policy Regulation)	1
Master of Science in Physical Science	MSc (Physical Science)	1
Master of Science in Statistical Science	MSc (Statistical Science)	1
Master of Clinical Pharmacy	M Clinical Pharmacy	2
Doctor of Pharmacy	DPharm	2
Doctor of Philosophy *	PhD*	2

DIPLOMAS

Postgraduate Diploma in Integrated Water Resource Management	PGDip (Integrated Water Resource Management)	1
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* Please refer to the programme information for specialisations.

EXPLANATION OF THE NATIONAL QUALIFICATIONS FRAMEWORK

The National Qualifications Framework (NQF) is “a *single integrated system for the classification, registration, publication and articulation of quality-assured national qualifications*” as stipulated in Section 4 of the NQF Act, 2008 (Act No 67 of 2008).

The National Qualifications Framework (NQF) has ten levels of which Higher Education qualifications occupy 6 levels of the NQF, namely Levels 5 to 10.

Levels 5 to 7 comprise the undergraduate qualifications (with the exception of the Professional Bachelor's degree at Level 8) and Levels 8 to 10 comprise the postgraduate qualifications.

NQF LEVELS	QUALIFICATION TYPES
5	Higher Certificate
6	Advanced Certificate Diploma
7	Advanced Diploma Bachelor's Degree
8	Honours Degree Postgraduate Diploma Professional Bachelor's Degree
9	Master's Degree Professional Master's Degree
10	Doctoral Degree Professional Doctoral Degree

As cited in the Higher Education Qualifications Sub-Framework (CHE, 2013)

FACULTY BOARD AND FACULTY OFFICE STAFF

Ex Officio Members: The Rector, Vice Rector/s, Registrar and Dean

Professors: RF Ajayi, SS Appunni, C Arendse, M Aucamp, R Appadu, O Azu, R Bailie, P Baker, M Benjeddou, DR Beukes, E Beukes, JS Boatwright, SS Botha, B Bladegroen, RJ Blignaut, A Christoffels, CH Conana, V Couldridge, F Cummings, M D'Amato, A Dube, T Dube, N Ebrahim, S Egieyeh, EC Elson, A Engelbrecht, D Fisher, F Fisher, D Frei, M Ghaziasgar, MJ Gibbons, M Grenfell, S Halindintwali, U Hesse, DC Hiss, D Holgate (Chairperson), OE Isafiade, E Iwuoha, MN Jahed, J Joubert, N Jovanovic, T Kanyerere, E Kapangaziwiri, B Kirby, M Keyster, L Khotseng, S Khoza, AJ Klein, T Kohn, L Leenen, V Linkov, M Lochner, R Luus, R Maartens, S Macfarlane, A Madiehe, R Madjoe, SF Malan, GW Maneveldt, L Maree, B Maritz, D Marshall, JA McCartney, M Meyer, E Mehdihezad, AK Mishra, TK Monsees, T Mulaudzi-Masuku, J Muniyakazi, ECS Mwambene, C Nyirenda, K Obikeze, A Oelofse, MO Onani, C Opuwari, M Opuwari, JN Orce, K Patidar, P Pillay, K Pokpas, DJR Pugh, A Rajkaran, N Ross, H Samsodien, MGR Santos, M Shaw, AJ Smit, N Stiegler, R Taylor, M Tchokonte, N Tiffin, S Triambak, IM Trindade, M Viljoen, KL Ward, T Waryo

Drs: S Abdul-Rasool, P Abrantes, O Alexander, D Anderson, M Bennett, TD Bhaga, S Brigg, H Brydon, S Clarke, R Cloete, C Cupido, J de Smidt, E Egieyeh, N Haines-Arries, A Henney, M Herbert, A Hitzeroth, N Ikegwuoha, C Ikpo, R Jacobs, E Kapp, M Kasu, J Keddie, B Langa, W Lee, K Makhatini, C Makasu, S Maku-Vyambwera, GF Malgas, T Maphanga, S Matshawule, L Mbonile, L Mciteka, M Milne, SJ Mkhize, N Mohamed, T Morris, R Muir, A Mukhtar, GE Muller, TFG Muller, F Muya, MJ Norman, P Nsengiyumya, M Nsumai, P O'Farrell, OM Oluyide, MB Parker, S Pasupathi, K Pearce, A Razafindrakoto, D Solomons, G van Schalkwyk, Y Sukai, TS Volkwyn, C Willemse

Messrs: C Baker, H Harribhai, A Latief, T Lesch, R Maclons, MA Maneli, C Mupfiga, N Naidoo, M Ongansie, G Williams

Mmes: LT Dube, RC Eager, C Greyling, N Keuler, R Lombard-Jacobus, M Laubscher, M Levendal, GR Raitt, TC Ras, SA Short, Y van Heerden, T Wannenberg, Y Williams

Representatives from the Faculties of:

Education:	Dr B May
Arts and Humanities:	Dr M van Heerden
Community and Health Sciences:	Mrs J Wilkenon
Economic and Management Sciences:	Dr J Breytenbach

FACULTY OFFICE STAFF

Dean:	Prof D Holgate, PhD (UCT)
Deputy Deans:	Prof RF Ajayi, BSc (Hons) PhD (UWC) Prof A Dube, BPharm (Zimbabwe) MPharm (UWC) PhD (Monash) Prof K Patidar, PhD (IIT Kanpur)
Learning & Teaching Specialist:	Prof CH Conana, MEd (SU) PhD (UWC)
Acting Faculty Manager:	Ms K Parker, BSc (Hons) (UNISA) PGDip (UCT) MEd (UWC)
Administrator to the Dean:	Ms T Parker, Dip (PC Training and Business College)
Senior Faculty Officer:	Ms K Parker, BSc (Hons) (UNISA) PGDip (UCT) MEd (UWC)
Faculty Officers:	Ms C Carolus, BA (Hons) (UWC) Mr D Du Plooy, BAdmin (UWC) Ms N Isaacs

Health and Safety Officer:
Administrative Officer:
Administrative Assistants:

Ms F Pathan, BA PGCE (UWC)
 Mr R Williams, BA (UNISA) PGDip (UWC)
 Mr A Snell
 Ms C Hendricks, BSc (UWC)
 Ms S Gxagxama, BSc (Hons) (UWC)
 Ms O Conana, BSc (UWC) PGDip (Varsity College)
 Mr P Sifile, BA (Hons) (UWC)

LECTURING AND TECHNICAL STAFF

BIODIVERSITY AND CONSERVATION BIOLOGY

Departmental Chairperson:	Prof A Rajkaran, PhD (NMU)
Senior Professor:	Prof MJ Gibbons, BSc (Hons) (Liverpool) PhD (UCT)
Professors:	Prof JS Boatwright, PhD (UJ) Prof GW Maneveldt, PhD (UWC) Prof AJ Smit, MSc (UPE) PhD (UCT) Prof V Couldridge, PhD (Wits)
Associate Professors:	Prof A Engelbrecht, PhD (SU) Prof B Maritz, PhD (Wits) Prof A Rajkaran, PhD (NMU)
Emeritus Professors:	Prof A Channing, PhD UED (UKZN) Prof L Raitt, PhD (SU)
Extra-Ordinary Professors:	Prof EM Moll, PhD (UKZN) Prof P Barham, PhD (Bristol)
Extra-Ordinary Researchers:	Dr J Arnolds, PhD (CPUT) PhD (NWU) Dr M Brown, PhD (UWC) Dr K Christison, PhD (UFS) Ms JC Coetzee, MSc (UCT) Dr P Cotiyane-Pondo, PhD (NMU) Dr A de Lecea, PhD (UKZN) Dr MGJ Hendricks, PhD (UWC) Dr A le Roux, PhD (UNISA) Dr S Naidoo, PhD (UKZN) Dr RA Maritz, PhD (Cornell) Dr JS McManus, PhD (Wits) Dr T Samaai, PhD (UWC) Dr MI Samuels, PhD (UCT) Dr MPE Schurch, PhD (Southampton) Dr B Smuts, MBChB (UCT) Dr AA Turner, PhD (UWC) Dr CD van der Lingen, PhD (UCT) Dr E van Jaarsveld, PhD (UP) Dr L Waller, PhD (UCT) Dr P O'Farrell, PhD (UCT)
Senior Lecturer:	Ms RC Eager, MSc (UWC)
Lecturers:	Ms M Laubscher, MSc (SU) Ms GR Raitt, MSc (SU) Ms SA Short, MSc (SU)
Chief Officers:	Vacant

Senior Officers:

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Mr BA Flynn, BSc (Hons) (UWC)
Ms L Gelderbloem, BSc (Hons) (UWC)
Ms A King, BSc MPA (UWC)
Ms P Mpange, BSc (UKZN)
Mr A Rakgoale, MSc (UCT)
Ms L van Heerden, Dipl (Varsity College)

Officers:**Administrative Officer:****BIOTECHNOLOGY****Departmental Chairperson:****Professors:**

Vacant
Prof M Benjeddou, PhD (UWC)
Prof M D'Amato, PhD (Buenos Aires)
Prof M Keyster, MSc (UWC) PhD (SU)
Prof A Madiehe, PhD (Louisiana State)
Prof M Meyer, PhD (UWC)

Associate Professors:

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Prof AJ Klein, PhD (UWC)
Prof T Mulaudzi-Masuku, PhD (UWC)
Prof DJR Pugh, DPhil (Oxon)
Prof DA Cowan, PhD (Waikato)
Prof N Ludidi, PhD (UWC)
Prof B Ndimba, PhD (Durham)

Emeritus Professor:**Extra-Ordinary Professors:**

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Dr Y Sukai, PhD (UKZN)
Dr M Kasu, BSc (Hons) MPhil (UCT) PhD (UWC)

Lecturer:**Research Fellows:**

Dr O Caleb, PhD (SU)
Dr A Fadaka, PhD (UWC)
Dr K Gabuza, PhD (UWC)
Dr S Meyer, PhD (UWC)
Dr N Sibuyi, PhD (UWC)

Chief Officers:

Dr Z Abrahams-October, PhD (UWC)
Mrs S Grové, MSc (SU)
Dr L Husselman, PhD (UWC)

Senior Officers:

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Ms E Anthony, BSc ADM (UWC) BSc (Hons) (UCT) BA (Hons) (UWC)
Mr C Gelderbloem, BSc (Hons) (UWC)
Mrs P Lebenya, MTech (TUT) MSc (SU)
Ms S Nene, BSc (Zululand) BSc (Hons) (UKZN) BA (Hons) (UWC)
Ms L Barnies, MDP (UWC)
Ms M Pool

Administrative Officer:**Administrative Assistant:****CHEMISTRY****Departmental Chairperson:****SARCHI Chair (NanoElectrochemistry and Sensor Technology):**

Prof MO Onani, MSc (Kenyatta) PhD (UKZN)
Prof E Iwuoha, BSc (Hons) (Nigeria) PhD (Ibadan)
FRSC (United Kingdom)

SARCHI Chair (Analytical Systems and Processes for Priority and Emerging Contaminants (ASPPEC):

Senior Professors:

Prof P Baker, BSc (UCT) MSc (UWC) PhD (SU)
 Prof P Baker, BSc (UCT) MSc (UWC) PhD (SU)
 Prof E Iwuoha, BSc (Hons) (Nigeria) PhD (Ibadan)
 FRCS (United Kingdom)

Professors:

Prof RF Ajayi, BSc (Hons) PhD (UWC)
 Prof L Khotseng, PhD (SU)
 Prof AK Mishra, MSc PhD (Delhi)

Associate Professors:

Prof MO Onani, MSc (Kenyatta) PhD (UKZN)
 Prof E Beukes, BSc (Hons) MSc PhD (RU)
 Prof MN Jahed, MTech (CPUT) PhD (UWC)
 Prof K Pokpas, BSc (Hons) MSc PhD (UWC)
 Prof N Ross, BSc (Hons) MSc PhD (UWC)
 Prof T Waryo, BSc MSc (Addis Ababa) PhD (Graz)

Emeritus Professors:

Prof M Davies-Coleman, BSc (Hons) PhD (RU)
 Prof IR Green, BSc (Hons) PhD (UCT)
 Prof L Petrik, BA (UNISA) PhD (UWC)
 Prof JJS Titinchi, MSc (Basrah) PhD (Indian IT Roorkee)

Extra-Ordinary Professors:

Prof WT Mabusela, BSc (UFH) MSc (RU) PhD (UCT)
 Prof P Makgwane, BSc (Hons) MSc (UP) PhD (NMU)
 Prof D Shallcross, DPhil (Oxon)

Senior Lecturers:

Prof B Owaga, BSc (Egerton) MSc (Wits) PhD (UJ)
 Dr O Alexander, BSc (Hons) MSc PhD (UFS)
 Dr C Cupido, BSc (Hons) MSc (URV Spain) PhD (UWC)
 Dr C Ikpo, BSc (Hons) (Owerri) MSc (Lagos) PhD (UWC)

Lecturers:

Dr F Muya, BSc (Hons) MSc PhD (UWC)
 Dr S Brigg, BSc (Hons) MSc PhD (SU)
 Dr L Mciteka, PhD (RU)
 Dr SJ Mkhize, BSc (Hons) (UKZN) MSc (UCT) PhD (Chiba University)
 Dr N Mohamed, HDE MSc PhD (UWC)

Chief Officers:

Vacant

Senior Officers:

Mr T Lesch, BTech (CPUT)
 Mr B de Wet, BSc (UWC)
 Mr A Mantyi, BCom BSc (UWC)

**Senior Technical Officer:
 Officer:**

Vacant
 Mr BD Makhoba, BSc (UWC)
 Mr N Nako, BSc (Hons) MSc (UWC)

Administrator:

Ms J Santon, Dip (False Bay College)

Administrative Assistant:

Ms N Mkokeli, Dip (Buffalo City College)

COMPUTER SCIENCE

Departmental Chairperson:

Prof A Bagula, MEng (Belgium) MSc (SU) PhD
(Royal Institute of Technology Sweden)

Extra-Ordinary Professors:

Prof JM Blackledge, BSc ARCS PhD DIC (London)
PhD (Jyvaskyla Finland)
Prof SK Das, PhD (Central Florida) MS (Indian
Institute of Science)
Prof T Dimphl, Habil (Eberhard Krls, Germany)
PhD (Erfurt, Germany)
Prof A Gerber, BEng (UP) MEng (WITS), PhD
(UNISA)
Prof R Millham, PhD (De Montford, England) MSc
(Abertay, Scotland)
Prof C Thron, PhD (Kentucky) PhD (Wisconsin)
BA (Princeton)

Associate Professors:

Prof WD Tucker, BA (Trinity) MS (ASU) PhD (UCT)
Prof M Ghaziasgar, MSc (UWC) PhD (UWC)
Prof OE Isafiade, PGDip (UWC) MSc PhD (UCT)
Prof L Leenen, BSc (Hons) MSc (UJ) PhD
(Wollongong Australia)
Prof C Nyirenda, PhD (Tokyo Tech Japan)
Dr F Mouton, BSc (Hons) MSc PhD (UP)
Prof A Radovanovic, PhD (UWC)
Dr O Oluwaghem, PhD (Covenant University
Nigeria)

Extraordinary Senior Lecturer:

Research Fellow:

Dr MJ Norman, BSc (Hons) PhD (UWC) MSc
(UCT)

Senior Lecturer:

Lecturers:

Dr A Henney, MSc PhD (UWC)
Dr OM Oluyide, MSc, PhD (UKZN)
Mr CK Baker, BSc (Hons) (UWC) MSc (UCT)
Mr N Naidoo, MSc (UWC)
Mr MA Maneli, MSc (UWC)
Mr DJD Leenderts

nGAP Lecturer:

Senior Officers:

Mr MT Allie, BSc (Hons) (UWC)

Administrators:

Ms R Abbott
Ms F Jacobs, BCom (UWC)

EARTH SCIENCES

Departmental Chairperson:

Professors:

Prof M Opuwari, MSc (Port Harcourt) PhD (UWC)
Prof T Dube, MSc (Twente, Netherlands) PhD
(UKZN)

UNESCO Chair (Hydrogeology):

Associate Professors:

Prof D Frei, MSc (TU Berlin) Dr rer nat (Göttingen)
Prof M Opuwari, MSc (Port Harcourt) PhD (UWC)
Prof T Kanyerere, BEd MSc (Malawi) PhD (UWC)
Prof R Bailie, MSc (Wits) PhD (UJ) Pr Nat Sci
Prof M Grenfell, MSc (UKZN) PhD (Exeter)
Prof N Jovanovic, MSc (CIHEAM, Italy) PhD (UP)
Prof T Kanyerere, BEd MSc (Malawi) MPhil
(Norway) PhD (UWC)

Emeritus Professors:	Prof E Kapangaziwiri MSc PhD (Rhodes University) Prof C Okujeni, Dr rer nat (TU Berlin) Prof D Mazvimavi, BSc (Hons) (Zimbabwe) MSc (Vrije U Brussels) PhD (Wageningen) Prof J van Bever Donker, Drs (Leiden) PhD (UCT) Prof Y Xu, MSc (Chengdu) PhD (UFS)
Extra-Ordinary Professors:	Prof E Braune, MSc (Imperial College London) PhD (UWC) Prof T Chatterjee, MSc PhD (India School of Mines) Prof J Day, PhD (UCT) Prof J King, PhD (UCT) Prof J Goldin, BA (RU) PhD (UCT) Prof AD Levine, BA (Bates) MS (Tulane) PhD (California) Prof PJ Oberholster, PhD (UP) Prof L Swatuk, PhD (Dalhousie) Prof A Turton, PhD (UNISA)
Senior Lecturers:	Dr S Clarke, MSc (UWC) PhD (SU) Dr T Maphanga, MSc (Limpopo) PhD (UWC) Dr R Muir, PhD (UCT)
Lecturers:	Ms TD Bhaga, BSc MSc (UWC) Ms Y Williams, BSc MSc (UWC)
Adjunct Lecturer:	Dr S Mohammed, MSc (Uniben, Nigeria) PhD (UWC)
Graduate Teaching Assistants:	Ms V Silinda, BSc (Hons) (UWC)
Chief Officers:	Mr EJ Swartbooi, MSc (UWC) Mr H Solomon, BSc (Hons) (Eritrea) MSc (UWC)
Senior Officer:	Mr MY Hendricks, BSc MSc (UWC) Mr CW Andries, BSc MSc (UWC)
Department Administrator:	Ms W Davids
Department Administrator:	Ms M Naidoo
Technical Assistants:	Ms J Becorney Mr S Davids
Administrative Assistant:	Ms C Carnow

MATHEMATICS AND APPLIED MATHEMATICS

Departmental Chairperson:	Dr D Solomons, PhD (UCT)
Senior Professors:	Prof D Holgate, PhD (UCT) Prof K Patidar, PhD (IIT Kanpur)
Professors:	Prof R Appadu, PhD (Mauritius) Prof J Munyakazi, PhD (UWC) Prof ECS Mwambene, BSc (Hons) (Malawi) MSc (London) MSc (Zimbabwe) Dr rer nat (Vienna)
Associate Professors:	Prof E Mehdinezhad, PhD (UCT) Prof P Pillay, PhD (UDW)
Emeritus Professor:	Prof RL Fray, MSc (UWC) MSc PhD (SU)
Senior Lecturers:	Dr W Lee, PhD (UP) Dr C Makasu, DPhil (Zimbabwe)

Lecturers:	Dr A Razafindrakoto, PhD (SU) Dr D Solomons, PhD (UCT) Dr S Maku-Vyambwera, BSc (WSU) BSc (Hons) MSc PhD (UWC)
ECP Lecturers:	Dr GE Muller, PhD (UWC) Dr A Mukhtar, PhD (UWC) Dr M Nsumai, PhD (UWC) Dr G van Schalkwyk, PhD (UWC) Mr M Ongansie, BSc (Hons) (UWC)
Chief Officer:	Mr A Taylor, HDE MSc (UWC)
Senior Officers:	Ms L Adams, MSc (UCT) Ms F Karriem, MSc (UWC)
Officer:	Mr T Gongxeka, B Tech (NMU) PGDip (UWC)
Administrative Officers:	Ms GL Hendricks, MDP (UWC) Ms N Lukas, BTech (CPUT)

MEDICAL BIOSCIENCES

Departmental Chairperson:	Prof A Oelofse, MSc (SU) PhD (Wageningen)
Professors:	Prof O Azu, DPhil (College of Medicine, Lagos) Prof D Fisher, MSc (UCT) PhD (UWC) Prof T Kohn, ADM (UCT) BSc (Hons) (SU) PhD (SU) Prof TK Monsees, BSc MSc PhD (Bremen) Habil (Giessen) Prof A Oelofse, MSc (SU) PhD (Wageningen) Prof M Shaw, BSc (Hons) (UCT) PhD (Glasgow)
Associate Professors:	Prof F Fisher Rahiman, PhD (UWC) Prof DC Hiss, PhD (UCT) Prof B Kirby, PhD (UCT) Prof L Maree, MSc (SU) PhD (UWC) Prof C Opuwari, PhD (UWC)
Emeritus Professors:	Prof C Africa, PhD (London) Prof R Henkel, PhD (Marburg) Habil (Giessen) Prof G van der Horst, PhD (SU) PhD (UPE) EM Cert (Zeiss, Germany)
Senior Lecturers:	Prof GS Maritz, BSc (Hons) PhD MBA (SU) Dr S Abdul-Rasool, MBChB (Baghdad) MSc (Al-Nahrain) PhD (UCT) Dr P Abrantes, PhD (UWC) PGDip (Edinburgh) Dr J de Smidt, BSc MBChB (SU) Dipl (UWC) PhD (UWC) Dr K Makhathini, PhD (UKZN) Dr L Mbonile, MBChB (Dar) MPH (UP) PGDip (ITM-Antwerp)
Lecturers:	Dr M Bennett, PhD (UWC) Dr N Haines-Arries, MSc (UWC) Mr H Harribhai, BMedSci (Hons) (UDW) MSc (UWC) Dr B Langa, MSc (UWC)

**Associate Lecturer:
Chief Officers:**

Senior Officers:

Administrator (Finance):

Officers:

Administrative Officers:

Dr T Morris, MSc PhD (UWC)
Mr C Mupfiga, MSc (UWC)
Mrs TC Ras, MSc (UWC)
Dr K Pearce, PhD (UWC)
Dr C Willemse, MSc PhD (UWC) PGCE (UNISA)
Mr G Williams, MSc (SU)
Ms V Jooste, MSc (UWC)
Mr Y Mnyamana, MSc (UWC)
Ms M Smith, BSc (Hons) MSc (UWC)
Mr A Braaf, BSc (Hons) (UWC)
Ms C McKay, MSc (SU)
Ms I Messeir, BSc (Hons) MSc (UWC)
Ms TB Shabalala, BSc (Hons) (UKZN)
Mr D Solomons
Ms N Schoeman, BAdmin (Hons) HCED ADPA (UWC)
Mr M Ntsabo
Ms C Davids, Dipl (INTEC)
Ms N Louw, Dipl (Financial Management Institute of SA) Cert (AGATE)

SCHOOL OF PHARMACY

Director:

Professors:

Emeritus Professors:

Associate Professors:

**Extra-Ordinary Professor:
Adjunct Professors:**

Prof SF Malan, DTE BPharm MSc PhD (NWU)
Prof M Aucamp, BPharm MSc PhD (NWU)
Prof DR Beukes, BSc BSc (Hons) PhD (RU)
Prof A Dube, BPharm (Zimbabwe) MPharm (UWC) PhD (Monash)
Prof J Joubert, BPharm MSc PhD (NWU)
Prof NC Butler, MPharm (UWC) BSc (Hons) (SU) PhD (Minnesota)
Prof P Mugabo, BSc MBChB (Rwanda) MMed (Louvain) PhD (Ghent)
Prof N Ebrahim, BPharm MPharm PhD (UWC)
Prof S Egieyeh, BPharm (Lagos) MPharm PhD (UWC)
Prof S Khoza, BPharm MSc (Zimbabwe) PhD (Texas, Austin)
Prof JA McCartney, DipPharm MSc (UPE) PhD (NMU)
Prof K Obikeze, BPharm (Nigeria) MPharm PhD (UWC)
Prof H Samsodien, BPharm MPharm (UWC) PhD (UCT)
Prof M Viljoen, BPharm MSc PhD (NWU)
Prof KL Ward, BPharm MPharm PhD (UWC)
Prof WR Folk, BA (Rice, Texas) PhD (Stanford)
Dr RD Crutchley, PharmD (UNC-Chapel Hill)
Dr L Gwaza, BS Pharm (Honors) MS (Zimbabwe) PhD (Utrecht, Netherlands)

Senior Lecturers:

Lecturers:

Adjunct Lecturers:

Research Fellow:

Chief Officer:

Senior Officers:

Administrative Officers:

Prof GL Muntingh, BPharm (UP) MSc PhD (Medunsa)
Prof P Valodia, BPharm MPharm (UWC) PhD (UCT)
Dr E Kapp, BPharm MSc (NWU) PhD (UWC)
Dr MB Parker, BPharm MPharm PhD (UWC)
Ms LT Dube, BPharm (RU) MPH (UWC)
Dr E Egieyeh, BPharm (Lagos) MPharm PhD (UWC)
Mrs C Greyling, BPharm (NWU) MPharm (NMU)
Ms Y van Heerden, BPharm MPharm (NMU)
Ms N Keuler, BPharm (NWU) MSc PhD (UWC)
Dr M Milne, BPharm (NMU) MSc PhD (NWU)
Dr I Okeyo, BPharm MPharm (RU) PhD (UWC)
Dr TW Laudone, PharmD (MCPHSU, Boston, MA)
Mr J Whitty, BEng (SETU, Ireland) MSc (Trinity College, Ireland)
Dr E Tommelein, BSc MSc PhD (Gent, Belgium)
Ms EMR Braaf, BSc (SU) BSc (Hons) (UWC)
Mr Y Kippie, BTech (CPUT)
Ms S Mini, BSc (Hons) MSc (UWC)
Ms A Ramplin, BSc (Hons) (UWC)
Ms S Blake, BA HDE (UWC)
Mr MA Kamedien, BCom (Hons) (UWC)
Mr R Maart, BSc (Hons) (UWC) MSc (UWC)
Ms R Symonds
Ms BL van Rooyen

PHYSICS AND ASTRONOMY

Departmental Chairperson:

SARCHI Chairs (Astrophysics):

Senior Professor:

Professors:

Associate Professors:

Emeritus Professors:

Prof S Halindintwali, PhD (UWC)
Prof MGR Santos, MA (Cantab) PhD (Oxon)
Prof R Maartens, PhD (UCT)
Prof C Arendse, PhD (UWC)
Prof D Marshall, BSc (Hons) (UCT) DPhil (Oxon)
Prof JN Orce, PhD (Brighton)
Prof MGR Santos, MA (Cantab) PhD (Oxon)
Prof S Triambak, MSc (Pune) PhD (Notre Dame)
Prof EC Elson, PhD (UCT)
Prof S Halindintwali, MSc (Rwanda) PhD (UWC)
Prof M Lochner, PhD (UCT)
Prof S Macfarlane, PhD (Radboud University)
Prof R Madjoe, HDE MSc (UWC) MSc PhD (Louisiana State)
Prof M Tchokonte, DEA (Yaounde) MSc PhD (Wits)
Prof R Bharuthram, PhD (UKZN)
Prof D Knoesen, PhD (SU)
Prof CJ Linder, BSc (Hons) (RU) EdM (Rutgers)
HDE (RU) EdD (British Columbia)
Prof R Lindsay, BSc (Hons) (SU) DPhil (Oxon)

Extra-Ordinary Professors:

Prof A Baker, PhD (Caltech)
 Prof C Clarkson, PhD (Glasgow)
 Prof R Dave, PhD (California)
 Prof P Garrett, PhD (McMaster)
 Prof S Guha, PhD (Arizona State)
 Prof M Jarvis, DPhil (Oxon)
 Prof D Jenkins, DPhil (York)
 Prof D Kilkenney, PhD (St Andrews)
 Prof M Kosch, PhD (UKZN)
 Prof E Lawrie, PhD (Universite Paris-Sud)
 Prof J Lawrie, (Stellenbosch)
 Prof T Lynas-Gray, (St Andrews)
 Prof AK Mahapatro, PhD (Jawaharlal Nehru University, India)
 Prof S Maharaj, PhD (UKZN)
 Prof P Miceli, DPhil (Illinois at Urbana-Champaign)
 Prof M Snow, PhD (Colorado)
 Prof R Taylor, PhD (British Columbia)
 Prof M Vaccari, MSc PhD (Padova, Italy)
 Dr M Herbert, HDE PhD (UCT)
 Dr G Malgas, PhD (UWC)
 Dr TFG Muller, HDE PhD (UWC)
 Dr TS Volkwyn, PhD (Uppsala)
 Dr S Matshawule, MSc (UWC) PhD (UWC)
 Mr R Maclons, BSc (Hons) (UWC)
 Mr IN Hlatwayo, MSc (Zululand)
 Dr L Makhathini, PhD (SU)
 Mr I Schroeder, BSc (Hons) PGDip (UCT)
 Ms N Peterson, MSc (UWC)
 Mr M van Rooyen, BSc (UWC)
 Mr Z Bester
 Ms A Adams
 Ms M Gabler
 Ms J Engel

Senior Lecturers:**Lecturer:****ECP Lecturers:****Chief Officers:****Senior Officer:****Officers:****PC Lab Manager:****Administrative Officer:****Administrative Officer (Astrophysics):****Administrative Assistant:****STATISTICS AND POPULATION STUDIES****Departmental Chairperson:****Professors:**

Dr H Brydon, MSc PhD (UWC)
 Prof SS Appunni, PhD (Kerela)
 Prof RJ Blignaut, BSc (UJ) BSc (Hons) (UP) MSc (UCT) PhD (UP)
 Prof N Stiegler, PhD (UWC) DEA Analyse Démographique (Bordeaux Montesquieu) Maitrise Administration Economique et Sociale (Bordeaux IV) Licence (Bordeaux I)
 Prof R Luus, MCom PhD (SU)
 Prof C Koen, PhD (UCT) PhD (UJ)
 Prof D Kotze, DCom (SU) MSc MBL (UNISA)
 Prof A Fauve-Chamoux, PhD (Ecole des Hautes Etudes en Sciences Sociales Paris)
 Prof M Kruger, PhD (NWU)

Associate Professor:**Emeritus Professors:****Extra-Ordinary Professors:**

	Prof S Steel, PhD (SU)
	Prof K Oguiuba, PhD (Nigeria) PhD (UWC)
	Prof I Mantsurov, DSc PhD (KNEU, Ukraine)
	Prof JP Bouchard, PhD (Toulouse, France)
	PhD (UPPA, France)
	Prof M Garenne, PhD (PENN)
	Prof R Fernandez-Castilla, PhD (London)
Senior Lecturers:	Dr H Brydon, MSc PhD (UWC)
	Dr J Keddie, MSc BSc PhD (NMU)
	Dr P Nsengiyumva, PhD (UWC)
Lecturers:	Dr R Jacobs, MSc PhD (UWC)
	Mr A Latief, MSc (UWC)
	Ms R Lombard-Jacobus, MSc (UWC)
	Ms T Wannenberg, MSc (UWC)
Extra-Ordinary Senior Lecturers:	Dr E Omoluabi, PhD (IEDUP Paris)
Extra-Ordinary Researcher:	Mrs E Jordaan, MSc (UWC)
Chief Officer:	Mrs C Morkel, MPhil (UWC)
Senior Officer:	Mr M Valentine, MCom (UWC)
Administrative Officer:	Vacant
Administrative Assistant:	Ms S Pieters

ARC (UNIT OF THE RANGE AND FORAGE INST)

Researchers:	Mr C Cupido, MSc (SU)
	Dr MI Samuels, MSc (UWC) PhD (UCT)
Research Assistant:	Vacant

ELECTRON MICROSCOPY UNIT (EMU)

HOD:	Vacant
Associate Professors:	Prof F Cummings, PhD (UWC)
	Prof SS Botha, PhD (UWC)
Senior Technical Officer:	Mr AB Josephs, BSc (UWC)
Administrative Officer:	Ms N Kensley

ENVIRONMENTAL EDUCATION AND RESOURCES UNIT (EERU)

Manager:	Ms M Levendal, BSc (Hons) (UWC) MSc (SU)
Horticulturalist:	Ms Z Maphanga, BTech (CPUT)
Education and Research Officer:	Ms L van Breda, ND (CPUT)
Nature Conservator:	Mr R Adams, BTech (CPUT)
Reserve Staff:	Mr M Kapiera, ND (CPUT)
	Mr J Ntiliza

INSTITUTE FOR MICROBIAL BIOTECHNOLOGY AND METAGENOMICS (IMBM)

Director	Prof IM Trindade, PhD (UCT)
SARCHI Chair (Microbial Genomics):	Prof IM Trindade, PhD (UCT)

Professor:	Prof IM Trindade, PhD (UCT)
Associate Professor:	Vacant
Lecturer:	Dr N Ikegwuoha, PhD (UWC)
Researcher and Innovation Manager:	Dr A Burger, PhD (SU)
Chief Officer:	Dr L van Zyl, PhD (UWC)
Technical Officer:	Ms S Lawrence, MSc (UWC)
Administrator:	Mrs C Rensburg, BTech (CPUT)
General Assistants:	Ms C Cupido

INSTITUTE FOR WATER STUDIES (IWS)

Director:	Prof T Dube, MSc (Twente, Netherlands) PhD (UKZN)
Emeritus Professor:	Prof D Mazvimavi, BSc (Hons) (Zimbabwe) MSc (Vrije U Brussels) PhD (Wageningen)
Extra-Ordinary Professors:	Prof C Brown, BSc (Hons) (UKZN) MSc (UPE) PhD (UCT) Prof JA Day, PhD (UCT) Prof J Goldin, BA (RU) PhD (UCT)
Administrator:	Ms M Naidoo

INTER-UNIVERSITY INSTITUTE FOR DATA INTENSIVE ASTRONOMY (IDIA)

Director: SKA Chair (UWC-UCT)	Prof R Taylor, PhD (British Columbia)
Professors:	Prof R Taylor, PhD (British Columbia)
Associate Professor:	Prof S Macfarlane, PhD (Radboud University)
Data Scientist:	Mr E Hussein, MSc (UWC)
Administrative/Project Officer:	Ms L Hendricks

MASTERS NANOSCIENCE PROGRAMME

Director:	Vacant
Coordinator:	Ms Z Xauka
Administrator:	Ms CH Manuel

SOUTH AFRICAN INSTITUTE FOR ADVANCED MATERIAL CHEMISTRY (SAIAMC)

Director: Senior Professor:	Prof V Linkov, PhD (SU)
HySA Systems Director:	Dr S Pasupathi, PhD (Pisa, Italy)
Associate Professor:	Prof B Bladergroen, PhD (UWC)
Extra-Ordinary Professor:	Prof M Mdleleni, PhD (California) MBA (Southern Queensland)
HySA Programme Manager:	Dr M Lototsky, PhD (LVIV)
Key Technology Specialist:	Dr W Davids
Engineer:	Dr Y Klochko, PhD (Ukraine)
Officer:	Mr C Cornelius
Research Assistant:	Mr U March
Technician:	Mr DJ Philander

Administrator:	Vacant
Safety Manager:	Dr S Chidziva, PhD (Ulster UK)
Coordinator: Energy Storage Training Programme	Mr R Manuel
Administrative Officer:	Mr N Jacobs
Administrative Assistant HySA Systems:	Vacant

SOUTH AFRICAN NATIONAL BIO-INFORMATICS INSTITUTE (SANBI)

Director:	Prof A Christoffels, PhD (UWC)
Senior Professor:	Prof A Christoffels, PhD (UWC)
Associate Professors:	Prof N Tiffin, PhD (London) MPH (UCT)
Extra-Ordinary Professors:	Prof C Verma, PhD (York)
Senior Lecturer:	Dr R Cloete, PhD (UWC)
Senior Researcher:	Dr D Anderson, PhD (UWC)
Researchers:	Mr P van Heusden, BSc (UCT) Ms I Muchada, MSc (UCT)
Personal Assistant of Director:	Vacant
Admin Assistants:	Ms F Mpithi Ms J Williams
Financial Administrator:	Ms F Mullins, Dipl (Varsity College)
Student Administrator:	Vacant
System Administrator:	Mr S Wentzel, BSc (UWC)
Software Developer:	Mr Q Coert, Dip (CPUT)
Web content Developer:	Mr C Rae
Research Associate:	Ms A Bedeker, BPsych (UWC)
Project Administrator (PHA4GE):	Ms N Nacerodien Weitz, BA (UWC)
Communication Specialist:	Mr R Matima, MPH (UCT)
Technical Managers:	Dr A Naidu, PhD (UWC) Mr J Southgate, BSc (Hons) (UWC)
Business Dev Manager:	Mr M Bridger, MBA (SU)

RULES FOR MAINSTREAM PROGRAMMES

BACHELOR OF SCIENCE IN APPLIED GEOLOGY (3214)

B.1 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Applied Geology Degree – BSc (Applied Geology)**:

B.1.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 33 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 4 (50-59%) in Mathematics, and
 - Level 4 (50-59%) in Physical Sciences

B.1.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Unless Senate decides otherwise, an applicant shall not be admitted to any of the Bachelor of Science programmes without having obtained a full matriculation or age exemption. Furthermore, an applicant shall not be admitted to any of the Bachelor of Science programmes if they have not obtained a pass in Mathematics and either Physical Science or Biology at Higher grade with at least a D symbol or at Standard grade with at least a C symbol.

OR

- (b) A qualification or level of competence that the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.1.3 Alternative admission

Alternative admission does not apply to this programme.

B.2 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.3 DURATION

Unless Senate decides otherwise, the duration of the degree is three years.

B.4 CURRICULUM

B.4.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all groups)		
Group 1 (select all modules)		
Earth Science 111	ESC111	15
Earth Science 121	ESC121	15
Group 2 (select one module)		
Computing Fundamentals 114	COS114	15
Computing Fundamentals 124	COS124	15
Group 3 (select one module)		
English for Educational Development 117	EED117	15
English for Educational Development 127	EED127	15
Group 4 (select all modules)		
Chemistry 114 (N)	CHE114	15
Chemistry 124 (N)	CHE124	15
Group 5 (select all modules)		
Physics 116	PHY116	15
Physics 126	PHY126	15
Group 6		
Mathematics 115 (N)	MAM115	15
Total		135

B.4.2 Level 2

Module Name	Module Code	Credits
Compulsory (select all modules)		
Geology 232	APG232	20
Geology 234	APG234	20
Geology 235	APG235	20
Chemistry 211	CHM211	20
Chemistry 212	CHM212	20
	Sub-total	100
Electives (select one module)		
Environmental and Water Science 211	EWP211	20
Environmental and Water Science 221	EWP221	20
	Sub-total	20
	Total	120

B.4.3 Level 3

Module Name	Module Code	Credits
Compulsory (select all modules)		
Geology 312	APG312	30
Geology 323	APG323	30
Geology 324	APG324	30
Geology 325	APG325	30
	Total	120
	FINAL TOTAL	375

B.5 ASSESSMENT

B.5.1 Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.5.2 Sub–minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.5.3 Weighting of practical component

B.5.3.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.5.3.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.5.4 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.5.5 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.5.6 Attendance of practicals

Where there is a practical component in a programme a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.6 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.6.1 Level 1

A student shall promote to Level 2 of study when 105 credits have been obtained of the 135 credits required in Level 1 of study, and when all pre-requisites for admission to Level 2 modules have been met.

B.6.2 Level 2

A student shall promote to Level 3 of study when at least a total of 215 credits have been obtained and when all 135 credits of Level 1 of study have been obtained.

B.7 ADVANCE REGISTRATION

B.7.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.7.2 Level 1

A student who has obtained at least 75 credits at Level 1 of study may be allowed to register for a maximum of 40 credits at Level 2.

B.7.3 Level 2

B.7.3.1 A student who has obtained at least 80 credits at Level 2 of study may be allowed to register for a maximum of 60 credits at Level 3.

Please Note:

A student must complete all Level 1 modules before registering for Level 3 modules.

B.8 RENEWAL OF REGISTRATION

B.8.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.8.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.9 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.9.1 Modules from other faculties

B.9.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for

Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.9.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.9.1.3 Computing Fundamentals (COS114 **or** COS124) and English for Educational Development (EED117 **or** EED127): These are compulsory, credit-bearing modules, comprising one semester module for COS and one semester module for EED.

B.9.2 Exemption from COS and/or EED

A student, who has developed these skills, may apply to the faculty for exemption from this requirement, subject to the general rules of the University. The process to determine their level of competency in these modules may involve one or more assessments.

When exempted, such a student must replace these modules with other modules of their choice to the same credit value.

B.9.3 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.9.3.1 Chemistry: (CHE114 and CHE124) **or** (CHE116 and CHM126).

B.9.3.2 Physics: (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).

B.9.3.3 Mathematics: MAT105 **or** MAM115 **or** (MAM151 and MAM152).

B.9.3.4 Statistics: STA111 **or** STA125 **or** STA151 **or** BUS131 **or** BUS132 **or** STA141 **or** STA142 **or** STC101.

B.9.3.5 A student may not claim credits for modules passed prior to 1999.

B.9.4 Endorsed Degree

B.9.4.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.9.4.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
CHE114	Chemistry 114 (N)	C	1	15	5	S1	None	None	CFA	
CHE124	Chemistry 124 (N)	C	1	15	5	S2	None	None	CFA	
COS114	Computing Fundamentals 114	E	1	15	5	S1	None	None	CA	
COS124	Computing Fundamentals 124	E	1	15	5	S2	None	None	CA	
EED117	English for Educational Development 117	E	1	15	5	S1	None	None	CFA	SM
EED127	English for Educational Development 127	E	1	15	5	S2	None	None	CFA	SM
ESC111	Earth Science 111	C	1	15	5	S1	None	None	CFA	
ESC121	Earth Science 121	C	1	15	5	S2	None	None	CFA	
MAM115	Mathematics 115 (N)	C	1	15	5	S1	None	None	CA	
PHY116	Physics 116	C	1	15	5	S1	None	None	CFA	
PHY126	Physics 126	C	1	15	5	S2	None	None	CFA	
APG232	Geology 232	C	2	20	6	S1	ESC111, ESC121	None	CFA	
APG234	Geology 234	C	2	20	6	S2	ESC111, ESC121	None	CFA	
APG235	Geology 235	C	2	20	6	S2	ESC111, ESC121	None	CFA	
CHM211	Chemistry 211	C	2	20	6	S1	CHE114, CHE124	None	CFA	
CHM212	Chemistry 212	C	2	20	6	S2	CHE114, CHE124	None	CFA	
EWP211	Environmental and Water Science 211	E	2	20	6	S1	ESC111, ESC121	None	CFA	
EWP221	Environmental and Water Science 221	E	2	20	6	S2	ESC111, ESC121	None	CFA	
APG312	Geology 312	C	3	30	7	S2	APG232, APG234	None	CFA	
APG323	Geology 323	C	3	30	7	S1	APG232, APG234, APG235	None	CFA	
APG324	Geology 324	C	3	30	7	S1	APG232, APG234, APG235	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
APG325	Geology 325	C	3	30	7	S2	APG232, APG234, APG235	None	CFA	

BACHELOR OF SCIENCE IN BIODIVERSITY AND CONSERVATION BIOLOGY (3217)

B.10 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Biodiversity and Conservation Biology Degree - BSc (Biodiversity and Conservation Biology)**:

B.10.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 33 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 4 (50-59%) in Mathematics, and
 - Level 4 (50-59%) in Physical Sciences, or
 - Level 4 (50-59%) in Life Sciences

B.10.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Unless Senate decides otherwise, an applicant shall not be admitted to any of the Bachelor of Science programmes without having obtained a full matriculation or age exemption. Furthermore, an applicant shall not be admitted to any of the Bachelor of Science programmes if they have not obtained a pass in Mathematics and either Physical Science or Biology at Higher grade with at least a D symbol or at Standard grade with at least a C symbol.

OR

- (b) A qualification or level of competence that the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.10.3 Alternative admission

Alternative admission does not apply to this programme.

B.11 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.12 DURATION

Unless Senate decides otherwise, the duration for the degree is three years.

B.13 CURRICULUM

B.13.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all groups)		
Group 1 (select both modules)		
Life Sciences 141	LSC141	15
Life Sciences 142	LSC142	15
Group 2 (select two modules)		
*A student who intends to select CHM211 in Level 2 must select CHE114 and CHE124 in Level 1.		
*Chemistry 114 (N) or Chemistry 116 (N)	CHE114/ CHE116	15
AND		
*Chemistry 124 (N) or Chemistry 126 (N)	CHE124/ CHM126	15
Group 3 (select one module)		
English for Educational Development 117	EED117	15
English for Educational Development 127	EED127	15
Group 4 (select both modules)		
Earth Science 111	ESC111	15
Earth Science 121	ESC121	15
Group 5 (select one module)		
Statistics 111	STA111	15
Statistics 125	STA125	15
	Total	120

B.13.2 Level 2

Module Name	Module Code	Credits
Compulsory (select all modules)		
Biodiversity and Conservation Biology 211	BDC211	20
Biodiversity and Conservation Biology 221	BDC221	20
Biodiversity and Conservation Biology 222	BDC222	20
Biodiversity and Conservation Biology 223	BDC223	20
	Sub-total	80
Electives (select one group)		
Group 1 (select both modules)		
Environmental and Water Science 211	EWP211	20
Environmental and Water Science 221	EWP221	20
	Sub-total	40
Group 2 (select both modules)		
Chemistry 211	CHM211	20
Chemistry 212	CHM212	20
	Sub-total	40
	Total	120

B.13.3 Level 3

Module Name	Module Code	Credits
Compulsory (select four modules)		
Biodiversity and Conservation Biology 332	BDC332	30
Biodiversity and Conservation Biology 333	BDC333	30
Biodiversity and Conservation Biology 334	BDC334	30
Biodiversity and Conservation Biology 335	BDC335	30
Environmental and Water Science 321	EWP321	30
Total		120
FINAL TOTAL		360

B.14 ASSESSMENT

B.14.1 Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.14.2 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.14.3 Weighting of practical component

B.14.3.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.14.3.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.14.4 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.14.5 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.14.6 Attendance of practicals

Where there is a practical component in a programme a minimum attendance of 80% of the practicals is required to gain access to the examination in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.15 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.15.1 Level 1

A student shall promote to Level 2 of study when 90 credits have been obtained of the 120 credits required in Level 1 of study and when all pre-requisites for admission to Level 2 modules have been met.

B.15.2 Level 2

A student shall promote to Level 3 of study when at least a total of 200 credits have been obtained and when all 120 credits of Level 1 of study have been obtained.

B.16 ADVANCE REGISTRATION

Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.16.1 Level 1

A student who has obtained at least 75 credits at Level 1 of study may be allowed to register for a maximum of 40 credits at Level 2.

B.16.2 Level 2

B.16.2.1 A student who has obtained at least 80 credits at Level 2 of study may be allowed to register for a maximum of 60 credits at Level 3.

Please Note: A student must complete all Level 1 modules before registering for Level 3 modules.

B.17 RENEWAL OF REGISTRATION

B.17.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.17.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.18 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.18.1 Modules from other faculties

B.18.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.18.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.18.1.3 Computing Fundamentals (COS114 **or** COS124) and English for Educational Development (EED117 **or** EED127): These are compulsory, credit-bearing modules, comprising one semester module for COS and one semester module for EED.

B.18.2 Exemption from COS and/or EED

A student, who has developed these skills, may apply to the faculty for exemption from this requirement, subject to the general rules of the University. The process to determine their level of competency in these modules may involve one or more assessments. When exempted, such a student must replace these modules with other modules of their choice to the same credit value.

B.18.3 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.18.3.1 Chemistry: (CHE114 and CHE124) **or** (CHE116 and CHM126).

B.18.3.2 Physics: (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).

B.18.3.3 Mathematics: MAT105 **or** MAM115 **or** (MAM151 and MAM152).

B.18.3.4 Statistics: STA111 **or** STA125 **or** STA151 **or** BUS131 **or** BUS132 **or** STA141 **or** STA142 **or** STC101.

B.18.3.5 A student may not claim credits for modules passed prior to 1999.

B.18.4 Endorsed Degree

B.18.4.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.18.4.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
CHE124 or CHM126	Chemistry 124 (N) or Chemistry 126 (N)	C	1	15	5	S2	None	None	CFA	
EED117	English for Educational Development 117	E	1	15	5	S1	None	None	CFA	SM
EED127	English for Educational Development 127	E	1	15	5	S2	None	None	CFA	SM
LSC141	Life Sciences 141	C	1	15	5	S1	None	None	CFA	
LSC142	Life Sciences 142	C	1	15	5	S2	None	None	CFA	
STA111	Statistics 111	E	1	15	5	S1	None	None	CFA	
STA125	Statistics 125	E	1	15	5	S2	None	None	CFA	
CHE114 or CHE116	Chemistry 114 (N) or Chemistry 116 (N)	C	1	15	5	S1	None	None	CFA	
ESC111	Earth Science 111	C	1	15	5	S1	None	None	CFA	
ESC121	Earth Science 121	C	1	15	5	S2	None	None	CFA	
BDC211	Biodiversity and Conservation Biology 211	C	2	20	6	S2	LSC142 or LFS152	None	CFA	
BDC221	Biodiversity and Conservation Biology 221	C	2	20	6	S1	LSC142 or LFS152	None	CFA	
BDC222	Biodiversity and Conservation Biology 222	C	2	20	6	S1	LSC141/142 or LFS151/152	None	CFA	
BDC223	Biodiversity and Conservation Biology 223	C	2	20	6	S2	LSC141 and LSC142, or LFS151 and LFS152; and CHE114 or CHE124 or CHE116 or CHM126	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
CHM211	Chemistry 211	C	2	20	6	S1	CHE114, CHE124	None	CFA	
CHM212	Chemistry 212	C	2	20	6	S2	CHE114, CHE124	None	CFA	
EWP211	Environmental and Water Science 211	E	2	20	6	S1	ESC111, ESC121	None	CFA	
EWP221	Environmental and Water Science 221	E	2	20	6	S2	ESC111, ESC121	None	CFA	
BDC332	Biodiversity and Conservation Biology 332	C	3	30	7	S2	BDC222	None	CFA	
BDC333	Biodiversity and Conservation Biology 333	C	3	30	7	S1	BDC222, BDC211	None	CFA	
BDC334	Biodiversity and Conservation Biology 334	C	3	30	7	S2	BDC222, BDC211, BDC223	None	CFA	
BDC335	Biodiversity and Conservation Biology 335	C	3	30	7	S1	BDC222, BDC211, BDC221	None	CFA	
EWP321	Environmental and Water Science 321	C	3	30	7	S2	EWP211, EWP221	None	CFA	

BACHELOR OF SCIENCE IN BIOTECHNOLOGY (3211)

B.19 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Biotechnology Degree – BSc (Biotechnology)**:

B.19.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 33 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 4 (50-59%) in Mathematics, and
 - Level 4 (50-59%) in Physical Sciences, or
 - Level 4 (50-59%) in Life Sciences

B.19.2 Admission requirements for applicants who matriculated before 2008

- (a) Unless Senate decides otherwise, an applicant shall not be admitted to any of the Bachelor of Science programmes without having obtained a full matriculation or age exemption. Furthermore, an applicant shall not be admitted to any of the Bachelor of Science programmes if they have not obtained a pass in Mathematics and either Physical Science or Biology at Higher grade with at least a D symbol or at Standard grade with at least a C symbol.

OR

- (b) A qualification or level of competence that the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.19.3 Alternative admission

Alternative admission does not apply to this programme.

B.20 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.21 DURATION

Unless Senate decides otherwise, the duration of the degree is three years.

B.22 CURRICULUM

B.22.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all groups)		
Group 1 (select all modules)		
Chemistry 114 (N)	CHE114	15
Chemistry 124 (N)	CHE124	15
Life Sciences 141	LSC141	15
Life Sciences 142	LSC142	15
Group 2 (select one module)		
Computing Fundamentals 114	COS114	15
Computing Fundamentals 124	COS124	15
Group 3 (select one module)		
English for Educational Development 117	EED117	15
English for Educational Development 127	EED127	15
	Sub-total	90
Electives (select one group)		
Group 1 (select both modules)		
Physics 116 (N)	PHY116	15
Physics 126 (N)	PHY126	15
Group 2 (select both modules)		
Mathematics 115 (N)	MAM115	15
Statistics 125	STA125	15
	Sub-total	30
	Total	120

B.22.2 Level 2

Module Name	Module Code	Credits
Compulsory (select all modules)		
Biotechnology 211	BTN211	20
Biotechnology 215	BTN215	20
Biotechnology 222	BTN222	20
Biotechnology 223	BTN223	20
	Sub-total	80
Electives (select two modules)		
Biotechnology 213	BTN213	20
Biotechnology 218	BTN218	20
Chemistry 211	CHM211	20
Chemistry 212	CHM212	20
	Sub-total	40
	Total	120

B.22.3 Level 3

Module Name	Module Code	Credits
Compulsory (select all modules)		
Biotechnology 312	BTN312	30
Biotechnology 315	BTN315	30
Biotechnology 316	BTN316	30
Biotechnology 322	BTN322	30
	Total	120
	FINAL TOTAL	360

B.23 ASSESSMENT

B.23.1 Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.23.2 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.23.3 Weighting of practical component

B.23.3.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.23.3.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.23.4 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.23.5 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.23.6 Attendance of practicals

Where there is a practical component in a programme a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.24 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.24.1 Level 1

A student shall promote to Level 2 of study when 90 credits have been obtained of the 120 credits required in Level 1 of study and when all pre-requisites for admission to Level 2 modules have been met.

B.24.2 Level 2

A student shall promote to Level 3 of study when at least a total of 200 credits have been obtained and when all 120 credits of Level 1 of study have been obtained.

B.25 ADVANCE REGISTRATION

B.25.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.25.2 Level 1

A student who has obtained at least 75 credits at Level 1 of study may be allowed to register for a maximum of 40 credits at Level 2.

B.25.3 Level 2

B.25.3.1 A student who has obtained at least 80 credits at Level 2 of study may be allowed to register for a maximum of 60 credits at Level 3.

Please Note: A student must complete all Level 1 modules before registering for Level 3 modules.

B.26 RENEWAL OF REGISTRATION

B.26.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.26.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.27 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.27.1 Modules from other faculties

B.27.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.27.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.27.1.3 Computing Fundamentals (COS114 or COS124) and English for Educational Development (EED117 or EED127): These are compulsory, credit-bearing modules, comprising one semester module for COS and one semester module for EED.

B.27.2 Exemption from COS and/or EED

A student, who has developed these skills, may apply to the faculty for exemption from this requirement, subject to the general rules of the University. The process to determine their level of competency in these modules may involve one or more assessments.

When exempted, such a student must replace these modules with other modules of their choice to the same credit value.

B.27.3 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.27.3.1 Chemistry: (CHE114 and CHE124) or (CHE116 and CHM126).

B.27.3.2 Physics: (PHY111 and PHY121) or (PHY116 and PHY126) or (PHY112 and PHY113 and PHY116) or (PHY151 and PHY152).

B.27.3.3 Mathematics: MAT105 or MAM115 or (MAM151 and MAM152).

B.27.3.4 Statistics: STA111 or STA125 or STA151 or BUS131 or BUS132 or STA141 or STA142 or STC101.

B.27.3.5 A student may not claim credits for modules passed prior to 1999.

B.27.4 Endorsed Degree

B.27.4.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.27.4.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requirement	Co-Requirement	Assessment Type	Service Module
LSC141	Life Sciences 141	C	1	15	5	S1	None	None	CFA	
LSC142	Life Sciences 142	C	1	15	5	S2	None	None	CFA	
COS114	Computing Fundamentals 114	E	1	15	5	S1	None	None	CA	
COS124	Computing Fundamentals 124	E	1	15	5	S2	None	None	CA	
EED117	English for Educational Development 117	E	1	15	5	S1	None	None	CFA	SM
EED127	English for Educational Development 127	E	1	15	5	S2	None	None	CFA	SM
PHY116	Physics 116 (N)	C	1	15	5	S1	None	None	CFA	
PHY126	Physics 126 (N)	C	1	15	5	S2	None	None	CFA	
MAM115	Mathematics 115 (N)	C	1	15	5	S1	None	None	CA	
STA125	Statistics 125	C	1	15	5	S2	None	None	CFA	
CHE114	Chemistry 114 (N)	C	1	15	5	S1	None	None	CFA	
CHE124	Chemistry 124 (N)	C	1	15	5	S2	None	None	CFA	
BTN211	Biotechnology 211	C	2	20	6	S1	CHE114/ CHE124 LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121	None	CA	
BTN215	Biotechnology 215	C	2	20	6	S1	CHE114/ CHE124 LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121	None	CA	
BTN222	Biotechnology 222	C	2	20	6	S1	CHE114/ CHE124 LSC141/ LSC142	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
							LFS151/ LFS152 LSC141/ MBS111/ MBS121			
BTN223	Biotechnology 223	C	2	20	6	S2	CHE114/ CHE124 LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121	None	CA	
BTN213	Biotechnology 213	E	2	20	6	S2	CHE114 & CHE124 & LSC141 & LSC142. LFS151 & LFS152 / LSC141 & MBS111 & MBS121	None	CFA	
BTN218	Biotechnology 218	E	2	20	6	S2	CHE114 & CHE124 & LSC141 & LSC142. LFS151 & LFS152 / LSC141 & MBS111 & MBS121	None	CFA	
CHM211	Chemistry 211	E	2	20	6	S1	CHE114, CHE124	None	CFA	
CHM212	Chemistry 212	E	2	20	6	S2	CHE114, CHE124	None	CFA	
BTN312	Biotechnology 312	C	3	30	6	S2	BTN211, BTN215, BTN222, BTN223	None	CFA	
BTN315	Biotechnology 315	C	3	30	7	S1	BTN211, BTN215, BTN222, BTN223	None	CFA	
BTN316	Biotechnology 316	C	3	30	7	S1	BTN211, BTN215, BTN222, BTN223	None	CFA	
BTN322	Biotechnology 322	C	3	30	7	S2	BTN211, BTN215, BTN222, BTN223	None	CFA	

BACHELOR OF SCIENCE IN CHEMICAL SCIENCES (3220)

B.28 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Chemical Sciences Degree - BSc (Chemical Sciences)**:

B.28.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 33 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 4 (50-59%) in Mathematics, and
 - Level 4 (50-59%) in Physical Sciences

B.28.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Unless Senate decides otherwise, an applicant shall not be admitted to any of the Bachelor of Science programmes without having obtained a full matriculation or age exemption. Furthermore, an applicant shall not be admitted to any of the Bachelor of Science programmes if they have not obtained a pass in Mathematics and either Physical Science or Biology at Higher grade with at least a D symbol or at Standard grade with at least a C symbol.

OR

- (b) A qualification or level of competence that the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.28.3 Alternative admission

Alternative admission does not apply to this programme.

B.29 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.30 DURATION

Unless Senate decides otherwise, the duration of the degree is three years.

B.31 CURRICULUM

B.31.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all modules)		
Chemistry 114 (N)	CHE114	15
Chemistry 124 (N)	CHE124	15
Physics 111	PHY111	15
Physics 121	PHY121	15
Life Sciences 141	LSC141	15
Life Sciences 142	LSC142	15
Mathematics 105	MAT105	30
	Total	120

Please Note:

A student who has passed CHE116 and CHM126 may proceed with the programme provided that they have obtained at least 60% in each of these modules, subject to approval by the HoD and the Dean.

B.31.2 Level 2

Module Name	Module Code	Credits
Compulsory (select all modules)		
Chemistry 211	CHM211	20
Chemistry 212	CHM212	20
Chemistry 221	CHM221	20
	Sub-total	60

Electives (select from Group 1 and Group 2)

Group 1 (select one sub-group)

Group 1.1 (select both modules)

Physics 212	PHY212	20
Physics 222	PHY222	20

Group 1.2 (select both modules)

Physics 217	PHY217	20
Physics 227	PHY227	20

Group 1.3 (select both modules)

Mathematics 211	MAT211	20
Mathematics 221	MAT221	20

Group 1.4 (select both modules)

Mathematics 211	MAT211	20
Mathematics 212	MAT212	20

Group 1.5 (select two modules, one from each semester)

First Semester

Biotechnology 211	BTN211	20
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or

Biotechnology 215	BTN215	20
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Second Semester

Biotechnology 213	BTN213	20
or		
Biotechnology 223	BTN223	20

Group 2**Select one module not selected in Group 1**

Physics 212	PHY212	20
Physics 217	PHY217	20
Mathematics 211	MAT211	20
Biotechnology 211 or	BTN211/	
Biotechnology 215	BTN215	20
	Sub-total	60
	Total	120

B.31.3 Level 3

Module Name	Module Code	Credits
Compulsory (select all modules)		
Chemistry 311	CHM311	30
Chemistry 312	CHM312	30
Chemistry 321	CHM321	30
Chemistry 322	CHM322	30
	Total	120
	FINAL TOTAL	360

B.32 ASSESSMENT

B.32.1 Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.32.2 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.32.3 Weighting of practical component

B.32.3.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.32.3.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.32.4 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.32.5 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.32.6 Attendance of practicals

Where there is a practical component in a programme a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.33 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.33.1 Level 1

A student shall promote to Level 2 of study when 90 credits have been obtained of the 120 credits required in Level 1 of study and when all pre-requisites for admission to Level 2 modules have been met.

B.33.2 Level 2

A student shall promote to Level 3 of study when at least a total of 200 credits have been obtained and when all 120 credits of Level 1 of study have been obtained.

B.34 ADVANCE REGISTRATION

B.34.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.34.2 Level 1

A student who has obtained at least 75 credits at Level 1 of study may be allowed to register for a maximum of 40 credits at Level 2.

B.34.3 Level 2

A student who has obtained at least 80 credits at Levels 1 and 2 of study may be allowed to register for a maximum of 60 credits at Level 3.

Please Note: A student must complete all Level 1 modules before registering for Level 3 modules.

B.35 RENEWAL OF REGISTRATION

B.35.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

- B.35.2** Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.36 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.36.1 Modules from other faculties

- B.36.1.1** A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.
- B.36.1.2** At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.
- B.36.1.3** Computing Fundamentals (COS114 **or** COS124) and English for Educational Development (EED117 **or** EED127): These are compulsory, credit-bearing modules, comprising one semester module for COS and one semester module for EED.

B.36.2 Exemption from COS and/or EED

A student, who has developed these skills, may apply to the faculty for exemption from this requirement, subject to the general rules of the University. The process to determine their level of competency in these modules may involve one or more assessments.

When exempted, such a student must replace these modules with other modules of their choice to the same credit value.

B.36.3 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

- B.36.3.1 Chemistry:** (CHE114 and CHE124) **or** (CHE116 and CHM126).
- B.36.3.2 Physics:** (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).
- B.36.3.3 Mathematics:** MAT105 **or** MAM115 **or** (MAM151 and MAM152).
- B.36.3.4 Statistics:** STA111 **or** STA125 **or** STA151 **or** BUS131 **or** BUS132 **or** STA141 **or** STA142 **or** STC101.
- B.36.3.5** A student may not claim credits for modules passed prior to 1999.

B.36.4 Endorsed Degree

B.36.4.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.36.4.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
CHE114	Chemistry 114 (N)	C	1	15	5	S1	None	None	CFA	
CHE124	Chemistry 124 (N)	C	1	15	5	S2	None	None	CFA	
LSC141	Life Sciences 141	C	1	15	5	S1	None	None	CFA	
LSC142	Life Sciences 142	C	1	15	5	S2	None	None	CFA	
MAT105	Mathematics 105	E	1	30	5	YR	None	None	CFA	
PHY111	Physics 111	C	1	15	5	S1	None	MAT103 MAT105	CFA	
PHY121	Physics 121	C	1	15	5	S2	None	MAT103 MAT105	CFA	
BTN211	Biotechnology 211	E	2	20	6	S1	CHE114/124 LSC141/142 LFS151/152 LSC141/ MBS111/121	None	CA	
BTN213	Biotechnology 213	E	2	20	6	S2	CHE114/124 LSC141/142 LFS151/152 LSC141/ MBS111/121	None	CFA	
BTN215	Biotechnology 215	E	2	20	6	S1	CHE114/124 LSC141/142 LFS151/152 LSC141/ MBS111/121	None	CA	
BTN223	Biotechnology 223	E	2	20	6	S2	CHE114/124 LSC141/142 LFS151/152	None	CA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
							LSC141/ MBS111/121			
CHM211	Chemistry 211	C	2	20	6	S1	CHE114, CHE124	None	CFA	
CHM212	Chemistry 212	C	2	20	6	S2	CHE114, CHE124	None	CFA	
CHM221	Chemistry 221	C	2	20	6	S2	CHE114, CHE124	None	CFA	
MAT211	Mathematics 211	E	2	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	None	CFA	
MAT212	Mathematics 212	E	2	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	MAT211	CA	
MAT221	Mathematics 221	E	2	20	6	YR	None	MAT211	CFA	
PHY212	Physics 212	E	2	20	6	S1	PHY111/ PHY121 OR PHY151/ PHY152	MAT211/ MAT104/ MAT152	CFA	
PHY217	Physics 217	E	2	20	6	S1	None	None	CFA	
PHY222	Physics 222	E	2	20	6	S2	PHY111/ PHY121 OR PHY151/ PHY152 AND MAT105/ MAT103/ MAM152	None	CFA	
PHY227	Physics 227	E	2	20	6	S2	PHY116/ PHY126 OR PHY111/ PHY151	None	CFA	
CHM311	Chemistry 311	C	3	30	7	S1	CHM211/ CHM212	None	CFA	
CHM312	Chemistry 312	C	3	30	7	S2	CHM211/ CHM212	None	CFA	
CHM321	Chemistry 321	C	3	30	7	S1	CHM211/ CHM212/ CHM221	None	CFA	
CHM322	Chemistry 322	C	3	30	7	S1	CHM211/ CHM212/ CHM221	None	CFA	

BACHELOR OF SCIENCE IN COMPUTER SCIENCE (3221)

B.37 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Computer Science Degree - BSc (Computer Science)**:

B.37.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 33 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 5 (60-69%) in Mathematics, and
 - Level 4 (50-59%) in Physical Sciences, or
 - Level 4 (50-59%) in Life Sciences, or
 - Level 4 (50-59%) in Information Technology

B.37.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Unless Senate decides otherwise, an applicant shall not be admitted to any of the Bachelor of Science programmes without having obtained a full matriculation or age exemption. Furthermore, an applicant shall not be admitted to any of the Bachelor of Science programmes if they have not obtained a pass in Mathematics and either Physical Science or Biology at Higher grade with at least a D symbol or at Standard grade with at least a C symbol.

OR

- (b) A qualification or level of competence that the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.37.3 Alternative admission

Alternative admission does not apply to this programme.

B.38 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.39 DURATION

Unless Senate decides otherwise, the duration of the degree is three years.

B.40 CURRICULUM

B.40.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all modules)		
Computer Science 101	COS101	30
Computing Fundamentals 114	COS114	15
English for Educational Development 127	EED127	15
Mathematics 105	MAT105	30
	Sub-total	90
Electives (select one group)		
Group 1 (select both modules)		
Information Systems 132	IFS132	15
Statistics 111	STA111	15
Group 2 (select both modules)		
Physics 111	PHY111	15
Physics 121	PHY121	15
Group 3 (select both modules)		
Statistics 111	STA111	15
Statistics 121	STA121	15
	Sub-total	30
	Total	120

B.40.2 Level 2

Module Name	Module Code	Credits
Compulsory (select both modules)		
Computer Science 211	CSC211	20
Computer Science 212	CSC212	20
	Sub-total	40
Electives (select two groups)		
Group 1 (select both modules)		
Information Systems 231	IFS231	20
Information Systems 233	IFS233	20
Group 2.1 (select both modules)		
Mathematics 211 (pre-requisite for Honours in Statistics)	MAT211	20
Mathematics 212	MAT212	20
Group 2.2 (select both modules)		
Mathematics 211 (pre-requisite for Honours in Statistics)	MAT211	20
Mathematics 221 (pre-requisite for Honours in Statistics)	MAT221	20
Group 3 (select both modules)		
Statistics 211	STA211	20
Statistics 221	STA221	20

Group 4 (select both modules)

Physics 212	PHY212	20
Physics 222	PHY222	20

Group 5 (select both modules)

Physics 217	PHY217	20
Physics 227	PHY227	20
	Sub-total	80
	Total	120

B.40.3 Level 3

Module Name	Module Code	Credits
Compulsory (select both modules)		
Computer Science 311	CSC311	30
Computer Science 312	CSC312	30
	Sub-total	60

Electives (select one group)**Group 1 (select two modules)**

Mathematics 311	MAT311	30
Mathematics 312	MAT312	30
Mathematics 321	MAT321	30
Mathematics 322	MAT322	30

Group 2 (select both modules)

Statistics 331 (pre-requisite for Honours in Statistics)	STA331	30
Statistics 332 (pre-requisite for Honours in Statistics)	STA332	30

Group 3 (select both modules)

Information Systems 316	IFS316	30
Information Systems 317	IFS317	30

Group 4 (select both modules)

Physics 314	PHY314	30
Physics 324	PHY324	30

Group 5 (select both modules)

Physics 317	PHY317	30
Physics 327	PHY327	30
	Sub-total	60
	Total	120

FINAL TOTAL 360**B.41 ASSESSMENT**

B.41.1 Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.41.2 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.41.3 Weighting of practical component

The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.41.4 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.41.5 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.41.6 Attendance of practicals

Where there is a practical component in a programme, a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.42 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.42.1 Level 1

A student shall promote to Level 2 of study when 90 credits have been obtained of the 120 credits required in Level 1 of study and when all pre-requisites for admission to Level 2 modules have been met.

B.42.2 Level 2

A student shall promote to Level 3 of study when at least a total of 200 credits have been obtained and when all 120 credits of Level 1 of study have been obtained.

B.43 ADVANCE REGISTRATION

B.43.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.43.2 Level 1

A student who has obtained at least 75 credits at Level 1 of study may be allowed to register for a maximum of 40 credits at Level 2.

B.43.3 Level 2

B.43.3.1 A student who has obtained at least 80 credits at Level 2 of study may be allowed to register for a maximum of 60 credits at Level 3.

Please Note: A student must complete all Level 1 modules before registering for Level 3 modules.

B.44 RENEWAL OF REGISTRATION

B.44.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.44.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.45 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.45.1 Modules from other faculties

B.45.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.45.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.45.1.3 Computing Fundamentals (COS114 **or** COS124) and English for Educational Development (EED117 **or** EED127): These are compulsory, credit-bearing modules, comprising one semester module for COS and one semester module for EED.

B.45.2 Exemption from COS and/or EED

A student, who has developed these skills, may apply to the faculty for exemption from this requirement, subject to the general rules of the University. The process to determine their level of competency in these modules may involve one or more assessments.

When exempted, such a student must replace these modules with other modules of their choice to the same credit value.

B.45.3 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.45.3.1 Chemistry: (CHE114 and CHE124) **or** (CHE116 and CHM126).

B.45.3.2 Physics: (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).

B.45.3.3 Mathematics: MAT105 **or** MAM115 **or** (MAM151 and MAM152).

B.45.3.4 Statistics: STA111 **or** STA125 **or** STA151 **or** BUS131 **or** BUS132 **or** STA141 **or** STA142 **or** STC101.

B.45.3.5 A student may not claim credits for modules passed prior to 1999.

B.45.4 Endorsed Degree

B.45.4.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.45.4.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
COS114	Computing Fundamentals 114	C	1	15	5	S1	None	None	CA	
EED127	English for Educational Development 127	C	1	15	5	S2	None	None	CFA	SM
MAT105	Mathematics 105	C	1	30	5	YR	None	None	CFA	
STA111	Statistics 111	E	1	15	5	S1	None	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
STA121	Statistics 121	E	1	15	5	S2	None	STA111/125/ 151 or BUS131/132 or STA141/142	CFA	
PHY111	Physics 111	E	1	15	5	S1	None	None	CFA	
PHY121	Physics 121	E	1	15	5	S2	None	None	CFA	
COS101	Computer Science 101	C	1	30	5	YR	None	None	CA	
IFS132	Information Systems 132	E	1	15	5	S2	None	None	CA	
CSC211	Computer Science 211	C	2	20	6	S1	COS101	None	CA	
CSC212	Computer Science 212	C	2	20	6	S2	COS101	None	CA	
IFS231	Information Systems 231	E	2	20	6	S1	IFS131/ IFS132, COS101/ COS114, COS124, EED127, ISC153, MAT151/ MAT103, MAT104, STA111, IFS131/ IFS132, COS101, PHY151, MAT151	None	CA	SM
IFS233	Information Systems 233	E	2	20	6	S2	IFS231	None	CA	SM
MAT211	Mathematics 211 (pre- requisite for Honours in Statistics)	E	2	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	None	CFA	
MAT212	Mathematics 212	E	2	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	MAT211	CA	
MAT221	Mathematics 221 (pre- requisite for Honours in Statistics)	E	2	20	6	YR	None	MAT211	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
PHY212	Physics 212	E	2	20	6	S1	PHY111/ PHY121 OR PHY151/ PHY152	MAT211/ MAT104/ MAT152	CFA	
PHY217	Physics 217	E	2	20	6	S1	None	None	CFA	
PHY222	Physics 222	E	2	20	6	S2	PHY111/ PHY121 OR PHY151/ PHY152 AND MAT105/ MAT103/ MAM152	None	CFA	
PHY227	Physics 227	E	2	20	6	S2	PHY116/ PHY126 OR PHY111/ PHY151	None	CFA	
STA211	Statistics 211	E	2	20	6	S1	MAT105 or (MAM151 + MAM152) & STA111/ STA125/ STA151	None	CFA	
STA221	Statistics 221	E	2	20	6	S2	STA211 (at least qualified to write the exam for STA211)	None	CFA	
CSC311	Computer Science 311	C	3	30	7	S1	CSC211 and CSC212	None	CA	
CSC312	Computer Science 312	C	3	30	7	S2	CSC211 and CSC212	None	CA	
IFS316	Information Systems 316	E	3	30	7	S1	IFS233 and IFS244	None	CA	SM
IFS317	Information Systems 317	E	3	30	7	S2	IFS316	None	CA	SM
MAT311	Mathematics 311	E	3	30	7	YR	MAT211	None	CFA	
MAT312	Mathematics 312	E	3	30	7	YR	MAT211, MAT212/ MAT221	None	CA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
MAT321	Mathematics 321	E	3	30	7	YR	MAT221	None	CFA	
MAT322	Mathematics 322	E	3	30	7	YR	MAT211, MAT212/ MAT221	None	CA	
PHY314	Physics 314	E	3	30	7	S1	PHY212 PHY222 MAT211 MAT221	None	CFA	
PHY317	Physics 317	E	3	30	7	S1	PHY227	None	CFA	
PHY324	Physics 324	E	3	30	7	S2	PHY212 PHY222 MAT211 MAT221	None	CFA	
PHY327	Physics 327	E	3	30	7	S2	PHY212/ PHY217, MAT105/ MAT103/ MAM152	None	CA	
STA331	Statistics 331 (pre-requisite for Honours in Statistics)	E	3	30	7	S1	STA211, STA221	None	CFA	
STA332	Statistics 332 (pre-requisite for Honours in Statistics)	E	3	30	7	S2	STA211, STA221	None	CFA	

BACHELOR OF SCIENCE IN ENVIRONMENT AND WATER SCIENCE **(3331)**

B.46 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Environment and Water Science Degree - BSc (Environment and Water Science)**:

B.46.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 33 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First additional Language), and
 - Level 4 (50-59%) in Mathematics, and
 - Level 4 (50-59%) in Physical Sciences, or
 - Level 4 (50-59%) in Life Sciences

B.46.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Unless Senate decides otherwise, an applicant shall not be admitted to any of the Bachelor of Science programmes without having obtained a full matriculation or age exemption. Furthermore, an applicant shall not be admitted to any of the Bachelor of Science programmes if they have not obtained a pass in Mathematics and either Physical Science or Biology at Higher grade with at least a D symbol or at Standard grade with at least a C symbol.

OR

- (b) A qualification or level of competence that the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) and (b) above.

B.46.3 Alternative admission

Alternative admission does not apply to this programme.

B.47 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.48 DURATION

Unless Senate decides otherwise, the duration of the degree is three years.

B.49 CURRICULUM

B.49.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all groups)		
Group 1 (select both modules)		
Earth Science 111	ESC111	15
Earth Science 121	ESC121	15
Group 2 (select one module)		
Chemistry 116 (N)	CHE116	15
Chemistry 126 (N)	CHM126	15
Group 3 (select one module)		
English for Educational Development 117	EED117	15
English for Educational Development 127	EED127	15
Group 4 (select one module)		
Statistics 111	STA111	15
Statistics 125	STA125	15
	Sub-total	75
Electives (select two groups)		
Group 1 (select both modules)		
Life Science 141	LSC141	15
Life Science 142	LSC142	15
Group 2 (select both modules)		
Physics 116 (N)	PHY116	15
Physics 126 (N)	PHY126	15
	Sub-total	60
	Total	135

B.49.2 Level 2

Compulsory Modules

Module Name	Module Code	Credits
Environmental and Water Science 211	EWP211	20
Environmental and Water Science 221	EWP221	20
Environmental and Water Science 222	EWP222	20
Geology 232	APG232	20
	Sub-total	80

Electives (select a total of 40 credits from below)

Group 1 (Select one module)

Biodiversity and Conservation Biology 211	BDC211	20
Biodiversity and Conservation Biology 221	BDC221	20
Biodiversity and Conservation Biology 222	BDC222	20
Biodiversity and Conservation Biology 223	BDC223	20

Group 2 (select one module)

Physics 217	PHY217	20
Physics 227	PHY227	20
	Sub-total	40
	Total	120

B.49.3 Level 3

Module Name	Module Code	Credits
Compulsory (select all modules)		
Geographical Information Systems 311	EWP311	30
Environmental and Water Science 312	EWP312	30
Environmental and Water Science 321	EWP321	30
Environmental and Water Science 322	EWP322	30
	Total	120
	FINAL TOTAL	375

B.50 ASSESSMENT

B.50.1 Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.50.2 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.50.3 Weighting of practical component

B.50.3.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.50.3.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.50.4 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.50.5 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.50.6 Attendance of practicals

Where there is a practical component in a programme, a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.51 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.51.1 Level 1

A student shall promote to Level 2 of study when 105 credits have been obtained of the 135 credits required in Level 1 of study and when all pre-requisites for admission to Level 2 modules have been met.

B.51.2 Level 2

A student shall promote to Level 3 of study when at least a total of 215 credits have been obtained and all pre-requisites for admission to Level 3 modules, including all Level 1 credits, have been met.

B.52 ADVANCE REGISTRATION

B.52.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.52.2 Level 1

A student who has obtained at least 75 credits at Level 1 of study may be allowed to register for a maximum of 40 credits at Level 2.

B.52.3 Level 2

B.52.3.1 A student who has obtained at least 80 credits at Level 2 of study may be allowed to register for a maximum of 60 credits at Level 3.

Please Note: A student must complete all Level 1 modules before registering for Level 3 modules.

B.53 RENEWAL OF REGISTRATION

B.53.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.53.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.54 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.54.1 Modules from other faculties

B.54.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.54.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.54.1.3 Computing Fundamentals (COS114 **or** COS124) and English for Educational Development (EED117 **or** EED127): These are compulsory, credit-bearing modules, comprising one semester module for COS and one semester module for EED.

B.54.2 Exemption from COS and/or EED

A student, who has developed these skills, may apply to the faculty for exemption from this requirement, subject to the general rules of the University. The process to determine their level of competency in these modules may involve one or more assessments.

When exempted, such a student must replace these modules with other modules of their choice to the same credit value.

B.54.3 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.54.3.1 Chemistry: (CHE114 and CHE124) **or** (CHE116 and CHM126).

B.54.3.2 Physics: (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).

B.54.3.3 Mathematics: MAT105 **or** MAM115 **or** (MAM151 and MAM152).

B.54.3.4 Statistics: STA111 **or** STA125 **or** STA151 **or** BUS131 **or** BUS132 **or** STA141 **or** STA142 **or** STC101.

B.54.3.5 A student may not claim credits for modules passed prior to 1999.

B.54.4 Endorsed Degree

B.54.4.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.54.4.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
EED117	English for Educational Development 117	E	1	15	5	S1	None	None	CFA	SM
EED127	English for Educational Development 127	E	1	15	5	S2	None	None	CFA	SM
STA111	Statistics 111	C/E	1	15	5	S1	None	None	CFA	
STA125	Statistics 125	C/E	1	15	5	S2	None	None	CFA	
LSC141	Life Science 141	E	1	15	5	S1	None	None	CFA	
LSC142	Life Science 142	E	1	15	5	S2	None	None	CFA	
PHY116	Physics 116 (N)	E	1	15	5	S1	None	None	CFA	
PHY126	Physics 126 (N)	E	1	15	5	S2	None	None	CFA	
PHY217	Physics 217	E	2	20	6	S1	None	None	CFA	
PHY227	Physics 227	E	2	20	6	S2	PHY116/ 126 OR PHY111/ 151	None	CFA	
ESC111	Earth Science 111	C	1	15	5	S1	None	None	CFA	
ESC121	Earth Science 121	C	1	15	5	S2	None	None	CFA	
CHE116	Chemistry 116 (N)	C/E	1	15	5	S1	None	None	CFA	
CHM126	Chemistry 126 (N)	C/E	1	15	5	S2	None	None	CFA	
EWP211	Environmental and Water Science 211	C	2	20	6	S1	ESC111, ESC121	None	CFA	
EWP221	Environmental and Water Science 221	C	2	20	6	S2	ESC111 ESC121	None	CFA	
EWP222	Environmental and Water Science 222	C	2	20	6	S1	ESC111 ESC121	EWP211 EWP221	CA	
APG232	Geology 232	C	2	20	6	S1	ESC111 ESC121	None	CFA	
BDC211	Biodiversity and Conservation Biology 211	E	2	20	6	S2	LSC142 or LFS152	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
BDC221	Biodiversity and Conservation Biology 221	E	2	20	6	S1	LSC142 or LFS152	None	CFA	
BDC222	Biodiversity and Conservation Biology 222	E	2	20	6	S1	LSC141/ 142 or LFS151/ 152	None	CFA	
BDC223	Biodiversity and Conservation Biology 223	E	2	20	6	S2	LSC141 and LSC142, or LFS151 and LFS152; and CHE114 or CHE124 or CHE116 or CHM126	None	CFA	
EWP311	Geographical Information Systems 311	C	3	30	7	S1	EWP211 EWP221 EWP222	None	CFA	
EWP312	Environmental and Water Science 312	C	3	30	7	S1	EWP211 EWP221 EWP222	None	CFA	
EWP321	Environmental and Water Science 321	C	3	30	7	S2	EWP211 EWP221	None	CFA	
EWP322	Environmental and Water Science 322	C	3	30	7	S2	EWP211 EWP221 EWP222	None	CFA	

BACHELOR OF SCIENCE IN MATHEMATICS AND STATISTICAL SCIENCES (3227)

B.55 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Mathematics and Statistical Sciences Degree - BSc (Mathematics and Statistical Sciences)**:

B.55.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 33 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 5 (60-69%) in Mathematics, and
 - Level 4 (50-59%) in Physical Sciences, or
 - Level 4 (50-59%) in Life Sciences, or
 - Level 4 (50-59%) in Information Technology

B.55.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Unless Senate decides otherwise, an applicant shall not be admitted to any of the Bachelor of Science programmes without having obtained a full matriculation or age exemption. Furthermore, an applicant shall not be admitted to any of the Bachelor of Science programmes if they have not obtained a pass in Mathematics and either Physical Science or Biology at Higher grade with at least a D symbol or at Standard grade with at least a C symbol.

OR

- (b) A qualification or level of competence that the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.55.3 Alternative admission

Alternative admission does not apply to this programme.

B.56 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.57 DURATION

Unless Senate decides otherwise, the duration of the degree is three years.

B.58 CURRICULUM

Mathematics Stream

B.58.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all groups)		
Group 1 (select all modules)		
Mathematics 105	MAT105	30
Statistics 111	STA111	15
Statistics 121	STA121	15
Group 2 (select one module)		
Computing Fundamentals 114	COS114	15
Computing Fundamentals 124	COS124	15
Group 3 (select one module)		
English for Educational Development 117	EED117	15
English for Educational Development 127	EED127	15
	Sub-total	90
Electives (select one group)		
Group 1 (select both modules)		
Mathematics 112	APM112	15
Applied Mathematics 123	APM123	15
Group 2 (select both modules)		
Physics 111	PHY111	15
Physics 121	PHY121	15
Group 3 (select both modules)		
Physics 116 (N)	PHY116	15
Physics 126 (N)	PHY126	15
	Sub-total	30
	Total	120

B.58.2 Level 2

Module Name	Module Code	Credits
Compulsory (select both modules)		
Mathematics 211	MAT211	20
Mathematics 221	MAT221	20
	Sub-total	40
Electives (select four modules)		
Mathematics 212	MAT212	20
Mathematics 232	MAT232	20
Statistics 211	STA211	20
Statistics 221	STA221	20
Physics 212	PHY212	20
Physics 222	PHY222	20

Physics 217	PHY217	20
Physics 227	PHY227	20
	Sub-total	80
	Total	120

B.58.3 Level 3

Module Name	Module Code	Credits
Compulsory (select all groups)		
Group 1		
Mathematics 311	MAT311	30
Group 2 (select one module)		
Mathematics 321	MAT321	30
Mathematics 322	MAT322	30
Mathematics 312	MAT312	30
	Sub-total	60
Electives (select two modules)		
Mathematics 321 (only if not selected in the compulsory list)	MAT321	30
Mathematics 322 (only if not selected in the compulsory list)	MAT322	30
Mathematics 312 (only if not selected in the compulsory list)	MAT312	30
Statistics 331	STA331	30
Statistics 332	STA332	30
	Sub-total	60
	Total	120
	FINAL TOTAL	360

Statistics Stream

B.58.4 Level 1

Module Name	Module Code	Credits
Compulsory (select all groups)		
Group 1 (select all modules)		
Mathematics 105	MAT105	30
Statistics 111	STA111	15
Statistics 121	STA121	15
Group 2 (select one module)		
Computing Fundamentals 114	COS114	15
Computing Fundamentals 124	COS124	15
Group 3 (select one module)		
English for Educational Development 117	EED117	15
English for Educational Development 127	EED127	15
	Sub-total	90

Electives (select electives to the value of at least 30 credits)**Group 1 (select both modules)**

Physics 116	PHY116	15
Physics 126	PHY126	15

Group 2 (select all modules)

Economics 151	ECO151	10
Economics 152	ECO152	10
Management 131 or Management 132	MAN131/132	15

Group 3

Computer Science 101	COS101	30
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Group 4 (select both modules)

Mathematics 112	APM112	15
Applied Mathematics 123	APM123	15
	Sub-total	30-35
	Total	120-125

B.58.5 Level 2**Module Name****Module Code****Credits****Compulsory (select all modules)**

Statistics 211	STA211	20
Statistics 221	STA221	20
Mathematics 211	MAT211	20
	Sub-total	60

Electives (select modules to the value of 60 credits)**Group 1**

Computer Science 211	CSC211	20
Computer Science 212	CSC212	20

Group 2

Mathematics 221 (pre-requisite for Honours in Statistics)	MAT221	20
Mathematics 212	MAT212	20
Mathematics 232	MAT232	20
Physics 217	PHY217	20
Physics 227	PHY227	20

Group 3

Economics 231	ECO231	10
Economics 232	ECO232	10
Economics 235	ECO235	10
Economics 242 (pre-requisite for Economics 311)	ECO242	10
	Sub-total	60
	Total	120

B.58.6 Level 3

Module Name	Module Code	Credits
Compulsory		
Statistics 331 (pre-requisite for Honours in Statistics)	STA331	30
Statistics 332 (pre-requisite for Honours in Statistics)	STA332	30
	Sub-total	60
Electives (select one group)		
Group 1 (select two modules)		
Mathematics 311	MAT311	30
Mathematics 321	MAT321	30
Mathematics 312	MAT312	30
Mathematics 322	MAT322	30
Group 2 (select both modules)		
Computer Science 311	CSC311	30
Computer Science 312	CSC312	30
Group 3 (select four modules)		
Compulsory		
Economics 331	ECO331	15
Economics 332	ECO332	15
and		
Select two of:		
Economics 335	ECO335	15
Economics 336	ECO336	15
Economics 311 (pre-requisite for Honours in Economics)	ECO311	15
Economics 334	ECO334	15
	Sub-total	60
	Total	120
	FINAL TOTAL	360-365

B.59 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.59.1 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.59.2 Weighting of practical component

B.59.2.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.59.2.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.59.3 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.59.4 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.60 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.60.1 Level 1

A student shall promote to Level 2 of study when 90 credits have been obtained of the 120 credits required in Level 1 of study and when all pre-requisites for admission to Level 2 modules have been met.

B.60.2 Level 2

A student shall promote to Level 3 of study when at least a total of 200 credits have been obtained and when all 120 credits of Level 1 of study have been obtained.

B.61 ADVANCE REGISTRATION

B.61.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.61.2 Level 1

A student who has obtained at least 75 credits at Level 1 of study may be allowed to register for a maximum of 40 credits at Level 2.

B.61.3 Level 2

B.61.3.1 A student who has obtained at least 80 credits at Level 2 of study may be allowed to register for a maximum of 60 credits at Level 3.

Please Note: A student must complete all Level 1 modules before registering for Level 3 modules.

B.62 RENEWAL OF REGISTRATION

B.62.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar

- B.62.2** Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.63 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.63.1 Modules from other faculties

- B.63.1.1** A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme or in the case of Statistics students who shall be allowed to take Economics as a module in their programme.

- B.63.1.2** At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

- B.63.1.3** Computing Fundamentals (COS114 or COS124) and English for Educational Development (EED117 or EED127): These are compulsory, credit-bearing modules, comprising one semester module for COS and one semester module for EED.

B.63.2 Exemption from COS and/or EED

A student, who has developed these skills, may apply to the faculty for exemption from this requirement, subject to the general rules of the University. The process to determine their level of competency in these modules may involve one or more assessments.

When exempted, such a student must replace these modules with other modules of their choice to the same credit value.

B.63.3 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

- B.63.3.1 Chemistry:** (CHE114 and CHE124) or (CHE116 and CHM126).

- B.63.3.2 Physics:** (PHY111 and PHY121) or (PHY116 and PHY126) or (PHY112 and PHY113 and PHY116) or (PHY151 and PHY152).

- B.63.3.3 Mathematics:** MAT105 or MAM115 or (MAM151 and MAM152).

- B.63.3.4 Statistics:** STA111 or STA125 or STA151 or BUS131 or BUS132 or STA141 or STA142 or STC101.

- B.63.3.5** A student may not claim for modules passed prior to 1999.

B.63.4 Endorsed Degree

B.63.4.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.63.4.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
APM112	Mathematics 112	E	1	15	5	S1	None	MAT105/ MAT104/ MAM152	CFA	
APM123	Applied Mathematics 123	E	1	15	5	YR	None	MAT105/ MAT104/ MAM152	CA	
COS101	Computer Science 101	E	1	30	5	YR	None	None	CA	
COS114	Computing Fundamentals 114	C	1	15	5	S1	None	None	CA	
COS124	Computing Fundamentals 124	C	1	15	5	S2	None	None	CA	
ECO151	Economics 151	E	1	10	5	S1	None	None	CFA	SM
ECO152	Economics 152	E	1	10	5	S2	None	None	CFA	SM
EED117	English for Educational Development 117	E	1	15	5	S1	None	None	CFA	SM
EED127	English for Educational Development 127	E	1	15	5	S2	None	None	CFA	SM
MAN131/ 132	Management 131 or Management 132	E	1	15	5	S1/ S2	None	None	CFA	SM
MAT105	Mathematics 105	C	1	30	5	YR	None	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
PHY111	Physics 111	E	1	15	5	S1	None	MAT103 MAT105	CFA	
PHY116	Physics 116 (N)	E	1	15	5	S1	None	None	CFA	
PHY121	Physics 121	E	1	15	5	S2	None	MAT103 MAT105	CFA	
PHY126	Physics 126 (N)	E	1	15	5	S2	None	None	CFA	
STA111	Statistics 111	C	1	15	5	S1	None	None	CFA	
STA121	Statistics 121	C	1	15	5	S2	None	STA111/125/ 151 or BUS131/132 or STA141/142	CFA	
CSC211	Computer Science 211	E	2	20	6	S1	COS101	None	CA	
CSC212	Computer Science 212	E	2	20	6	S2	COS101	None	CA	
ECO231	Economics 231	E	2	10	6	S2	ECO241/ ECO151	None	CFA	SM
ECO232	Economics 232	E	2	10	6	S1	ECO133/ ECO134/ ECO152	None	CFA	SM
ECO235	Economics 235	E	2	10	6	S2	ECO241/ ECO151	None	CFA	SM
ECO242	Economics 242 (pre-requisite for Economics 311)	E	2	10	6	S1	ECO241/ ECO151/ ECO152	ECO231	CFA	SM
MAT211	Mathematics 211	C	2	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	None	CFA	
MAT212	Mathematics 212	E	2	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	MAT211	CA	
MAT221	Mathematics 221	C	2	20	6	YR	None	MAT211	CFA	
MAT232	Mathematics 232	E	2	20	6	S1	APM112, MAT105/ MAT103	None	CFA	
PHY212	Physics 212	E	2	20	6	S1	PHY111/ PHY121 OR PHY151/ PHY152	MAT211/ MAT104/ MAT152	CFA	
PHY217	Physics 217	E	2	20	6	S1	None	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
PHY222	Physics 222	E	2	20	6	S2	PHY111/ PHY121 OR PHY151/ PHY152 AND MAT105/ MAT103/ MAM152	None	CFA	
PHY227	Physics 227	E	2	20	6	S2	PHY116/ PHY126 OR PHY111/ PHY151	None	CFA	
STA211	Statistics 211	E	2	20	6	S1	MAT105 or (MAM151 + MAM152) & STA111/ STA125/ STA151	None	CFA	
STA221	Statistics 221	E	2	20	6	S2	STA211 (at least qualified to write the exam for STA211)	None	CFA	
CSC311	Computer Science 311	E	3	30	7	S1	CSC211 and CSC212	None	CA	
CSC312	Computer Science 312	E	3	30	7	S2	CSC211 and CSC212	None	CA	
ECO311	Economics 311 (pre-requisite for Honours in Economics)	E	3	15	7	S1	ECO242	None	CFA	SM
ECO331	Economics 331	E	3	15	7	S1	ECO231	None	CFA	SM
ECO332	Economics 332	E	3	15	7	S2	ECO232	None	CFA	SM
ECO334	Economics 334	E	3	15	7	S1	ECO231, ECO232	None	CFA	SM
ECO335	Economics 335	E	3	15	7	S2	ECO231, ECO232	None	CFA	SM
ECO336	Economics 336	E	3	15	7	S2	ECO232	None	CFA	SM
MAT311	Mathematics 311	C/E	3	30	7	YR	MAT211	None	CFA	
MAT312	Mathematics 312	C/E	3	30	7	YR	MAT211, MAT212/ MAT221	None	CA	
MAT321	Mathematics 321	C/E	3	30	7	YR	MAT221	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
MAT322	Mathematics 322	C/E	3	30	7	YR	MAT211, MAT212/ MAT221	None	CA	
STA331	Statistics 331 (pre-requisite for Honours in Statistics)	E	3	30	7	S1	STA211, STA221	None	CFA	
STA332	Statistics 332 (pre-requisite for Honours in Statistics)	E	3	30	7	S2	STA211, STA221	None	CFA	

BACHELOR OF SCIENCE IN MEDICAL BIOSCIENCE (3230)

B.64 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Medical Bioscience Degree – BSc (Medical Bioscience)**:

B.64.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 33 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 4 (50-59%) in Mathematics, and
 - Level 4 (50-59%) in Life Sciences

B.64.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Unless Senate decides otherwise, an applicant shall not be admitted to any of the Bachelor of Science programmes without having obtained a full matriculation or age exemption. Furthermore, an applicant shall not be admitted to any of the Bachelor of Science programmes if they have not obtained a pass in Mathematics and either Physical Science or Biology at Higher grade with at least a D symbol or at Standard grade with at least a C symbol.

OR

- (b) A qualification or level of competence that the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.64.3 Alternative admission

Alternative admission does not apply to this programme.

B.65 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.66 DURATION

Unless Senate decides otherwise, the duration of the degree is three years.

B.67 CURRICULUM

B.67.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all groups)		
Group 1 (select all modules)		
Life Sciences 141	LSC141	15
Statistics 125	STA125	15
Chemistry 116 (N)	CHE116	15
Chemistry 126 (N)	CHM126	15
Medical Biosciences 111	MBS111	15
Medical Biosciences 121	MBS121	15
Group 2 (select one module)		
Computing Fundamentals 114	COS114	15
Computing Fundamentals 124	COS124	15
Group 3 (select one module)		
English for Educational Development 117	EED117	15
English for Educational Development 127	EED127	15
	Total	120

B.67.2 Level 2

Module Name	Module Code	Credits
Compulsory (select all modules)		
Medical Biosciences 231	MBS231	20
Medical Biosciences 232	MBS232	20
Medical Microbiology 251	MIC251	20
Medical Microbiology 252	MIC252	20
	Sub-total	80
Electives (select two modules)		
Biotechnology 211	BTN211	20
Biotechnology 223	BTN223	20
Biotechnology 215	BTN215	20
Biotechnology 213	BTN213	20
	Sub-total	40
	Total	120

B.67.3 Level 3

Module Name	Module Code	Credits
Compulsory (select all modules)		
Medical Biosciences 331	MBS331	30
Medical Biosciences 332	MBS332	30
Medical Microbiology 353	MIC353	30
Medical Microbiology 354	MIC354	30
	Total	120
	FINAL TOTAL	360

B.68 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.68.1 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.68.2 Weighting of practical component

B.68.2.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.68.2.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.68.3 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.68.4 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.68.5 Attendance of practicals

Where there is a practical component in a programme, a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.69 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.69.1 Level 1

A student shall promote to Level 2 of study when 90 credits have been obtained of the 120 credits required in Level 1 of study and when all pre-requisites for admission to Level 2 modules have been met.

B.69.2 Level 2

A student shall promote to Level 3 of study when at least a total of 200 credits have been obtained and when all 120 credits of Level 1 of study have been obtained.

B.70 ADVANCE REGISTRATION

Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.70.1 Level 1

A student who has obtained at least 75 credits at Level 1 of study may be allowed to register for a maximum of 40 credits at Level 2.

B.70.2 Level 2

B.70.2.1 A student who has obtained at least 80 credits at Level 2 of study may be allowed to register for a maximum of 60 credits at Level 3.

Please Note: A student must complete all Level 1 modules before registering for Level 3 modules.

B.71 RENEWAL OF REGISTRATION

B.71.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.71.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.72 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.72.1 Modules from other faculties

B.72.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.72.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.72.1.3 Computing Fundamentals (COS114 or COS124) and English for Educational Development (EED117 or EED127): these are compulsory, credit-bearing modules, comprising one semester module for COS and one semester module for EED.

B.72.2 Exemption from COS and/or EED

A student, who has developed these skills, may apply to the faculty for exemption from this requirement, subject to the general rules of the University. The process to determine their level of competency in these modules may involve one or more assessments.

When exempted, such a student must replace these modules with other modules of their choice to the same credit value.

B.72.3 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.72.3.1 Chemistry: (CHE114 and CHE124) **or** (CHE116 and CHM126).

B.72.3.2 Physics: (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).

B.72.3.3 Mathematics: MAT105 **or** MAM115 **or** (MAM151 and MAM152).

B.72.3.4 Statistics: STA111 **or** STA125 **or** STA151 **or** BUS131 **or** BUS132 **or** STA141 **or** STA142 **or** STC101.

B.72.3.5 A student may not claim credits for modules passed prior to 1999.

B.72.4 Endorsed Degree

B.72.4.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.72.4.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
LSC141	Life Sciences 141	C	1	15	5	S1	None	None	CFA	
STA125	Statistics 125	C	1	15	5	S2	None	None	CFA	
COS114	Computing Fundamentals 114	C/E	1	15	5	S1	None	None	CA	
COS124	Computing Fundamentals 124	C/E	1	15	5	S2	None	None	CA	
EED117	English for Educational Development 117	E	1	15	5	S1	None	None	CFA	SM
EED127	English for Educational Development 127	E	1	15	5	S2	None	None	CFA	SM
CHE116	Chemistry 116 (N)	C	1	15	5	S1	None	None	CFA	
CHM126	Chemistry 126 (N)	C	1	15	5	S2	None	None	CFA	
MBS111	Medical Biosciences 111	C	1	15	5	S1	None	None	CFA	
MBS121	Medical Biosciences 121	C	1	15	5	S2	None	None	CFA	
MBS231	Medical Biosciences 231	C	2	20	6	S1	LSC141/ MBS111, MBS121	None	CFA	
MBS232	Medical Biosciences 232	C	2	20	6	S2	LSC141/ MBS111, MBS121	None	CFA	
MIC251	Medical Microbiology 251	C	2	20	6	S1	LSC141	None	CFA	
MIC252	Medical Microbiology 252	C	2	20	6	S2	MIC251	None	CFA	
BTN211	Biotechnology 211	E	2	20	6	S1	CHE114/ CHE124 LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121	None	CA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
BTN223	Biotechnology 223	E	2	20	6	S2	CHE114/ CHE124 LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121	None	CA	
BTN215	Biotechnology 215	E	2	20	6	S1	CHE114/ CHE124 LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121	None	CA	
BTN213	Biotechnology 213	E	2	20	6	S2	CHE114/ CHE124 LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121	None	CFA	
MBS331	Medical Biosciences 331	C	3	30	7	S1	MBS 231 and MBS232	None	CFA	
MBS332	Medical Biosciences 332	C	3	30	7	S2	MBS231, MBS232 and MBS331	None	CFA	
MIC353	Medical Microbiology 353	C	3	30	7	S1	MIC251 and MIC252	None	CFA	
MIC354	Medical Microbiology 354	C	3	30	7	S2	MIC251 and MIC252	None	CFA	

BACHELOR OF SCIENCE IN PHYSICAL SCIENCE (3233)

B.73 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Physical Science Degree - BSc (Physical Science)**:

B.73.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 33 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 5 (60-69%) in Mathematics, and
 - Level 4 (50-59%) in Physical Sciences

B.73.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Unless Senate decides otherwise, an applicant shall not be admitted to any of the Bachelor of Science programmes without having obtained a full matriculation or age exemption. Furthermore, an applicant shall not be admitted to any of the Bachelor of Science programmes if they have not obtained a pass in Mathematics and either Physical Science or Biology at Higher grade with at least a D symbol or at Standard grade with at least a C symbol.

OR

- (b) A qualification or level of competence that the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.73.3 Alternative admission

Alternative admission does not apply to this programme.

B.74 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.75 DURATION

Unless Senate decides otherwise, the duration of the degree is three years.

B.76 CURRICULUM

B.76.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all groups)		
Group 1 (select all modules)		
Mathematics 105	MAT105	30
Physics 111	PHY111	15
Physics 121	PHY121	15
	Sub-total	60
Electives (select two groups)		
Group 2 (select one EED module and one COS module)		
Computing Fundamentals 114	COS114	15
Computing Fundamentals 124	COS124	15
English for Educational Development 117	EED117	15
English for Educational Development 127	EED127	15
Group 3 (select both modules)		
Earth Science 111	ESC111	15
Earth Science 121	ESC121	15
Group 4 (select both modules)		
Chemistry 114 (N)	CHE114	15
Chemistry 124 (N)	CHE124	15
Group 5 (select both modules)		
Mathematics 112	APM112	15
Applied Mathematics 123	APM123	15
Group 6		
Computer Science 101	COS101	30
	Sub-total	60
	Total	120

B.76.2 Level 2

Module Name	Module Code	Credits
Compulsory (select all modules)		
Physics 212	PHY212	20
Physics 222	PHY222	20
Mathematics 211	MAT211	20
Mathematics 221	MAT221	20
	Sub-total	80
Electives (select any two modules)		
Physics 217	PHY217	20
Physics 227	PHY227	20
Mathematics 212	MAT212	20
Mathematics 232	MAT232	20

Chemistry 211	CHM211	20
Chemistry 212	CHM212	20
Computer Science 211	CSC211	20
Computer Science 212	CSC212	20
	Sub-total	40
	Total	120

B.76.3 Level 3

Module Name	Module Code	Credits
Compulsory (select both modules)		
Physics 314	PHY314	30
Physics 324	PHY324	30
	Sub-total	60
Electives (select two modules)		
Chemistry 311	CHM311	30
Chemistry 312	CHM312	30
Computer Science 311	CSC311	30
Computer Science 312	CSC312	30
Mathematics 311	MAT311	30
Mathematics 321	MAT321	30
Mathematics 312	MAT312	30
Mathematics 322	MAT322	30
Physics 317	PHY317	30
Physics 327	PHY327	30
	Sub-total	60
	Total	120
	FINAL TOTAL	360

B.77 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.77.1 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.77.2 Weighting of practical component

B.77.2.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.77.2.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.77.3 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of

continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.77.4 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.77.5 Attendance of practicals

Where there is a practical component in a programme, a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.78 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.78.1 Level 1

A student shall promote to Level 2 of study when 90 credits have been obtained of the 120 credits required in Level 1 of study and when all pre-requisites for admission to Level 2 modules have been met.

B.78.2 Level 2

A student shall promote to Level 3 of study when at least a total of 200 credits have been obtained and when all 120 credits of Level 1 of study have been obtained.

B.79 ADVANCE REGISTRATION

B.79.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.79.2 Level 1

A student who has obtained at least 75 credits at Level 1 of study may be allowed to register for a maximum of 40 credits at Level 2.

B.79.3 Level 2

B.79.3.1 A student who has obtained at least 80 credits at Level 2 of study may be allowed to register for a maximum of 60 credits at Level 3.

Please Note: A student must complete all Level 1 modules before registering for Level 3 modules.

B.80 RENEWAL OF REGISTRATION

B.80.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar

B.80.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.81 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.81.1 Modules from other faculties

B.81.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.81.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.81.1.3 Computing Fundamentals (COS114 **or** COS124) and English for Educational Development (EED117 **or** EED127): These are compulsory, credit-bearing modules, comprising one semester module for COS and one semester module for EED.

B.81.2 Exemption from COS and/or EED

A student, who has developed these skills, may apply to the faculty for exemption from this requirement, subject to the general rules of the University. The process to determine their level of competency in these modules may involve one or more assessments.

When exempted, such a student must replace these modules with other modules of their choice to the same credit value.

B.81.3 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.81.3.1 Chemistry: (CHE114 and CHE124) **or** (CHE116 and CHM126).

B.81.3.2 Physics: (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).

B.81.3.3 Mathematics: MAT105 **or** MAM115 **or** (MAM151 and MAM152).

B.81.3.4 Statistics: STA111 or STA125 or STA151 or BUS131 or BUS132 or STA141 or STA142 or STC101.

B.81.3.5 A student may not claim credits for modules passed prior to 1999.

B.81.4 Endorsed Degree

B.81.4.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.81.4.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requirement	Co-Requirement	Assessment Type	Service Module
MAT105	Mathematics 105	C	1	30	5	YR	None	None	CFA	
PHY111	Physics 111	C	1	15	5	S1	None	MAT103 MAT105	CFA	
PHY121	Physics 121	C	1	15	5	S2	None	MAT103 MAT105	CFA	
COS114	Computing Fundamentals 114	E	1	15	5	S1	None	None	CA	
COS124	Computing Fundamentals 124	E	1	15	5	S2	None	None	CA	
EED117	English for Educational Development 117	E	1	15	5	S1	None	None	CFA	SM
EED127	English for Educational Development 127	E	1	15	5	S2	None	None	CFA	SM
ESC111	Earth Science 111	E	1	15	5	S1	None	None	CFA	
ESC121	Earth Science 121	E	1	15	5	S2	None	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
APM112	Mathematics 112	E	1	15	5	S1	None	MAT105/ MAT104/ MAM152	CFA	
APM123	Applied Mathematics 123	E	1	15	5	YR	None	MAT105/ MAT104/ MAM152	CA	
CHE114	Chemistry 114 (N)	E	1	15	5	S1	None	None	CFA	
CHE124	Chemistry 124 (N)	E	1	15	5	S2	None	None	CFA	
CHM211	Chemistry 211	E	2	20	6	S1	CHE114, CHE124	None	CFA	
CHM212	Chemistry 212	E	2	20	6	S2	CHE114, CHE124	None	CFA	
CHM311	Chemistry 311	E	3	30	7	S1	CHM211/ CHM212	None	CFA	
CHM312	Chemistry 312	E	3	30	7	S2	CHM211/ CHM212	None	CFA	
COS101	Computer Science 101	E	1	30	5	YR	None	None	CA	
CSC211	Computer Science 211	E	2	20	6	S1	COS101	None	CA	
CSC212	Computer Science 212	E	2	20	6	S2	COS101	None	CA	
CSC311	Computer Science 311	E	3	30	7	S1	CSC211 and CSC212	None	CA	
CSC312	Computer Science 312	E	3	30	7	S2	CSC211 and CSC212	None	CA	
MAT211	Mathematics 211	C	2	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	None	CFA	
MAT212	Mathematics 212	E	2	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	MAT211	CA	
MAT221	Mathematics 221	C	2	20	6	YR	None	MAT211	CFA	
MAT232	Mathematics 232	E	2	20	6	S1	APM112, MAT105/ MAT103	None	CFA	
MAT311	Mathematics 311	E	3	30	7	YR	MAT211	None	CFA	
MAT312	Mathematics 312	E	3	30	7	YR	MAT211, MAT212/ MAT221	None	CA	
MAT321	Mathematics 321	E	3	30	7	YR	MAT221	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
MAT322	Mathematics 322	E	3	30	7	YR	MAT211, MAT212/ MAT221	None	CA	
PHY212	Physics 212	C	2	20	6	S1	PHY111/ PHY121 OR PHY151/ PHY152	MAT211/ MAT104/ MAT152	CFA	
PHY217	Physics 217	E	2	20	6	S1	None	None	CFA	
PHY222	Physics 222	C	2	20	6	S2	PHY111/ PHY121 OR PHY151/ PHY152 AND MAT105/ MAT103/ MAM152	None	CFA	
PHY227	Physics 227	E	2	20	6	S2	PHY116/ PHY126 OR PHY111/ PHY151	None	CFA	
PHY314	Physics 314	C	3	30	7	S1	PHY212 PHY222 MAT211 MAT221	None	CFA	
PHY317	Physics 317	E	3	30	7	S2	PHY227	None	CFA	
PHY324	Physics 324	C	3	30	7	S1	PHY212 PHY222 MAT211 MAT221	None	CFA	
PHY327	Physics 327	E	3	30	7	S2	PHY212/ PHY217, MAT105/ MAT103/ MAM152	None	CA	

BACHELOR OF PHARMACY (3305)

B.82 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Pharmacy Degree – BPharm**:

B.82.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 38 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First additional Language), and
 - Level 4 (50-59%) in Mathematics, and
 - Level 4 (50-59%) in Physical Sciences, and
 - Level 4 (50-59%) in Life Sciences

B.82.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) A Matriculation Exemption with the following subject requirements:
- HG (50%, D symbol) Mathematics or SG (60%, C symbol) Mathematics, and
 - HG (50%, D symbol) Biology or SG (60%, C symbol) Biology, and
 - HG (50%, D symbol) Physical Sciences or SG (60%, C symbol) Physical Sciences

OR

- (b) A qualification or level of competence that the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.83 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.84 DURATION

The degree shall extend over four years full-time.

B.85 CURRICULUM

B.85.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all modules)		
Human Biology 113	HUB113	15
Human Biology 123	HUB123	15
Chemistry 114 (N)	CHE114	15
Chemistry 124 (N)	CHE124	15

Pharmacy 116
 Pharmacology and Clinical Pharmacy 123
 Physics 112
 Mathematics 115 (N)

PHA116	15
PHC123	15
PHY112	15
MAM115	15
Sub-total	120

Group 2 (select one module)

Introduction to Afrikaans 002 (SC)
 Introduction to Xhosa 002 (SC)

AFR002	10
XHO002	10
Sub-total	10
Total	130

B.85.2 Level 2

Module Name

Compulsory (select all modules)

Pharmaceutics 231
 Pharmaceutics 232
 Pharmaceutical Chemistry 232
 Pharmaceutical Chemistry 233
 Pharmacology and Clinical Pharmacy 213
 Pharmacology and Clinical Pharmacy 223
 Biotechnology 217
 Pharmacy Practice 214
 Pharmacy Practice 224

Module Code	Credits
PCE231	15
PCE232	15
PCH232	15
PCH233	15
PHC213	15
PHC223	15
BTN217	15
PPR214	10
PPR224	15
Total	130

B.85.3 Level 3

Module Name

Compulsory (select all modules)

Pharmaceutics 331
 Pharmaceutics 333
 Pharmaceutical Chemistry 312
 Pharmaceutical Chemistry 322
 Pharmacology and Clinical Pharmacy 313
 Pharmacology and Clinical Pharmacy 323
 Pharmacy Practice 314
 Pharmacy Practice 324

Module Code	Credits
PCE331	15
PCE333	15
PCH312	15
PCH322	15
PHC313	15
PHC323	15
PPR314	15
PPR324	15
Total	120

B.85.4 Level 4

Module Name

Compulsory (select all modules)

Pharmaceutics 431
 Pharmacy 408
 Pharmacy 426
 Pharmacology and Clinical Pharmacy 405
 Pharmacology and Clinical Pharmacy 423
 Pharmacy Practice 414

Module Code	Credits
PCE431	15
PHA408	15
PHA426	15
PHC405	30
PHC423	15
PPR414	15
Sub-total	105

Electives (select one module)

Pharmaceutics 401	PCE401	15
Pharmaceutical Chemistry 402	PCH402	15
Pharmacology and Clinical Pharmacy 403	PHC403	15
Pharmacy Practice 404 (not offered in 2026)	PPR404	15
Pharmacy Practice 434	PPR434	15
Rational Medicine Use 425	PHC425	15
Community Engagement 427 (not offered in 2026)	PHA427	15
*Global Pharmacy 424 (not offered in 2026)	PHC424	15
Paediatrics for Pharmacists 426	PHC426	15
	Sub-total	15
	Total	120
	FINAL TOTAL	500

***Please Note:** *This module may only be selected by USA exchange or affiliated students.*

B.86 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.86.1 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.86.2 Weighting of practical examinations

The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.86.3 Attendance of practicals

Where there is a practical or experiential learning component in a programme, a minimum attendance of 80% of the practicals or experiential learning sessions are required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.87 PROMOTION RULES

B.87.1 Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.87.2 No student shall be admitted to the next level of study unless all the modules of the previous level of study have been successfully completed after no more than two attempts.

B.87.3 A student shall promote to Level 3 of study on passing all Level 2 modules.

B.87.4 A student shall promote to Level 4 of study on passing all Level 3 modules.

B.88 ADVANCE REGISTRATION

B.88.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

A student registering advance modules may not register for more than 90 credits in total for that year.

B.88.2 A student who has obtained at least 60 credits at Level 1 of study may be allowed to register for a maximum of 45 credits at Level 2.

B.88.3 A student who has obtained at least 210 credits at Level 2 of study may be allowed to register for a maximum of 60 credits at Level 3.

B.88.4 A student who has obtained at least 310 credits at Level 3 of study and has completed all Level 2 modules may be allowed to register for a maximum of 60 credits at Level 4.

B.89 RENEWAL OF REGISTRATION

Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

B.90 SPECIAL REQUIREMENTS FOR THE PROGRAMME

The curriculum shall extend over four years, followed by one year of internship under the auspices of the South African Pharmacy Council (SAPC). The internship may be done in a community, hospital, industrial practice or at an approved university.

Please Note: A foreign student completing the BPharm degree at a South African university does not automatically qualify for internship or for registration as a pharmacist in South Africa.

A student will have to register with the SAPC from their first year and pay a fee as determined by the SAPC.

A student will be required to provide proof of the necessary immunisation and personal indemnity insurance as required by the SAPC to perform the acts pertaining to a pharmacy student at experiential sites.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
AFR002	Introduction to Afrikaans 002 (SC)	E	1	10	5	YR	None	None	CFA	SM
CHE114	Chemistry 114 (N)	C	1	15	5	S1	None	None	CFA	
CHE124	Chemistry 124 (N)	C	1	15	5	S2	None	None	CFA	
HUB113	Human Biology 113	C	1	15	5	S1	None	None	CFA	
HUB123	Human Biology 123	C	1	15	5	S2	None	None	CFA	
MAM115	Mathematics 115 (N)	C	1	15	5	S1	None	None	CA	
PHA116	Pharmacy 116	C	1	15	5	S1	None	None	CA	
PHC123	Pharmacology and Clinical Pharmacy 123	C	1	15	5	S2	HUB123	None	CFA	
PHY112	Physics 112	C	1	15	5	S2	None	None	CFA	
XHO002	Introduction to Xhosa 002 (SC)	E	1	10	5	YR	None	None	CFA	SM
BTN217	Biotechnology 217	C	2	15	6	S1	None	None	CFA	
PCE231	Pharmaceutics 231	C	2	15	6	S1	MAM115, PHC123, PHY112	None	CFA	
PCE232	Pharmaceutics 232	C	2	15	6	S2	MAM115, PHC123, PHY112	None	CFA	
PCH232	Pharmaceutical Chemistry 232	C	2	15	6	S1	CHE114, CHE124, PCH233/ min 40% in PCH233	None	CFA	
PCH233	Pharmaceutical Chemistry 233	C	2	15	6	S2	CHE114, CHE124	None	CFA	
PHC213	Pharmacology and Clinical Pharmacy 213	C	2	15	6	S1	PHC123, HUB113, HUB123, PHA116	None	CFA	
PHC223	Pharmacology and Clinical Pharmacy 223	C	2	15	6	S2	HUB113, HUB123, PHC123, PHA116/ min 40% for PHC213	None	CFA	

Module Code	Module Name	Compulsory/	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requirement	Assessment Type	Service Module
PPR214	Pharmacy Practice 214	C	2	10	6	S1	PHA116, PHC123	None	CFA	
PPR224	Pharmacy Practice 224	C	2	15	6	YR	PCE231, PHC213, PPR214/ min 40% in each of these level 2 modules	PHC223	CFA	
PCE331	Pharmaceutics 331	C	3	15	7	S1	PCE231, PCE232	None	CFA	
PCE333	Pharmaceutics 333	C	3	15	7	S2	PHC123, PCE231, PCE232	None	CFA	
PCH312	Pharmaceutical Chemistry 312	C	3	15	7	S1	PCH232, PCH233, BTN216	None	CFA	
PCH322	Pharmaceutical Chemistry 322	C	3	15	7	S2	PCH232, PCH233, PCH312/ min 40% in PCH312	None	CFA	
PHC313	Pharmacology and Clinical Pharmacy 313	C	3	15	7	S1	PHC213, PHC223	None	CFA	
PHC323	Pharmacology and Clinical Pharmacy 323	C	3	15	7	S2	PHC213, PHC223, PHC313/ min 40% for PHC313	None	CFA	
PPR314	Pharmacy Practice 314	C	3	15	7	S1	PPR214, PPR224	None	CA	
PPR324	Pharmacy Practice 324	C	3	15	7	S2	PPR214, PPR224, PPR314 (min 40%)	None	CA	
PCE401	Pharmaceutics 401	E	4	15	8	YR	PCE331, PCE333	None	CFA	
PCE431	Pharmaceutics 431	C	4	15	8	S1	PCE331, PCE333	None	CFA	
PCH402	Pharmaceutical Chemistry 402	E	4	15	8	YR	PCH312, PCH322	None	CFA	
PHA408	Pharmacy 408	C	4	15	8	YR	PCE331, PCE333, PCH312, PCH322, PHC313, PHC323, PPR314, PPR324	None	CA	
PHA426	Pharmacy 426	C	4	15	8	S1	None	None	CFA	

Module Code	Module Name	Compulsory/	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
PHA427	Community Engagement 427	E	4	15	8	S2	PCH312, PCH322, PCE331, PCE333	None	CA	
PHC403	Pharmacology and Clinical Pharmacy 403	E	4	15	8	YR	PHC313, PHC323	None	CFA	
PHC405	Pharmacology and Clinical Pharmacy 405	C	4	30	8	S2	PCE331, PCE333, PCH312, PCH322, PHC313, PHC323, PPR314, PPR324	None	CA	
PHC423	Pharmacology and Clinical Pharmacy 423	C	4	15	8	S1	PHC313, PHC323, PPR314, PPR324	None	CA	
PHC424	Global Pharmacy 424	E	4	15	8	S1	Determined by home university	None	CA	
PHC425	Rational Medicine Use 425	E	4	15	8	YR	PHC323, PPR324	None	CA	
PHC426	Paediatrics for Pharmacists 426	E	4	15	8	YR	PHC313, PHC323	None	CA	
PPR404	Pharmacy Practice 404	E	4	15	8	YR	PPR314, PPR324	None	CFA	
PPR414	Pharmacy Practice 414	C	4	15	8	YR	PPR314, PPR324	None	CFA	
PPR434	Pharmacy Practice 434	E	4	15	8	YR	PPR314, PPR324	None	CFA	

RULES FOR EXTENDED PROGRAMMES

BACHELOR OF SCIENCE IN APPLIED GEOLOGY (3011)

B.91 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Applied Geology Degree- BSc (Applied Geology)**:

B.91.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 27 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional language), and
 - Level 3 (40-49%) in Mathematics, and
 - Level 3 (40-49%) in Physical Sciences

B.91.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Senate, upon recommendation of the faculty, may admit applicants with a full Matriculation exemption with Mathematics and either Science or Biology at Higher Grade with an E-symbol or Standard Grade with a D-symbol.

OR

- (b) A qualification or level of competence which the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.91.3 Alternative admission

Alternative admission requirements for applicants in terms of Recognition of Prior Learning (RPL)

An applicant who completed the NSC in 2008 or thereafter, but has not obtained an endorsement, shall be considered for alternative admission after the age of 23. Such an applicant shall be required to complete an RPL portfolio development course and submit to a process where relevant learning and/or experience shall be assessed.

An applicant who is 23 years or older and does not have a matriculation certificate or NSC, but who may have obtained other qualifications or experience that may be deemed to be equivalent to admission criteria for the particular study programmes shall also be required to complete an RPL portfolio development course or an RPL portfolio process as agreed upon, and to submit to a process where such learning, qualification, and/or experience shall be assessed.

B.92 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.93 DURATION

Unless Senate decides otherwise, the duration of the degree is four years.

B.94 CURRICULUM

B.94.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all modules)		
Introduction to Science 153	ISC153	30
Mathematics 151 (SF)	MAM151	15
Physics 151 (SF)	PHY151	15
Total		60

B.94.2 Level 2

Module Name	Module Code	Credits
Compulsory (select all modules)		
Chemistry 114 (N)	CHE114	15
Chemistry 124 (N)	CHE124	15
Earth Science 111	ESC111	15
Earth Science 121	ESC121	15
Physics 152 (SF)	PHY152	15
Total		75

B.94.3 Level 3

Module Name	Module Code	Credits
Compulsory (select all modules)		
Geology 232	APG232	20
Geology 234	APG234	20
Geology 235	APG235	20
Chemistry 211	CHM211	20
Chemistry 212	CHM212	20
Sub-total		100
Electives (select one module)		
Environmental and Water Science 211	EWP211	20
Environmental and Water Science 221	EWP221	20
Sub-total		20
Total		120

B.94.4 Level 4

Module Name	Module Code	Credits
Compulsory (select all modules)		
Geology 312	APG312	30
Geology 323	APG323	30
Geology 324	APG324	30
Geology 325	APG325	30
Total		120
FINAL TOTAL		375

B.95 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.95.1 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.95.2 Weighting of practical component

The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%. For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.95.3 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.95.4 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.95.5 Attendance of practicals

Where there is a practical component in a programme, a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.96 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.96.1 Level 1

A student shall promote to Level 2 of study on obtaining 45 credits and when all pre-requisites for Level 2 modules have been met.

B.96.2 Level 2

A student shall promote to Level 3 of study on obtaining 105 credits at Level 2 and when all pre-requisites for Level 3 modules have been met.

B.96.3 Level 3

A student shall promote to Level 4 of study on obtaining 215 credits and when all Level 1 and 2 modules have been passed, and all Level 4 pre-requisites have been met.

B.97 ADVANCE REGISTRATION

B.97.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.97.1.1 Level 1

A student who has passed 2 of the 3 modules at Level 1 of study may be allowed to register for a maximum of 30 credits or one year-long module at Level 2.

B.97.1.2 Level 2

A student who has obtained 30 credits at Level 2 of study may be allowed to register for a maximum of 40 credits at Level 3 and completing all modules at Level 1. Total credits registered may not exceed 90.

B.97.1.3 Level 3

A student who has obtained 80 credits at Level 3 of study may be allowed to register for a maximum of 60 credits at Level 4 and completing all modules at Levels 1 and 2. Total credits registered may not exceed 90.

Please Note: A student must complete all Level 1 and 2 modules before registering for Level 4 modules.

B.98 RENEWAL OF REGISTRATION

B.98.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.98.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.99 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.99.1 Modules from other faculties

B.99.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except

for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.99.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.99.2 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.99.2.1 Chemistry: (CHE114 and CHE124) **or** (CHE116 and CHM126).

B.99.2.2 Physics: (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).

B.99.2.3 Mathematics: MAT105 **or** MAM115 **or** (MAM151 and MAM152).

B.99.2.4 Statistics: STA111 **or** STA125 **or** STA151 **or** BUS131 **or** BUS132 **or** STA141 **or** STA142 **or** STC101.

B.99.2.5 A student may not claim credits for modules passed prior to 1999.

B.99.3 Endorsed Degree

B.99.3.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.99.3.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
ISC153	Introduction to Science 153	C	1	30	5	YR	None	PHY151/ LFS151; MAM151/ MAM150	CA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
MAM151	Mathematics 151 (SF)	C	1	15	5	YR	None	None	CFA	
PHY151	Physics 151 (SF)	C	1	15	5	YR	None	MAM151	CFA	
PHY152	Physics 152 (SF)	C	2	15	5	YR	None	None	CFA	
CHE114	Chemistry 114 (N)	C	2	15	5	S1	None	None	CFA	
CHE124	Chemistry 124 (N)	C	2	15	5	S2	None	None	CFA	
ESC111	Earth Science 111	C	2	15	5	S1	None	None	CFA	
ESC121	Earth Science 121	C	2	15	5	S2	None	None	CFA	
APG232	Geology 232	C	3	20	6	S1	ESC111, ESC121	None	CFA	
APG234	Geology 234	C	3	20	6	S2	ESC111, ESC121	None	CFA	
APG235	Geology 235	C	3	20	6	S2	ESC111, ESC121	None	CFA	
CHM211	Chemistry 211	C	3	20	6	S1	CHE114, CHE124	None	CFA	
CHM212	Chemistry 212	C	3	20	6	S2	CHE114, CHE124	None	CFA	
EWP211	Environmental and Water Science 211	E	3	20	6	S1	ESC111, ESC121	None	CFA	
EWP221	Environmental and Water Science 221	E	3	20	6	S2	ESC111, ESC121	None	CFA	
APG312	Geology 312	C	4	30	7	S2	APG232, APG234	None	CFA	
APG323	Geology 323	C	4	30	7	S1	APG232, APG234, APG235	None	CFA	
APG324	Geology 324	C	4	30	7	S1	APG232, APG234, APG235	None	CFA	
APG325	Geology 325	C	4	30	7	S2	APG232, APG234, APG235	None	CFA	

BACHELOR OF SCIENCE IN BIODIVERSITY AND CONSERVATION BIOLOGY (3015)

B.100 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Biodiversity and Conservation Biology Degree - BSc (Biodiversity and Conservation Biology)**:

B.100.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 27 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 3 (40-49%) in Mathematics, and
 - Level 3 (40-49%) in Physical Sciences, or
 - Level 3 (40-49%) in Life Sciences

B.100.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Senate, upon recommendation of the faculty, may admit applicants with a full Matriculation Exemption with Mathematics and either Science or Biology at Higher Grade with an E-symbol or Standard Grade with a D-symbol.

OR

- (b) A qualification or level of competence which the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.100.3 Alternative admission

Alternative admission requirements for applicants in terms of Recognition of Prior Learning (RPL)

An applicant who completed the NSC in 2008 or thereafter, but has not obtained an endorsement, shall be considered for alternative admission after the age of 23. Such an applicant shall be required to complete an RPL portfolio development course and submit to a process where relevant learning and/or experience shall be assessed.

An applicant who is 23 years or older and does not have a matriculation certificate or NSC, but who may have obtained other qualifications or experience that may be deemed to be equivalent to admission criteria of the particular study programmes shall also be required to complete an RPL portfolio development course or an RPL portfolio process as agreed upon, and to submit to a process where such learning, qualification, and/or experience shall be assessed.

B.101 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.102 DURATION

Unless Senate decides otherwise, the duration of the degree is four years.

B.103 CURRICULUM

B.103.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all modules)		
Life Sciences 151(SF)	LFS151	15
Introduction to Mathematics 150	MAM150	15
Introduction to Science 153	ISC153	30
	Sub-total	60

B.103.2 Level 2

Module Name	Module Code	Credits
Compulsory		
Group 1 (select all modules)		
Life Sciences 152(SF)	LFS152	15
Statistics 151 (SF)	STA151	15
*Chemistry 114 (N) or Chemistry 116 (N)	CHE114/ CHE116	15

Group 2

Electives (select two modules)

Where two Earth Science modules are chosen in Level 2, Environmental and Water Science must be taken as electives in Level 3.

Where two Chemistry modules are chosen in Level 2, Chemistry must be taken as electives in Level 3.

***Students wanting to enrol for CHM211 must select the CHE114/CHE124 combination.**

Earth Science 111	ESC111	15
Earth Science 121	ESC121	15
*Chemistry 124 (N) or Chemistry 126 (N)	CHE124/ CHM126	15
	Sub-total	75

B.103.3 Level 3

Module Name	Module Code	Credits
Compulsory (select all modules)		
Biodiversity and Conservation Biology 211	BDC211	20
Biodiversity and Conservation Biology 221	BDC221	20

Biodiversity and Conservation Biology 222	BDC222	20
Biodiversity and Conservation Biology 223	BDC223	20
	Sub-total	80

Electives (select one group)

Group 1 (select both modules)

Environmental and Water Science 211	EWP211	20
Environmental and Water Science 221	EWP221	20

Group 2 (select both modules)

Chemistry 211	CHM211	20
Chemistry 212	CHM212	20
	Sub-total	40
	Total	120

B.103.4 Level 4

Module Name	Module Code	Credits
Compulsory (select four modules)		
Biodiversity and Conservation Biology 332	BDC332	30
Biodiversity and Conservation Biology 333	BDC333	30
Biodiversity and Conservation Biology 334	BDC334	30
Biodiversity and Conservation Biology 335	BDC335	30
Environmental and Water Science 321	EWP321	30
	Total	120
	FINAL TOTAL	375

B.104 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.104.1 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.104.2 Weighting of practical component

The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%. For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.104.3 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.104.4 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.104.5 Attendance of practicals

Where there is a practical component in a programme, a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.105 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.105.1 Level 1

A student shall promote to Level 2 of study on obtaining 45 credits and when all pre-requisites for Level 2 modules have been met.

B.105.2 Level 2

A student shall promote to Level 3 of study on obtaining 105 credits at Levels 1 and 2 and when all pre-requisites for Level 3 modules have been met.

B.105.3 Level 3

A student shall promote to Level 4 of study on obtaining 215 credits and when all Level 1 and 2 modules have been passed.

B.106 ADVANCE REGISTRATION

B.106.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.106.1.1 Level 1

A student who has passed 2 of the 3 modules at Level 1 of study may be allowed to register for a maximum of 30 credits or one year-long module at Level 2.

B.106.1.2 Level 2

A student who has obtained 30 credits at Level 2 of study may be allowed to register for a maximum of 40 credits at Level 3 and completing all modules at Level 1. Total credits registered may not exceed 90.

B.106.1.3 Level 3

A student who has obtained 80 credits at Level 3 of study may be allowed to register for a maximum of 60 credits at Level 4 and completing all modules at Levels 1 and 2. Total credits registered may not exceed 90.

Please Note: A student must complete all Level 1 and 2 modules before registering for Level 4 modules.

B.107 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.107.1 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.108 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.108.1 Modules from other faculties

B.108.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.108.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.108.2 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.108.2.1 Chemistry: (CHE114 and CHE124) **or** (CHE116 and CHM126).

B.108.2.2 Physics: (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).

B.108.2.3 Mathematics: MAT105 **or** MAM115 **or** (MAM151 and MAM152).

B.108.2.4 Statistics: STA111 **or** STA125 **or** STA151 **or** BUS131 **or** BUS132 **or** STA141 **or** STA142 **or** STC101.

B.108.2.5 A student may not claim credits for modules passed prior to 1999.

B.108.3 Endorsed Degree

B.108.3.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.108.3.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Prerequisite	Co-Prerequisite	Assessment Type	Service Module
ISC153	Introduction to Science 153	C	1	30	5	YR	None	PHY151/ LFS151; MAM151/ MAM150	CA	
LFS151	Life Sciences 151(SF)	C	1	15	5	YR	None	None	CFA	
MAM150	Introduction to Mathematics 150	C	1	15	5	YR	None	None	CFA	
CHE114 or CHE116	Chemistry 114 (N) or Chemistry 116 (N)	C	2	15	5	S1	None	None	CFA	
CHE124 or CHM126	Chemistry 124 (N) or Chemistry 126 (N)	C	2	15	5	S2	None	None	CFA	
ESC111	Earth Science 111	C	2	15	5	S1	None	None	CFA	
ESC121	Earth Science 121	C	2	15	5	S2	None	None	CFA	
LFS152	Life Sciences 152(SF)	C	2	15	5	YR	None	None	CFA	
STA151	Statistics 151 (SF)	C	2	15	5	YR	None	None	CFA	
BDC211	Biodiversity and Conservation Biology 211	C	3	20	6	S2	LSC142 or LFS152	None	CFA	
BDC221	Biodiversity and Conservation Biology 221	C	3	20	6	S1	LSC142 or LFS152	None	CFA	
BDC222	Biodiversity and Conservation Biology 222	C	3	20	6	S1	LSC141/ 142 or LFS151/ 152	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
BDC223	Biodiversity and Conservation Biology 223	C	3	20	6	S2	LSC141 and LSC142, or LFS151 and LFS152; and CHE114 or CHE124 or CHE116 or CHM126	None	CFA	
CHM211	Chemistry 211	C	3	20	6	S1	CHE114, CHE124	None	CFA	
CHM212	Chemistry 212	C	3	20	6	S2	CHE114, CHE124	None	CFA	
EWP211	Environmental and Water Science 211	E	3	20	6	S1	ESC111, ESC121	None	CFA	
EWP221	Environmental and Water Science 221	E	3	20	6	S2	ESC111, ESC121	None	CFA	
BDC332	Biodiversity and Conservation Biology 332	C	4	30	7	S2	BDC222	None	CFA	
BDC333	Biodiversity and Conservation Biology 333	C	4	30	7	S1	BDC222, BDC211	None	CFA	
BDC334	Biodiversity and Conservation Biology 334	C	4	30	7	S2	BDC222, BDC211, BDC223	None	CFA	
BDC335	Biodiversity and Conservation Biology 335	C	4	30	7	S1	BDC222, BDC211, BDC221	None	CFA	
EWP321	Environmental and Water Science 321	C	4	30	7	S2	EWP211, EWP221	None	CFA	

BACHELOR OF SCIENCE IN BIOTECHNOLOGY (3007)

B.109 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Biotechnology Degree - BSc (Biotechnology)**:

B.109.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 27 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 3 (40-49%) in Mathematics, and
 - Level 3 (40-49%) in Physical Sciences, or
 - Level 3 (40-49%) in Life Sciences

B.109.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Senate, upon recommendation of the faculty, may admit an applicant with a full Matriculation Exemption with Mathematics and either Science or Biology at Higher Grade with an E-symbol or Standard Grade with a D-symbol.

OR

- (b) A qualification or level of competence which the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.109.3 Alternative admission

Alternative admission requirements for applicants in terms of Recognition of Prior Learning (RPL)

An applicant who completed the NSC in 2008 or thereafter, but has not obtained an endorsement, shall be considered for alternative admission after the age of 23. Such an applicant shall be required to complete an RPL portfolio development course and submit to a process where relevant learning and/or experience shall be assessed.

An applicant who is 23 years or older and does not have a matriculation certificate or NSC, but who may have obtained other qualifications or experience that may be deemed to be equivalent to admission criteria for the particular study programmes shall also be required to complete an RPL portfolio development course or an RPL portfolio process as agreed upon, and to submit to a process where such learning, qualification, and/or experience shall be assessed.

B.110 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.111 DURATION

Unless Senate decides otherwise, the duration of the degree is four years.

B.112 CURRICULUM

B.112.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all modules)		
Introduction to Science 153	ISC153	30
Life Sciences 151(SF)	LFS151	15
Introduction to Mathematics 150	MAM150	15
Total		60

B.112.2 Level 2

Module Name	Module Code	Credits
Compulsory (select all modules)		
Chemistry 114 (N)	CHE114	15
Chemistry 124 (N)	CHE124	15
Life Sciences 152 (SF)	LFS152	15
Statistics 151 (SF)	STA151	15
Total		60

B.112.3 Level 3

Module Name	Module Code	Credits
Compulsory (select all modules)		
Biotechnology 211	BTN211	20
Biotechnology 215	BTN215	20
Biotechnology 222	BTN222	20
Biotechnology 223	BTN223	20
Sub-total		80
Electives (select two modules)		
Biotechnology 213	BTN213	20
Biotechnology 218	BTN218	20
Chemistry 211	CHM211	20
Chemistry 212	CHM212	20
Sub-total		40
Total		120

B.112.4 Level 4

Module Name	Module Code	Credits
Compulsory (select all modules)		
Biotechnology 312	BTN312	30
Biotechnology 315	BTN315	30
Biotechnology 316	BTN316	30
Biotechnology 322	BTN322	30
Total		120
FINAL TOTAL		360

B.113 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.113.1 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.113.2 Weighting of practical component

B.113.2.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.113.2.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.113.3 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.113.4 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.113.5 Attendance of practicals

Where there is a practical component in a programme, a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.114 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.114.1 Level 1

A student shall promote to Level 2 of study on obtaining 45 credits and when all pre-requisites for Level 2 modules have been met.

B.114.2 Level 2

A student shall promote to Level 3 of study on obtaining 90 credits at Level 2 and when all pre-requisites for Level 3 modules have been met.

B.114.3 Level 3

A student shall promote to Level 4 of study on obtaining 200 credits and when all Level 1 and 2 modules have been passed.

B.115 ADVANCE REGISTRATION

B.115.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.115.2 Level 1

A student who has passed 2 of the 3 modules at Level 1 of study may be allowed to register for a maximum of 30 credits or one year-long module at Level 2.

B.115.3 Level 2

A student who has obtained 30 credits at Level 2 of study may be allowed to register for a maximum of 40 credits at Level 3 and completing all modules at Level 1. Total credits registered may not exceed 90.

B.115.4 Level 3

A student who has obtained 80 credits at Level 3 of study may be allowed to register for a maximum of 60 credits at Level 4 and completing all modules at Levels 1 and 2. Total credits registered may not exceed 90.

Please Note: A student must complete all Level 1 and 2 modules before registering for Level 4 modules.

B.116 RENEWAL OF REGISTRATION

B.116.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.116.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.117 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.117.1 Modules from other faculties

B.117.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except

for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.117.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.117.2 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.117.2.1 Chemistry: (CHE114 and CHE124) **or** (CHE116 and CHM126).

B.117.2.2 Physics: (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).

B.117.2.3 Mathematics: MAT105 **or** MAM115 **or** (MAM151 and MAM152).

B.117.2.4 Statistics: STA111 **or** STA125 **or** STA151 **or** BUS131 **or** BUS132 **or** STA141 **or** STA142 **or** STC101.

B.117.2.5 A student may not claim credits for modules passed prior to 1999.

B.117.3 Endorsed Degree

B.117.3.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.117.3.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
ISC153	Introduction to Science 153	C	1	30	5	YR	None	PHY151/ LFS151; MAM151/ MAM150	CA	
LFS151	Life Sciences 151(SF)	C	1	15	5	YR	None	None	CFA	
MAM150	Introduction to Mathematics 150	C	1	15	5	YR	None	None	CFA	
CHE114	Chemistry 114 (N)	C	2	15	5	S1	None	None	CFA	
CHE124	Chemistry 124 (N)	C	2	15	5	S2	None	None	CFA	
LFS152	Life Sciences 152 (SF)	C	2	15	5	YR	None	None	CFA	
STA151	Statistics 151 (SF)	C	2	15	5	YR	None	None	CFA	
BTN211	Biotechnology 211	C	3	20	6	S1	CHE114/ CHE124 LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121	None	CA	
BTN215	Biotechnology 215	C	3	20	6	S1	CHE114/ CHE124 LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121	None	CA	
BTN222	Biotechnology 222	C	3	20	6	S1	CHE114/ CHE124 LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121	None	CFA	
BTN223	Biotechnology 223	C	3	20	6	S2	CHE114/ CHE124	None	CA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
							LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121			
BTN213	Biotechnology 213	E	3	20	6	S2	CHE114 & CHE124 & LSC141 & LSC142. LFS151 & LFS152 / LSC141 & MBS111 & MBS121	None	CFA	
BTN218	Biotechnology 218	E	3	20	6	S2	CHE114 & CHE124 & LSC141 & LSC142. LFS151 & LFS152 / LSC141 & MBS111 & MBS121	None	CFA	
CHM211	Chemistry 211	E	3	20	6	S1	CHE114, CHE124	None	CFA	
CHM212	Chemistry 212	E	3	20	6	S2	CHE114, CHE124	None	CFA	
BTN312	Biotechnology 312	C	4	30	6	S2	BTN211, BTN215, BTN222, BTN223	None	CFA	
BTN315	Biotechnology 315	C	4	30	7	S1	BTN211, BTN215, BTN222, BTN223	None	CFA	
BTN316	Biotechnology 316	C	4	30	7	S1	BTN211, BTN215, BTN222, BTN223	None	CFA	
BTN322	Biotechnology 322	C	4	30	7	S2	BTN211, BTN215, BTN222, BTN223	None	CFA	

BACHELOR OF SCIENCE IN CHEMICAL SCIENCES (3019)

B.118 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Chemical Sciences Degree - BSc (Chemical Sciences)**:

B.118.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 27 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional language), and
 - Level 3 (40-49%) in Mathematics, and
 - Level 3 (40-49%) in Physical Sciences

B.118.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Senate, upon recommendation of the faculty, may admit an applicant with a full Matriculation Exemption with Mathematics and either Science or Biology at Higher Grade with an E-symbol or Standard Grade with a D-symbol.

OR

- (b) A qualification or level of competence which the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.118.3 Alternative admission

Alternative admission requirements for applicants in terms of Recognition of Prior Learning (RPL)

An applicant who completed the NSC in 2008 or thereafter, but has not obtained an endorsement, shall be considered for alternative admission after the age of 23. Such an applicant shall be required to complete an RPL portfolio development course and submit to a process where relevant learning and/or experience shall be assessed.

An applicant who is 23 years or older and does not have a matriculation certificate or NSC, but who may have obtained other qualifications or experience that may be deemed to be equivalent to admission criteria for the particular study programmes shall also be required to complete an RPL portfolio development course or an RPL portfolio process as agreed upon, and to submit to a process where such learning, qualification, and/or experience shall be assessed.

B.119 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.120 DURATION

Unless Senate decides otherwise, the duration of the degree is four years.

B.121 CURRICULUM

B.121.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all modules)		
Physics 151(SF)	PHY151	15
Mathematics 151 (SF)	MAM151	15
Introduction to Science 153	ISC153	30
	Sub-total	60

B.121.2 Level 2

Module Name	Module Code	Credits
Compulsory (select all modules)		
Physics 152 (SF)	PHY152	15
Mathematics 152 (SF)	MAM152	15
Chemistry 114 (N)	CHE114	15
Chemistry 124 (N)	CHE124	15
	Sub-total	60

Please Note: A student who has passed CHE116 and CHM126 may proceed with the programme provided that they have obtained at least 60% in each of these modules, subject to approval by the HoD and the Dean.

B.121.3 Level 3

Module Name	Module Code	Credits
Compulsory (select all modules)		
Chemistry 211	CHM211	20
Chemistry 212	CHM212	20
Chemistry 221	CHM221	20
	Sub-total	60

Electives (select from Group 1 and Group 2)

Group 1 (select one sub-group)

Group 1.1 (select both modules)

Physics 212	PHY212	20
Physics 222	PHY222	20

Group 1.2 (select both modules)

Physics 217	PHY217	20
Physics 227	PHY227	20

Group 1.3 (select both modules)

Mathematics 211	MAT211	20
Mathematics 221	MAT221	20

Group 1.4 (select both modules)

Mathematics 211	MAT211	20
Mathematics 212	MAT212	20

Group 2**Select one module not selected in Group 1**

Physics 212	PHY212	20
Physics 217	PHY217	20
Mathematics 211	MAT211	20
	Sub-total	60
	Total	120

B.121.4 Level 4**Module Name****Compulsory (select all modules)**

Chemistry 311	CHM311	30
Chemistry 312	CHM312	30
Chemistry 321	CHM321	30
Chemistry 322	CHM322	30
	Total	120

FINAL TOTAL 360**B.122 ASSESSMENT**

Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.122.1 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.122.2 Weighting of practical component

B.122.2.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.122.2.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.122.3 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.122.4 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.122.5 Attendance of practicals

Where there is a practical component in a programme, a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.123 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.123.1 Level 1

A student shall promote to Level 2 of study on obtaining 45 credits and when all pre-requisites for Level 2 modules have been met.

B.123.2 Level 2

A student shall promote to Level 3 of study on obtaining 90 credits at Level 2 and when all pre-requisites for Level 3 modules have been met.

B.123.3 Level 3

A student shall promote to Level 4 of study on obtaining 200 credits and when all Level 1 and 2 modules have been passed.

B.124 ADVANCE REGISTRATION

B.124.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.124.2 Level 1

A student who has passed 2 of the 3 modules at Level 1 of study may be allowed to register for a maximum of 30 credits or one year-long module at Level 2.

B.124.3 Level 2

A student who has obtained 30 credits at Level 2 of study may be allowed to register for a maximum of 40 credits at Level 3 and completing all modules at Level 1. Total credits registered may not exceed 90.

B.124.4 Level 3

A student who has obtained 80 credits at Level 3 of study may be allowed to register for a maximum of 60 credits at Level 4 and completing all modules at Levels 1 and 2. Total credits registered may not exceed 90.

Please Note: A student must complete all Level 1 and 2 modules before registering for Level 4 modules.

B.125 RENEWAL OF REGISTRATION

B.125.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.125.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.126 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.126.1 Modules from other faculties

B.126.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.126.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.126.2 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.126.2.1 Chemistry: (CHE114 and CHE124) or (CHE116 and CHM126).

B.126.2.2 Physics: (PHY111 and PHY121) or (PHY116 and PHY126) or (PHY112 and PHY113 and PHY116) or (PHY151 and PHY152).

B.126.2.3 Mathematics: MAT105 or MAM115 or (MAM151 and MAM152).

B.126.2.4 Statistics: STA111 or STA125 or STA151 or BUS131 or BUS132 or STA141 or STA142 or STC101.

B.126.2.5 A student may not claim credits for modules passed prior to 1999.

B.126.3 Endorsed Degree

B.126.3.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.126.3.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
ISC153	Introduction to Science 153	C	1	30	5	YR	None	PHY151/ LFS151; MAM151/ MAM150	CA	
MAM151	Mathematics 151 (SF)	C	1	15	5	YR	None	None	CFA	
PHY151	Physics 151 (SF)	C	1	15	5	YR	None	MAM151	CFA	
MAM152	Mathematics 152 (SF)	C	2	15	5	YR	MAM151	None	CFA	
PHY152	Physics 152 (SF)	C	2	15	5	YR	None	None	CFA	
CHE114	Chemistry 114 (N)	C	2	15	5	S1	None	None	CFA	
CHE124	Chemistry 124 (N)	C	2	15	5	S2	None	None	CFA	
CHM211	Chemistry 211	C	3	20	6	S1	CHE114, CHE124	None	CFA	
CHM212	Chemistry 212	C	3	20	6	S2	CHE114, CHE124	None	CFA	
CHM221	Chemistry 221	C	3	20	6	S2	CHE114, CHE124	None	CFA	
MAT211	Mathematics 211	E	3	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	None	CFA	
MAT212	Mathematics 212	E	3	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	MAT211	CA	
MAT221	Mathematics 221	E	3	20	6	YR	None	MAT211	CFA	
PHY212	Physics 212	E	3	20	6	S1	PHY111/ PHY121 OR PHY151/ PHY152	MAT211/ MAT104/ MAT152	CFA	
PHY217	Physics 217	E	3	20	6	S1	None	None	CFA	
PHY222	Physics 222	E	3	20	6	S2	PHY111/ PHY121 OR	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
							PHY151/ PHY152 AND MAT105/ MAT103/ MAM152			
PHY227	Physics 227	E	3	20	6	S2	PHY116/ PHY126 OR PHY111/ PHY151	None	CFA	
CHM311	Chemistry 311	C	4	30	7	S1	CHM211/ CHM212	None	CFA	
CHM312	Chemistry 312	C	4	30	7	S2	CHM211/ CHM212	None	CFA	
CHM321	Chemistry 321	C	4	30	7	S1	CHM211/ CHM212/ CHM221	None	CFA	
CHM322	Chemistry 322	C	4	30	7	S1	CHM211/ CHM212/ CHM221	None	CFA	

BACHELOR OF SCIENCE IN COMPUTER SCIENCE (3023)

B.127 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Computer Science Degree - BSc (Computer Science)**:

B.127.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 27 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 3 (40-49%) in Mathematics, and
 - Level 3 (40-49%) in Physical Sciences, or
 - Level 3 (40-49%) in Life Sciences, or
 - Level 4 (50-59%) in Information Technology

B.127.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Senate, upon recommendation of the faculty, may admit an applicant with a full Matriculation Exemption with Mathematics and either Science or Biology at Higher Grade with an E-symbol or Standard Grade with a D-symbol.

OR

- (b) A qualification or level of competence which the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.127.3 Alternative admission

Alternative admission requirements for applicants in terms of Recognition of Prior Learning (RPL)

An applicant who completed the NSC in 2008 or thereafter, but has not obtained an endorsement, shall be considered for alternative admission after the age of 23. Such an applicant shall be required to complete an RPL portfolio development course and submit to a process where relevant learning and/or experience shall be assessed.

An applicant who is 23 years or older and does not have a matriculation certificate or NSC, but who may have obtained other qualifications or experience that may be deemed to be equivalent to admission criteria for the particular study programmes shall also be required to complete an RPL portfolio development course or an RPL portfolio process as agreed upon, and to submit to a process where such learning, qualification, and/or experience shall be assessed.

B.128 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.129 DURATION

Unless Senate decides otherwise, the duration of the degree is four years.

B.130 CURRICULUM

B.130.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all modules)		
Introduction to Science 153	ISC153	30
Mathematics 151(SF)	MAM151	15
Physics 151(SF)	PHY151	15
Sub-total		60

B.130.2 Level 2

Module Name	Module Code	Credits
Compulsory (select all modules)		
Computer Science 101	COS101	30
Mathematics 152 (SF)	MAM152	15
Sub-total		45
Electives (select one module)		
Mathematics 112	APM112	15
Applied Mathematics 123	APM123	15
Information Systems 131 or Information Systems 132	IFS131/132	15
Physics 152	PHY152	15
Statistics 151 (SF)	STA151	15
Sub-total		15
Total		60

B.130.3 Level 3

Module Name	Module Code	Credits
Compulsory (select both modules)		
Computer Science 211	CSC211	20
Computer Science 212	CSC212	20
Sub-total		40
Electives (select two groups)		
Group 1 (select both modules)		
Information Systems 231	IFS231	20
Information Systems 233	IFS233	20
Group 2 (select one sub-group)		
Group 2.1 (select all modules)		
Mathematics 211 (pre-requisite for Honours in Statistics)	MAT211	20
Mathematics 212	MAT212	20

Group 2.2 (select both modules)

Mathematics 211 (pre-requisite for Honours in Statistics)	MAT211	20
Mathematics 221 (pre-requisite for Honours in Statistics)	MAT221	20

Group 3 (select both modules)

Statistics 211	STA211	20
Statistics 221	STA221	20

Group 4 (select both modules)

Physics 212	PHY212	20
Physics 222	PHY222	20

Group 5 (select both modules)

Physics 217	PHY217	20
Physics 227	PHY227	20
	Sub-total	80
	Total	120

B.130.4 Level 4

Module Name	Module Code	Credits
Compulsory (select both modules)		
Computer Science 311	CSC311	30
Computer Science 312	CSC312	30
	Sub-total	60
Electives (select one group)		
Group 1 (select two modules)		
Mathematics 311	MAT311	30
Mathematics 312	MAT312	30
Mathematics 321	MAT321	30
Mathematics 322	MAT322	30
Group 2 (select one sub-group)		
Group 2.1 (select both modules)		
Statistics 331 (pre-requisite for Honours in Statistics)	STA331	30
Statistics 332 (pre-requisite for Honours in Statistics)	STA332	30
Group 3 (select both modules)		
Information Systems 316	IFS316	30
Information Systems 317	IFS317	30
Group 4 (select both modules)		
Physics 314	PHY314	30
Physics 324	PHY324	30
Group 5 (select both modules)		
Physics 317	PHY317	30
Physics 327	PHY327	30
	Sub-total	60
	Total	120
	FINAL TOTAL	360

B.131 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.131.1 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.131.2 Weighting of practical component

B.131.2.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.131.2.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.131.3 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.131.4 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.131.5 Attendance of practicals

Where there is a practical component in a programme, a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.132 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.132.1 Level 1

A student shall promote to Level 2 of study on obtaining 45 credits and when all pre-requisites for Level 2 modules have been met.

B.132.2 Level 2

A student shall promote to Level 3 of study on obtaining 90 credits at Level 2 and when all pre-requisites for Level 3 modules have been met.

B.132.3 Level 3

A student shall promote to Level 4 of study on obtaining 200 credits and when all Level 1 and 2 modules have been passed.

B.133 ADVANCE REGISTRATION

B.133.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.133.2 Level 1

A student who has passed 2 of the 3 modules at Level 1 of study may be allowed to register for a maximum of 30 credits or one year-long module at Level 2.

B.133.3 Level 2

A student who has obtained 30 credits at Level 2 of study may be allowed to register for a maximum of 40 credits at Level 3 and completing all modules at Level 1. Total credits registered may not exceed 90.

B.133.4 Level 3

A student who has obtained 80 credits at Level 3 of study may be allowed to register for a maximum of 60 credits at Level 4 and completing all modules at Levels 1 and 2. Total credits registered may not exceed 90.

Please Note: A student must complete all Levels 1 and 2 modules before registering for Level 4 modules.

B.134 RENEWAL OF REGISTRATION

B.134.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.134.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.135 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.135.1 Modules from other faculties

B.135.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.135.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.135.2 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.135.2.1 Chemistry: (CHE114 and CHE124) **or** (CHE116 and CHM126).

B.135.2.2 Physics: (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).

B.135.2.3 Mathematics: MAT105 **or** MAM115 **or** (MAM151 and MAM152).

B.135.2.4 Statistics: STA111 **or** STA125 **or** STA151 **or** BUS131 **or** BUS132 **or** STA141 **or** STA142 **or** STC101.

B.135.2.5 A student may not claim credits for modules passed prior to 1999.

B.135.3 Endorsed Degree

B.135.3.1 In order to obtain an "endorsed degree" a student must obtain at least 240 credits in the compulsory modules of the degree.

B.135.3.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
ISC153	Introduction to Science 153	C	1	30	5	YR	None	PHY151/ LFS151; MAM151/ MAM150	CA	
MAM151	Mathematics 151 (SF)	C	1	15	5	YR	None	None	CFA	
PHY151	Physics 151 (SF)	C	1	15	5	YR	None	MAM151	CFA	
APM112	Mathematics 112	E	2	15	5	S1	None	MAT105/ MAT104/ MAM152	CFA	
APM123	Applied Mathematics 123	E	2	15	5	YR	None	MAT105/ MAT104/ MAM152	CA	
COS101	Computer Science 101	C	2	30	5	YR	None	None	CA	
IFS131	Information Systems 131	E	2	15	5	S1	None	None	CA	SM
IFS132	Information Systems 132	E	2	15	5	S2	None	None	CA	SM
MAM152	Mathematics 152 (SF)	C	2	15	5	YR	MAM151	None	CFA	
PHY152	Physics 152 (SF)	C	2	15	5	YR	None	None	CFA	
STA151	Statistics 151 (SF)	E	2	15	5	YR	None	None	CFA	
CSC211	Computer Science 211	C	3	20	6	S1	COS101	None	CA	
CSC212	Computer Science 212	C	3	20	6	S2	COS101	None	CA	
IFS231	Information Systems 231	E	3	20	6	S1	IFS131/ IFS132, COS101/ COS114, COS124, EED127, ISC153, MAT151/ MAT103, MAT104, STA111, IFS131/ IFS132, COS101,	None	CA	SM

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
							PHY151, MAT151			
IFS233	Information Systems 233	E	3	20	6	S2	IFS231	None	CA	SM
MAT211	Mathematics 211 (pre-requisite for Honours in Statistics)	E	3	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	None	CFA	
MAT212	Mathematics 212	E	3	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	MAT211	CA	
MAT221	Mathematics 221 (pre-requisite for Honours in Statistics)	E	3	20	6	YR	None	MAT211	CFA	
PHY212	Physics 212	E	3	20	6	S1	PHY111/ PHY121 OR PHY151/ PHY152	MAT211/M AT104/ MAT152	CFA	
PHY217	Physics 217	E	3	20	6	S1	None	None	CFA	
PHY222	Physics 222	E	3	20	6	S2	PHY111/ PHY121 OR PHY151/ PHY152 AND MAT105/ MAT103/ MAM152	None	CFA	
PHY227	Physics 227	E	3	20	6	S2	PHY116/ PHY126 OR PHY111/ PHY151	None	CFA	
STA211	Statistics 211	E	3	20	6	S1	MAT105 or (MAM151 + MAM152) & STA111/ STA125/ STA151	None	CFA	
STA221	Statistics 221	E	3	20	6	S2	STA211 (at least qualified to write the exam for STA211)	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
CSC311	Computer Science 311	C	4	30	7	S1	CSC211 and CSC212	None	CA	
CSC312	Computer Science 312	C	4	30	7	S2	CSC211 and CSC212	None	CA	
IFS316	Information Systems 316	E	4	30	7	S1	IFS233 and IFS244	None	CA	SM
IFS317	Information Systems 317	E	4	30	7	S2	IFS316	None	CA	SM
MAT311	Mathematics 311	E	4	30	7	YR	MAT211	None	CFA	
MAT312	Mathematics 312	E	4	30	7	YR	MAT211, MAT212/ MAT221	None	CA	
MAT321	Mathematics 321	E	4	30	7	YR	MAT221	None	CFA	
MAT322	Mathematics 322	E	4	30	7	YR	MAT211, MAT212/ MAT221	None	CA	
PHY314	Physics 314	E	4	30	7	S1	PHY212 PHY222 MAT211 MAT221	None	CFA	
PHY317	Physics 317	E	4	30	7	S1	PHY227	None	CFA	
PHY324	Physics 324	E	4	30	7	S2	PHY212 PHY222 MAT211 MAT221	None	CFA	
PHY327	Physics 327	E	4	30	7	S2	PHY212/ PHY217, MAT105/ MAT103/ MAM152	None	CA	
STA331	Statistics 331 (pre-requisite for Honours in Statistics)	E	4	30	7	S1	STA211, STA221	None	CFA	
STA332	Statistics 332 (pre-requisite for Honours in Statistics)	E	4	30	7	S2	STA211, STA221	None	CFA	

BACHELOR OF SCIENCE IN ENVIRONMENT AND WATER SCIENCE (3027)

B.136 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Environment and Water Science Degree - BSc (Environment and Water Science)**:

B.136.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 27 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 3 (40-49%) in Mathematics, and
 - Level 3 (40-49%) in Physical Sciences, or
 - Level 3 (40-49%) in Life Sciences

B.136.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Senate, upon recommendation of the faculty, may admit an applicant with a full Matriculation Exemption with Mathematics and either Science or Biology at Higher Grade with an E-symbol or Standard Grade with a D-symbol.

OR

- (b) A qualification or level of competence which the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.136.3 Alternative admission

Alternative admission requirements for applicants in terms of Recognition of Prior Learning (RPL)

An applicant who completed the NSC in 2008 or thereafter, but has not obtained an endorsement, shall be considered for alternative admission after the age of 23. Such an applicant shall be required to complete an RPL portfolio development course and submit to a process where relevant learning and/or experience shall be assessed.

An applicant who is 23 years or older and does not have a matriculation certificate or NSC, but who may have obtained other qualifications or experience that may be deemed to be equivalent to admission criteria for the particular study programmes shall also be required to complete an RPL portfolio development course or an RPL portfolio process as agreed upon, and to submit to a process where such learning, qualification, and/or experience shall be assessed.

B.137 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.138 DURATION

Unless Senate decides otherwise, the duration of the degree is four years.

B.139 CURRICULUM

B.139.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all modules)		
Introduction to Science 153	ISC153	30
Life Sciences 151 (SF)	LFS151	15
Introduction to Mathematics 150	MAM150	15
	Sub-total	60

B.139.2 Level 2

Module Name	Module Code	Credits
Compulsory (select all groups)		
Group 1 (select both modules)		
Life Sciences 152 (SF)	LFS152	15
Statistics 151 (SF)	STA151	15
Group 2 (select both modules)		
Earth Science 111	ESC111	15
Earth Science 121	ESC121	15
Group 3 (select one module)		
Chemistry 116 (N)	CHE116	15
Chemistry 126 (N)	CHM126	15
	Sub-total	75

B.139.3 Level 3

Module Name	Module Code	Credits
Compulsory (selected all modules)		
Applied Geology 232	APG232	20
Environmental and Water Science 211	EWP211	20
Environmental and Water Science 221	EWP221	20
Environmental and Water Science 222	EWP222	20
	Sub-total	80
Electives (select a total of 40 credits below)		
Biodiversity and Conservation Biology 211	BDC211	20
Biodiversity and Conservation Biology 221	BDC221	20
Biodiversity and Conservation Biology 222	BDC222	20
Biodiversity and Conservation Biology 223	BDC223	20
	Sub-total	40
	Total	120

B.139.4 Level 4**Module Name****Compulsory (select all modules)**

Geographical Information Systems 311
 Environmental and Water Science 312
 Environmental and Water Science 321
 Environmental and Water Science 322

Module Code

EWP311
 EWP312
 EWP321
 EWP322

Credits

30
 30
 30
 30

Total**120****FINAL TOTAL****375****B.140 ASSESSMENT**

Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.140.1 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.140.2 Weighting of practical component

B.140.2.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.140.2.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.140.3 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.140.4 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.140.5 Attendance of practicals

Where there is a practical component in a programme, a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.141 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.141.1 Level 1

A student shall promote to Level 2 of study on obtaining 45 credits and when all pre-requisites for Level 2 modules have been met.

B.141.2 Level 2

A student shall promote to Level 3 of study on obtaining 105 credits at Level 2 and when all pre-requisites for Level 3 modules have been met.

B.141.3 Level 3

A student shall promote to Level 4 of study on obtaining 215 credits and when all Level 1 and 2 modules have been passed and when all pre-requisites for admission to Level 4 modules have been met.

B.142 ADVANCE REGISTRATION

B.142.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.142.2 Level 1

A student who has passed 2 of the 3 modules at Level 1 of study may be allowed to register for a maximum of 30 credits or one year-long module at Level 2.

B.142.3 Level 2

A student who has obtained 30 credits at Level 2 of study may be allowed to register for a maximum of 40 credits at Level 3 and completing all modules at Level 1. Total credits registered may not exceed 90.

B.142.4 Level 3

A student who has obtained 80 credits at Level 3 of study may be allowed to register for a maximum of 60 credits at Level 4 and completing all modules at Levels 1 and 2. Total credits registered may not exceed 90.

Please Note: A student must complete all Level 1 and 2 modules before registering for Level 4 modules.

B.143 RENEWAL OF REGISTRATION

B.143.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.143.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic

performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.144 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.144.1 Modules from other Faculties

B.144.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.144.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.144.2 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.144.2.1 Chemistry: (CHE114 and CHE124) **or** (CHE116 and CHM126).

B.144.2.2 Physics: (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).

B.144.2.3 Mathematics: MAT105 **or** MAM115 **or** (MAM151 and MAM152).

B.144.2.4 Statistics: STA111 **or** STA125 **or** STA151 **or** BUS131 **or** BUS132 **or** STA141 **or** STA142 **or** STC101.

B.144.2.5 A student may not claim credits for modules passed prior to 1999.

B.144.3 Endorsed Degree

B.144.3.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.144.3.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
ISC153	Introduction to Science 153	C	1	30	5	YR	None	PHY151/ LFS151 MAM151/ MAM150	CA	
LFS151	Life Sciences 151 (SF)	C	1	15	5	YR	None	None	CFA	
MAM150	Introduction to Mathematics 150	C	1	15	5	YR	None	None	CFA	
CHE116	Chemistry 116 (N)	C/E	2	15	5	S1	None	None	CFA	
CHM126	Chemistry 126 (N)	C/E	2	15	5	S2	None	None	CFA	
ESC111	Earth Science 111	C	2	15	5	S1	None	None	CFA	
ESC121	Earth Science 121	C	2	15	5	S2	None	None	CFA	
LFS152	Life Sciences 152 (SF)	C	2	15	5	YR	None	None	CFA	
STA151	Statistics 151 (SF)	C	2	15	5	YR	None	None	CFA	
APG232	Geology 232	C	3	20	6	S1	ESC111, ESC121	None	CFA	
BDC211	Biodiversity and Conservation Biology 211	E	3	20	6	S2	LSC142 or LFS152	None	CFA	
BDC221	Biodiversity and Conservation Biology 221	E	3	20	6	S1	LSC142 or LFS152	None	CFA	
BDC222	Biodiversity and Conservation Biology 222	E	3	20	6	S1	LSC141/142 or LFS151/ 152	None	CFA	
BDC223	Biodiversity and Conservation Biology 223	E	3	20	6	S2	LSC141 & LSC142, or LFS151 & LFS152; & CHE114 or CHE124 or CHE116 or CHM126	None	CFA	
EWP211	Environmental and Water Science 211	C	3	20	6	S1	ESC111, ESC121	None	CFA	
EWP221	Environmental and Water Science 221	C	3	20	6	S2	ESC111, ESC121	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
EWP222	Environmental and Water Science 222	C	3	20	6	S1	ESC111, ESC121	EWP211, EWP221	CA	
EWP311	Geographical Information Systems 311	C	4	30	7	S1	EWP211, EWP221, EWP222	None	CFA	
EWP312	Environmental and Water Science 312	C	4	30	7	S1	EWP211, EWP221, EWP222	None	CFA	
EWP321	Environmental and Water Science 321	C	4	30	7	S2	EWP211, EWP221	None	CFA	
EWP322	Environmental and Water Science 322	C	4	30	7	S2	EWP211, EWP221, EWP222	None	CFA	

BACHELOR OF SCIENCE IN MATHEMATICS AND STATISTICAL SCIENCES (3031)

B.145 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Mathematics and Statistical Sciences Degree - BSc (Mathematics and Statistical Sciences)**:

B.145.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 27 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 3 (40-49%) in Mathematics, and
 - Level 3 (40-49%) in Physical Sciences, or
 - Level 3 (40-49%) in Life Sciences, or
 - Level 4 (50-59%) in Information Technology

B.145.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Senate, upon recommendation of the faculty, may admit an applicant with a full Matriculation Exemption with Mathematics and either Science or Biology at Higher Grade with an E-symbol or Standard Grade with a D-symbol.

OR

- (b) A qualification or level of competence which the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.145.3 Alternative admission

Alternative admission requirements for applicants in terms of Recognition of Prior Learning (RPL)

An applicant who completed the NSC in 2008 or thereafter, but has not obtained an endorsement, shall be considered for alternative admission after the age of 23. Such an applicant shall be required to complete an RPL portfolio development course and submit to a process where relevant learning and/or experience shall be assessed.

An applicant who is 23 years or older and does not have a matriculation certificate or NSC, but who may have obtained other qualifications or experience that may be deemed to be equivalent to admission criteria for the particular study programmes shall also be required to complete an RPL portfolio development course or an RPL portfolio process as agreed upon, and to submit to a process where such learning, qualification, and/or experience shall be assessed.

B.146 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.147 DURATION

Unless Senate decides otherwise, the duration of the degree is four years.

B.148 CURRICULUM

Mathematics Stream

B.148.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all modules)		
Introduction to Science 153	ISC153	30
Mathematics 151(SF)	MAM151	15
Physics 151(SF)	PHY151	15
	Sub-total	60

B.148.2 Level 2

Module Name	Module Code	Credits
Compulsory (select all modules)		
Mathematics 152 (SF)	MAM152	15
Statistics 111	STA111	15
Statistics 121	STA121	15
Electives (select one module)		
Mathematics 112	APM112	15
Applied Mathematics 123	APM123	15
Physics 152 (SF)	PHY152	15
	Sub-total	60

B.148.3 Level 3

Module Name	Module Code	Credits
Compulsory (select both modules)		
Mathematics 211	MAT211	20
Mathematics 221	MAT221	20
	Sub-total	40
Electives (select four modules)		
Mathematics 212	MAT212	20
Mathematics 232	MAT232	20
Physics 212	PHY212	20
Physics 217	PHY217	20
Physics 222	PHY222	20
Physics 227	PHY227	20
Statistics 211	STA211	20
Statistics 221	STA221	20
	Sub-total	80
	Total	120

B.148.4 Level 4

Module Name	Module Code	Credits
Compulsory (select all groups)		
Group 1		
Mathematics 311	MAT311	30
Group 2 (select one module)		
Mathematics 312	MAT312	30
Mathematics 321	MAT321	30
Mathematics 322	MAT322	30
	Sub-total	60
Electives (select two modules)		
Mathematics 312 (only if not selected in the compulsory list)	MAT312	30
Mathematics 321 (only if not selected in the compulsory list)	MAT321	30
Mathematics 322 (only if not selected in the compulsory list)	MAT322	30
Statistics 331	STA331	30
Statistics 332	STA332	30
	Sub-total	60
	Total	120
	FINAL TOTAL	360

Statistics Stream**B.148.5 Level 1**

Module Name	Module Code	Credits
Compulsory (select all modules)		
Mathematics 151(SF)	MAM151	15
Introduction to Science 153	ISC153	30
Statistics 151 (SF)	STA151	15
	Sub-total	60

B.148.6 Level 2

Module Name	Module Code	Credits
Compulsory (select all modules)		
Mathematics 152 (SF)	MAM152	15
Statistics 121	STA121	15
Electives (select 30 credits)		
Mathematics 112	APM112	15
Applied Mathematics 123	APM123	15
Computer Science 101	COS101	30
	Sub-total	60

B.148.7 Level 3

Module Name	Module Code	Credits
Compulsory (select all modules)		
Statistics 211	STA211	20
Statistics 221	STA221	20
	Sub-total	40
Electives (select four modules)		
Computer Science 211	CSC211	20
Computer Science 212	CSC212	20
Mathematics 211 (pre-requisite for Honours in Statistics)	MAT211	20
Mathematics 212	MAT212	20
Mathematics 221 (pre-requisite for Honours in Statistics)	MAT221	20
Mathematics 232	MAT232	20
	Sub-total	80
	Total	120

B.148.8 Level 4

Module Name	Module Code	Credits
Compulsory (select all modules)		
Statistics 331 (pre-requisite for Honours in Statistics)	STA331	30
Statistics 332 (pre-requisite for Honours in Statistics)	STA332	30
	Sub-total	60
Electives (select one group)		
Group 1 (select two modules)		
Mathematics 311	MAT311	30
Mathematics 312	MAT312	30
Mathematics 321	MAT321	30
Mathematics 322	MAT322	30
Group 2 (select both modules)		
Computer Science 311	CSC311	30
Computer Science 312	CSC312	30
	Sub-total	60
	Total	120
	FINAL TOTAL	360

B.149 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.149.1 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.149.2 Weighting of practical component

B.149.2.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.149.2.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.149.3 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.149.4 Re-evaluation of practical modules

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.150 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.150.1 Level 1

A student shall promote to Level 2 of study on obtaining 45 credits and when all pre-requisites for Level 2 modules have been met.

B.150.2 Level 2

A student shall promote to Level 3 of study on obtaining 90 credits at Level 2 and when all pre-requisites for Level 3 modules have been met.

B.150.3 Level 3

A student shall promote to Level 4 of study on obtaining 200 credits and when all Level 1 and 2 modules have been passed.

B.151 ADVANCE REGISTRATION

B.151.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.151.2 Level 1

A student who has passed 2 of the 3 modules at Level 1 of study may be allowed to register for a maximum of 30 credits or one year-long module at Level 2.

B.151.3 Level 2

A student who has obtained 30 credits at Level 2 of study may be allowed to register for a maximum of 40 credits at Level 3 and completing all modules at Level 1. Total credits registered may not exceed 90.

B.151.4 Level 3

A student who has obtained 80 credits at Level 3 of study may be allowed to register for a maximum of 60 credits at Level 4 and completing all modules at Levels 1 and 2. Total credits registered may not exceed 90.

Please Note: A student must complete all Level 1 and 2 modules before registering for Level 4 modules.

B.152 RENEWAL OF REGISTRATION

B.152.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.152.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.153 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.153.1 Modules from other Faculties

B.153.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.153.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.153.2 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.153.2.1 Chemistry: (CHE114 and CHE124) **or** (CHE116 and CHM126).

B.153.2.2 Physics: (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).

B.153.2.3 Mathematics: MAT105 **or** MAM115 **or** (MAM151 and MAM152).

B.153.2.4 Statistics: STA111 or STA125 or STA151 or BUS131 or BUS132 or STA141 or STA142 or STC101.

B.153.2.5 A student may not claim credits for modules passed prior to 1999.

B.154. Endorsed Degree

B.154.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.154.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
MATHEMATICS STREAM										
ISC153	Introduction to Science 153	C	1	30	5	YR	None	PHY151/ LFS151; MAM151/ MAM150	CA	
MAM151	Mathematics 151(SF)	C	1	15	5	YR	None	None	CFA	
PHY151	Physics 151 (SF)	C	1	15	5	YR	None	MAM151	CFA	
APM112	Mathematics 112	E	2	15	5	S1	None	MAT105/ MAT104/ MAM152	CFA	
APM123	Applied Mathematics 123	E	2	15	5	YR	None	MAT105/ MAT104/ MAM152	CA	
MAM152	Mathematics 152 (SF)	C	2	15	5	YR	MAM151	None	CFA	
PHY152	Physics 152 (SF)	C	2	15	5	YR	None	None	CFA	
STA111	Statistics 111	C	2	15	5	S1	None	None	CFA	
STA121	Statistics 121	C	2	15	5	S2	None	STA111/ 125/151 or BUS131/ 132 or STA141/ 142	CFA	
MAT211	Mathematics 211	C	3	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
MAT212	Mathematics 212	E	3	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	MAT211	CA	
MAT221	Mathematics 221	C	3	20	6	YR	None	MAT211	CFA	
MAT232	Mathematics 232	E	3	20	6	S1	APM112, MAT105/ MAT103	None	CFA	
PHY212	Physics 212	E	3	20	6	S1	PHY111/ PHY121 OR PHY151/ PHY152	MAT211/ MAT104/ MAT152	CFA	
PHY217	Physics 217	E	3	20	6	S1	None	None	CFA	
PHY222	Physics 222	E	3	20	6	S2	PHY111/ PHY121 OR PHY151/ PHY152 AND MAT105/ MAT103/ MAM152	None	CFA	
PHY227	Physics 227	E	3	20	6	S2	PHY116/ PHY126 OR PHY111/ PHY151	None	CFA	
STA211	Statistics 211	E	3	20	6	S1	MAT105 or (MAM151 + MAM152) & STA111/ STA125/ STA151	None	CFA	
STA221	Statistics 221	E	3	20	6	S2	STA211 (at least qualified to write the exam for STA211)	None	CFA	
MAT311	Mathematics 311	C/E	4	30	7	YR	MAT211	None	CFA	
MAT312	Mathematics 312	C/E	4	30	7	YR	MAT211, MAT212/ MAT221	None	CA	
MAT321	Mathematics 321	C/E	4	30	7	YR	MAT221	None	CFA	
MAT322	Mathematics 322	C/E	4	30	7	YR	MAT211, MAT212/ MAT221	None	CA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
STATISTICS STREAM										
ISC153	Introduction to Science 153	C	1	30	5	YR	None	PHY151/ LFS151; MAM151/ MAM150	CA	
MAM151	Mathematics 151(SF)	C	1	15	5	YR	None	None	CFA	
STA151	Statistics 151 (SF)	C	1	15	5	YR	None	None	CFA	
APM112	Mathematics 112	E	2	15	5	S1	None	MAT105/ MAT104/ MAM152	CFA	
APM123	Applied Mathematics 123	E	2	15	5	YR	None	MAT105/ MAT104/ MAM152	CA	
COS101	Computer Science 101	E	2	30	5	YR	None	None	CA	
CSC211	Computer Science 211	E	2	20	6	S1	COS101	None	CA	
CSC212	Computer Science 212	E	2	20	6	S2	COS101	None	CA	
MAM152	Mathematics 152 (SF)	C	2	15	5	YR	MAM151	None	CFA	
STA121	Statistics 121	C	2	15	5	S2	None	STA111/ 125/ 151 or BUS131/13 2 or STA141/14 2	CFA	
CSC311	Computer Science 311	E	3	30	7	S1	CSC211 and CSC212	None	CA	
CSC312	Computer Science 312	E	3	30	7	S2	CSC211 and CSC212	None	CA	
MAT211	Mathematics 211	C	3	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	None	CFA	
MAT212	Mathematics 212	E	3	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	MAT211	CA	
MAT221	Mathematics 221	C	3	20	6	YR	None	MAT211	CFA	
MAT232	Mathematics 232	E	3	20	6	S1	APM112, MAT105/ MAT103	None	CFA	
STA211	Statistics 211	E	3	20	6	S1	MAT105 or (MAM151 + MAM152) & STA111/ STA125/ STA151	None	CFA	
STA221	Statistics 221	E	3	20	6	S2	STA211 (at least qualified to write the exam for STA211)	None	CFA	
MAT311	Mathematics 311	C/E	4	30	7	YR	MAT211	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
MAT312	Mathematics 312	C/E	4	30	7	YR	MAT211, MAT212/ MAT221	None	CA	
MAT321	Mathematics 321	C/E	4	30	7	YR	MAT221	None	CFA	
MAT322	Mathematics 322	C/E	4	30	7	YR	MAT211, MAT212/ MAT221	None	CA	
STA331	Statistics 331 (pre- requisite for Honours in Statistics)	E	4	30	7	S1	STA211, STA221	None	CFA	
STA332	Statistics 332 (pre- requisite for Honours in Statistics)	E	4	30	7	S2	STA211, STA221	None	CFA	

BACHELOR OF SCIENCE IN MEDICAL BIOSCIENCE (3035)

B.155 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Medical Bioscience Degree - BSc (Medical Bioscience)**:

B.155.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 27 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional Language), and
 - Level 3 (40-49%) in Mathematics, and
 - Level 3 (40-49%) in Life Sciences

B.155.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Senate, upon recommendation of the faculty, may admit an applicant with a full Matriculation Exemption with Mathematics and either Science or Biology at Higher Grade with an E-symbol or Standard Grade with a D-symbol.

OR

- (b) A qualification or level of competence which the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.155.3 Alternative admission

Alternative admission requirements for applicants in terms of Recognition of Prior Learning (RPL)

An applicant who completed the NSC in 2008 or thereafter, but has not obtained an endorsement, shall be considered for alternative admission after the age of 23. Such an applicant shall be required to complete an RPL portfolio development course and submit to a process where relevant learning and/or experience shall be assessed.

An applicant who is 23 years or older and does not have a matriculation certificate or NSC, but who may have obtained other qualifications or experience that may be deemed to be equivalent to admission criteria for the particular study programmes shall also be required to complete an RPL portfolio development course or an RPL portfolio process as agreed upon, and to submit to a process where such learning, qualification, and/or experience shall be assessed.

B.156 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.157 DURATION

Unless Senate decides otherwise, the duration of the degree is four years.

B.158 CURRICULUM

B.158.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all modules)		
Life Sciences 151 (SF)	LFS151	15
Introduction to Mathematics 150	MAM150	15
Introduction to Science 153	ISC153	30
	Sub-total	60

B.158.2 Level 2

Module Name	Module Code	Credits
Compulsory (select all modules)		
Statistics 151 (SF)	STA151	15
Medical Biosciences 111	MBS111	15
Medical Biosciences 121	MBS121	15
Chemistry 116 (N)	CHE116	15
Chemistry 126 (N)	CHM126	15
	Sub-total	75

B.158.3 Level 3

Module Name	Module Code	Credits
Compulsory (select all modules)		
Medical Biosciences 231	MBS231	20
Medical Biosciences 232	MBS232	20
Medical Microbiology 251	MIC251	20
Medical Microbiology 252	MIC252	20
	Sub-total	80
Electives (select two modules)		
Biotechnology 211	BTN211	20
Biotechnology 223	BTN223	20
Biotechnology 213	BTN213	20
	Sub-total	40
	Total	120

B.158.4 Level 4

Module Name	Module Code	Credits
Compulsory (select all modules)		
Medical Biosciences 331	MBS331	30
Medical Biosciences 332	MBS332	30
Medical Microbiology 353	MIC353	30
Medical Microbiology 354	MIC354	30
	Total	120
	FINAL TOTAL	375

B.159 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.159.1 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.159.2 Weighting of practical component

B.159.2.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.159.2.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.159.3 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.159.4 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.159.5 Attendance of practicals

Where there is a practical component in a programme, a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.160 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.160.1 Level 1

A student shall promote to Level 2 of study on obtaining 45 credits and when all pre-requisites for Level 2 modules have been met.

B.160.2 Level 2

A student shall promote to Level 3 of study on obtaining 105 credits at Level 2 and when all pre-requisites for Level 3 modules have been met.

B.160.3 Level 3

A student shall promote to Level 4 of study on obtaining 200 credits and when all Level 1 and 2 modules have been passed.

B.161 ADVANCE REGISTRATION

B.161.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.161.2 Level 1

A student who has passed 2 of the 3 modules at Level 1 of study may be allowed to register for a maximum of 30 credits or one year-long module at Level 2.

B.161.3 Level 2

A student who has obtained 30 credits at Level 2 of study may be allowed to register for a maximum of 40 credits at Level 3 and completing all modules at Level 1. Total credits registered may not exceed 90.

B.161.4 Level 3

A student who has obtained 80 credits at Level 3 of study may be allowed to register for a maximum of 60 credits at Level 4 and completing all modules at Levels 1 and 2. Total credits registered may not exceed 90.

Please Note: A student must complete all Level 1 and 2 modules before registering for Level 4 modules.

B.162 RENEWAL OF REGISTRATION

B.162.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.162.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.163 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.163.1 Modules from other faculties

B.163.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.163.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.163.2 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.163.2.1 Chemistry: (CHE114 and CHE124) **or** (CHE116 and CHM126).

B.163.2.2 Physics: (PHY111 and PHY121) **or** (PHY116 and PHY126) **or** (PHY112 and PHY113 and PHY116) **or** (PHY151 and PHY152).

B.163.2.3 Mathematics: MAT105 **or** MAM115 **or** (MAM151 and MAM152).

B.163.2.4 Statistics: STA111 **or** STA125 **or** STA151 **or** BUS131 **or** BUS132 **or** STA141 **or** STA142 **or** STC101.

B.163.2.5 A student may not claim credits for modules passed prior to 1999.

B.163.3 Endorsed Degree

B.163.3.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.163.3.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requirement	Assessment Type	Service Module
ISC153	Introduction to Science 153	C	1	30	5	YR	None	PHY151/ LFS151; MAM151/ MAM150	CA	
LFS151	Life Sciences 151 (SF)	C	1	15	5	YR	None	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
MAM150	Introduction to Mathematics 150	C	1	15	5	YR	None	None	CFA	
CHE116	Chemistry 116 (N)	C	2	15	5	S1	None	None	CFA	
CHM126	Chemistry 126 (N)	C	2	15	5	S2	None	None	CFA	
MBS111	Medical Biosciences 111	C	2	15	5	S1	None	None	CFA	
MBS121	Medical Biosciences 121	C	2	15	5	S2	None	None	CFA	
STA151	Statistics 151 (SF)	C	2	15	5	YR	None	None	CFA	
BTN211	Biotechnology 211	E	3	20	6	S1	CHE114/ CHE124 LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121	None	CA	
BTN213	Biotechnology 213	E	3	20	6	S2	CHE114/ CHE124 LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121	None	CFA	
BTN223	Biotechnology 223	E	3	20	6	S2	CHE114/ CHE124 LSC141/ LSC142 LFS151/ LFS152 LSC141/ MBS111/ MBS121	None	CA	
MBS231	Medical Biosciences 231	C	3	20	6	S1	LSC141/ MBS111, MBS121	None	CFA	
MBS232	Medical Biosciences 232	C	3	20	6	S2	LSC141/ MBS111, MBS121	None	CFA	
MIC251	Medical Microbiology 251	C	3	20	6	S1	LSC141	None	CFA	
MIC252	Medical Microbiology 252	C	3	20	6	S2	MIC251	None	CFA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
MBS331	Medical Biosciences 331	C	4	30	7	S1	MBS 231 and MBS232	None	CFA	
MBS332	Medical Biosciences 332	C	4	30	7	S2	MBS231, MBS232 and MBS331	None	CFA	
MIC353	Medical Microbiology 353	C	4	30	7	S1	MIC251 and MIC252	None	CFA	
MIC354	Medical Microbiology 354	C	4	30	7	S2	MIC251 and MIC252	None	CFA	

BACHELOR OF SCIENCE IN PHYSICAL SCIENCE (3120)

B.164 ADMISSION

Unless Senate decides otherwise, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science in Physical Science Degree - BSc (Physical Science)**:

B.164.1 Minimum admission requirements for applicants who matriculated from 2008

- (a) The National Senior Certificate for Bachelor's Degree study with a score of no fewer than 27 points calculated according to the University's approved points system, as well as the following specific subject requirements:
- Level 4 (50-59%) in English (Home or First Additional Language), and
 - Level 3 (40-49%) in another Language (Home or First Additional language), and
 - Level 3 (40-49%) in Mathematics, and
 - Level 3 (40-49%) in Physical Sciences

B.164.2 Minimum admission requirements for applicants who matriculated before 2008

- (a) Senate, upon recommendation of the faculty, may admit an applicant with a full Matriculation Exemption with Mathematics and either Science or Biology at Higher Grade with an E-symbol or Standard Grade with a D-symbol.

OR

- (b) A qualification or level of competence which the Senate of the University has deemed to be equivalent to the requirements stipulated in (a) above.

B.164.3 Alternative admission

Alternative admission requirements for applicants in terms of Prior Learning (RPL)

An applicant who completed the NSC in 2008 or thereafter, but has not obtained an endorsement, shall be considered for alternative admission after the age of 23. Such an applicant shall be required to complete an RPL portfolio development course and submit to a process where relevant learning and/or experience shall be assessed.

An applicant who is 23 years or older and does not have a matriculation certificate or NSC, but who may have obtained other qualifications or experience that may be deemed to be equivalent to admission criteria for the particular study programmes shall also be required to complete an RPL portfolio development course or an RPL portfolio process as agreed upon, and to submit to a process where such learning, qualification, and/or experience shall be assessed.

B.165 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.166 DURATION

Unless Senate decides otherwise, the duration of the degree is four years.

B.167 CURRICULUM

B.167.1 Level 1

Module Name	Module Code	Credits
Compulsory (select all modules)		
Mathematics 151 (SF)	MAM151	15
Physics 151(SF)	PHY151	15
Introduction to Science 153	ISC153	30
	Sub-total	60

B.167.2 Level 2

Module Name	Module Code	Credits
Compulsory (select both modules)		
Mathematics 152 (SF)	MAM152	15
Physics 152 (SF)	PHY152	15
	Sub-total	30

Electives (select one group)

Group 1 (select both modules)

Chemistry 114 (N)	CHE114	15
Chemistry 124 (N)	CHE124	15

Group 2 (select both modules)

Mathematics 112	APM112	15
Applied Mathematics 123	APM123	15

Group 3

Computer Science 101	COS101	30
	Sub-total	30
	Total	60

B.167.3 Level 3

Module Name	Module Code	Credits
Compulsory (select all modules)		
Physics 212	PHY212	20
Physics 222	PHY222	20
Mathematics 211	MAT211	20
Mathematics 221	MAT221	20
	Sub-total	80

Electives (select one group)

Group 1 (select two modules)

Chemistry 211	CHM211	20
Chemistry 212	CHM212	20
Mathematics 212	MAT212	20
Mathematics 232	MAT232	20
Physics 217	PHY217	20
Physics 227	PHY227	20

Group 2 (select both modules)

Computer Science 211	CSC211	20
Computer Science 212	CSC212	20
	Sub-total	40
	Total	120

B.167.4 Level 4

Module Name	Module Code	Credits
Compulsory (select both modules)		
Physics 314	PHY314	30
Physics 324	PHY324	30
	Sub-total	60
Electives (select two modules)		
Chemistry 311	CHM311	30
Chemistry 312	CHM312	30
Computer Science 311	CSC311	30
Computer Science 312	CSC312	30
Mathematics 311	MAT311	30
Mathematics 321	MAT321	30
Mathematics 312	MAT312	30
Mathematics 322	MAT322	30
Physics 317	PHY317	30
Physics 327	PHY327	30
	Sub-total	60
	Total	120
	FINAL TOTAL	360

B.168 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the General Calendar: General Information Part 1 in addition to the following:

B.168.1 Sub-minimum requirement

A sub-minimum of 40% is required in the practical part for modules which include a practical component.

B.168.2 Weighting of practical component

B.168.2.1 The percentage of the total marks allocated to the practical examination where it forms part of the theory evaluation shall not exceed 50%.

B.168.2.2 For the purpose of promotion, the practical component shall not be credited independently of the theory component of a particular module.

B.168.3 Final mark

Except where otherwise indicated, the final mark shall be obtained by means of a system of continuous evaluation, which will differ from module to module, combined with examinations, as indicated in the University's Assessment Policy.

B.168.4 Re-evaluation of practical components

No re-evaluation will be allowed for practical components that are evaluated on a continuous basis.

B.168.5 Attendance of practicals

Where there is a practical component in a programme, a minimum attendance of 80% of the practicals is required to gain access to the examination, in addition to the 40% sub-minimum requirement needed for the coursework mark.

B.169 PROMOTION RULES

Unless Senate decides otherwise and subject to Rule A.3.2.3 or as provided for in the Faculty rules:

B.169.1 Level 1

A student shall promote to Level 2 of study on obtaining 45 credits and when all pre-requisites for Level 2 modules have been met.

B.169.2 Level 2

A student shall promote to Level 3 of study on obtaining 90 credits at Level 2 and when all pre-requisites for Level 3 modules have been met.

B.169.3 Level 3

A student shall promote to Level 4 of study on obtaining 200 credits and when all Levels 1 and 2 modules have been passed.

B.170 ADVANCE REGISTRATION

B.170.1 Notwithstanding the above, in consultation with the faculty a student who has not obtained sufficient credits to promote may register for modules at the next level, provided all pre-requisites for the next level module have been met and no timetable clashes occur as a result of this registration.

B.170.2 Level 1

A student who has passed 2 of the 3 modules at Level 1 of study may be allowed to register for a maximum of 30 credits or one year-long module at Level 2.

B.170.3 Level 2

A student who has obtained 30 credits at Level 2 of study may be allowed to register for a maximum of 40 credits at Level 3 and completing all modules at Level 1. Total credits registered may not exceed 90.

B.170.4 Level 3

A student who has obtained 80 credits at Level 3 of study may be allowed to register for a maximum of 60 credits at Level 4 and completing all modules at Levels 1 and 2. Total credits registered may not exceed 90.

Please Note: A student must complete all Level 1 and 2 modules before registering for Level 4 modules.

B.171 RENEWAL OF REGISTRATION

B.171.1 Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.171.2 Unless Senate decides otherwise, a student who is academically refused in terms of Rule A.3.2.3 may apply in writing to the Dean for renewal of registration if they can provide evidence of extenuating circumstances which impacted on their academic performance, and can show that these circumstances have changed to the extent that success upon re-registration can reasonably be expected.

Please Note: This rule may only be invoked once in the entire duration of the degree programme.

B.172 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.172.1 Modules from other faculties

B.172.1.1 A maximum of 30 credits in modules from other faculties shall be permitted, except for Computer Science students who shall be allowed to take Information Systems as a module in their programme.

B.172.1.2 At least 100 of the total number of credits obtained at Levels 2 and 3 of study shall be from the same programme.

B.172.2 Restriction on choice of modules

The following limitations apply to modules that can be credited towards obtaining a degree or certificate. Credit shall not be granted for:

B.172.2.1 Chemistry: (CHE114 and CHE124) or (CHE116 and CHM126).

B.172.2.2 Physics: (PHY111 and PHY121) or (PHY116 and PHY126) or (PHY112 and PHY113 and PHY116) or (PHY151 and PHY152).

B.172.2.3 Mathematics: MAT105 or MAM115 or (MAM151 and MAM152).

B.172.2.4 Statistics: STA111 or STA125 or STA151 or BUS131 or BUS132 or STA141 or STA142 or STC101.

B.172.2.5 A student may not claim credits for modules passed prior to 1999.

B.172.3 Endorsed Degree

B.172.3.1 In order to obtain an “endorsed degree” a student must obtain at least 240 credits in the compulsory modules of the degree.

B.172.3.2 A student who does not fulfil this criterion may be awarded a BSc degree (not endorsed) provided they have completed 130 credits in a major module (i.e. at Levels 1, 2 and 3 of study) and has accumulated a total of 360 credits, with at least 120 credits at each year level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
ISC153	Introduction to Science 153	C	1	30	5	YR	None	PHY151/ LFS151; MAM151/ MAM150	CA	
MAM151	Mathematics 151 (SF)	C	1	15	5	YR	None	None	CFA	
PHY151	Physics 151 (SF)	C	1	15	5	YR	None	MAM151	CFA	
APM112	Mathematics 112	E	2	15	5	S1	None	MAT105/ MAT104/ MAM152	CFA	
APM123	Applied Mathematics 123	E	2	15	5	YR	None	MAT105/ MAT104/ MAM152	CA	
CHE114	Chemistry 114 (N)	E	2	15	5	S1	None	None	CFA	
CHE124	Chemistry 124 (N)	E	2	15	5	S2	None	None	CFA	
COS101	Computer Science 101	E	2	30	5	YR	None	None	CA	
MAM152	Mathematics 152 (SF)	C	2	15	5	YR	MAM151	None	CFA	
PHY152	Physics 152 (SF)	C	2	15	5	YR	None	None	CFA	
CHM211	Chemistry 211	E	3	20	6	S1	CHE114, CHE124	None	CFA	
CHM212	Chemistry 212	E	3	20	6	S2	CHE114, CHE124	None	CFA	
CSC211	Computer Science 211	E	3	20	6	S1	COS101	None	CA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
CSC212	Computer Science 212	E	3	20	6	S2	COS101	None	CA	
MAT211	Mathematics 211	C	3	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	None	CFA	
MAT212	Mathematics 212	E	3	20	6	YR	MAT105 or MAT103/ MAT104 or MAM151/ MAM152	MAT211	CA	
MAT221	Mathematics 221	C	3	20	6	YR	None	MAT211	CFA	
MAT232	Mathematics 232	E	3	20	6	S1	APM112, MAT105/ MAT103	None	CFA	
PHY212	Physics 212	C	3	20	6	S1	PHY111/ PHY121 OR PHY151/ PHY152	MAT211/M AT104/ MAT152	CFA	
PHY217	Physics 217	E	3	20	6	S1	None	None	CFA	
PHY222	Physics 222	C	3	20	6	S2	PHY111/ PHY121 OR PHY151/ PHY152 AND MAT105/ MAT103/ MAM152	None	CFA	
PHY227	Physics 227	E	3	20	6	S2	PHY116/ PHY126 OR PHY111/ PHY151	None	CFA	
CHM311	Chemistry 311	E	4	30	7	S1	CHM211/ CHM212	None	CFA	
CHM312	Chemistry 312	E	4	30	7	S2	CHM211/ CHM212	None	CFA	
CSC311	Computer Science 311	E	4	30	7	S1	CSC211 and CSC212	None	CA	
CSC312	Computer Science 312	E	4	30	7	S2	CSC211 and CSC212	None	CA	
MAT311	Mathematics 311	E	4	30	7	YR	MAT211	None	CFA	
MAT312	Mathematics 312	E	4	30	7	YR	MAT211, MAT212/ MAT221	None	CA	
MAT321	Mathematics 321	E	4	30	7	YR	MAT221	None	CFA	
MAT322	Mathematics 322	E	4	30	7	YR	MAT211, MAT212/ MAT221	None	CA	

Module Code	Module Name	Compulsory/ Elective	Level	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
PHY314	Physics 314	C	4	30	7	S1	PHY212 PHY222 MAT211 MAT221	None	CFA	
PHY317	Physics 317	E	4	30	7	S2	PHY227	None	CFA	
PHY324	Physics 324	C	4	30	7	S1	PHY212 PHY222 MAT211 MAT221	None	CFA	
PHY327	Physics 327	E	4	30	7	S2	PHY212/ PHY217, MAT105/ MAT103/ MAM152	None	CA	

RULES FOR POSTGRADUATE PROGRAMMES

POSTGRADUATE DIPLOMA IN INTEGRATED WATER RESOURCE MANAGEMENT (3880)

(not offered to new students in 2026)

B.173 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Postgraduate Diploma in Integrated Water Resource Management - PGDip (Integrated Water Resource Management)**:

B.173.1 An applicant must be in possession of a Bachelor's Degree in any relevant discipline or a National Diploma in a relevant discipline with at least two years appropriate experience (Rule A.2.1.4 applies).

B.174 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.175 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study and two years part-time study.

B.176 CURRICULUM

Module Name	Module Code	Credits
Compulsory		
Group 1 (select all modules)		
Integrated Water Resources Management in Practice 731	WAT731	20
Water Resources Management 732	WAT732	20
Spatial Analysis for Integrated Water Resources Management 733	WAT733	20
Environmental Management and Water Regulations 734	WAT734	20
	Sub-total	80
Electives (select two modules)		
Water Quality Management 735	WAT735	20
Aquatic Ecosystems Management 736	WAT736	20
Environmental Education and Water Security 737	WAT737	20
Water Demand Management for Sustainable Development 738	WAT738	20
	Sub-total	40
	FINAL TOTAL	120

B.177 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.178 PROGRESSION RULES

B.178.1 Full-time

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme in the following year.

B.178.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 40 credits per annum to proceed with their studies. A student who has accumulated 80 credits within two years may be allowed to proceed to the following year to complete the programme.

B.179 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.180 SPECIAL REQUIREMENTS FOR THE PROGRAMME

There are no special requirements for this programme.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
WAT733	Spatial Analysis for Integrated Water Resources Management 733	C	20	8	S1	None	None	CFA
WAT731	Integrated Water Resources Management in Practice 731	C	20	8	S1	None	None	CFA
WAT732	Water Resources Management 732	C	20	8	S1	None	None	CFA
WAT734	Environmental Management and Water Regulations 734	C	20	8	S1	None	None	CFA
WAT735	Water Quality Management 735	E	20	8	S2	None	None	CFA
WAT736	Aquatic Ecosystems Management 736	E	20	8	S2	None	None	CFA

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
WAT737	Environmental Education and Water Security 737	E	20	8	S2	None	None	CFA
WAT738	Water Demand Management for Sustainable Development 738	E	20	8	S2	None	None	CFA

BACHELOR OF SCIENCE HONOURS IN APPLIED GEOLOGY (3710)

B.181 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science Honours in Applied Geology Degree - BScHons (Applied Geology)**:

B.181.1 At least a 60% pass in Level 3 Geology with a minimum of 55% per semester (or its equivalent modules at other universities). If less than 55% was obtained in Level 3 Geology, an applicant may be subjected to an assessment of skills obtained or to repeat one or more appropriate modules at Level 3 or both.

B.182 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.183 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study and two years part-time study.

B.184 CURRICULUM

Module Name	Module Code	Credits
Compulsory (select all modules)		
Research Project 709	APG709	30
Advanced Geology 721	APG721	30
Applied Geochemistry 722	APG722	20
Economic Geology and Exploration Techniques 723	APG723	20
Petroleum Geology 724	APG724	20
FINAL TOTAL		120

B.185 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.186 PROGRESSION RULES

B.186.1 Full-time

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme in the following year.

B.186.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 60 credits per annum to proceed with their studies. A

student who has accumulated 90 credits within two years may be allowed to proceed to the following year to complete the programme.

B.187 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.3.2.3, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar

B.188 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.188.1 Please note not all modules may be offered in a specific year and modules may be offered in either semester.

B.188.2 The Honours programme shall extend over a period of ten months. A period of eight months is required to complete taught compulsory modules, after which a period of two months is required for the Honours project.

B.188.3 Contact time entails two hours for theoretical/conceptual work and three hours for practical work per day.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
APG709	Research Project 709	C	30	8	YR	None	None	FA
APG721	Advanced Geology 721	C	30	8	S1	None	None	CFA
APG722	Applied Geochemistry 722	C	20	8	S2	None	None	CA
APG723	Economic Geology and Exploration Techniques 723	C	20	8	S1	APG321	None	CFA
APG724	Petroleum Geology 724	C	20	8	S2	None	None	CFA

BACHELOR OF SCIENCE HONOURS IN ASTROPHYSICS (3793)

(not offered in 2026)

B.189 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science Honours in Astrophysics Degree - BScHons (Astrophysics)**:

B.189.1 An applicant must have obtained a Bachelor of Science Degree with an average of 60% in Physics (Mathematics 211, 221 and 212 or their equivalent are recommended modules for admission to this programme).

B.190 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.191 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study and two years part-time study.

B.192 CURRICULUM

The Honours programme in Astrophysics is presented in conjunction with the National Astrophysics and Space Science Programme (NASSP) at the University of Cape Town under a Memorandum of Agreement.

B.192.1 Astrophysics

Module Name	Module Code	Credits
Compulsory		
Research Module 701	ASP701	30
	Sub-total	30
Electives (select 90 credits)		
Spectroscopy 702	ASP702	10
General Astrophysics 703	ASP703	10
Computational Methods in Astronomy 704	ASP704	20
Advanced Topics in Physics 706	ASP706	10
Geospace 707	ASP707	10
General Relativity 708	ASP708	10
Observational Techniques 709	ASP709	10
Cosmology 710	ASP710	10
Space Science Topics 711	ASP711	20
Advanced Topics in Astrophysics 712	ASP712	20
Electrodynamics 713	ASP713	10
	Sub-total	90
	FINAL TOTAL	120

B.193 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.194 PROGRESSION RULES

B.194.1 Full-time

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme in the following year.

B.194.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 60 credits per annum to proceed with their studies. A student who has accumulated 90 credits within two years may be allowed to proceed to the following year to complete the programme.

B.195 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.196 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.196.1 Full-time students

The final mark for each theory module will be a combination of marks obtained in tests and assignments.

B.196.2 Part-time students

As above, except that the student takes two theory modules per year over two years (in total 4 modules plus the Research module). The Honours Research module can be completed at any time during the two years.

B.196.3 Contact Time

Lectures per week: 10

Research module / Practical work: An average of six hours per week.

BACHELOR OF SCIENCE HONOURS IN BIODIVERSITY AND CONSERVATION BIOLOGY (3731)

B.197 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science Honours in Biodiversity and Conservation Biology Degree - BScHons (Biodiversity and Conservation Biology)**:

- (a) An applicant must have a first degree in whole organism biology and have attained at least 60% in either zoological or botanical subjects in the final year,

OR

- (b) With the permission of Senate, an applicant without a degree but with demonstrable and equivalent experience in whole organism biology.

B.198 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.199 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study or two years part-time study.

B.200 CURRICULUM

Module Name	Module Code	Credits
Compulsory (select all modules)		
Research Project 700	BCB700	40
Biostatistics 744	BCB744	20
Research Methods 742	BCB742	20
	Sub-total	80
Electives (select two modules)		
Biodiversity Information Management 736	BCB736	20
Marine Biology 738	BCB738	20
Herpetology 739	BCB739	20
Systematics and Evolution 731	BCB731	20
Quantitative Ecology 743	BCB743	20
	Sub-total	40
	FINAL TOTAL	120

B.201 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.202 PROGRESSION RULES

B.202.1 Full-time

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme in the following year.

B.202.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 60 credits per annum to proceed with their studies. A student who has accumulated 100 credits within two years may be allowed to proceed to the following year to complete the programme.

B.203 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.204 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.204.1 Please note not all modules may be offered in a specific year and modules may be offered in either semester.

B.204.2 With a continuous assessment system in place, the assessment will be appropriate to the module and may take the form of tests and/or written assignments and/or oral presentations and/or practical work. A final oral examination may be required.

B.204.3 The elective modules are to be selected in conjunction with the project supervisor. All students shall complete a minimum of 100 credits from the programme. The final module selection is dependent on the availability of academic staff and the absence of timetable clashes.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
BCB700	Research Project 700	C	40	8	YR	None	None	CA
BCB744	Biostatistics 744	C	20	8	S1	None	None	CA
BCB742	Research Methods 742	C	20	8	S1	None	None	CA

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
BCB736	Biodiversity Information Management 736	E	20	8	S2	None	None	CA
BCB738	Marine Biology 738	E	20	8	S2	None	None	CA
BCB739	Herpetology 739	E	20	8	S2	None	None	CA
BCB731	Systematics and Evolution 731	E	20	8	S2	None	None	CA
BCB743	Quantitative Ecology 743	E	20	8	S2	None	BCB744	CA

BACHELOR OF SCIENCE HONOURS IN BIOTECHNOLOGY (3707)

B.205 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science Honours in Biotechnology Degree - BSCHons (Biotechnology)**:

B.205.1 An applicant should obtain at least a 60% pass at Level 3 in an appropriate subject and with the approval of the Departmental Postgraduate Committee is required for admission into the programme.

B.206 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.207 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study or two years part-time study.

B.208 CURRICULUM

Module Name	Module Code	Credits
Compulsory (select all modules)		
Research Project 703	BTY703	40
Molecular Biology and Bioinformatics Practical 701	BTY701	15
Literature Review 702	BTY702	15
	Sub-total	70
Electives (select five modules)		
Advanced Plant Biotechnology 704	BTY704	10
Genomics 709	BTY709	10
Apoptosis 710	BTY710	10
Advanced Food Microbiology 711	BTY711	10
Forensic DNA Analysis 714	BTY714	10
Special Topics 717	BTY717	10
Proteomics 718	BTY718	10
Biofuels 719	BTY719	10
Molecular Mechanisms of Cancer 721	BTY721	10
Next Generation Sequencing 722	BTY722	10
	Sub-total	50
	FINAL TOTAL	120

B.209 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.210 PROGRESSION RULES

B.210.1 Full-time

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme in the following year.

B.210.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 60 credits per annum to proceed with their studies. A student who has accumulated 90 credits within two years may be allowed to proceed to the following year to complete the programme.

B.211 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.212 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.212.1 Please note not all modules may be offered in a specific year and modules may be offered in either semester.

B.212.2 With a continuous assessment system in place, the assessment will be appropriate to the module and may take the form of tests and/or written assignments and/or oral presentations and/or practical work. A final oral examination may be required.

B.212.3 In addition to the above modules, a student shall be required to present one or more seminars. The Research Project may be defended in an oral examination.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
BTY701	Molecular Biology and Bioinformatics Practical 701	C	15	8	S1	None	None	CFA
BTY702	Literature Review 702	C	15	8	YR	None	None	CA
BTY703	Research Project 703	C	40	8	YR	None	None	CA
BTY704	Advanced Plant Biotechnology 704	E	10	8	S1	60% in GEO 3 RD YR	None	FA
BTY709	Genomics 709	E	10	8	YR	None	None	CA
BTY710	Apoptosis 710	E	10	8	YR	None	None	CFA
BTY711	Advanced Food Microbiology 711	E	10	8	S2	None	None	CFA
BTY714	Forensic DNA Analysis 714	E	10	8	YR	None	None	CFA
BTY717	Special Topics 717	E	10	8	YR	None	None	CA
BTY718	Proteomics 718	E	10	8	YR	None	None	CFA
BTY719	Biofuels 719	E	10	8	YR	None	None	CFA
BTY721	Molecular Mechanisms of Cancer 721	E	10	8	S1	None	None	CFA
BTY722	Next Generation Sequencing 722	E	10	8	S1	None	None	CFA

BACHELOR OF SCIENCE HONOURS IN CHEMISTRY (3734)

B.213 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science Honours in Chemistry Degree - BScHons (Chemistry)**:

B.213.1 At least 60% pass at Level 3 in Chemistry and with the approval of the Departmental Postgraduate Committee is required for admission into the programme.

B.214 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.215 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study or two years part-time study.

B.216 CURRICULUM

Module Name	Module Code	Credits
Compulsory (select all modules)		
Analytical Chemistry 701	CHM701	15
Organic Chemistry 702	CHM702	15
Inorganic Chemistry 703	CHM703	15
Physical Chemistry 704	CHM704	15
Applied Chemistry 705	CHM705	15
Research Project 707	CHM707	30
Chemistry Advanced Specialist Topics 708	CHM708	15
FINAL TOTAL		120

B.217 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.218 PROGRESSION RULES

B.218.1 Full-time

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme in the following year.

B.218.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 60 credits per annum to proceed with their studies. A

student who has accumulated 90 credits within two years may be allowed to proceed to the following year to complete the programme.

B.219 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.220 SPECIAL REQUIREMENTS FOR THE PROGRAMME

There are no special requirements for this programme.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre- Requisite	Co-Requisite	Assessment Type
CHM701	Analytical Chemistry 701	C	15	8	S1	None	None	CFA
CHM702	Organic Chemistry 702	C	15	8	S2	None	None	CFA
CHM703	Inorganic Chemistry 703	C	15	8	S2	None	None	CFA
CHM704	Physical Chemistry 704	C	15	8	S1	None	None	CFA
CHM705	Applied Chemistry 705	C	15	8	S1	None	None	CFA
CHM707	Research Project 707	C	30	8	S2	None	None	FA
CHM708	Chemistry Advanced Specialist Topics 708	C	15	8	S2	None	None	CA

BACHELOR OF SCIENCE HONOURS IN COMPUTATIONAL FINANCE

(3739)

(not offered in 2026)

B.221 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science Honours in Computational Finance Degree - BSchHons (Computational Finance)**:

- (a) a Bachelor's degree with majors in Mathematics and Statistics with at least a 60% average in STA331 and STA332 and Mathematics Level 3 modules and the approval of the Departmental Postgraduate Committee is required for admission into the programme.
- (b) an applicant with extensive experience in a financial field will be considered on an individual basis.
- (c) an applicant entering the programme through the RPL route is required to complete a testing phase as determined by departments.

B.222 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.223 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study or two years part-time study.

B.224 CURRICULUM

Module Name	Module Code	Credits
Compulsory (select all modules)		
Research Project 748	MAM748	
or		30
Research Project 761	STA761	
Probability and Stochastic Processes 712 (not offered in 2026)	COF712	15
Stochastic Calculus for Finance 714	COF714	15
Financial Management 756	MAN756	15
Accounting and Finance for Managers 762	MAN762	15
	Sub-total	90
Electives (select modules to the value of 30 credits)		
Financial Accounting 781	ACC781	15
Simulation 710 (not offered in 2026)	COF710	15
Time Series Analysis 711	COF711	15
Financial Engineering 716	COF716	15
Financial Risk Management 726 (not offered in 2026)	COF726	15
Computational Linear Algebra 728	MAM728	15

Research Methodology 702	STA702	15
Theoretical Statistics 705 (not offered in 2026)	STA705	15
	Sub-total	30
	FINAL TOTAL	120

Please note: Modules offered according to the availability of staff

B.224.1 A student may take an equivalent of two Honours elective modules, not exceeding 30 credits from cognate or associate disciplines, provided that they have permission from both HoDs.

B.225 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.226 PROGRESSION RULES

B.226.1 Full-time

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme the following year.

B.226.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 60 credits per annum to proceed with their studies. A student who has accumulated 90 credits within two years may be allowed to proceed to the following year to complete the programme.

B.227 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.228 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.228.1 Please note not all modules may be offered in a specific year and modules may be offered in either semester.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
MAM748	Research Project 748	C	30	8	YR	None	None	CFA
STA761	Research Project 761	C	30	8	YR	None	None	FA
COF712	Probability and Stochastic Processes 712	C	15	8	S1	STA331 or equivalent	None	FA
COF714	Stochastic Calculus for Finance 714	C	15	8	S1	None	None	CFA
MAN756	Financial Management 756	C	15	8	S2	MDP580 or ADM668 or ADM616 or MAN748 or equivalent	None	CFA
MAN762	Accounting and Finance for Managers 762	C	15	8	S1	None	None	CFA
ACC781	Financial Accounting 781	C	15	8	S2	MDP553/ ADM619/ MAN753 or equivalent	None	CFA
COF710	Simulation 710 (	E	15	8	S1	STA331; STA332; MAT211; MAT221; Matlab program or equiv	None	CFA
COF711	Time Series Analysis 711	E	15	8	S2	STA211; STA221; MAT211; MAT221 or equiv	None	FA
COF716	Financial Engineering 716	E	15	8	S2	None	None	CFA
COF726	Financial Risk Management 726	E	15	8	S1	STA331 or equivalent	None	CFA
MAM728	Computational Linear Algebra 728	E	15	8	S1/ S2	None	None	CFA
STA702	Research Methodology 702	E	15	8	S1	None	STA501	CFA
STA705	Theoretical Statistics 705	E	15	8	S1	STA331; STA332; MAT211; or equiv	None	CFA

BACHELOR OF SCIENCE HONOURS IN COMPUTER SCIENCE (3735)

B.229 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science Honours in Computer Science Degree - BScHons (Computer Science)**:

B.229.1 an applicant must have a Bachelor of Science Degree in Computer Science or another degree with a major in Computer Science, and must have obtained a 60% average for the Level 3 Computer Science modules. In addition, the applicant is required to have completed English for Educational Development (EED) or its equivalent, or shall complete it as a co-requisite.

B.229.2 The approval of the Departmental Postgraduate Committee is required for admission into the programme.

B.230 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.231 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study or two years part-time study.

B.232 CURRICULUM

B.232.1 A student may take up to an equivalent of two Honours modules, not exceeding 40 credits, from cognate or associate disciplines, provided they have permission from the HoDs.

Module Name	Module Code	Credits
Compulsory (select all modules)		
Project Analysis and Design 731	COS731	15
Project Implementation and Testing 732	COS732	15
	Sub-total	30

Electives (select six modules from Group 1, or select four to five modules from Group 1 and one to two modules from Group 2)

Group 1		
Advanced Software Engineering 726	COS726	15
ICT4D	COS734	15
An Introduction to the Internet of Things 735	COS735	15
Big Data Engineering 736	COS736	15
Artificial Intelligence 737	COS737	15
Cybersecurity 738	COS738	15
4IR Special Topics 739	COS739	15
Cloud Computing 740	COS740	15
Applied Artificial Intelligence 741	COS741	15

Immersive Technologies 742	COS742	15
Advance Immersive Technologies 743	COS743	15
	Sub-total	60-90
Group 2		
Mathematics		
Computational Linear Algebra 728	MAM728	15
Coding Theory 747 (not offered in 2026)	MAM747	15
Statistics		
Statistical Modelling 737	STA737	15
Data Mining I 760	STA760	15
Physics		
Applied and Computational Physics 720	PHY720	30
Advanced Analytical Techniques 726	PHY726	30
Information Systems		
Delivery Information Systems Benefits 735	IFS735	20
Information Systems Procurement 736	IFS736	20
Strategic Information Systems Management 740	IFS740	20
	Sub-total	15-30
	FINAL TOTAL	120

B.233 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.234 PROGRESSION RULES

B.234.1 Full-time

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme in the following year.

B.234.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 60 credits per annum to proceed with their studies. A student who has accumulated 90 credits within two years may be allowed to proceed to the following year to complete the programme.

B.235 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.236 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.236.1 Special Topics presented by visiting lecturers, industry partners, etc., may be offered as an additional module.

B.236.2 Not all modules are offered every year.

B.236.3 A full-time Honours module may be delivered as a block to allow part-time students to attend.

B.236.4 The semester wherein a module may be offered could change depending on staffing, visiting lecturers or special topics offered.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
COS726	Advanced Software Engineering 726	E	15	8	S1	None	None	CA	
COS731	Project Analysis and Design 731	C	15	8	S1	None	None	CA	
COS732	Project Implementation and Testing 732	C	15	8	S2	None	None	CA	
COS734	ICT4D	E	15	8	S2	None	None	CA	
COS735	An Introduction to the Internet of Things 735	E	15	8	S1	None	None	CA	
COS736	Big Data Engineering 736	E	15	8	S1	None	None	CA	
COS737	Artificial Intelligence 737	E	15	8	S2	None	None	CA	
COS738	Cybersecurity 738	E	15	8	S2	None	None	CA	
COS739	4IR Special Topics 739	E	15	8	S1/S2	None	None	CA	
COS740	Cloud Computing 740	E	15	8	S2	None	None	CA	
COS741	Applied Artificial Intelligence 741	E	15	8	S1	CSC311, CSC312	None	CA	
COS742	Immersive Technologies 742	E	15	8	S1	None	None	CA	
COS743	Advance Immersive Technologies 743	E	15	8	S2	None	None	CA	
IFS735	Delivery Information Systems Benefits 735	E	20	8	S1	None	None	CFA	SM
IFS736	Information Systems Procurement 736	E	20	8	S1	None	None	CFA	SM

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type	Service Module
IFS740	Strategic Information Systems Management 740	E	20	8	S1	None	None	CA	SM
MAM728	Computational Linear Algebra 728	E	15	8	S1/ S2	None	None	CFA	
MAM747	Coding Theory 747	E	15	8	S1	None	None	CFA	
PHY720	Applied and Computational Physics 720	E	30	8	S1	None	None	CFA	
PHY726	Advanced Analytical Techniques 726	E	30	8	S2	PHY724	None	CFA	
STA737	Statistical Modelling 737	E	15	8	S1	None	None	FA	
STA760	Data Mining I 760	E	15	8	S2	STA331, STA332 OR EQUI	None	FA	

BACHELOR OF SCIENCE HONOURS IN ENVIRONMENT AND WATER SCIENCE (3780)

B.237 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science Honours in Environment and Water Science Degree - BScHons (Environment and Water Science)**:

- B.237.1** an applicant must have a Bachelor of Science Degree in Environmental and Water Science with an average mark of at least 60% in the exit level modules.
- B.237.2** an applicant with a Bachelor of Science Degree in a related field can on recommendation of the Postgraduate Committee of the Department and approval of Faculty be considered for selection.

B.238 SELECTION

Final selection shall be based on an applicants meeting the criteria as determined by the Faculty.

B.239 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study or two years part-time study.

B.240 CURRICULUM

Module Name	Module Code	Credits
Compulsory (select all modules)		
Environmental Assessment and Management 701	EWP701	20
Advanced Geographical Information Systems 702	EWP702	20
Groundwater Hydrology 703	EWP703	20
Surface Water Hydrology 704	EWP704	20
Research Project or Internship 705	EWP705	30
Anthropocene Rivers 707	EWP707	20
	FINAL TOTAL	130

B.241 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.242 PROGRESSION RULES

B.242.1 Full-time

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme in the following year.

B.242.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 60 credits per annum to proceed with their studies. A student who has accumulated 90 credits within two years may be allowed to proceed to the following year to complete the programme.

B.243 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.244 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.244.1 Please note not all modules may be offered in a specific year and modules may be offered in either semester.

B.244.2 With a continuous assessment system in place, the assessment will be appropriate to the module and may take the form of tests and/or written assignments and/or oral presentations and/or practical work. A final oral examination may be required.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre- Requisite	Co-Requisite	Assessment Type
EWP701	Environmental Assessment and Management 701	C	20	8	S1	None	None	CFA
EWP702	Advanced Geographical Information Systems 702	C	20	8	S1	None	None	CFA
EWP703	Groundwater Hydrology 703	C	20	8	S1	None	None	CFA
EWP704	Surface Water Hydrology 704	C	20	8	S2	None	None	CFA
EWP705	Research Project or Internship 705	C	30	8	YR	None	None	CFA
EWP707	Anthropocene Rivers 707	C	20	8	S2	None	None	CA

BACHELOR OF SCIENCE HONOURS IN MATHEMATICS (3736)

B.245 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science Honours in Mathematics Degree - BScHons (Mathematics)**:

B.245.1 At least a 60% pass in Level 3 Mathematics or its equivalent and the approval of the Departmental Postgraduate Committee.

B.245.2 If less than 60% was obtained in Level 3 Mathematics the applicant may be required to take an entry examination (test) or repeat one or more appropriate modules at Level 3.

B.246 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.247 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study or two years part-time study.

B.248 CURRICULUM

Module Name	Module Code	Credits
Compulsory		
Research Project 748	MAM748	30
	Sub-total	30
Electives (select six modules)		
Ordinary Differential Equations 707	MAM707	15
Number Theory 709	MAM709	15
Graph Theory 710	MAM710	15
Partial Differential Equations 717	MAM717	15
Measure and Integration 727	MAM727	15
Computational Linear Algebra 728	MAM728	15
Mathematical Modelling in Epidemiology 730 (not offered in 2026)	MAM730	15
Topology 737	MAM737	15
Functions of a Complex Variable 738 (not offered in 2026)	MAM738	15
Group Theory 739	MAM739	15
Introduction to Optimal Control 740 (not offered in 2026)	MAM740	15
Coding Theory 747 (not offered in 2026)	MAM747	15
Rings and Modules 757 (not offered in 2026)	MAM757	15
Functional Analysis 767	MAM767	15
Numerical Analysis 777 (not offered in 2026)	MAM777	15
Mathematical Physics 731	MAM731	15
Category Theory and Order Theory 732	MAM732	15
Stochastic Calculus for Finance 714	COF714	15
Financial Engineering 716	COF716	15
	Sub-total	90
	FINAL TOTAL	120

B.248.1 A student may take an equivalent of two Honours modules, not exceeding 30 credits from cognate or associate disciplines, provided that they have permission from both HoDs.

B.249 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.250 PROGRESSION RULES

B.250.1 Full-time

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme in the following year.

B.250.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 60 credits per annum to proceed with their studies. A student who has accumulated 90 credits within two years may be allowed to proceed to the following year to complete the programme.

B.251 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.252 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.252.1 Please note not all modules may be offered in a specific year and modules may be offered in either semester.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
COF714	Stochastic Calculus for Finance 714	E	15	8	S1	None	None	CFA
COF716	Financial Engineering 716	E	15	8	S2	None	None	CFA

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
MAM707	Ordinary Differential Equations 707	E	15	8	S1	None	None	CFA
MAM709	Number Theory 709	E	15	8	S2	None	None	CFA
MAM710	Graph Theory 710	E	15	8	S1/ S2	None	None	CFA
MAM717	Partial Differential Equations 717	E	15	8	S2	None	None	CFA
MAM727	Measure and Integration 727	E	15	8	S2	None	None	CFA
MAM728	Computational Linear Algebra 728	E	15	8	S1/ S2	None	None	CFA
MAM730	Mathematical Modelling in Epidemiology 730	E	15	8		None	None	CFA
MAM731	Mathematical Physics 731	E	15	8	S1	None	None	CFA
MAM732	Category Theory and Order Theory 732	E	15	8	S2	None	None	CA
MAM737	Topology 737	E	15	8	S1	None	None	CA
MAM738	Functions of a Complex Variable 738	E	15	8	YR	None	None	CFA
MAM739	Group Theory 739	E	15	8	S2	None	None	CFA
MAM740	Introduction to Optimal Control 740	E	15	8		None	None	CFA
MAM747	Coding Theory 747	E	15	8	S1	None	None	CFA
MAM748	Research Project 748	C	30	8	YR	None	None	CFA
MAM757	Rings and Modules 757	E	15	8	S1/ S2	None	None	CFA
MAM767	Functional Analysis 767	E	15	8	S2	None	None	CFA
MAM777	Numerical Analysis 777	E	15	8	S1/ S2	None	None	CFA

BACHELOR OF SCIENCE HONOURS IN MEDICAL BIOSCIENCE (3721)

B.253 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science Honours in Medical Bioscience Degree – BScHons (Medical Bioscience)**:

B.253.1 at least 60% pass at Level 3 in one of the Bachelor of Science (Medical Bioscience) modules, or related field with the approval of the Departmental Postgraduate Committee.

B.254 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.255 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study or two years part-time study.

B.256 CURRICULUM

Module Name	Module Code	Credits
Compulsory (select all modules)		
Advanced Technique Studies 702	MBS702	20
Research Project 704	MBS704	50
Applied Statistics 501	STA501	15
	Sub-total	85
Electives (select four modules)		
Forensic DNA Analysis 714	BTY714	10
Specialist Module in Medical Bioscience 701	MBS701	10
Specialist Module in Anatomical Science 705	MBS705	10
The Microbiology of Female Morbidity 707 (not offered in 2026)	MBS707	10
Apoptosis and Cell Signalling 709	MBS709	10
Advanced Cardiovascular Physiology 711 (not offered in 2026)	MBS711	10
Foetal Origins of Adult Disease 712	MBS712	10
Advanced Studies in Reproduction and Health 713	MBS713	10
Applied Immunology 714	MBS714	10
Applied Exercise Science 715	MBS715	10
Nutrition in Health and Disease 716	MBS716	10
	Sub-total	40
	FINAL TOTAL	125

B.257 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.258 PROGRESSION RULES

B.258.1 Full-time

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme in the following year.

B.258.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 60 credits per annum to proceed with their studies. A student who has accumulated 90 credits within two years may be allowed to proceed to the following year to complete the programme.

B.259 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.260 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.260.1 Please note not all modules may be offered in a specific year and modules may be offered in either semester.

B.260.2 With a continuous assessment system in place, the assessment will be appropriate to the module and may take the form of tests and/or written assignments and/or oral presentations and/or practical work. A final oral examination may be required.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
BTY714	Forensic DNA Analysis 714	E	10	8	YR	None	None	CFA
MBS701	Specialist Module in Medical Bioscience 701	E	10	8	YR	None	None	CFA
MBS702	Advanced Technique Studies 702	C	20	8	YR	None	None	CFA
MBS704	Research Project 704	C	50	8	YR	None	None	CA
MBS705	Specialist Module in Anatomical Science 705	E	10	8	S2	None	None	CFA

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
MBS707	The Microbiology of Female Morbidity 707	E	10	8	S2	None	None	CFA
MBS709	Apoptosis and Cell Signalling 709	E	10	8	S1	None	None	CFA
MBS711	Advanced Cardiovascular Physiology 711	E	10	8		None	None	CFA
MBS712	Foetal Origins of Adult Disease 712	E	10	8	S1/ S2	None	None	CFA
MBS713	Advanced Studies in Reproduction and Health 713	E	10	8	S2	None	None	CFA
MBS714	Applied Immunology 714	E	10	8	S1	None	None	CFA
MBS715	Applied Exercise Science 715	E	10	8	S1/ S2	None	None	CFA
MBS716	Nutrition in Health and Disease 716	E	10	8	S2	None	None	CFA
STA501	Applied Statistics 501	C	15	8	S1	None	None	FA

BACHELOR OF SCIENCE HONOURS IN PHYSICAL SCIENCE (3714)

B.261 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science Honours in Physical Science Degree - BScHons (Physical Science)**:

B.261.1 a Bachelor of Science Degree with an average of 60% in Physical Sciences exit level modules and Mathematics or its equivalent at second level.

B.262 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.263 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study or two years part-time study.

B.264 CURRICULUM

The following streams are offered in the programme:

- Material Science (MATSCI)
- Accelerator and Nuclear Physics (MANUS)

B.264.1 Material Science (MATSCI)

Module Name	Module Code	Credits
Compulsory (select all modules)		
Research Module 709	PHY709	30
Applied and Computational Physics 720	PHY720	30
Theoretical Materials Science 724	PHY724	20
Applied Materials Science 725	PHY725	20
Electives (select modules to the value of 30 credits)		
Advanced Analytical Techniques 726	PHY726	30
General Astrophysics 703	ASP703	10
Advanced Topics in Astrophysics 712	ASP712	20
FINAL TOTAL		130

B.264.2 Accelerator and Nuclear Physics (MANUS)

Module Name	Module Code	Credits
Compulsory (select all modules)		
Research Module 709	PHY709	30
Applied and Computational Physics 720	PHY720	30
Accelerator Science 721	PHY721	20
Quantum and Statistical Physics 722	PHY722	20

Electives (select modules to the value of 30 credits)

Nuclear Physics 723	PHY723	30
General Astrophysics 703	ASP703	10
Advanced Topics in Astrophysics 712	ASP712	20
FINAL TOTAL		130

B.265 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.266 PROGRESSION RULES**B.266.1 Full-time**

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme in the following year.

B.266.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 60 credits per annum to proceed with their studies. A student who has accumulated 90 credits within two years may be allowed to proceed to the following year to complete the programme.

B.267 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.268 SPECIAL REQUIREMENTS FOR THE PROGRAMME**B.268.1 Full-time students**

The final mark for each theory module will be a combination of marks obtained in tests and assignments.

B.268.2 Part-time students

As above, except that the student takes two theory modules per year over two years (in total 4 modules plus the Research module). The Honours Research module can be completed at any time during the two years.

B.268.3 Contact Time

Lectures per week: 10

Research module / Practical work: An average of six hours per week

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
ASP703	General Astrophysics 703	E	10	8	S1	None	None	CFA
ASP712	Advanced Topics in Astrophysics 712	E	20	8	S2	None	None	CFA
PHY709	Research Module 709	C	30	8	S2	None	None	CA
PHY720	Applied and Computational Physics 720	C	30	8	S1	None	None	CFA
PHY721	Accelerator Science 721	C	20	8	S1	None	None	CFA
PHY722	Quantum and Statistical Physics 722	C	20	8	S2	None	None	CFA
PHY723	Nuclear Physics 723	E	30	8	S1	None	PHY720	CFA
PHY724	Theoretical Materials Science 724	C	20	8	S1	None	None	CFA
PHY725	Applied Materials Science 725	C	20	8	S2	PHY724	None	CFA
PHY726	Advanced Analytical Techniques 726	E	30	8	S2	PHY724	None	CFA

BACHELOR OF SCIENCE HONOURS IN POPULATION STUDIES (3738)

B.269 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science Honours in Population Studies Degree - BScHons (Population Studies)**:

- B.269.1** a Bachelor's degree with at least 60% average in the appropriate Level 3 modules, formal training in numeracy and the approval of the Departmental Postgraduate Committee.
- B.269.2** an applicant who has majored in Demography, Statistics, Anthropology, Computer Science, Economics, Geography, Information Systems, Management, Public Health, Psychology, Sociology or its equivalent may be considered for admission.

B.270 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.271 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study or two years part-time study.

B.272 CURRICULUM

Module Name	Module Code	Credits
Compulsory (select all groups)		
Group 1 (select all modules)		
Official Statistics 703	POP703	15
Survey Methods 706	POP706	15
Demographic Analysis 707	POP707	15
Research Project 761	POP761	30
	Sub-total	75
Group 2 (select one module)		
Applied Statistics 501 (Non-Statistics majors)	STA501	15
Biostatistics 733 (Statistics majors) (not offered in 2026)	STA733	15
	Sub-total	15
Electives (select two modules)		
Gender Issues and Reproductive Health 708	POP708	15
Social Demography 709 (not offered in 2026)	POP709	15
Mathematical Demography and Population Modelling 710 (not offered in 2026)	POP710	15
Population Projections 790	STA790	15
	Sub-total	30
	FINAL TOTAL	120

B.272.1 A student may take an equivalent of two Honours modules, not exceeding 30 credits from cognate or associate disciplines, provided that they have permission from both HoDs.

B.273 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.274 PROGRESSION RULES

B.274.1 Full-time

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme in the following year.

B.274.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 60 credits per annum to proceed with their studies. A student who has accumulated 90 credits within two years may be allowed to proceed to the following year to complete the programme.

B.275 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.276 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.276.1 Please note not all modules may be offered in a specific year and modules may be offered in either semester.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
POP703	Official Statistics 703	C	15	8	S1	None	None	CFA
POP706	Survey Methods 706	C	15	8	S1	None	None	CFA
POP707	Demographic Analysis 707	C	15	8	YR	None	None	CFA

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
POP708	Gender Issues and Reproductive Health 708	E	15	8	S2	POP707	None	CFA
POP709	Social Demography 709	E	15	8	S2	POP707	None	CFA
POP710	Mathematical Demography and Population Modelling 710	E	15	8	S2	POP707	None	CFA
POP761	Research Project 761	C	30	8	YR	None	None	CA
STA501	Applied Statistics 501 (Non-Statistics majors)	C	15	8	S1	None	None	FA
STA733	Biostatistics 733 (Statistics majors)	C	15	8	S1	STA331, STA332	None	CFA
STA790	Population Projections 790	E	15	8	S1	POP707	None	CFA

BACHELOR OF SCIENCE HONOURS IN STATISTICAL SCIENCE (3737)

B.277 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.3, a student shall be required to meet the following criteria to be enrolled for the **Bachelor of Science Honours in Statistical Science Degree - BScHons (Statistical Science)**:

- B.277.1** At least a 60% pass in exit level Statistics with a minimum of 55% per semester or its equivalent.
- B.277.2** If less than 60% was obtained in exit level Statistics the applicant may be required to take an entry examination (test) or repeat one or more appropriate modules at Level 3 or both.
- B.277.3** In addition to the above admission to an Honours in Statistics Sciences, Level 2 Mathematics is also required.
- B.277.4** Some elective modules in the Data Science stream require Level 3 Computer Science and one elective requires Level 3 Mathematics.

B.278 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.279 DURATION

Unless Senate decides otherwise, the duration of the programme shall extend over one year full-time study or two years part-time study.

B.280 CURRICULUM

Stream 1: Statistical Science Stream

Module Name	Module Code	Credits
Compulsory (select all modules)		
Multivariate Analysis 701	STA701	15
Research Project 761	STA761	30
Statistical Modelling 737	STA737	15
Introductory Multivariate Analysis 739	STA739	15
	Sub-total	75
Electives (select three modules)		
Simulation 710 (not offered in 2026)	COF710	15
Probability and Stochastic Processes 712 (not offered in 2026)	COF712	15
Financial Risk Management 726 (not offered in 2026)	COF726	15
Theoretical Statistics 705 (not offered in 2026)	STA705	15
Experimental Design 710 (not offered in 2026)	STA710	15
Biostatistics 733 (not offered in 2026)	STA733	15
Statistical Genetics 736 (not offered in 2026)	STA736	15
Time Series Analysis 711	COF711	15

Data Mining I 760	STA760	15
Applied Analytics 738	STA738	15
	Sub-total	45
	FINAL TOTAL	120

Stream 2: Data Science Stream

Module Name	Module Code	Credits
Compulsory (select all modules)		
Multivariate Analysis 701	STA701	15
Research Project 761	STA761	30
Statistical Modelling 737	STA737	15
Introductory Multivariate Analysis 739	STA739	15
	Sub-total	75
Electives (select three modules)		
*Artificial Intelligence 737	COS737	15
Data Mining I 760	STA760	15
Time Series Analysis 711	COF711	15
Big Data Engineering 736	COS736	15
**Graph Theory 710	MAM710	15
	Sub-total	45
	FINAL TOTAL	120

* Module offered by the Department of Computer Sciences, Level 3 Computer Science required.

** Module offered by the Department of Mathematics and Applied Mathematics; Level 3 Mathematics required.

B.280.1 A student may take an equivalent of two Honours modules, not exceeding 30 credits in total from cognate or associate disciplines, provided that they have permission from both HoDs.

B.281 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.282 PROGRESSION RULES

B.282.1 Full-time

Unless Senate decides otherwise, a full-time student shall complete the programme in one year. A student who has passed at least 60 credits may proceed with their studies to complete the programme in the following year.

B.282.2 Part-time

Unless Senate decides otherwise, a part-time student shall complete the programme in two consecutive years and accumulate at least 60 credits per annum to proceed with their studies. A

student who has accumulated 90 credits within two years may be allowed to proceed to the following year to complete the programme.

B.283 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.284 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.284.1 Please note not all modules may be offered in a specific year and modules may be offered in either semester.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requirement	Assessment Type
COF710	Simulation 710	E	15	8	S1	STA331; STA332; MAT211; MAT221; Matlab program or equivalent	None	CFA
COF711	Time Series Analysis 711	E	15	8	S2	STA211; STA221; MAT211; MAT221 or equivalent	None	FA
COF712	Probability and Stochastic Processes 712	E	15	8	S1	STA331 or equivalent	None	FA
COF726	Financial Risk Management 726	E	15	8	S1	STA331 or equivalent	None	CFA
COS736	Big Data Engineering 736	E	15	8	S1	None	None	CA
COS737	Artificial Intelligence 737	E	15	8	S2	None	None	CA
STA701	Multivariate Analysis 701	C	15	8	S2	STA331; STA332; (STA739) or equivalent; knowledge of SAS	None	FA
STA705	Theoretical Statistics 705	E	15	8	S1	STA331; STA332; MAT211; or equivalent	None	CFA
STA710	Experimental Design 710	E	15	8	S1	STA331; STA332; or equivalent	None	CFA
STA733	Biostatistics 733	E	15	8	S1	STA331; STA332; or equivalent	None	CFA
STA736	Statistical Genetics 736	E	15	8	S2	STA701, STA733, COF712	None	CFA

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Duration	Pre-Requisite	Co-Requisite	Assessment Type
STA737	Statistical Modelling 737	C	15	8	S1	None	None	FA
STA738	Applied Analytics 738	E	15	8	S1	None	None	FA
STA739	Introductory Multivariate Analysis 739	C	15	8	S1	None	None	FA
STA760	Data Mining I 760	E	15	8	S2	STA331; STA332 or equivalent	None	FA
STA761	Research Project 761	C	30	8	YR	None	None	FA
MAM710	Graph Theory 710	E	15	8	S1/ S2	None	None	CFA

MASTER OF PHARMACY (3851)

B.285 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Pharmacy Degree - MPharm**:

- a) an applicant must have obtained a 4-year Pharmacy degree or diploma (e.g. BPharm, BSc (Pharm) or Dip.Pharm).
- b) Factors considered in selection include academic merit, appropriate experience, reports from referees, and compatibility of research and career interest with available research projects and/or supervisors and results of interview.

B.286 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.287 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.288 CURRICULUM

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Pharmaceuticae Master's Thesis 801	PHA801	
2nd Enrolment Code		
Pharmaceuticae Master's Thesis 802	PHA802	180
	FINAL TOTAL	180

B.289 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.290 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion, adequate progress has been made during the current year.

B.291 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.292 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.292.1 A thesis dealing with an original research topic must be presented.

B.292.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee, as a component of the programme when deemed necessary.

B.292.3 A thesis on a research topic must be presented. A student may be required to do extra module which is not examinable if required by the supervisor.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
PHA801	Pharmaceuticae Master's Thesis 801	C	180	9	1	YR	None	None	FA
PHA802	Pharmaceuticae Master's Thesis 802	C		9	2	YR	None	None	FA

MASTER OF PHILOSOPHY IN POPULATION STUDIES (3921)

B.293 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Philosophy in Population Studies Degree - MPhil (Population Studies)**:

- B.293.1** an applicant must be in possession of an Honours degree in Population Studies, Demography, Statistics or relevant field with at least 60% average.
- B.293.2** an applicant with the required Population Studies knowledge and research skills may be accepted into the full-thesis Master's programme.
- B.293.3** Departmental approval is subject to the availability of a supervisor.

B.294 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.295 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.296 CURRICULUM

A student must select one mode of study:

B.296.1 Mode 1 – Thesis

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Population Studies Master's Thesis 801	POP801	
2nd Enrolment Code		180
Population Studies Master's Thesis 802	POP802	
	FINAL TOTAL	180

B.296.2 Mode 2 – Mini-Thesis and Coursework (not offered in 2026)

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Population Studies Mini-Thesis 803	POP803	
2nd Enrolment Code		120
Population Studies Mini-Thesis 804	POP804	
	Sub-total	120

Electives (select four modules)

*Demographic Analysis 807	POP807	15
Official Statistics 809	POP809	15
*Survey Methods 811	POP811	15
*Population Projections 812	POP812	15
African Historical Demography 813	POP813	15
Social Demography 814	POP814	15
Gender Issues and Reproductive Health 815	POP815	15
Mathematical Demography and Population Modelling 848	POP848	15
	Sub-total	60
	FINAL TOTAL	180

*Modules compulsory if not completed at Honours level

B.297 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.298 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion adequate progress has been made during the current year.

B.299 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.300 SPECIAL REQUIREMENTS FOR THE PROGRAMME

- B.300.1** Modules indicated with an asterisk must be completed if not completed in the Honours level.
- B.300.2** Modules may not be repeated at Master's level if completed at Honours level.
- B.300.3** The modules will be taught depending on availability of staff.
- B.300.4** Electives at Master's level (maximum 30 credits) may be chosen from Anthropology, Economics, History, Management, Public Health or Sociology, with the approval of both HoDs, if these modules enhance the knowledge needed for Population Studies.
- B.300.5** Assessment methods could also include written research projects, oral presentations and other activities.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
POP801	Population Studies Master's Thesis 801	C	180	9	1	YR	None	None	FA
POP802	Population Studies Master's Thesis 802	C		9	2	YR	None	None	FA
POP803	Population Studies Mini- Thesis 803	C	180	9	1	YR	None	None	FA
POP804	Population Studies Mini- Thesis 804	C		9	2	YR	None	None	FA
POP807	Demographic Analysis 807	E	15	9	1/2	S1	None	None	FA
POP809	Official Statistics 809	E	15	9	1/2	S1	None	None	FA
POP811	Survey Methods 811	E	15	9	1/2	S1	None	None	CFA
POP812	Population Projections 812	E	15	9	1/2	S2	POP807	None	FA
POP813	African Historical Demography 813	E	15	9	1/2	S2	POP807	None	FA
POP814	Social Demography 814	E	15	9	1/2	S2	POP807	None	FA
POP815	Gender Issues and Reproductive Health 815	E	15	9	1/2	S2	POP807	None	FA
POP848	Mathematical Demography and Population Modelling 848	E	15	9	1/2	S2	POP807	None	FA

MASTER OF SCIENCE IN APPLIED GEOLOGY (3891)

B.301 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Applied Geology Degree - MSc (Applied Geology)**:

- B.301.1** an applicant must have obtained an Honours degree with 60% average in Geology and extensive practical experience.
- B.301.2** Departmental approval is subject to the availability of a supervisor.
- B.301.3** an applicant without an Honours degree may require additional information including:
- A curriculum vitae detailing all previous work experience and the period of employment
 - Referee reports from current and previous employers
 - Any other information as may be required.

B.302 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.303 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.304 CURRICULUM

B.304.1 Thesis

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Geology Master's Thesis 801	GEL801	
2nd Enrolment Code		180
Geology Master's Thesis 802	GEL802	
	FINAL TOTAL	180

B.305 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.306 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion adequate progress has been made during the current year.

B.307 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.308 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.308.1 A thesis dealing with an original research topic must be presented.

B.308.2 The Chairperson of the Department may prescribe additional undergraduate and postgraduate modules in Geology or related subjects as required for the intended research.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
GEL801	Geology Master's Thesis 801	C	180	9	1	YR	None	None	FA
GEL802	Geology Master's Thesis 802	C		9	2	YR	None	None	FA

MASTER OF SCIENCE IN BIODIVERSITY AND CONSERVATION BIOLOGY (3824)

B.309 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Biodiversity and Conservation Biology Degree - MSc (Biodiversity and Conservation Biology)**:

B.309.1 an applicant must have obtained an appropriate Honours degree.

B.309.2. Departmental approval is subject to the availability of a supervisor.

B.310 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.311 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.312 CURRICULUM

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Biodiversity Master's Thesis 801	BCB801	
2nd Enrolment Code		180
Biodiversity Master's Thesis 802	BCB802	
	FINAL TOTAL	180

B.313 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.314 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion, adequate progress has been made during the current year.

B.315 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.316 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.316.1 A thesis dealing with an original research topic must be presented.

B.316.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee, as a component of the programme when deemed necessary.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
BCB801	Biodiversity Master's Thesis 801	C	180	9	1	YR	None	None	FA
BCB802	Biodiversity Master's Thesis 802	C		9	2	YR	None	None	FA

MASTER OF SCIENCE IN BIOINFORMATICS (3895)

B.317 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Bioinformatics Degree - MSc (Bioinformatics)**:

- B.317.1** an applicant should be in possession of a Bachelor of Science Honours degree with an average of at least 60% in a related scientific subject area.
- B.317.2** an applicant with an undergraduate degree or diploma (M+3) having at least 3-5 years relevant experience may be considered.
- B.317.3** an applicant who has shown significant potential, but did not achieve an average of 60% may be considered by the Departmental Postgraduate Committee. The applicant may be required to write a well motivated essay on bioinformatics and their role and expectations from the programme. An applicant may be interviewed by the Departmental Postgraduate Committee.
- B.317.4** Departmental approval is subject to the availability of a supervisor.

B.318 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.319 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.320 CURRICULUM

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Bioinformatics Master's Thesis 801	SAN801	
2nd Enrolment Code		180
Bioinformatics Master's Thesis 802	SAN802	
	FINAL TOTAL	180

B.321 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.322 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion adequate progress has been made during the current year.

B.323 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.324 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.324.1 A thesis dealing with an original research topic must be presented.

B.324.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee, as a component of the programme when deemed necessary.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
SAN801	Bioinformatics Master's Thesis 801	C	180	9	1	YR	None	None	FA
SAN802	Bioinformatics Master's Thesis 802	C		9	2	YR	None	None	FA

MASTER OF SCIENCE IN BIOTECHNOLOGY (3185)

B.325 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Biotechnology Degree - MSc (Biotechnology)**:

B.325.1 an applicant should be in possession of a Bachelor of Science Honours degree with an average of at least 60% in the relevant subject/field.

B.325.2 Departmental approval is subject to the availability of a supervisor.

B.326 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.327 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.328 CURRICULUM

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Biotechnology Master's Thesis 801	BTY801	
2nd Enrolment Code		180
Biotechnology Master's Thesis 802	BTY802	
	FINAL TOTAL	180

B.329 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.330 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion adequate progress has been made during the current year.

B.331 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.332 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.332.1 A thesis dealing with an original research topic must be presented.

B.332.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee, as a component of the programme when deemed necessary.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
BTY801	Biotechnology Master's Thesis 801	C	180	9	1	YR	None	None	FA
BTY802	Biotechnology Master's Thesis 802	C		9	2	YR	None	None	FA

MASTER OF SCIENCE IN CHEMICAL SCIENCES (3823)

B.333 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Chemical Sciences Degree - MSc (Chemical Sciences)**:

B.333.1 an applicant should be in possession of a Bachelor of Science Honours degree with an average of at least 60% in Chemistry or relevant subject/field.

B.333.2 Departmental approval is subject to the availability of a supervisor.

B.334 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.335 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.336 CURRICULUM

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Chemistry Master's Thesis 801	CHE801	
2nd Enrolment Code		180
Chemistry Master's Thesis 802	CHE802	
	FINAL TOTAL	180

B.337 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.338 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion adequate progress has been made during the current year.

B.339 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.340 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.340.1 A thesis dealing with an original research topic must be presented.

B.340.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee, as a component of the programme when deemed necessary.

B.340.3 Modules may not be repeated at Master's level if completed at Honours level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
CHE801	Chemistry Master's Thesis 801	C	180	9	1	YR	None	None	FA
CHE802	Chemistry Master's Thesis 802	C		9	2	YR	None	None	FA

MASTER OF SCIENCE IN COMPUTATIONAL FINANCE (3093)

B.341 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Computational Finance Degree - MSc (Computational Finance)**:

B.341.1 an applicant should be in possession of a Bachelor of Science Honours degree with an average of at least 60% in the subject/field.

B.341.2 Departmental approval is subject to the availability of a supervisor.

B.342 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.343 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.344 CURRICULUM

A student must select one mode of study.

B.344.1 Mode 1 – Mini–Thesis and Coursework (not offered to new students in 2026)

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Mathematics 803	MAM803	
Statistics Mini-Thesis 803	STA803	
2nd Enrolment Code		120
Mathematics 804	MAM804	
Statistics Mini-Thesis 804	STA804	
	Sub-total	120
Electives (select four modules)		
Simulation 817 (not offered in 2026)	COF817	15
International Finance 818 (not offered in 2026)	COF818	15
Financial Computing 821 (not offered in 2026)	COF821	15
Theoretical Statistics 826 (not offered in 2026)	COF826	15
Financial Risk Management 827	COF827	15
Probability and Stochastic Processes 828 (not offered in 2026)	COF828	15
Time Series Analysis 829 (not offered in 2026)	COF829	15
Statistical Arbitrage 832 (not offered in 2026)	COF832	15
Credit Derivatives 835 (not offered in 2026)	COF835	15
Linear Financial Models 836 (not offered in 2026)	COF836	15
Advanced Financial Management 704	MAN704	15
Investment Management 705	MAN705	15
Profit Deter and Financial Analysis 706	MAN706	15

Small Business Development and Entrepreneur 711	MAN711	15
Finance Seminar 742	MAN742	15
Advanced Management Accounting 745	MAN745	15
Principles of Management Accounting 753	MAN753	15
Investment Analysis 780	MAN780	15
Statistics for Investments 787	MAN787	15
Sub-total		60
FINAL TOTAL		180

B.344.2 Mode 2 – Research Project and Coursework

Module Name	Module Code	Credits
Compulsory		
Industry Research Project 839	STA839	90
Electives (select six modules)	Sub-total	90
Simulation 817 (not offered in 2026)	COF817	15
International Finance 818 (not offered in 2026)	COF818	15
Financial Computing 821 (not offered in 2026)	COF821	15
Theoretical Statistics 826 (not offered in 2026)	COF826	15
Financial Risk Management 827	COF827	15
Probability and Stochastic Processes 828 (not offered in 2026)	COF828	15
Time Series Analysis 829 (not offered in 2026)	COF829	15
Statistical Arbitrage 832 (not offered in 2026)	COF832	15
Credit Derivatives 835 (not offered in 2026)	COF835	15
Linear Financial Models 836 (not offered in 2026)	COF836	15
Advanced Financial Management 704	MAN704	15
Investment Management 705	MAN705	15
Profit Deter and Financial Analysis 706	MAN706	15
Small Business Development and Entrepreneur 711	MAN711	15
Finance Seminar 742	MAN742	15
Advanced Management Accounting 745	MAN745	15
Principles of Management Accounting 753	MAN753	15
Investment Analysis 780	MAN780	15
Statistics for Investments 787	MAN787	15
Sub-total	Sub-total	90
FINAL TOTAL	FINAL TOTAL	180

B.344.3 Mode 3 – Thesis

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Mathematics 801	MAM801	
Statistics Master's Thesis 801	STA801	180
2nd Enrolment Code		
Mathematics 802	MAM802	
Statistics Master's Thesis 802	STA802	
FINAL TOTAL	FINAL TOTAL	180

B.345 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.346 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion adequate progress has been made during the current year.

B.347 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.348 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.348.1 A thesis dealing with an original research topic must be presented.

B.348.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee.

B.348.3 Electives will be taught depending on availability of staff.

B.348.4 Modules completed at Honours level may not be repeated at Master's level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
COF817	Simulation 817	E	15	9	1/2	S1	STA211, STA221, MAT211, MAT221	None	FA
COF818	International Finance 818	E	15	9	1/2	S1	None	None	FA
COF821	Financial Computing 821	E	15	9	1/2	S1	STA331 or equivalent	None	FA
COF826	Theoretical Statistics 826 4	E	15	9	1/2	S1	STA331; STA332; MAT211; MAT221 or equiv	None	FA

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
COF827	Risk Management 827	E	15	9	1/2	S1	STA331 or equivalent	None	FA
COF828	Probability and Stochastic Processes 828	E	15	9	1/2	S1	STA331 or equivalent	None	FA
COF829	Time Series Analysis 829	E	15	9	1/2	S2	STA211, STA221, MAT211, MAT221	None	FA
COF832	Statistical Arbitrage 832	E	15	9	1/2	S1	MAN781; MAN756 or equivalent.	None	FA
COF835	Credit Derivatives 835	E	15	9	1/2	S2	STA331 or equivalent	None	FA
COF836	Linear Financial Models 836	E	15	9	1/2	S2	STA331 or equivalent	None	FA
MAM801	Mathematics 801	C	180	9	1	YR	None	None	FA
MAM802	Mathematics 802	C		9	2	YR	None	None	FA
MAM803	Mathematics Master's Thesis 803	C	120	9	1	YR	None	None	FA
MAM804	Mathematics Master's Thesis 804	C		9	2	YR	None	None	FA
MAN704	Advanced Financial Management 704	E	15	9	1/2	S1	None	None	CFA
MAN705	Investment Management 705	E	15	9	1/2	S2	None	None	CFA
MAN706	Profit Deter and Financial Analysis 706	E	15	9	1/2	S2	MAN604/ MAN704 and ADM681/A CC781 or equivalent level 6 module	None	CFA
MAN711	Small Business Development and Entrepreneur 711	E	15	9	1/2	S2	None	None	CFA
MAN742	Finance Seminar 742	E	15	9	1/2	S2	None	None	CFA
MAN745	Advanced Management Accounting 745	E	15	9	1/2	S1	None	None	CFA
MAN753	Principles of Management Accounting 753	E	15	9	1/2	S1	None	None	CFA
MAN780	Investment Analysis 780	E	15	9	1/2	S2	ADM605/ MAN705/ MAN307	None	CFA
MAN787	Statistics for Investments 787	E	15	9	1/2	S2	None	None	CFA

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
STA801	Statistics Master's Thesis 801	C	180	9	1	YR	STA810 or equivalent	None	FA
STA802	Statistics Master's Thesis 802	C		9	2	YR	STA810 or equivalent	None	FA
STA803	Medical Bioscience Mini-Thesis 803	C	60	9	1	YR	STA810	None	FA
STA804	Medical Bioscience Mini-Thesis 804	C		9	2	YR	STA810	None	FA
STA839	Industry Research Project 839	C	90	9	1/2	YR	STA810 or equivalent	None	FA

MASTER OF SCIENCE IN COMPUTER SCIENCE (3848)

B.349 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Computer Science Degree - MSc (Computer Science)**:

- B.349.1** an applicant should be in possession of a Bachelor of Science Honours degree with an average of at least 60% in Computer Science, or
- B.349.2** an equivalent 4-year Bachelor's degree with a Computer Science major, and
- B.349.3** departmental approval is subject to the availability of a supervisor.

B.350 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.351 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.352 CURRICULUM

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Computer Science Master's Thesis 801	COS801	
2nd Enrolment Code		180
Computer Science Master's Thesis 802	COS802	
	FINAL TOTAL	180

B.353 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.354 PROGRESS RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion adequate progress has been made during the current year.

B.355 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.356 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.356.1 A thesis dealing with an original research topic must be presented.

B.356.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee, as a component of the programme when deemed necessary.

B.356.3 Modules may not be repeated at Master's level if completed at Honours level.

B.356.4 Current research areas available for supervision include Machine Learning, Computational Intelligence, Internet Computing, Computer Networks and Mobile Computing and Information and Communication Technologies for Development (ICT4D). Details may be found on the Departmental website.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
COS801	Computer Science Master's Thesis 801	C	180	9	1	YR	None	None	FA
COS802	Computer Science Master's Thesis 802	C		9	2	YR	None	None	FA

MASTER OF SCIENCE IN ENVIRONMENT AND WATER SCIENCE (3085)

B.357 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Environment and Water Science Degree – MSc (Environment and Water Science)**:

B.357.1 an Honours degree in Environment and Water Science, or

B.357.2 an applicant with a Bachelor of Science Honours degree in a related field can be considered by the Departmental Postgraduate Committee, and

B.357.3 departmental approval is subject to the availability of a supervisor.

B.358 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.359 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.360 CURRICULUM

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Environmental Science Master's Thesis 801	ENV801	
2nd Enrolment Code		180
Environmental Science Master's Thesis 802	ENV802	
	FINAL TOTAL	180

B.361 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.362 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion adequate progress has been made during the current year.

B.363 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.364 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.364.1 A thesis dealing with an original research topic must be presented.

B.364.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee, as a component of the programme when deemed necessary.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
ENV801	Environmental Science Master's Thesis 801	C	180	9	1	YR	None	None	FA
ENV802	Environmental Science Master's Thesis 802	C		9	2	YR	None	None	FA

MASTER OF SCIENCE IN MATHEMATICAL SCIENCE (3849)

B.365 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Mathematical Science Degree - MSc (Mathematical Science)**:

- B.365.1** a 4-year Bachelors degree in Mathematics with an average of at least 60%, or a science or engineering subject with a significant mathematics component, or
- B.365.2** an Honours degree in Mathematics with an average of at least 60%, or a science or engineering subject with a significant mathematics component, or
- B.365.3** a BEd degree with an average of 60% in Mathematics as a major subject for the undergraduate degree, or any equivalent degree.
- B.365.4** Departmental approval is subject to the availability of a supervisor.

B.366 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.367 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.368 CURRICULUM

A student must select one mode of study.

B.368.1 Mode 1- Mathematics – Thesis

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Mathematics Master's Thesis 801	MAM801	
2nd Enrolment Code		
Mathematics Master's Thesis 802	MAM802	180
	FINAL TOTAL	180

B.368.2 Mode 2 - Mathematics – Mini-Thesis and Coursework

(not offered in 2026)

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Mathematics Mini-Thesis 803	MAM803	
2nd Enrolment Code		
Mathematics Mini-Thesis 804	MAM804	120
	Sub-total	120

Electives (select four modules)

Algebraic Coding Theory 820	MAM820	15
Algebraic Number Theory 821	MAM821	15
Advanced Numerical Analysis 822	MAM822	15
Algebraic Topology 823	MAM823	15
Category Theory 824	MAM824	15
Design Theory 825	MAM825	15
Graph Theory 826	MAM826	15
Group Theory 827	MAM827	15
Representation Theory of Finite Groups 828	MAM828	15
	Sub-total	60

FINAL TOTAL 180

B.368.3 Mode 3 - Mathematics Education – Thesis

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Mathematics Education Master's Thesis 807	MAM807	
2nd Enrolment Code		
Mathematics Education Master's Thesis 808	MAM808	180
	FINAL TOTAL	180

B.368.4 Mode 4 - Mathematics Education – Mini-Thesis and Coursework

(not offered in 2026)

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Mathematics Education Mini-Thesis 805	MAM805	
2nd Enrolment Code		
Mathematics Education Mini-Thesis 806	MAM806	120
	Sub-total	120
Electives (select four modules)		
Mathematics Education 811	MAM811	15
Mathematics Education 812	MAM812	15
Mathematics Education 813	MAM813	15
Mathematics Education 814	MAM814	15
Mathematics Education 815	MAM815	15
Mathematics Education 816	MAM816	15
	Sub-total	60
	FINAL TOTAL	180

B.368.5 Mode 5 - *AIMS Programme

Module Name	Module Code	Credits
Compulsory		
Mathematics Mini-Thesis / Research Project 803	AIM803	60
Communication Skills, Computing and LaTeX 811	AIM811	7

Mathematical Problem Solving 812	AIM812	7
Problem Solving in Physics 813	AIM813	7
Scientific Software Development in Python 814	AIM814	7
Experimental Mathematics with SAGE 815	AIM815	7
Selected Topics in Mathematical Sciences 816	AIM816	7
Professional Development Modules 817	AIM817	7
	Sub-total	109

Electives (select one module)

Advanced Topics in Mathematics Science A 818	AIM818	77
Advanced Topics in Mathematics Science B 819	AIM819	77
Advanced Topics in Mathematics Science C 820	AIM820	77
Advanced Topics in Mathematics Science D 821	AIM821	77
Advanced Topics in Mathematics Science E 822	AIM822	77
	Sub-total	77

FINAL TOTAL 186

*African Institute for Mathematical Sciences

B.369 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.370 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion adequate progress has been made during the current year.

B.371 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1.

B.372 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.372.1 A thesis dealing with an original research topic must be presented.

B.372.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee as a component of the programme when deemed necessary.

B.372.3 Modules may not be repeated at Master's level if completed at Honours level.

B.372.4 The academic assessment of a student for the Master's degree in Mathematical Sciences for mode 5 (AIMS Programme) is completed via:

- (i) Continuous assessment through written assignments, tutorials, short tests and presentations set by the lecturers, and

- (ii) A written mini-thesis which the student is required to present (orally) to a panel of examiners. This panel includes an internal examiner, and an external examiner, and
- (iii) Integrated assessment - a portfolio for each student is compiled, containing the grades achieved for each of the courses attended as well as observations on their presentations, all their assignments, completed exercises and their final mini-thesis, and
- (iv) External evaluation of a student's performance and all aspects of the programme is conducted by six senior academics representing the different mathematical sciences disciplines (including Physics).

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
MAM801	Mathematics Master's Thesis 801	C	180	9	1	YR	None	None	FA
MAM802	Mathematics Master's Thesis 802	C	180	9	2	YR	None	None	FA
MAM803	Mathematics Master's Thesis 803 (Mathematics – Mini-Thesis and Coursework)	C	120	9	1	YR	None	None	FA
MAM804	Mathematics Master's Thesis 804 (Mathematics – Mini-Thesis and Coursework)	C	120	9	2	YR	None	None	FA
MAM807	Mathematics Education Master's Thesis 807 (Mathematics Education – Thesis)	C	180	9	1	YR	None	None	FA
MAM808	Mathematics Education Master's Thesis 808 (Mathematics Education – Thesis)	C	180	9	2	YR	None	None	FA
AIM803	Mathematics Mini-Thesis / Research Project 803 (*AIMS Programme)	C	60	9	1	S1	None	None	FA
AIM811	Communication Skills, Computing and LaTeX 811	C	7	9	1/2	S1	None	None	CA
AIM812	Mathematical Problem Solving 812	C	7	9	1/2	S1	None	None	CA
AIM813	Problem Solving in Physics 813	C	7	9	1/2	S1	None	None	CA
AIM814	Scientific Software Development in Python 814	C	7	9	1/2	S1	None	None	CA
AIM815	Experimental Mathematics with SAGE 815	C	7	9	1/2	S1	None	None	CA

Module Code	Module Name	Compulsory	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
AIM816	Selected Topics in Mathematical Sciences 816	C	7	9	1/2	S1	None	None	CA
AIM817	Professional Development Modules 817	C	7	9	1/2	S1	None	None	CA
AIM818	Advanced Topics in Mathematics Science A 818	E	77	9	1/2	S1	None	None	CA
AIM819	Advanced Topics in Mathematics Science B 819	E	77	9	1/2	S1	None	None	CA
AIM820	Advanced Topics in Mathematics Science C 820	E	77	9	1/2	S1	None	None	CA
AIM821	Advanced Topics in Mathematics Science D 821	E	77	9	1/2	S1	None	None	CA
AIM822	Advanced Topics in Mathematics Science E 822	E	77	9	1/2	S1	None	None	CA

MASTER OF SCIENCE IN MEDICAL BIOSCIENCES (3819)

B.373 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Medical Biosciences Degree - MSc (Medical Biosciences)**:

B.373.1 an applicant should have an Honours degree with an average of at least 60% pass in Medical Biosciences subjects or related field.

B.373.2 Departmental approval is subject to the availability of a supervisor.

B.373.3 an applicant without an Honours degree, additional information must be submitted:

- Curriculum vitae detailing all previous work experience and the period of employment;
- Referee reports from current and previous employers;
- Any other information that may be required.

B.374 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the faculty.

B.375 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.376 CURRICULUM

A student must select one mode of study.

B.376.1 Mode 1 – Thesis

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Medical Bioscience Master's Thesis 801	MBS801	
2nd Enrolment Code		180
Medical Bioscience Master's Thesis 802	MBS802	
	FINAL TOTAL	180

B.376.2 Mode 2 - Mini-Thesis and Coursework (not offered in 2026)

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Medical Bioscience Mini-Thesis 803	STA803	
2nd Enrolment Code		60
Medical Bioscience Mini-Thesis 804	STA804	

Advanced Technique Studies 805	MBS805	20
Biostatistics 835	STA835	10
	Sub-total	80
Electives (select four modules)		
Specialist Module in Anatomical Science 806	MBS806	10
The Microbiology of Female Morbidity and Mortality 808	MBS808	10
Advanced Studies in Reproduction and Health 809	MBS809	10
Advanced Nutritional Biosciences 810	MBS810	10
Apoptosis and Cell Signalling 811	MBS811	10
Physical Activity and the Environment 812	MBS812	10
Advanced Cardiovascular Physiology 813 (not offered in 2026)	MBS813	10
Specialist Module in Medical Biosciences 814	MBS814	10
	Sub-total	40
	FINAL TOTAL	120

B.377 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.378 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion adequate progress has been made during the current year.

B.379 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.380 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.380.1 A thesis dealing with an original research topic must be presented.

B.380.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee, as a component of the programme when deemed necessary.

B.380.3 Modules may not be repeated at Master's level if completed at Honours level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
MBS801	Medical Bioscience Master's Thesis 801	C	180	9	1	YR	None	None	CA
MBS802	Medical Bioscience Master's Thesis 802	C		9	2	YR	None	None	CA

MASTER OF SCIENCE IN NANOSCIENCE (3089)

B.381 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Nanoscience Degree - MSc (Nanoscience)**:

An applicant should have an Honours degree in either Physics, Chemistry, Biotechnology or Medical Bioscience or related field.

B.382 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the faculty.

B.383 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time students shall be two years.

B.384 CURRICULUM

The programme has three specialisation areas of which students should select only one.

B.384.1 Stream 1 – Nanophysics

Module Name	Module Code	Credits
Compulsory		
Advanced Nanophysics 831	NSS831	48
Experimental Techniques in Nanophysics 832	NSS832	16
Nanoscience Research Project 803	NSS803	15
Nanoscience Research Project 804	NSS804	85
	Sub-total	164
Electives (select all modules)		
Central Concepts in Nanoscience 809	NSS809	4
Management for Nanoscientists 810	NSS810	4
Foundations of Nanobiomedical Sciences for Non-Biologists 813	NSS813	4
Foundations of Nanochemistry for Non-Chemists 823	NSS823	4
	Sub-total	16
	FINAL TOTAL	180

B.384.2 Stream 2 – Nanobiomedical

Module Name	Module Code	Credits
Compulsory		
Nanoscience Research Project 803	NSS803	15
Nanoscience Research Project 804	NSS804	85
or		
Nanoscience Research Project 803 (Biotechnology Students)	NSB803	15
Nanoscience Research Project 804 (Biotechnology Students)	NSB804	85

or

Nanoscience Research Project 803 (Pharmacy Students)	NSP803	15
Nanoscience Research Project 804 (Pharmacy Students)	NSP804	85
Advanced Nanobiomedical Science 811	NSS811	48
Experimental Techniques in Nanobiomedical Science 812	NSS812	16
	Sub-total	164

Electives (select all modules)

Central Concepts in Nanoscience 809	NSS809	4
Management for Nanoscientists 810	NSS810	4
Foundations of Nanochemistry for Non-Chemists 823	NSS823	4
Foundations of Nanophysics for Non-Physicists 833	NSS833	4
	Sub-total	16

FINAL TOTAL 180

B.384.3 Stream 3 – Nanochemistry

Module Name

Module Code Credits

Compulsory

Advanced Nanochemistry 821	NSS821	48
Experimental Techniques in Nanochemistry 822	NSS822	16
Nanoscience Research Project 803	NSS803	15
Nanoscience Research Project 804	NSS804	85
	Sub-total	164

Electives (select all modules)

Central Concepts in Nanoscience 809	NSS809	4
Management for Nanoscientists 810	NSS810	4
Foundations of Nanobiomedical Sciences for Non-Biologists 813	NSS813	4
Foundations of Nanophysics for Non-Physicists 833	NSS833	4
	Sub-total	16

FINAL TOTAL 180

B.385 ASSESSMENT

Assessment is governed by Rule A.5.4 as stipulated in the University Calendar: General Information Part 1.

B.385.1 The electives (4 modules of 4 credits each) will be considered as one exit credit module for the purpose of awarding this structured Masters degree Cum Laude, Magna Cum Laude or Summa Cum Laude.

B.386 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion adequate progress has been made during the current year.

B.387 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.388 SPECIAL REQUIREMENTS FOR THE PROGRAMME

There are no special requirements for this programme.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requsite	Assessment Type
NSS/NSB/ NSP803	Nanoscience Research Project 803	C	100	9	1	YR	None	None	FA
NSS/NSB/ NSP804	Nanoscience Research Project 804	C		9	2	YR	None	None	FA
NSS809	Central Concepts in Nanoscience 809	E	4	8	1/2	YR	None	None	CFA
NSS810	Management for Nanoscientists 810	E	4	9	1/2	S1	None	None	CFA
NSS811	Advanced Nanobiomedical Science 811 (Stream 2 – Nanobiomedical)	C	48	9	1/2	YR	None	None	CFA
NSS812	Experimental Techniques in Nanobiomedical Science 812	C	16	9	1/2	YR	None	None	CFA
NSS813	Foundations of Nanobiomedical Sciences for Non-Biologists 813	E	4	9	1/2	S1	None	None	CFA
NSS821	Advanced Nanochemistry 821 (Stream 3 – Nanochemistry)	C	48	9	1/2	YR	CHE701, CHE704	None	CFA
NSS822	Experimental Techniques in Nanochemistry 822	C	16	9	1/2	YR	CHE701, CHE705	None	CFA
NSS823	Foundations of Nanochemistry for Non- Chemists 823	E	4	9	1/2	S1	None	None	CFA
NSS831	Advanced Nanophysics 831 (Stream 1 – Nanophysics)	C	48	9	1/2	YR	PHY724, PHY725, PHY726	None	CFA
NSS832	Experimental Techniques in Nanophysics 832	C	16	9	1/2	YR	PHY724, PHY725, PHY726	None	CFA

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
NSS833	Foundations of Nanophysics for Non-Physicists 833	E	4	8	1/2	YR	None	None	CFA

MASTER OF SCIENCE IN PHARMACEUTICAL SCIENCES (3181)

B.389 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Pharmaceutical Sciences Degree - MSc (Pharmaceutical Sciences)**:

- B.389.1** an applicant must have obtained either a 4-year Pharmacy degree [e.g. BPharm or BSc (Pharm)], or
- B.389.2** a relevant professional degree (e.g. MBChB, etc.), or
- B.389.3** a 4-year Science degree (e.g. BSc Hons), or
- B.389.4** an applicant with equivalent qualifications may be considered under special circumstances with suitable motivation. Factors considered include academic merit, appropriate experience, reports from referees, and compatibility of research, career interests and results of the interview.
- B.389.5** Departmental approval is subject to the availability of a supervisor.

B.390 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.391 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time students shall be two years.

B.392 CURRICULUM

B.392.1 Thesis

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Pharmaceutical Sciences Master's Thesis 801	PHS801	
2nd Enrolment Code		180
Pharmaceutical Sciences Master's Thesis 802	PHS802	
	FINAL TOTAL	180

A thesis on an approved topic in Pharmacy Practice, Pharmaceuticals, Pharmaceutical Chemistry or Pharmacology shall be presented.

B.393 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.394 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion adequate progress has been made during the current year.

B.395 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.396 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.396.1 A thesis dealing with an original research topic must be presented.

B.396.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee, as a component of the programme when deemed necessary.

B.396.3 A thesis on a research topic must be presented. A student may be required to do extra module work which is not examinable if required by the supervisor.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
PHS801	Pharmaceutical Sciences Master's Thesis 801	C	180	9	1	YR	None	None	FA
PHS802	Pharmaceutical Sciences Master's Thesis 802	C		9	2	YR	None	None	FA

MASTER OF SCIENCE IN PHARMACY ADMINISTRATION AND POLICY REGULATION (3859)

B.397 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Pharmacy Administration and Policy Regulation Degree - MSc (Pharmacy Administration and Policy Regulation)**:

- B.397.1** an applicant must have obtained either a 4-year Pharmacy degree (e.g. BPharm or BSc (Pharm)), or
- B.397.2** a 4-year Science degree (e.g. BSc Hons) with relevant experience in Pharmacy, or
- B.397.3** an applicant with equivalent qualifications may be considered under special circumstances with suitable motivation. Factors considered include academic merit, appropriate experience, reports from referees, and compatibility of research, career interests and results of the interview.

B.398 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.399 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum shall be two years.

B.400 CURRICULUM

- B.400.1** Online lectures and tutorials and block weeks.

Module Name	Module Code	Credits
Compulsory Module		
1st Enrolment		
Pharmacy Administration and Policy Mini-Thesis 803	PAR803	60
2nd Enrolment		
Pharmacy Administration and Policy Mini-Thesis 804	PAR804	
Electives (select all modules)		
Ethics 811	PAR811	10
Common Technical Document 812	PAR812	10
Regulatory Science 813	PAR813	10
Complementary and Traditional Medicines and their Regulation 814	PAR814	10
Regulatory Science and Medical Devices 815	PAR815	10
Pharmacovigilance 816	PAR816	10
Regulatory Affairs 817	PAR817	10
Research Methods 818	PAR818	10
Leadership and Negotiations 819	PAR819	10
Clinical Trials and Statistics 820	PAR820	20
Health Economics 821	PAR821	10
FINAL TOTAL		180

B.401 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.402 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion, adequate progress has been made during the current year.

B.403 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.404 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.404.1 A thesis dealing with an original research topic must be presented.

B.404.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee, as a component of the programme when deemed necessary.

B.404.3 A thesis on a research topic must be presented. A student may be required to do extra module which is not examinable if required by the supervisor.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requirement	Co-Requirement	Assessment Type
PAR803	Pharmacy Administration and Policy Mini-Thesis 803	C	60	9	1	S2	None	None	FA
PAR804	Pharmacy Administration and Policy Mini-Thesis 804	C		9	2	S2	None	None	FA
PAR811	Ethics 811	E	10	9	1/2	S1	None	None	CFA
PAR812	Common Technical Document 812	E	10	9	1/2	S1	None	None	CFA
PAR813	Regulatory Science 813	E	10	9	1/2	S1	None	None	CFA
PAR814	Complementary and Traditional Medicines and their Regulation 814	E	10	9	1/2	S1	None	None	CFA
PAR815	Regulatory Science and Medical Devices 815	E	10	9	1/2	S1	None	None	CFA

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
PAR816	Pharmacovigilance 816	E	10	9	1/2	S1	None	None	CFA
PAR817	Regulatory Affairs 817	E	10	9	1/2	S1	None	None	CFA
PAR818	Research Methods 818	E	10	9	1/2	S2	None	None	CFA
PAR819	Leadership and Negotiations 819	E	10	9	1/2	S1	None	None	CFA
PAR820	Clinical Trials and Statistics 820	E	20	9	1/2	S1	None	None	CFA
PAR821	Health Economics 821	E	10	9	1/2	S1	None	None	CFA

MASTER OF SCIENCE IN PHYSICAL SCIENCE (3081)

B.405 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Physical Science Degree - MSc (Physical Science)**:

An applicant should have an Honours degree in Physics or related field.

B.406 SELECTION

Final selection is based on an applicant meeting the criteria as determined by the faculty. An applicant who is part of the National Astrophysics and Space Science Programme (NASSP) needs to pass the NASSP Master's coursework for admission to this degree.

B.407 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.408 CURRICULUM

B.408.1 Mode 1 – Mini-Thesis and Coursework (not offered in 2026)

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Physics Mini-Thesis 803	PHY803	
2nd Enrolment Code		
Physics Mini-Thesis 804	PHY804	90
	Sub-total	90
Electives (select three modules)		
Quantum Mechanics 820	PHY820	30
Nuclear Physics 830	PHY830	30
Solid State Physics 840	PHY840	30
Physics Education 850	PHY850	30
Mathematical Methods of Physics 860	PHY860	30
	Sub-total	90
	FINAL TOTAL	180

B.408.2 Mode 2 – Thesis

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Physics Master's Thesis 801	PHY801	
2nd Enrolment Code		
Physics Master's Thesis 802	PHY802	180
	FINAL TOTAL	180

B.409 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.410 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion, adequate progress has been made during the current year.

B.411 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.412 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.412.1 A thesis dealing with an original research topic must be presented.

B.412.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee, as a component of the programme when deemed necessary.

B.412.3 Specific additional modules may be prescribed.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
PHY801	Physics Master's Thesis 801 (Mode 2 – Thesis)	C	180	9	1	YR	None	None	FA
PHY802	Physics Master's Thesis 802 (Mode 2 – Thesis)	C		9	2	YR	None	None	FA
PHY803	Physics Mini-Thesis 803 [Mode 1 – Mini-Thesis and Coursework]	C	90	9	1	YR	None	None	FA
PHY804	Physics Mini-Thesis 804 [Mode 1 – Mini-Thesis and Coursework]	C		9	2	YR	None	None	FA
PHY820	Quantum Mechanics 820	E	30	9	1/2	S1	None	None	CFA
PHY830	Nuclear Physics 830	E	30	9	1/2	S2	None	None	CFA
PHY840	Solid State Physics 840	E	30	9	1/2	S2	None	None	CFA

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
PHY850	Physics Education 850	E	30	9	1/2	S2	None	None	CFA
PHY860	Mathematical Methods of Physics 860	E	30	9	1/2	S2	None	None	CFA

MASTER OF SCIENCE IN STATISTICAL SCIENCE (3838)

B.413 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Science in Statistical Science Degree - MSc (Statistical Science)**:

- B.413.1** an applicant should have an Honours degree with an average of at least 60% in Statistics/Data Science and have passed an Honours level module with Statistical Learning content. An applicant may be required to take additional module(s) to strengthen background knowledge if needed.
- B.413.2** an applicant could have an Honours degree with an average of at least 60% in Computer Science or related field, but in addition the student should have passed an Honours level module with Statistical Learning content and have passed third year level Statistics. An applicant may be required to take additional module(s) to strengthen background knowledge if needed.
- B.413.3** an applicant may be required to take additional module(s) to strengthen background knowledge.

B.414 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.415 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum for full-time shall be one year or two years for part-time.

B.416 CURRICULUM

B.416.1 Mode 1 – Thesis

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Statistics Master's Thesis 801	STA801	
2nd Enrolment Code		180
Statistics Master's Thesis 802	STA802	
FINAL TOTAL		180

B.416.2 Mode 2 – Mini-Thesis and Coursework (not offered in 2026)

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Statistics Mini-Thesis 803	STA803	
2nd Enrolment Code		120
Statistics Mini-Thesis 804	STA804	
Sub-total		120

Electives (select four modules)

Simulation 817 (not offered in 2026)	COF817	15
Theoretical Statistics 826 (not offered in 2026)	COF826	15
Risk Management 827	COF827	15
Probability and Stochastic Processes 828 (not offered in 2026)	COF828	15
Time Series Analysis 829 (not offered in 2026)	COF829	15
Data Mining II 800	STA800	15
Research Methodology 810	STA810	15
Multivariate Analysis 829 (not offered in 2026)	STA829	15
Experimental Design 830 (not offered in 2026)	STA830	15
Biostatistics 835	STA835	15
Statistical Genetics 837 (not offered in 2026)	STA837	15
Matrix Methods 853	STA853	15
	Sub-total	60

FINAL TOTAL 180

B.416.3 Mode 3 – Research Project and Coursework (Specialisation in Data Science)

Module Name	Module Code	Credits
Compulsory		
Industry Research Project 839	STA839	90
	Sub-total	90

***Electives (select six modules)**

Risk Management 827	COF827	15
Data Mining II 800	STA800	15
Research Methodology 810	STA810	15
Business Intelligence 841	STA841	15
Contemporary Business Analysis 842	STA842	15
Multicriteria Decision Making 843	STA843	15
	Sub-total	90

FINAL TOTAL 180

*Please check with Department which elective modules will be taught.

B.417 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.418 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion, adequate progress has been made during the current year.

B.419 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.420 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.420.1 A thesis dealing with an original research topic must be presented.

B.420.2 Coursework may be included, as recommended by the Departmental Postgraduate Committee, as a component of the programme when deemed necessary.

B.420.3 Modules may not be repeated at Master's level if completed at Honours level.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
STA801	Statistics Master's Thesis 801 (Mode 1 – Thesis)	C	180	9	1	YR	STA810 or equivalent	None	FA
STA802	Statistics Master's Thesis 802 (Mode 1 – Thesis)	C		9	2	YR	STA810 or equivalent	None	FA
STA803	Statistics Mini-Thesis 803 (Mode 2 – Mini-Thesis and Coursework)	C	120	9	1	YR	STA810 or equivalent	None	FA
STA804	Statistics Mini-Thesis 804 (Mode 2 – Mini-Thesis and Coursework)	C		9	2	YR	STA810 or equivalent	None	FA
COF817	Simulation 817	E	15	9	1/2	S1	STA211, STA221, MAT211, MAT221	None	FA
COF826	Theoretical Statistics 826 4	E	15	9	1/2	S1	STA331; STA332; MAT211; MAT221 or equivalent	None	FA
COF827	Risk Management 827	E	15	9	1/2	S1	STA331 or equivalent	None	FA
COF828	Probability and Stochastic Processes 828	E	15	9	1/2	S1	STA331 or equivalent	None	FA
COF829	Time Series Analysis 829	E	15	9	1/2	S2	STA211, STA221, MAT211, MAT221	None	FA

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
STA800	Data Mining II 800	E	15	9	1/2	S2	STA331; STA332 or equivalent	None	FA
STA810	Research Methodology 810	E	15	9	1/2	S1	None	None	FA
STA829	Multivariate Analysis 829	E	15	9	1/2	S2	STA853 or equivalent and knowledge of SAS.	None	FA
STA830	Experimental Design 830	E	15	9	1/2	S1	STA331, STA332 or equivalent	None	CFA
STA835	Biostatistics 835	E	15	9	1/2	S1	STA331; STA332; or equivalent	None	FA
STA837	Statistical Genetics 837	E	15	9	1/2	S2	None	None	FA
STA839	Industry Research Project 839	C	90	9	1/2	YR	STA810 or equivalent	None	FA
STA841	Business Intelligence 841	E	15	9	1/2	S1	None	None	FA
STA842	Contemporary Business Analysis 842	E	15	9	1/2	S1	None	None	FA
STA843	Multicriteria Decision Making 843	E	15	9	1/2	S1	None	None	FA
STA853	Matrix Methods 853	E	15	9	1/2	S1	STA331; STA332; MAT211; MAT221 or equivalent	None	FA

MASTER OF CLINICAL PHARMACY (3852)

B.421 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.4, a student shall be required to meet the following criteria to be enrolled for the **Master of Clinical Pharmacy Degree – M Clinical Pharmacy**:

- B.421.1** an applicant must have obtained a 4-year Pharmacy degree or diploma (e.g. BPharm, BSc (Pharmacy) or Dip. Pharmacy).
- B.421.2** an applicant who does not have a South African qualification is required to send their results/qualifications to the South African Qualifications Authority (SAQA) in order to have them evaluated and equated to a South African qualification before admission can be considered.
- B.421.3** The rules and guidelines of UWC regarding Recognition of Prior Learning (RPL) shall apply.

B.422 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

An applicant should obtain at least an average of 60% in the relevant module(s) related to Clinical Pharmacy. In addition an applicant should:

- B.422.1** provide a portfolio of evidence of work done up to the time of application;
- B.422.2** attend an interview;
- B.422.3** be employed at a facility appropriate to achieve the learning objectives, e.g. a hospital.

B.423 DURATION

Unless Senate decides otherwise, the minimum duration of the curriculum shall be three years.

B.424 CURRICULUM

Module Name Compulsory	Module Code	Credits
Year Level 1 (select all modules)		
Introduction to Clinical Pharmacy and Pharmaceutical Care 811	PHC811	25
Evidence Based Medicine, Rational Medicine Use and Pharmacopidemiology 812	PHC812	30
The following modules may be selected in Year 1 or 2		
Applied Pharmacokinetics 814	PHC814	25
Pharmaceutical Care – Adult Medicine (1) 815	PHC815	30
Pharmaceutical Care - Adult Medicine (2) 820	PHC820	85

***Year Level 2**

Pharmaceutical Care – Paediatrics 816	PHC816	30
Mini-Thesis in Clinical Pharmacy 803	PHC803	120

Electives (select one module)

Advanced Pharmaceutical Care – HIV 821	PHC821	15
Advanced Pharmaceutical Care – Oncology 822	PHC822	15
Advanced Pharmaceutical Care – Critical Care 823	PHC823	15
	Sub-total	240

***Year Level 3**

Mini-Thesis in Clinical Pharmacy 804	PHC804	-
	Sub-total	

FINAL TOTAL 360

*Please register the mini-thesis in Year 2 and 3 of registration.

B.425 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.426 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion adequate progress has been made during the current year.

B.427 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.428 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.428.1 A student should complete 50% of the coursework, which should include the component the research will be conducted in, before the student can proceed with the prescribed research.

B.428.2 A thesis dealing with an original research topic must be presented. A student may be required to do extra module which is not examinable if required by the supervisor.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Enrolment Year	Duration	Pre-Requisite	Co-Requisite	Assessment Type
PHC803	Mini-Thesis in Clinical Pharmacy 803	C	120	9	2	YR	None	None	FA
PHC804	Mini-Thesis in Clinical Pharmacy 804	C		9	3	YR	None	None	FA
PHC811	Introduction to Clinical Pharmacy and Pharmaceutical Care 811	C	25	9	1	YR	BPharm	None	CA
PHC812	Evidence Based Medicine, Rational Medicine Use and Pharmacopidemiology 812	C	30	9	1	YR	BPharm	None	CA
PHC814	Applied Pharmacokinetics 814	C	25	9	1/2	YR	None	None	CA
PHC815	Pharmaceutical Care – Adult Medicine (1) 815	C	30	9	1/2	YR	None	None	CA
PHC816	Pharmaceutical Care – Paediatrics 816	C	30	9	2	YR	PHC811, PHC812	None	CA
PHC820	Pharmaceutical Care - Adult Medicine (2) 820	C	85	9	1/2	YR	None	None	CA
PHC821	Advanced Pharmaceutical Care – HIV 821	E	15	9	2	YR	PHC811, PHC812	None	CA
PHC822	Advanced Pharmaceutical Care – Oncology 822	E	15	9	2	YR	PHC811, PHC812	None	CA
PHC823	Advanced Pharmaceutical Care – Critical Care 823	E	15	9	2	YR	PHC811, PHC812	None	CA

DOCTOR OF PHARMACY (3951)

B.429 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.5, a candidate shall be required to meet the following criteria to be enrolled for the **Doctor of Pharmacy Degree - DPharm**:

An applicant should have obtained a MPharm degree or an equivalent qualification in order to be admitted to the DPharm degree.

B.430 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.431 DURATION

The duration of the degree is a minimum of two years subject to Rule A.4.4 as per University Calendar Part 1.

B.432 CURRICULUM

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Pharmaceutical Doctoral Thesis 901	PHA901	
2nd Enrolment Code		
Pharmaceutical Doctoral Thesis 902	PHA902	360
	FINAL TOTAL	360

B.433 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.434 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion, adequate progress has been made during the current year.

B.435 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.436 SPECIAL REQUIREMENTS FOR THE PROGRAMME

A thesis dealing with an original research topic must be presented.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (**C**), Electives (**E**), Year Module (**YR**), Semester Module (**S**), Pre-requisites and Co-requisites; Service Module (**SM**)

ASSESSMENT TYPE:

Continuous Assessment (**CA**) or Final Assessment (**FA**) or Continuous Assessment and Final Assessment (**CFA**)

Module Code	Module Name	Compulsory	Credits	NQF Level	Enrolment Year	Duration	Assessment Type
PHA901	Pharmaceutical Doctoral Thesis 901	C	360	10	1	YR	FA
PHA902	Pharmaceutical Doctoral Thesis 902	C		10	2	YR	FA

DOCTOR OF PHILOSOPHY (3961)

B.437 ADMISSION

Unless Senate decides otherwise and subject to Rule A.2.5, a candidate shall be required to meet the following criteria to be enrolled for the **Doctor of Philosophy Degree - PhD (as indicated in B.440 below)**:

B.437.1 A candidate shall have obtained a Master of Science degree or an equivalent qualification in the relevant or related field of study in order to be admitted to the PhD degree.

B.437.2 Proven research ability and acceptable research protocol in any of the specialisations.

B.437.3 Departmental approval is subject to the availability of a supervisor.

B.438 SELECTION

Final selection shall be based on an applicant meeting the criteria as determined by the Faculty.

B.439 DURATION

The duration of the degree is a minimum of two years subject to Rule A.4.4 as per University Calendar Part 1.

B.440 CURRICULUM

The PhD programme is offered in the following specialisation areas on an approved original research topic, in one of the following disciplines:

B.440.1 Applied Geology

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Geology Doctoral Thesis 901	GEL901	
2nd Enrolment Code		
Geology Doctoral Thesis 902	GEL902	360
	FINAL TOTAL	360

B.440.2 Biodiversity and Conservation Biology

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Biodiversity Doctoral Thesis 901	BCB901	
2nd Enrolment Code		
Biodiversity Doctoral Thesis 902	BCB902	360
	FINAL TOTAL	360

B.440.3 Bioinformatics

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Bioinformatics Doctoral Thesis 901	SAN901	
2nd Enrolment Code		
Bioinformatics Doctoral Thesis 902	SAN902	360
FINAL TOTAL		360

B.440.4 Biotechnology

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Biotechnology Doctoral Thesis 901	BTY901	
2nd Enrolment Code		
Biotechnology Doctoral Thesis 902	BTY902	360
FINAL TOTAL		360

B.440.5 Chemistry

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Chemistry Doctoral Thesis 901	CHE901	
2nd Enrolment Code		
Chemistry Doctoral Thesis 902	CHE902	360
FINAL TOTAL		360

B.440.6 Computer Science

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Computer Science Doctoral Thesis 901	COS901	
2nd Enrolment Code		
Computer Science Doctoral Thesis 902	COS902	360
FINAL TOTAL		360

B.440.7 Environmental Science

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Environmental and Water Science Doctoral Thesis 901	EWS901	
2nd Enrolment Code		
Environmental and Water Science Doctoral Thesis 902	EWS902	360
FINAL TOTAL		360

B.440.8 Mathematics

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Mathematics Doctoral Thesis 901	MAM901	
2nd Enrolment Code		
Mathematics Doctoral Thesis 902	MAM902	360
FINAL TOTAL		360

B.440.9 Mathematics Education

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Mathematics Doctoral Thesis 903	MAM903	
2nd Enrolment Code		
Mathematics Doctoral Thesis 904	MAM904	360
FINAL TOTAL		360

B.440.10 Medical Bioscience

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Medical Bioscience Doctoral Thesis 901	MBS901	
2nd Enrolment Code		
Medical Bioscience Doctoral Thesis 902	MBS902	360
FINAL TOTAL		360

B.440.11 Pharmaceutical Sciences

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Pharmaceutical Sciences Doctoral Thesis 901	PHS901	
2nd Enrolment Code		
Pharmaceutical Sciences Doctoral Thesis 902	PHS902	360
FINAL TOTAL		360

B.440.12 Physics

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Physics Doctoral Thesis 901	PHY901	
2nd Enrolment Code		
Physics Doctoral Thesis 902	PHY902	360
FINAL TOTAL		360

B.440.13 Population Studies

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Population Studies Doctoral Thesis 901	POP901	
2nd Enrolment Code		
Population Studies Doctoral Thesis 902	POP902	360
	FINAL TOTAL	360

B.440.14 Statistics

Module Name	Module Code	Credits
Compulsory		
1st Enrolment Code		
Statistics Doctoral Thesis 901	STA901	
2nd Enrolment Code		
Statistics Doctoral Thesis 902	STA902	360
	FINAL TOTAL	360

B.441 ASSESSMENT

Assessment is governed by Rule A.5 as stipulated in the University Calendar: General Information Part 1.

B.442 PROGRESSION RULES

Registration for the following year of study shall be recommended by the supervisor if in their opinion, adequate progress has been made during the current year.

B.443 RENEWAL OF REGISTRATION

Renewal of registration shall be governed by Rule A.4, as stipulated in the University Calendar: General Information Part 1 or as provided for in the faculty rules as contained in this Calendar.

B.444 SPECIAL REQUIREMENTS FOR THE PROGRAMME

B.444.1 Statistics

In addition to the general admissions rules an applicant must provide the Statistics and Population Studies department with a full curriculum vitae and at least three referee reports.

MODULE LIST PER YEAR LEVEL:

MODULE TYPE:

Compulsory (C), Electives (E), Year Module (YR), Semester Module (S), Pre-requisites and Co-requisites; Service Module (SM)

ASSESSMENT TYPE:

Continuous Assessment (CA) or Final Assessment (FA) or Continuous Assessment and Final Assessment (CFA)

Module Code	Module Name	Compulsory/ Elective	Credits	NQF Level	Entolment Year	Duration	Assessment Type
BCB901	Biodiversity Doctoral Thesis 901	C	360	10	1	YR	FA
BCB902	Biodiversity Doctoral Thesis 902	C		10	2	YR	FA
BTY901	Biotechnology Doctoral Thesis 901	C	360	10	1	YR	FA
BTY902	Biotechnology Doctoral Thesis 902	C		10	2	YR	FA
CHE901	Chemistry Doctoral Thesis 901	C	360	10	1	YR	FA
CHE902	Chemistry Doctoral Thesis 902	C		10	2	YR	FA
COS901	Computer Science Doctoral Thesis 901	C	360	10	1	YR	FA
COS902	Computer Science Doctoral Thesis 902	C		10	2	YR	FA
EWS901	Environmental and Water Science Doctoral Thesis 901	C	360	10	1	YR	FA
EWS902	Environmental and Water Science Doctoral Thesis 902	C		10	2	YR	FA
GEL901	Geology Doctoral Thesis 901	C	360	10	1	YR	FA
GEL902	Geology Doctoral Thesis 902	C		10	2	YR	FA
MAM901	Mathematics Doctoral Thesis 901	C	360	10	1	YR	FA
MAM902	Mathematics Doctoral Thesis 902	C		10	2	YR	FA
MAM903	Mathematics Doctoral Thesis 903	C	360	10	1	YR	FA
MAM904	Mathematics Doctoral Thesis 904	C		10	2	YR	FA
MBS901	Medical Bioscience Doctoral Thesis 901	C	360	10	1	YR	FA
MBS902	Medical Bioscience Doctoral Thesis 902	C		10	2	YR	FA
PHS901	Pharmaceutical Sciences Doctoral Thesis 901	C	360	10	1	YR	FA
PHS902	Pharmaceutical Sciences Doctoral Thesis 902	C		10	2	YR	FA
PHY901	Physics Doctoral Thesis 901	C	360	10	1	YR	FA
PHY902	Physics Doctoral Thesis 902	C		10	2	YR	FA
POP901	Population Studies Doctoral Thesis 901	C	360	10	1	YR	FA
POP902	Population Studies Doctoral Thesis 902	C		10	2	YR	FA
SAN901	Bioinformatics Doctoral Thesis 901	C	360	10	1	YR	FA
SAN902	Bioinformatics Doctoral Thesis 902	C		10	2	YR	FA
STA901	Statistics Doctoral Thesis 901	C	360	10	1	YR	FA
STA902	Statistics Doctoral Thesis 902	C		10	2	YR	FA

UNDERGRADUATE MODULE DESCRIPTORS

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Geology 232		Module Code			APG232	
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Applied Geology) (3214, 3011), BSc (Environment and Water Science) (3027, 3331)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Identify crystal forms and systems. - Describe compositional characteristics and physical properties of minerals. - Explain the basic principles of geochemistry as it relates to minerals and rocks. - Explain the basic principles / processes of sediment deposition and depositional environment. - Describe sedimentary rocks. - Describe the classification of sedimentary rocks and associated textures.						
Main Content	- Principles of crystallography - Composition- and structure- based classification of minerals - Atomic theory, electron configuration, periodic table - Bonding in minerals and element substitution - Identification of rocks and minerals in hand specimen - Geochemical principles – element substitution, geochemical analysis, major and trace elements, isotope systematics (stable and radiogenic), age dating of rocks - Classification of sedimentary rocks and associated textures.						
Pre-requisite	ESC111 and ESC121		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Geology 234		Module Code			APG234	
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit	20
Programme in which the module will be offered	BSc (Applied Geology) (3214, 3011)					Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Have a conceptual understanding of map construction. • Understand principles of rock deformation, stress and strain. • Develop skills for the structural analysis of deformed rocks.						

	• Recognize and describe faults, joints, folds, tectonic fabric and understand the mechanisms behind these structures • Have a conceptual understanding of tectonic settings and rock mechanics. • Describe the principles of interaction between light propagation in crystalline materials and apply these in the identification of minerals. • Use a polarized-light microscope to identify and describe minerals and rocks in thin section.			
Main Content	• Reading of geological maps and cross-sections. • Constructing simple cross-sections and block diagrams. • Understanding of topographic contours and structure contours. • Concepts of stress, strain and translation. • Description and analysis of deformed rocks. • Theory of light interaction with crystalline material. • Identification minerals under the polarized-light microscope using crystal-optical and morphological properties.			
Pre-requisite	ESC111 and ESC121	Prohibited module Combination		APG231
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		2	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Geology 235		Module Code			APG235	
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Applied Geology) (3214, 3011)			Total Learning Time		200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Observe and describe the occurrence and spatial arrangement of various types of rocks in the field. • Locate themselves in the field and record information collected in the field. • Produce geological maps at various scales and take geological measurements in the field. • Recognize igneous and metamorphic rocks and associated textures in hand specimen and under the microscope. • Relate the changes in the mineralogy of igneous rocks to the variation in the chemistry of the rocks. • Describe the origin of igneous and metamorphic rocks and associated textures, as well as the various igneous and metamorphic rock-forming processes.						
Main Content	• Geological mapping techniques • Structural analysis of geological data • Reporting geological data and report writing • Formation of igneous and metamorphic rocks • Description and classification of igneous and metamorphic rocks						

Pre-requisite	ESC111 and ESC121	Prohibited module Combination		APG233; EWP222	
Co-requisite	None				
Timetable Requirement per week		2	Lectures p.w.		
		1	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		60%
			FA		40%

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Geology 312		Module Code			APG312	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Applied Geology) (3214, 3011)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Outline the history of petroleum evolution • Assess the conditions to be fulfilled for commercial hydrocarbon accumulation • Categorize and distinguish different depositional environments • Analyze how natural sedimentary systems work and the extent to which it could be managed • Recognize facies, and evaluate facies associations and architectural elements • Outline the history of well logging and demonstrate its applications • Assess and compare the types of wireline logs and formats • Review logging terminology • Construct and interpret the correlation of logs • Identify and examine seismic reflectors • Identify and interpret parasequences and marine flooding surfaces • Evaluate sequence boundaries and system tracts • Identify and characterize reservoirs						
Main Content	• History of Petroleum evolution • Depositional environments • Facies Analysis • Depositional Systems and System tracts • Well logs and geological applications • Well log correlation and recognition of bounding discontinuities • Seismic exploration techniques • Introduction to reservoir modeling techniques.						
Pre-requisite	APG232 and APG234		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			
			3	Assignment p.w.			
			3	Tests & Exercises			
			10	Self-study			

Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%
Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Geology 323		Module Code			APG323	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered	BSc (Applied Geology) (3214, 3011)				Total Learning Time	300 hours	
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Evaluate and interpret the origin of igneous and metamorphic rocks and associated textures, as well as the various igneous and metamorphic rock-forming processes. • Compare and relate the changes in the mineralogy of igneous rocks to the variation in the chemistry and physical conditions of the rocks. • Evaluate and interpret phase diagrams for igneous rocks • Evaluate and propose the origin and evolution of various types of igneous rocks • Evaluate how different types of igneous rocks form in different tectonic settings. • Determine the original rocks that were metamorphosed to form various metamorphic rocks • Analyse and critique the physical conditions which will give rise to different types of metamorphic rocks • Determine the history of deformation and metamorphism of a metamorphic rock • Evaluate and critically reflect on how composition and physical conditions affect the mineralogy of a metamorphic rock • Demonstrate an integrated knowledge of metamorphic reactions						
Main Content	• Formation of igneous rocks • Introduction to thermodynamics • Phase diagrams and their application to understanding igneous rocks • Partial melting of the mantle and generation of basaltic rocks • Different types of magmas in different tectonic settings • Plate tectonics and its role in magmatism • Formation of deformational and textural features in metamorphic rocks • Chemographic diagrams – how composition and physical conditions influence a metamorphic rock • Metamorphic reactions • Pressure-temperature-time paths • Geothermobarometry • Metasomatism						
Pre-requisite	APG232; APG234; APG235		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	60%
						FA	40%
Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Geology 324		Module Code			APG324	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered	BSc (Applied Geology) (3214, 3011)					Total Learning Time	300 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Critically reflect on the large-scale dynamics of the Earth's interior with emphasis on the lithosphere dynamics, including the coupling of interior and surface processes.						
	• Evaluate the methods used for investigating large-scale tectonics.						
	• Recognize the geological processes on a range of scales, and analyze the close relationships between the outcrop-scale features and large-scale lithosphere dynamics.						
	• Utilize the stereographic projection methods for plotting, analysing and interpreting structural data.						
	• Illustrate the principles of concentration of mineral deposits in the context of various geological processes.						
	• Evaluate the occurrence of various genetic types of ore deposits in relation to associated host rocks, mineralogy of ores, ore grade and geological setting.						
	• Relate ore deposit formation and various geological processes to mineral exploration and resource evaluation.						
	• Evaluate the principles of mineral exploration geochemistry and their application to ore deposit estimation.						
	• Analyze and interpret geochemical data used in mineral exploration and production.						
Main Content	• Introduction to tectonics and plate tectonics.						
	• Wilson cycle.						
	• Processes on the early Earth.						
	• Plate movements.						
	• Geodynamics of plate margins, plate interiors, and the deep Earth.						
	• Mountain belts and sedimentary basins.						
	• Structural analysis by stereographic projection.						
	• Introduction to the genetic classification and formation of ore deposits.						
	• Magmatic, hydrothermal and sedimentary ore deposits and ore deposit models.						
	• Concepts of mineral exploration.						
• Introduction to exploration geochemistry							
Pre-requisite	APG232; APG234; APG235		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	60%
						FA	40%
Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Geology 325		Module Code			APG325	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered	BSc (Applied Geology) (3214, 3011)				Total Learning Time	300 hours	
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate an integrated knowledge of the principles and applications of spatial and related techniques in the field of geology. • Analyze and interpret geospatial data to construct geological maps • Demonstrate the application of spatial data query, data management and data analysis • Integrate geological data into computer-based models; process, analyse and evaluate resulting images. • Evaluate geological data through geostatistical techniques • Develop geological models in the search for various Earth resources. • Acquire, analyze and interpret remotely sensed geological data • Analyze and characterize igneous and metamorphic rocks in the field. • Synthesise a detailed and comprehensive geological map of an area based on field observations and field measurements. • Evaluate and report on observations, measurements and interpretations of rocks mapped in the field.						
Main Content	• Geographical information systems concepts and application • Principles of cartography and geospatial data output • Spatial data models and data structure • Principles of coordinate systems and map projection • Development and management of geodatabases • Vector and raster analysis • Application of statistical techniques for geological data analysis • 2D and 3D geological and geochemical modelling • Principles of Satellite Imagery/Principles of electromagnetic radiation and their role in remote sensing • Principles of remote sensing and their application in Earth science. • Image correction, interpretations and classification • Construction of Base Maps and Geological Maps. Observation of, characterization and mapping of igneous and metamorphic rocks in the field. • Recording structural measurements of rocks in the field. • Production of a geological report and geological maps of a field area						
Pre-requisite	APG232; APG234; APG235		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3		Lectures p.w.		
			2		Practicals p.w.		
			0		Tutorials p.w.		

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	70% 30%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Mathematics 112		Module Code			APM112	
Module Type	Elective	NQF Level	5	Year Level	1/2	NQF Credit	15
Programmes in which the module will be offered		BSc (Mathematics and Statistical Sciences) (3227, 3031), BSc Physical Science (3233, 3120)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Use notation and discuss basic concepts in the contexts of logic, set theory and elementary number theory • Analyse arguments in the above contexts in order to distinguish between logically valid and logically invalid arguments • Compare and contrast various methods of proof in the above contexts • Construct counter examples in the above contexts • Structure logically valid arguments in the above contexts						
Main Content	• The logic of Compound statements. • The logic of Quantitative Statements. • Elementary Number Theory and Methods of Proof. • Sequences and Mathematical Induction. • Set theory.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	MAT105 or MAT104 or MAM152						
Timetable Requirement per week			4	Lectures p.w.			
			0	Practicals p.w.			
			2	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Applied Mathematics 123		Module Code			APM123	
Module Type	Elective	NQF Level	5	Year Level	1/2	NQF Credit	15
Programmes in which the module will be offered		BSc (Mathematics and Statistical Science) (3227), BSc (Physical Science) (3233), BSc (Computer Science) (3023)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Use available programming software to support their basic understanding of topics covered in first year mathematics.						

	• Understand how to simulate basic functions using programming. • Have an essential understanding of first order ordinary differential equations and its various applications.				
Main Content	• Introduction to arithmetic operations • Basics of algorithm development Array operations (Vectors and Matrices} • Mathematical operations with one-dimensional arrays • Mathematical operations with two-dimensional arrays • Two and three-dimensional plots • Programming using appropriate software language/ packages (e.g., MATLAB/ SCILAB/ Octave/ Julia/ etc.) • Solution of first order ordinary differential equations (Application problems to be covered from applied sciences such as Physics, Chemistry, and other biological sciences).				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	MAT105 or MAT104 or MAM152				
Timetable Requirement per week		2	Lectures p.w.		
		1	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Animal Diversity 211		Module Code			BDC211	
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217,3015), BSc (Environment and Water Science) (3331, 3027)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the evolutionary origins and relationships of animal Taxa and understand which major changes took place over evolutionary time. • Identify and classify all major invertebrate and vertebrate taxa. • List and compare the structure and function of at least three animal systems and understand how bodies of all major invertebrate and vertebrate groups work. • Observe and dissect specimens to investigate the external and internal structure of animals. • Curate and display invertebrate specimens. • Read and conceptualise scientific ideas. • Interpret and summarise information to write essays.						
Main Content	• Evolution, classification and identification of invertebrate and vertebrate animals; a comparative study of the structure and function of three of the following systems: skeletal, circulatory, reproductive, endocrine, digestive, respiratory and excretory.						

Pre-requisite	LSC142 or LFS152	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		2	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		60%
			FA		40%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Plant Diversity 221		Module Code			BDC221	
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217,3015), BSc (Environment and Water Science) (3331, 3027)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Recall, classify and illustrate a suite of lower photosynthetic organisms. - Explain the evolutionary relationship between lower photosynthetic organisms. - Know how lower photosynthetic organisms are anatomically organized and what factors influence their growth and development. - Distinguish between different lower photosynthetic organisms to the higher taxonomic levels. - Compare reproduction and life cycles in various photosynthetic organisms. - Recognize the diversity and functional significance of morphological and reproductive structure found amongst flowering plants. - Identify and classify key fynbos plant families and genera. - Explain critically how fynbos plants are grouped together. - Apply plant cladistics methodology to trace phylogenetic histories of plant families and genera; and - Construct plant identification keys.						
Main Content	- The diversity of photosynthetic organisms and lower plants. - Evolutionary relationships among photosynthetic organisms and lower plants. - The origin of organelles by endosymbiosis. - Growth and organization; Life cycles and their environmental correlates. - Identification and classification systems. - Plant characters of use in cladistic methods. - Sources of taxonomic information. - A review of flowering plant diversity in the fynbos, with special attention to the most species rich and endemic taxa.						
Pre-requisite	LSC142 or LFS152		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week	3	Lectures p.w.		
	2	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Population Biology and Evolution 222		Module Code			BDC222	
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217,3015), BSc (Environment and Water Science) (3331, 3027)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Describe methods used to collect field data and use them to estimate population sizes of different organisms. • Identify and appraise the factors and processes that affect population size and growth rate. • Develop simple computer models of populations and use them to determine the most appropriate method to manage populations sustainably. • Select and calculate appropriate statistics for summarizing simple biological data, and for evaluating phylogenetic trees. • Evaluate the evidence for speciation and evolution. • Demonstrate a detailed knowledge of the theory of evolution and phylogenetics. • Apply cladistic methodology and software to various kinds of data in order to discover the historical relationships of organisms. • Apply the rules of nomenclature. • Create a new species description. • Take responsibility for organising and managing their time and workload. • Work effectively with others, for mutual benefit, when appropriate. • Confidently present their work orally to an audience. • Prepare and write scientific reports within the sphere of accepted norms, using rubrics. • Use a personal computer for simple data analysis;						
Main Content	• Basic population dynamics and life-tables. Age-structured population models. Population census techniques. Intraspecific competition and derivation of the logistic growth equation. Modelling populations showing logistic growth. Harvesting natural populations- deterministics and stochastic models. Population fluctuations-key factor analysis. Adaptation and the theory of natural selection. Speciation processes and patterns. Reconstruction of phylogeny, evolution and classification.						

	Concepts and principles of cladistics, modern systematics and taxonomy.			
Pre-requisite	(LSC141 and LSC142) or (LFS151 and LFS152)	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		2	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Ecophysiology 223		Module Code			BDC223	
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217,3015), BSc (Environment and Water Science) (3331, 3027)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the exchange of energy, heat and water between animals and their environment. • Critically evaluate the role that physiology plays in influencing an animal's survival in a changing environment. • Undertake basic physiological and behavioural experiments and apply appropriate concepts to analyse and interpret the data. • Use plant growth substances to manipulate plant growth. • Predict the effects of environmental change on plants with different photosynthetic pathways. • Compare the roles of various nutrients in plant growth and development. • Understand and evaluate the role that key environmental conditions have on a plant's physiology and the impact of this on its abundance and distribution. • Defend their position on the above in discussion or in written format.						
Main Content	• Interaction of animals with the environment in selected ecosystems with reference to energy, • Thermal physiology and thermoregulation in animals • Water relations and osmoregulation • Basic experimental methods in animal physiology • Plant growth and development, including an overview of selected plant hormones. • Nutrient and mineral assimilation, with emphasis on nitrogen. • Photosynthesis in C3 plants, with an exploration of alternative pathways. • Plant mineral nutrition. • Plant water relations.						

	• Environmental effects on plant growth and development. • Plant stress physiology.			
Pre-requisite	LSC141 and LSC142, or LFS151 and LFS152; and CHE114 or CHE124 or CHE116 or CHM126	Prohibited module Combination		BDC311 and BDC212
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		2	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Plant Diversity 231		Module Code			BDC231	
Module Type	Elective	NQF Level	6	Year Level	2	NQF Credit	20
Programme in which the module will be offered		BEd (Senior Phase and Further Education and Training Teaching) (4405)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Recall, classify and illustrate a suite of lower photosynthetic organisms. - Explain the evolutionary relationship between lower photosynthetic organisms. - Know how lower photosynthetic organisms are anatomically organized and what factors influence their growth and development. - Distinguish between different lower photosynthetic organisms to the higher taxonomic levels. - Compare reproduction and life cycles in various photosynthetic organisms. - Recognize the diversity and functional significance of morphological and reproductive structure found amongst flowering plants. - Identify and classify key fynbos plant families and genera. - Explain critically how fynbos plants are grouped together. - Apply plant cladistics methodology to trace phylogenetic histories of plant families and genera; and - Construct plant identification keys.						
Main Content	- The diversity of photosynthetic organisms and lower plants. - Evolutionary relationships among photosynthetic organisms and lower plants. - The origin of organelles by endosymbiosis. - Growth and organization; Life cycles and their environmental correlates. - Identification and classification systems. - Plant characters of use in cladistic methods. - Sources of taxonomic information. - A review of flowering plant diversity in the fynbos, with special attention to the most species rich and endemic taxa.						

Pre-requisite	LSC142 or LFS152	Prohibited module Combination	None		
Co-requisite	None				
Timetable Requirement per week			3	Lectures p.w.	
			2	Practicals p.w.	
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%	
			FA	50%	

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Animal Diversity 232		Module Code			BDC232	
Module Type	Elective	NQF Level	6	Year Level	2	NQF Credit	20
Programme in which the module will be offered		BEd (Senior Phase and Further Education and Training Teaching) (4405)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the evolutionary origins and relationships of animal Taxa and understand which major changes took place over evolutionary time. • Identify and classify all major invertebrate and vertebrate taxa. • List and compare the structure and function of at least three animal systems and understand how bodies of all major invertebrate and vertebrate groups work. • Observe and dissect specimens to investigate the external and internal structure of animals. • Curate and display invertebrate specimens. • Read and conceptualise scientific ideas. • Interpret and summarise information to write essays.						
Main Content	Evolution, classification and identification of invertebrate and vertebrate animals; a comparative study of the structure and function of three of the following systems: skeletal, circulatory, reproductive, endocrine, digestive, respiratory and excretory.						
Pre-requisite	LSC142 or LFS152		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3		Lectures p.w.		
			2		Practicals p.w.		
			1		Tutorials p.w.		
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department		Biodiversity and Conservation Biology		
Module Name	Land-use and Conservation 332		Module Code		BDC332		
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30

Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217,3015)		Total Learning Time	300 hours	
Duration	Semester	Proposed Semester		Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to: - Analyse and identify landscapes from photography and satellite imagery. - Interpret human impacts on landscapes. - Conceptualise a Bioregional plan built on an existing Systematic Biodiversity Plan and identified Critical Biodiversity Areas. - Appraise a Biosphere Reserve as a regionally-based planning tool. - Prepare management reports reflecting the required structures for a Bio-regional plan, a Biodiversity Impact Assessment, and a Biodiversity Offset alternative. - Use standard conventions to write and illustrate professional scientific reports. - Use a personal computer for data analysis; and - Confidently present their work orally to an audience.					
Main Content	- Biodiversity concepts such a South African vegetation types and biomes, Disturbance ecology, fragmentation and use for developing biodiversity networks; - Non-equilibrium ecosystem dynamics (chaos models, state and transition models and tipping points); - Biodiversity Impact Assessments; Bioregional Planning which is based on a Systematic Biodiversity Plan and Critical Biodiversity Areas; - Restoration Ecology environmental monitoring.					
Pre-requisite	BDC222		Prohibited module Combination		None	
Co-requisite	None					
Timetable Requirement per week			3	Lectures p.w.		
			2	Practicals p.w.		
			0	Tutorials p.w.		
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	60%
					FA	40%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Animal Behaviour and Systematics 333		Module Code			BDC333	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered	BSc (Biodiversity and Conservation Biology) (3217, 3015)					Total Learning Time	300 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, the student should be able to: • Explain and compare different types of behaviour in animals within an evolutionary and ecological context. • Apply knowledge of an animal's behaviour to conservation issues. • Undertake basic behavioural experiments and apply appropriate concepts to analyse and interpret the data. • Apply genetic data to systematics.						

	• Explain the theory of laboratory procedure used to study molecular genetics. • Apply analytical methodology and software to various kinds of molecular data. • Present their work orally to an audience. • Use standard conventions to write and illustrate scientific reports. • Read, compare and contrast scientific papers.			
Main Content	• Function and evolution of selected animal behaviours, including foraging, reproductive, aggressive, social and signaling behaviours. • Experimental methods in animal behaviour. • Traditional and molecular characters in molecular systematic studies, techniques for acquiring molecular data in laboratory and internet databases. • Interpretation of data derived from various DNA sources. • Introduction to next-generation sequencing platforms and the application of genetic sequence data in population genetic studies and phylogenetic studies. • Critical revision of different DNA sequence data analysis techniques and pitfalls.			
Pre-requisite	BDC222 and BDC211	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		2	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Biogeography and Global Ecology 334		Module Code			BDC334	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217, 3015)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, the student should be able to: • Discuss the past, present and projected future patterns of global biogeography. • Examine the distribution of past floras, faunas and climate with respect to plate tectonics and compare them with current distributions. • Explain the role that the major environmental drivers play in driving these biogeographical patterns. • Understand the physical basis underpinning the components of global change. • Recognise the central importance that humans play in bringing about global change. • Understand the ecological, physiological and behavioural basis for biogeographical change.						

	• Contrast the fundamental differences between ecological biogeography and historical biogeography. • Consider the biogeography of key extant plant and animal lineages. • Apply the appropriate concepts to collect, analyse and interpret multivariate environmental and ecological data. • Present their position on the above in discussion or in written format.					
Main Content	• Global biogeography: key principles and concepts. • Continental drift and glaciation. • Theories of biogeography and biogeographic reconstruction. • Phylogeography • Latitudinal gradients in diversity. • Interactions of body and population size on diversity and distribution. • Island biogeography theory and its applications for conservation. • Earth as a system • The physical nature of environmental drivers of biogeography. • Global change: the distinction between natural variability and anthropogenically-driven change. • Overview of the biological responses to global change. • Basic data collection and analytical methods in biogeography.					
Pre-requisite	BDC211 and BDC221 and BDC223		Prohibited module Combination		None	
Co-requisite	None					
Timetable Requirement per week			3	Lectures p.w.		
			2	Practicals p.w.		
			1	Tutorials p.w.		
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	60%
					FA	40%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Community and Ecosystem Ecology 335		Module Code			BDC335	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217, 3015)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, the student should be able to: • Define and describe the attributes and composition of a community. • Develop simple computer models to explore how key factors and processes influence the structure and development of biological communities. • Explain central theories and discuss important ecological mechanisms that influence community ecological patterns in nature. • Explain and illustrate the concept of ecosystems and the way they function. • Differentiate between the characteristics of a suite of different ecosystems.						

	• Discuss critically the impacts of anthropogenic change on ecosystem integrity. • Develop skill at reading and evaluating research reports; • Prepare and write scientific reports within the sphere of accepted norms, using rubrics. • Treat the Internet as a mainstream learning resource.			
Main Content	• An overview of principles such as symbioses, pollination, inter-specific competition, herbivory, and succession that are used to describe community assemblages and regulation in ecological communities. • Numeral analysis of field data will be undertaken to allow students to see the importance of these models in supporting the ecological management of biological communities. • Conceptual overview of ecosystems, including their characteristics, components and drivers. • Exploration of selected marine, estuarine and terrestrial ecosystems. • Overview of anthropogenic and natural impacts on ecosystem integrity.			
Pre-requisite	BDC222 and BDC211 and BDC221	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		2	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Biotechnology 211		Module Code			BTN211	
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217, 3015), BSc (Chemical Sciences) (3220), BSc (Medical Bioscience) (3230,3035), BSc (Biotechnology) (3211, 3007)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, student should be able to: • Demonstrate an understanding of core concepts of microbiology, including the evolution and diversity of microbes; cell structure and function; metabolism; information flow and the role of microbes in ecosystems. • Explain the dynamics of commensal, opportunistic, and pathological relationships particularly between microbes and humans. • Carry out microbiological laboratory procedures in accordance with good laboratory practices. • Communicate the fundamental concepts of microbiology in both written and oral format.						
Main Content	• Introduction to microbiology, history and scope. • Diversity of microorganisms in terms of nutrition, growth and genetics.						

	• Hands-on introduction to safe handling of microorganisms. • Techniques for microbial culturing, identification, and genetic analysis. • Use of molecular and classical methods for the classification of microorganisms. • Role and significance of different groups of microorganisms in foods, plants, humans and animals.						
Pre-requisite	(a) (CHE114 and CHE124) and (b) (LSC141 and LSC142) or (LFS151 and LFS152) or (LSC141 and MBS111 and MBS121)			Prohibited module Combination	None		
Co-requisite	None						
Timetable Requirement per week	3		Lectures p.w.				
	2 x 3		Practicals p.w.				
	0		Tutorials p.w.				
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment		CA	100%	
Faculty	Natural Sciences		Home Department		Biotechnology		
Module Name	Biotechnology 213		Module Code		BTN213		
Module Type	Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Chemical Sciences) (3220), BSc (Medical Bioscience) (3230,3035), BSc (Biotechnology) (3211, 3007)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester		Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Characterize the importance of microorganisms in the environment • Describe the main factors that will influence the growth of microorganisms in food • Describe the characteristics and sources of predominant microorganisms in food • Discuss the causative agents, suspect foods, signs and symptoms of some major foodborne diseases • Discuss the methods used to isolate important spoilage microorganisms and food-borne pathogens and the role of food microbiologist in ensuring safe food products • Interpret laboratory results and decide what actions need to be taken to control the growth of a particular organism • Compare and contrast the pathological effects and detection methods for common food indicator microorganisms, foodborne pathogens • Explain the size and complexity of food safety management						
Main Content	• Basic microbial genetics • Key microbial groups in the environment • Factors that influence microbes in food • Food as substrate for microorganisms • Microbial diversity in foods • Microorganisms important in food • Principles of food preservation						

	• Food Fermentations • Food and public health • Microbial criteria and sampling plans • Food quality and safety • Foodborne pathogens • Food safety management • Food Biotechnology			
Pre-requisite	(a) (CHE114 and CHE124) and (b) (LSC141 and LSC142) or (LFS151 and LFS152) or (LSC141 and MBS111 and MBS121)		Prohibited module Combination	None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		2	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Biotechnology 214		Module Code			BTN214	
Module Type	Elective	NQF Level	6	Year Level	2	NQF Credit	20
Programme in which the module will be offered		BSc Dietetics (8231)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate an understanding of core concepts of microbiology, including the evolution and diversity of microbes; cell structure and function; metabolism; information flow and the role of microbes in ecosystems. • Explain the dynamics of commensal, opportunistic and pathological relationships particularly between microbes and humans. • Carry out microbiological laboratory procedures in accordance with good laboratory practices. • Communicate the fundamental concepts of microbiology in both written and oral format.						
Main Content	• Introduction to microbiology, history and scope. • Microscopy and specimen preparation. • Prokaryotes vs Eukaryotes: cell structure and function. • Diversity of microorganisms in terms of nutrition, growth and genetics. • Control of microorganisms • Introduction to Virology. • Introduction to environmental and water microbiology. • Introduction to food microbiology. • Role of microorganisms in food spoilage and foodborne diseases (food pathogens). • Intrinsic and extrinsic parameters. • Adaptations of microorganisms to food environments. • Food preservation						

	• Quality and safety management. • Hands-on introduction to safe handling of microorganisms. • Techniques for microbiological culturing, identification and genetic analysis. • Use of molecular and classical methods for the classification of microorganisms. • Role and significance of different groups of micro-organisms in foods, plants, humans and animals.			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		2	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department		Biotechnology		
Module Name	Biotechnology 215		Module Code		BTN215		
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217, 3015), BSc (Chemical Sciences) (3019, 3220), BSc (Medical Bioscience) (3230,3035), BSc (Biotechnology) (3211, 3007)			Total Learning Time		200 hours
Duration	Semester		Proposed Semester		First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Correctly perform calculations commonly encountered in the biotechnology laboratory. • Quantify the degree of uncertainty associated with an experimentally-determined result. • Use software tools to graphically represent data. • Research a topic using scientific papers.						
Main Content	• Significant digits • Units • Molarity and molecular weight • Calculate concentrations • Calculations common in the molecular biology laboratory • Calculations using the Michaelis Menten model of enzyme kinetics • Graphs and straight-line equation, by hand and using MS Excel • Introduction to statistics • Cataloging and reading scientific papers						
Pre-requisite	(a)(CHE114 and CHE124) and (b)(LSC141 and LSC142) or (LFS151 and LFS152) or (LSC141 and MBS111 and MBS121)			Prohibited module Combination		None	
Co-requisite	None						

Timetable Requirement per week	3	Lectures p.w.		
	0	Practicals p.w.		
	2	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Biotechnology 216		Module Code			BTN216	
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Medical Bioscience) (3230, 3035), BSc (Biodiversity and Conservation Biology) (3217, 3015), BSc (Dietetics) (8231), BSc (Sport and Exercise Science) (8051), BSc (Complementary Health Sciences) (8104, 8106)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • identify the major classes of biological molecules and their interactions • characterise the chemical and physical structure of DNA and RNA and how it stores and transfers genetic information • explain the role of enzymes in regulating biological processes and analyse their effects quantitatively • distinguish the major pathways by which carbohydrates are broken down to liberate energy for life						
Main Content	• Introduction to different classes of biomolecules (nucleic acids, proteins, lipids, carbohydrates), and their interactions • The energetics of life • DNA structure and function • Enzymes as biocatalysts • Carbohydrate metabolism						
Pre-requisite	a) CHE116 and CHM126) or (CHE114 and CHE124) and b) (LSC141 and LSC142) or (LFS151 and LFS152) or (LSC141 and MBS111 and MBS121)			Prohibited module Combination		None	
Co-requisite	None						
Timetable Requirement per week				3	Lectures p.w.		
				2	Practicals p.w.		
				1	Tutorials p.w.		
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		60%
					FA		40%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Biotechnology 217		Module Code			BTN217	
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	15
Programme in which the module will be offered		BPharmacy (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Identify the major classes of biological molecules and their interactions. - Characterise the chemical and physical structure of DNA and RNA and how it stores and transfers genetic information. - Explain the role of enzymes in regulating biological processes and analyse their effects quantitatively. - Distinguish the major pathways by which carbohydrates are broken down to liberate energy for life.						
Main Content	- Introduction to different classes of biomolecules (nucleic acids, proteins, lipids, carbohydrates), and their interactions. - The energetics of life - DNA structure and function - Enzymes as biocatalysts - Carbohydrate metabolism						
Pre-requisite	a) (CHE114 and CHE124); and (HUB113 and HUB123)			Prohibited module Combination		None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	60%
						FA	40%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Biotechnology 218		Module Code			BTN218	
Module Type	Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Biotechnology) (3007, 3211)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Discuss and apply principles and concepts in plant and agricultural biotechnology for achieving food security, sustainable agriculture and environmental sustainability. • Apply appropriate laboratory skills and protocols. • Evaluate the importance of the latest developments in specific themes of plant and agricultural biotechnology. • Retrieve relevant information from the internet and other sources. • Communicate effectively about the issues relevant to plant biotechnology, both orally and in writing.						
Main Content	• Plant cell and tissue culture. • Applications of plant cell and tissue culture in plant biotechnology. • Conventional plant breeding.						

	• Molecular markers in plant breeding. • Molecular diagnostics in agriculture. • Genetic engineering of plants and animals. • Production of therapeutics (proteins and vaccines) in plants and animals.			
Pre-requisite	(a)(CHE114 and CHE124); and (b)(LSC141 and LSC142) (LFS151 and LFS152) or (LSC141 and MBS111 and MBS121)	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week	3	Lectures p.w.		
	1	Practicals p.w.		
	1	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Biotechnology 222		Module Code			BTN222	
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Biotechnology) (3007, 3211)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the chemical and physical structure of DNA and RNA. • Characterize the mechanisms and regulation of converting genetic information to proteins in prokaryotic organisms. • Understand molecular methods for DNA extraction, quality control, sequencing and cloning. • Make use of standard instruments and methods used in molecular laboratories. • Explain the mechanisms of inheritance in eukaryotic organisms: from cells to groups or organisms. • Perform basic hypothesis testing with statistical analysis.						
Main Content	• Molecular Biology • Chemical and physical structure of DNA and RNA • DNA replication, transcription and translation in prokaryotes • Mechanisms for regulation of gene expression in Prokaryotes. • Molecular methods: (DNA extraction, quantification, PCR, sequencing, and recombinant DNA technologies (Theory and practice) • Eukaryotic molecular biology and basic genetic principles.						
Pre-requisite	(a)(CHE114 and CHE124); and (b)(LSC141 and LSC142) or (LFS151 and LFS152) or (LSC141 and MBS111 and MBS121)			Prohibited module Combination		None	
Co-requisite	None						

Timetable Requirement per week		3	Lectures p.w.				
		1	Practicals p.w.				
		1	Tutorials p.w.				
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	60%	
					FA	40%	
Faculty	Natural Sciences		Home Department		Biotechnology		
Module Name	Biotechnology 223		Module Code		BTN223		
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Biotechnology) (3007, 3211), BSc (Chemical Sciences) (3220), BSc (Medical Bioscience) (3230, 3035)			Total Learning Time		200 hours
Duration	Semester		Proposed Semester		Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to: - Describe the main functions of proteins within the cell. - Derive the chemical properties of amino acids and explain how these properties affect their roles in proteins. - Characterise the structure of proteins in terms of primary, secondary, tertiary and quaternary levels of organization. - Explain the role of enzymes in regulating biological processes and analyse their kinetic parameters quantitatively. - Discuss the major pathways by which carbohydrates are broken down to liberate energy for life.						
Main Content	- Introduction to amino acids and proteins; protein structure and function. - The importance of enzymes in cellular processes; the major classes of enzymes. - Introduction to enzyme kinetics using the Michaelis Menten model - Mechanisms of enzyme catalysis and regulation; thermodynamics, co-enzymes. - Carbohydrate metabolism: glycolysis, citric acid cycle, gluconeogenesis, glycogen.						
Pre-requisite	(a) (CHE114 and CHE124); and (b) (LSC141 and LSC142) or (LFS151 and LFS152) or (LSC141 and MBS111 and MBS121)			Prohibited module Combination		None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment		CA	100%	

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Environmental and Industrial Microbiology 312		Module Code			BTN312	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Biotechnology) (3007, 3211)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Determine and assess the diversity of microorganisms in different environments. • Compare different approaches and methods for gene discovery for biotechnological application. • Describe various approaches to engineer enzymes and organisms in order to exploit them for industrial use. • Design and build a microbial fuel cell for the synthesis of electricity. • Conduct experiments to generate data which will be analysed to assess enzyme activity and organism performance. • Use software tools to graphically present data, write essays and prepare oral and a poster presentation. • Research a topic using the internet and use a reference manager to catalogue research papers.						
Main Content	The following topics will be covered: • Diversity of microorganisms in different environments • Methods for assessing (both quantitatively and qualitatively) microbial diversity and community structure (including phylogenetic markers, 16S library analysis, DGGE, ARISA etc) • Methods for assessing microbial activity (including labeling technologies, transcriptomic analysis etc.) • The use of microorganisms and their products in industry, (including first, second and third generation processes), in marine biotechnology and in environmental microbiology • Fermentation processes • Downstream processing • Enzyme recovery, immobilization • Biocatalysis and biotransformations						
Pre-requisite	BTN211, BTN215, BTN222 and BTN223		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			2	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	60%
						FA	40%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Biochemistry 315		Module Code			BTN315	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30

Programmes in which the module will be offered		BSc (Biotechnology) (3007, 3211)		Total Learning Time	300 hours
Duration	Semester	Proposed Semester		First Semester	
Main Learning Outcomes	On completion of the module, students should be able to: • Discuss and analyse the elements of the eukaryotic translation, transcriptional and splicing machinery • Compare and contrast eukaryotic prokaryotic systems of gene regulation • Apply advanced techniques for manipulating DNA in the laboratory • Recognize and interrogate the structure and expression of genes and genomes • Analyze genes and genomes using computational tools and database				
Main Content	• Genes are DNA and code for proteins • Methods in molecular biology and genetic engineering • Eukaryotic gene structure • The content of the genome • Genome sequences and gene numbers • Chromosome structure and function • Mechanism and machinery of eukaryotic transcription • Eukaryotic translation • Eukaryotic transcription regulation • Regulatory RNA • Introduction to bioinformatics and databases • Sequences alignment and BLAST • Annotation of sequence data • Computational tools for gene and genome analysis				
Pre-requisite	BTN211, BTN215, BTN222 and BTN223	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		2	Practicals p.w.		
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment		CA	60%
				FA	40%

Faculty	Natural Sciences		Home Department		Biotechnology		
Module Name	Biochemistry 316		Module Code		BTN316		
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Biotechnology) (3007, 3211)			Total Learning Time		300 hours
Duration	Semester		Proposed Semester		First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Calculate the probability of DNA profile matching and prepare the data suitable for presentation as evidence in court. • Trace individual genealogies by using mitochondrial DNA and the Y-chromosomes. • Analyses the importance of viruses as independent genetic elements and how they can be exploited for the treatment of disease.						

	• Use the principle of gene therapy to design viral and non-viral gene delivery systems. • Apply an understanding of the mechanisms responsible for immunity to the development of diagnostic tests and vaccines.						
Main Content	• The genetic basis of identity • DNA typing system • Probability matching and use of DNA evidence • Tracing genealogies by mtDNA and the Y-chromosome • General characterization of viruses • Viral infections and antiviral agents • Viral disease and vaccines • AIDS and treatments • Prions • General principles of Gene Therapy • Viral delivery systems in gene therapy • Overview of immunity • Antibody diversity • Monoclonal antibodies • Antibody engineering • Immunological diagnostic tests • Immune memory and vaccination • Creating a vaccine						
Pre-requisite	BTN211, BTN215, BTN222 and BTN223	Prohibited module Combination		None			
Co-requisite	None						
Timetable Requirement per week		3	Lectures p.w.				
		2	Practicals p.w.				
		1	Tutorials p.w.				
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	60%	
					FA	40%	
Faculty	Natural Sciences		Home Department		Biotechnology		
Module Name	Biochemistry 322		Module Code		BTN322		
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Biotechnology) (3007, 3211)			Total Learning Time		300 hours
Duration	Semester		Proposed Semester		Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Express and purify proteins in the laboratory. • Visualise protein structures on a computer. • Explain the function of a number of proteins and cellular processes in terms of 3-dimensional structures. • Illustrate the relationship between protein structure and function in a number of diseases. • Conduct research using the library, the web and other sources of information. • Use the internet, MS Word, MS Excel, MS PowerPoint to complete tasks. • Present well-structured practical reports.						

Main Content	• Protein expression • Protein purification • Mechanisms of catalysis • Enzyme regulation and allostery; haemoglobin and oxygen transport; molecular motors • Membranes, membrane proteins, channels and pumps • Lipid biosynthesis • Lipid metabolism • Gene transcription • Gene translation • Protein turnover and the ubiquitin-proteasome system • Protein folding and chaperones • HIV protease and HIV reverse transcriptase • Cancer, pRb and p53					
	Pre-requisite	BTN211, BTN215, BTN222 and BTN223		Prohibited module Combination	None	
	Co-requisite	None				
	Timetable Requirement per week			3	Lectures p.w.	
				2	Practicals p.w.	
				1	Tutorials p.w.	
	Other teaching modes that does not require time-table					
	Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment	CA	60%
					FA	40%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Business Statistics 131/132		Module Code			BUS131/132	
Module Type	Elective	NQF Level	5	Year Level	1/2	NQF Credit	15
Programmes in which the module will be offered		BCom (1008) (1751), BCom (Accounting) (1021) (1753), BCom (Financial Accounting) (1175), BCom (Information Systems) (1033), BCom (Law) (7211)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			BUS131 First Semester BUS132 Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Recognize the importance of statistics in both private and public sectors • Summarize data into summary measures (e.g. mean and standard deviation) • Do simple statistical analysis • Use a computer to analyze data						
Main Content	• Descriptive statistics • Simple linear regression analysis • Probability • Sampling distributions • Inferential Statistics • Index numbers						

	• Time Series • Anova • Chi-square				
Pre-requisite	QSC131/ QSC132/ MAM126/ QSA131/ QSF141	Prohibited module Combination		STA111/ STA125/ STA142/ STA151/ STC101	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		0	Practicals p.w.		
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment		CA	60%
				FA	40%

Faculty	Natural Sciences		Home Department			Chemistry	
Module Name	Chemistry 114		Module Code			CHE114	
Module Type	Core/ Elective	NQF Level	5	Year Level	1/2	NQF Credit	15
Programmes in which the module will be offered		BSc (Chemical Sciences) (3019, 3220), BPharm (3305), BEd (Science Education) (4516, 4092), BSc (Applied Geology) (3011, 3214), BSc (Physical Science) (3120, 3233)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Apply the following chemical concepts and principles to qualitatively engage with real-world phenomena or examples: accepted symbolic conventions; models for understanding structure and bonding; links between electronic structure and reactivity; and mass and energy balance in chemical reactions. • Solve quantitative chemistry problems, both in familiar and novel contexts. • Conduct simple scientific investigations, including the collection, handling and interpretation of experimental data. • Conduct research using the library, the web and other sources of information. • Reference sources of information correctly. • Use the internet and computer-based word-processing, spreadsheet, and presentation software to complete selected tasks. • Recognise the relationship of chemistry to society, technology and the environment. • Begin to develop life-long learning capabilities and to see chemistry as discipline in a wider context. • Present a clear, well-structured oral presentation and well-structured practical reports. • Work productively in co-operative learning groups.						
Main Content	• Basic concepts of chemistry • Atoms, molecules and ions						

	• Chemical reactions • Quantitative information about chemical reactions (Stoichiometry) • Energy and chemical reactions • Atomic structure and periodic trends • Bonding and molecular structure • Gases and their properties • Intermolecular forces, liquids and solids, solutions and their behavior • Chemical kinetics: The rates of chemical reactions • Principles of reactivity: equilibria and electron transfer reactions			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		1	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Chemistry	
Module Name	Chemistry 116		Module Code			CHE116	
Module Type	Core/ Elective	NQF Level	5	Year Level	1/2	NQF Credit	15
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217, 3015), BSc (Biotechnology) (3211, 3007), BSc (Environment and Water Science) (3331, 3027), BSc (Medical Bioscience) (3230, 3035), BSc (Sport and Exercise Science) (8051), BSc (Complementary Health Sciences) (8104), BSc (Complementary Health Sciences-CMA) (8106), BSc (Dietetics) (8231)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Apply the following chemical concepts and principles to qualitatively engage with real-world phenomena or examples: accepted symbolic conventions; models for understanding structure and bonding; links between electronic structure and reactivity; and mass and energy balance in chemical reactions. • Solve quantitative chemistry problems, both in familiar and novel contexts. • Conduct simple scientific investigations, including the collection, handling and interpretation of experimental data. • Conduct research using the library, the web and other sources of information. • Reference sources of information correctly.						

	• Use the internet and computer-based word-processing, spreadsheet, and presentation software to complete selected tasks. • Recognise the relationship of chemistry to society, technology and the environment. • Begin to develop life-long learning capabilities and to see chemistry as discipline in a wider context. • Present a clear, well-structured oral presentation and well-structured practical reports. • Work productively in co-operative learning groups.			
Main Content	• Basic concepts of chemistry • Atoms, molecules and ions • Chemical reactions • Quantitative information about chemical reactions (Stoichiometry) • Energy and chemical reactions • Atomic structure and periodic trends • Bonding and molecular structure • Gases and their properties • Electron transfer reactions • Intermolecular forces and liquids • Solutions and their behaviour			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		1	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Chemistry	
Module Name	Chemistry 118		Module Code			CHE118	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BDS (5101)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Apply the following chemical concepts and principles to qualitatively engage with real-world phenomena or examples: accepted symbolic conventions; models for understanding structure and bonding; links between electronic structure and reactivity; and mass and energy balance in chemical reactions. • Solve quantitative chemistry problems, both in familiar and novel contexts. • Conduct simple scientific investigations, including the collection, handling and interpretation of experimental data. • Conduct research using the library, the web and other sources of information. • Reference sources of information correctly. • Use the internet and computer-based word-processing, spreadsheet, and presentation software to complete selected tasks.						

	• Recognise the relationship of chemistry to society, technology and the environment. • Begin to develop life-long learning capabilities and to see chemistry as discipline in a wider context. • Present a clear, well-structured oral presentation and well-structured practical reports. • Work productively in co-operative learning groups.					
Main Content	• Basic concepts of chemistry • Atoms, molecules and ions • Chemical reactions • Quantitative information about chemical reactions (Stoichiometry) • Atomic structure and periodic trends • Bonding and molecular structure • Gases and their properties • Electron transfer reactions • The chemistry of acid and bases • Hydrocarbons, Alcohols and Ethers, Aldehydes and Ketones, • Carboxylic Acids and Esters, Amines and Amides, Carbohydrates, Proteins, Lipids					
Pre-requisite	None	Prohibited module Combination		None		
Co-requisite	None					
Timetable Requirement per week		3	Lectures p.w.			
		1	Practicals p.w.			
		1	Tutorials p.w.			
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		60%	
			FA		40%	
Faculty	Natural Sciences		Home Department		Chemistry	
Module Name	Chemistry 124		Module Code		CHE124	
Module Type	Core/ Elective	NQF Level	5	Year Level	1/2	
Programmes in which the module will be offered		BSc (Chemical Sciences) (3019, 3220), BPharm (3305), BSc (Applied Geology) (3011, 3214), BSc (Physical Science) (3120, 3233)			Total Learning Time	150 hours
Duration	Semester	Proposed Semester		Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Apply the following chemical concepts and principles to qualitatively engage with real-world phenomena or examples: the accepted language and conventions for naming simple compounds and communicating about their chemical properties and behaviour; the forces acting between particles in matter that influence properties and behaviour; the factors that influence chemical stability; the forces that drive chemical change; and the factors that control the rate at which chemical change takes place. • Solve quantitative chemistry problems, both in familiar and novel contexts.					

	• Conduct simple scientific investigations, including the collection, handling and interpretation of experimental data. • Conduct research using the library, the web and other sources of information. • Reference sources of information correctly. • Use the internet and computer-based word-processing, spreadsheet, and presentation software to complete tasks. • Begin to recognise the relationship of chemistry to society, technology and the environment. • Begin to develop life-long learning capabilities and to see learning about chemistry in a wider context. • Present clear, well-structured arguments using the content of this module to describe and explain the properties and behaviour of matter. • Work productively in co-operative learning groups.						
Main Content	• Introductory concepts in Organic Chemistry, including nomenclature, physical properties, and representation of structure of: alkanes, alkenes, alkynes, alkyl halides, alcohols, aldehydes and ketones, and carboxylic acids and their derivatives • An introduction to reactivity with reference to substitution, addition, elimination and oxidation of organic substrates. • An introduction to isomerism with reference to geometric, cis-trans, E,Z and R,S isomers/enantiomers. • An introduction to stability with reference to ring strain, and the relative stabilities of alkenes and carboniumions. • An introduction to organic reaction mechanisms. • An introduction to carbohydrates, lipids and proteins • An introduction to polymer chemistry						
Pre-requisite	None	Prohibited module Combination		None			
Co-requisite	None						
Timetable Requirement per week		3	Lectures p.w.				
		1	Practicals p.w.				
		1	Tutorials p.w.				
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	60%	
					FA	40%	
Faculty	Natural Sciences		Home Department		Chemistry		
Module Name	Chemistry (Nursing) ¹¹⁴		Module Code		CHM114		
Module Type	Core	NQF Level	5	Year Level	2	NQF Credit	15
Programme in which the module will be offered		BNurs (8313)			Total Learning Time		150 hours
Duration	Year		Proposed Semester		Both Semesters		
Main Learning Outcomes	On completion of this module, students should be able to: • Apply chemical concepts and principles to qualitatively engage with real-world phenomena • Use their understanding of atomic structure and bonding to balance chemical reactions. • Solve quantitative chemistry problems, both in familiar and novel contexts.						

	Recognize the relationship between chemistry and society and the environment.			
Main Content	- Introduction to chemistry, behavioural trends of elements as derived from periodic table - Atomic, ionic and molecular structures - Chemical formulae, the mole concept, chemical reactions, chemical bonds - Gases, solutions, acids and bases - Hydrocarbons, alcohols and ethers, aldehydes and ketones, carboxylic acids and esters, amines and amides, carbohydrates, proteins, lipids			
Pre-requisite	None	Prohibited module Combination		CHE118, CHE116
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		1	Practicals p.w.	
		2	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department				Chemistry	
Module Name	Chemistry (Nursing) 119		Module Code				CHM119	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	10	
Programme in which the module will be offered		BNurs (8312)				Total Learning Time	100 hours	
Duration	Semester		Proposed Semester			Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to: Understand the basics of inorganic, organic and general chemistry							
Main Content	Introduction to chemistry, periodic table, atoms molecules and ions, chemical formulae, mole concept, chemical reactions, chemical bonds, redox reactions, gases, solutions, acids and bases, hydrocarbons, alcohols and ethers, aldehydes and ketones, carboxylic acids and esters, amines and amides, carbohydrates, proteins, lipids.							
Pre-requisite	None		Prohibited module Combination			CHE114, CHE116		
Co-requisite	None							
Timetable Requirement per week			2	Lectures p.w.				
			0	Practicals p.w.				
			2	Tutorials p.w.				
Other teaching modes that does not require time-table								
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		60%	
					FA		40%	

Faculty	Natural Sciences		Home Department		Chemistry		
Module Name	Chemistry 126		Module Code		CHM126		
Module Type	Core/ Elective	NQF Level	5	Year Level	1/2	NQF Credit	15
Programmes in which the module will be offered	BSc (Biodiversity and Conservation Biology) (3217, 3015), BSc				Total Learning Time		150 hours

		(Biotechnology) (3211, 3007), BSc (Environmental and Water Science) (3331, 3027), BSc (Medical Bioscience) (3230, 3035)					
Duration	Semester	Proposed Semester			Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Apply the following chemical concepts and principles to qualitatively engage with real-world phenomena for examples: links between different types of equilibria and calculation the different parameters related to the reaction kinetics.						
	• Correlate between different organic function group and understanding the main differences between them.						
	• Solve quantitative chemistry problems, both in familiar and novel contexts.						
	• Conduct simple scientific investigations, including the collection, handling and interpretation of experimental data.						
	• Conduct research using the library, the web and other sources of information.						
	• Reference sources of information correctly.						
	• Use the internet and computer-based word-processing, spreadsheet, and presentation software to complete selected tasks.						
	• Recognise the relationship of chemistry to society, technology and the environment.						
	• Begin to develop life-long learning capabilities and to see chemistry as discipline in a wider context.						
Main Content	• Present a clear, well-structured oral presentation and well-structured practical reports.						
	• Work productively in co-operative learning groups.						
	• Reaction kinetics, equilibrium						
	• Acids and bases, acid ionization equilibria						
	• Solubility and solutions,						
	• Alkanes, alkenes, alkynes, alkyl halides, Alcohols, phenols, ethers, aldehydes and ketones, carboxylic acids and derivatives, carbohydrates, amines, amino acids and proteins, lipids and fats						
	Pre-requisite	None		Prohibited module Combination		CHE114, CHE116	
	Co-requisite	None					
	Timetable Requirement per week		3				
			1	Practicals p.w.			
1			Tutorials p.w.				
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	60%	
					FA	40%	
Faculty	Natural Sciences		Home Department		Chemistry		
Module Name	Chemistry 211		Module Code		CHM211		
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	
						20	
Programmes in which the module will be offered		BSc (Biotechnology) (3211, 3007), BSc (Chemical Sciences) (3220,			Total Learning Time	200 hours	

		3019), BSc (Physical Science) (3233, 3120)				
Duration	Semester	Proposed Semester		First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: - Describe the main atomic properties of elements and explain the major periodic trends observed for these properties. - Explain the major diagonal relationships in the periodic table. - Classify the main descriptive chemistry of the major classes of main group elements. - Describe the major physical properties of transition elements. - Explain and apply of the basic principles of organometallic chemistry of transition metals. - Identify and categorize the different reactions of aromatic compounds. - Examine and illustrate stereochemical relationships. - Identify and illustrate the basics of organometallic chemistry. - Identify and compare the contrasting reactions of organic and inorganic chemical compounds.					
Main Content	- Aromatic Chemistry, Stereochemistry, Bi-Functional Conjugated Systems. - Atomic properties of elements, diagonal relationships in the periodic table, descriptive chemistry of the important classes of main group elements, physical properties of transition metals, introduction to Organometallic chemistry.					
Pre-requisite	CHE114 and CHE124	Prohibited module Combination		None		
Co-requisite	None					
Timetable Requirement per week		3	Lectures p.w.			
		1	Practicals p.w.			
		1	Tutorials p.w.			
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	60%
					FA	40%

Faculty	Natural Sciences			Home Department			Chemistry	
Module Name	Chemistry 212			Module Code			CHM212	
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3		NQF Credit	20
Programmes in which the module will be offered		BSc (Biotechnology) (3211, 3007), BSc (Chemical Sciences) (3019), BSc (Physical Science) (3233, 3120)				Total Learning Time	200 hours	
Duration	Semester		Proposed Semester			Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Use gravimetric analysis to investigate and identify products from starting materials. • Interpret and illustrate the practice of titrimetric methods of analysis. • Construct and use pH titration curves to identify buffer zones associated with chemical reactions. • Mathematically translate and examine van der Waals equations for real gasses.							

	• Show a clear understanding of the laws of thermodynamics • Examine and use the theory and laws governing chemical kinetics in simple chemical systems.			
Main Content	• Gravimetry • Titrimetric methods for acids/bases • Redox, compleximetric and precipitation titrations • pH measurements and titration • Kinetic molecular theory • van der Waals equation for real gases • Laws of thermodynamics • Chemical kinetics • Theory of elementary reactions.			
Pre-requisite	CHE114 and CHE124	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		1	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Chemistry	
Module Name	Chemistry 221		Module Code			CHM221	
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered	BSc (Chemical Sciences) (3220, 3019)				Total Learning Time	200 hours	
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • To describe the importance of catalysis in the development of sustainable chemicals processes and applications • To relate key reactions in organometallic catalysis and their reaction mechanisms • To outline the different types of heterogeneous catalysts and the different strategies of immobilisation of molecular catalysts. • To explain the fundamentals of solid state chemistry • To be able to correlate the X-ray diffraction information with crystal structure. • To investigate the impact of chemistry on the environment • To show an extensive knowledge of nucleophilic substitution as well as nucleophilic addition reactions • To show an extensive knowledge of elimination reactions • To investigate the different reactions of aromatic hydrocarbons other than electrophilic aromatic substitution reactions. • To illustrate an understanding of advanced carbonyl chemistry						
Main Content	• Inorganic Chemistry • Introduction to Catalysis, Catalysis by Organometallic -Compounds: Study of the industrial processes and their mechanism: • Alkene hydrogenation (Wilkinsons Catalyst), Catalytic cycle • Molecular Symmetry: Symmetry and group theory- Point Groups, Character Tables, Application to Vibrational Spectroscopy						

	- The Structure of Solids- Electronic Materials - Band theory: metals, insulators, and semiconductors, Band gaps, doping, and devices. Crystalline Materials - Crystal structure: 7 crystal systems, 14 Bravais lattices, Miller indices. Properties of cubic crystals. X-ray diffraction. Defects: point, line, surface, bulk. Amorphous Materials - Inorganic glasses: silicates, other oxides, metallics. Aqueous Solutions - Liquids and solutions: solubility rules, acids, bases, pH. Solid Solutions - Phase stability: unary and binary phase diagrams. - Green Chemistry- Introduction to Green Chemistry, Fundamentals of Green Chemistry, Understand the importance of Green Chemistry, Benefits of Green Chemistry - Organic Chemistry - Nucleophilic Substitution Reactions (SN1 vs SN2 reaction mechanisms) - Elimination Reaction Mechanisms (E1 vs E2 reaction mechanisms) - Other Aromatic Hydrocarbon Reactions (nucleophilic aromatic, addition reactions of benzene derivatives, side-chain reactions of benzene derivatives, and reactions of phenols) - Carbonyl Chemistry (Aldol and related condensations, Claisen Ester Condensations, Alkylation of Enolate Ions, Crossed Claisen Condensations, Malonic Ester Synthesis, The Acetoacetic Ester Synthesis and Conjugate Additions: The Michael Addition)						
Pre-requisite	CHE114 and CHE124			Prohibited module Combination		None	
Co-requisite	CHM211						
Timetable Requirement per week				3		Lectures p.w.	
				1		Practicals p.w.	
				1		Tutorials p.w.	
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)			Method of Student Assessment		CA	60%
						FA	40%
Faculty	Natural Sciences			Home Department			Chemistry
Module Name	Chemistry 311			Module Code			CHM311
Module Type	Core/ Elective	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Chemical Sciences) (3220, 3019), BSc (Physical Science) (3233, 3120)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Perform experiments and solve problems involving thermodynamic and kinetic transformations of matter. - Construct and solve the Schrödinger equations for translational, rotational and vibrational motions and their application in simple atomic structure and molecular spectroscopy.						
Main Content	- Entropy and third law of thermodynamics, phase transformation phenomena, determination of reaction rates. - Theories of unimolecular gas phase reactions, and catalysed reactions. - Old quantum theories of oscillators, photoelectric effect, atomic spectra, and Compton effect. - postulates of quantum mechanics						

	• Schrödinger equation and applications to particle motion, rigid rotor, harmonic oscillator approximations, molecular spectroscopy and hydrogen atom.			
Pre-requisite	CHM211 and CHM212	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		5	Lectures p.w.	
		2	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences			Home Department			Chemistry		
Module Name	Chemistry 312			Module Code			CHM312		
Module Type	Core/ Elective	NQF Level	7	Year Level	3/4	NQF Credit		30	
Programmes in which the module will be offered		BSc (Chemical Sciences) (3220, 3019), BSc (Physical Science) (3233, 3120)				Total Learning Time		300 hours	
Duration	Semester		Proposed Semester			Second Semester			
Main Learning Outcomes	On completion of this module, students should be able to: • Use the principles, techniques, characteristics and applications of chemical processes and pharmaceutical industries and their products. • Demonstrate the nature and control of corrosion phenomena; as well as the role of electrochemistry chemical industries and in energy generation; electrochemical sensors and environmental electrochemistry.								
Main Content	• Characteristics and types of chemical industries. • Basic principles of process design. • Sources and properties of organic and inorganic raw materials. • Industrial chemistry processes including cracking, desulphurization, distillation. • Fischer Tropsch process, hydrogenation, liquefaction, and syn gas systems. • Pharmaceutical industry; chloro-alkali process; fuel cells, batteries and capacitors; electrolysis; electroplating, electrocatalysis and electrosynthesis. • Kinetics and technology of corrosion of metals and its prevention. • Potentiometric sensors or ion selective electrodes; amperometric sensors; environmental electrochemistry.								
Pre-requisite	CHM211 and CHM212			Prohibited module Combination			None		
Co-requisite	None								
Timetable Requirement per week				5		Lectures p.w.			
				2		Practicals p.w.			
				1		Tutorials p.w.			
Other teaching modes that does not require time-table									
Assessment Module Type	Continuous and Final Assessment (CFA)			Method of Student Assessment			CA		60%
							FA		40%

Faculty	Natural Sciences		Home Department			Chemistry		
Module Name	Chemistry 321		Module Code			CHM321		
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30	
Programmes in which the module will be offered		BSc (Chemical Sciences) (3220, 3019)			Total Learning Time		300 hours	
Duration	Semester		Proposed Semester			First Semester		
Main Learning Outcomes	On completion of this module, student should be able to:							
	• Show and classify the different organic rearrangements.							
	• Outline and illustrate the relationships between conformation and reactivity.							
	• Investigate and execute synthetic protocols of mono- and polysaccharides, heterocyclic aromatics and organometallic systems in organic chemistry.							
	• Examine and illustrate the bonding in co-ordination and inorganic chemistry.							
Main Content	• Implement and use methods to characterise co-ordination and organometallic compounds.							
	• Molecular rearrangements,							
	• applications of organometallic chemistry to organic synthesis, advance synthetic methodologies,							
	• non-benzenoid aromatics, heterocyclic aromatic systems, carbohydrate chemistry;							
	• Bonding theories in inorganic chemistry, introduction to co-ordination chemistry, stability of complex ions, physical methods in inorganic chemistry							
Pre-requisite	CHM211 and CHM212 and CHM221			Prohibited module Combination		None		
	Co-requisite							
Timetable Requirement per week			5		Lectures p.w.			
			2		Practicals p.w.			
			1		Tutorials p.w.			
Other teaching modes that does not require time-table								
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA		60%
						FA		40%

Faculty	Natural Sciences		Home Department			Chemistry	
Module Name	Chemistry 322		Module Code			CHM322	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Chemical Sciences) (3220, 3019)			Total Learning Time		300 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Describe and compare the various spectroscopic techniques • Describe and compare the various chromatographic techniques • Examine and classify various instrumental analysis techniques with respect to their principles of operation and instrument components.						

	- Justify and apply instrumental analysis techniques to identify and quantify analytes in unknown compounds. - Illustrate the application of electro analytical techniques of analysis - Apply instrumental analysis techniques for the structural elucidate of both organic and inorganic materials.			
Main Content	- Molecular spectroscopy (uv-visible and infrared spectroscopy, mass spectrometry, ¹H-NMR and ¹³C-NMR). - Atomic spectroscopy (atomic absorption, atomic emission spectroscopy). - Separation science (principles of chromatography, gas chromatography, liquid chromatography). - Electro analytical techniques (introduction to electro analytical chemistry, potentiometry, coulometry, voltammetry).			
Pre-requisite	CHM211 and CHM212 and CHM221	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		5	Lectures p.w.	
		2	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences			Home Department			Computer Science		
Module Name	Computer Science 101			Module Code			COS101		
Module Type	Core/ Elective	NQF Level	5	Year Level	1/2	NQF Credit		30	
Programmes in which the module will be offered		BSc (Computer Science) (3221, 3023), BSc (Physical Science) (3233, 3120), BSc (Mathematics and Statistical Sciences) (3227)				Total Learning Time		300 hours	
Duration	Year		Proposed Semester			Both Semesters			
Main Learning Outcomes	On completion of this module, students should be able to: • Read, understand and solve problems. • Apply problem solving techniques. • Have acquired improve analytical thinking. • Understand sequence, selection, repetition and control structures. • Design an algorithm to solve a given problem. • Express an algorithm in pseudocode/structured diagram. • Implement algorithms in high-level programming language (e.g. Java). • Understand the Software Development Life cycle. • To pass parameters and use arrays. • Understand and implement the Object Oriented (OO) paradigm and inheritance. • Design and present a team project.								
Main Content	• Problem solving; algorithm design; trace tables; constructs. • Abstraction; • Pseudocode. • Structured diagramming techniques (top-down design).								

	• Arrays and data structures. • Development and application of simple algorithms. • High level programming language fundamentals. • Basic data types and methods. • The Software Development Life Cycle. • Design tools and object oriented design. • Inter-modular communication: Parameter passing and variable scope. • Objects as parameters. • Introduction to inheritance. • Data modeling, File I/O and recursion.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Computing Fundamentals 114		Module Code			COS114	
Module Type	Core/ Elective	NQF Level	5	Year Level	1/2	NQF Credit	15
Programmes in which the module will be offered		BSc (Applied Geology) (3214), BSc (Biotechnology) (3211), BSc (Medical Bioscience) (3230), BSc (Computer Science) (3221), BSc (Mathematics and Statistical Sciences) (3227), BSc (Physical Science) (3233)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Use computers (including e-mail and Internet), Software and multimedia to create and edit documents. • Skillfully use of worksheets and databases. • Integrate different application software packages. • Understand programs, data, files and file organization on a computer system, computer architecture, the computer marketplace; local area networks, and e-mail; the Internet, data security and control.						
Main Content	Practical Concepts: • Introduction to an operating system, e-mail and the Internet. • Applications software including word processing and spreadsheets; introduction to database management; introduction to presentation graphics; and the integration of the above mentioned application programs. Theoretical Concepts: • System programs; data and data organization. • Files and records, file management. • Principles and usage of basic software.						

	• Systems programs; data and data organization. • Computer architecture; The computer marketplace. • Local area networks and e-mail. • The Internet, data security and control.			
Pre-requisite	None	Prohibited module Combination	Computer Literacy modules offered by other faculties and departments.	
Co-requisite	None			
Timetable Requirement per week	3	Lectures p.w.		
	3	Practicals p.w.		
	1	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Computing Fundamentals 124		Module Code			COS124	
Module Type	Core/ Elective	NQF Level	5	Year Level	1/2	NQF Credit	15
Programmes in which the module will be offered		BSc (Applied Geology) (3214), BSc (Biotechnology) (3211), BSc (Medical Bioscience) (3230), BSc (Computer Science) (3221), BSc (Mathematics and Statistical Sciences) (3227), BSc (Physical Science) (3233)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Use computers (including e-mail and Internet), Software and multimedia to create and edit documents. • Skillfully use of worksheets and databases. • Integrate different application software packages. • Understand programs, data, files and file organization on a computer system, computer architecture, the computer marketplace; local area networks, and e-mail; the Internet, data security and control.						
Main Content	Practical Concepts: • Introduction to an operating system, e-mail and the Internet. • Applications software including word processing and spreadsheets; introduction to database management; introduction to presentation graphics; and the integration of the above mentioned application programs. Theoretical Concepts: • System programs; data and data organization. • Files and records, file management. • Principles and usage of basic software. • Systems programs; data and data organization. • Computer architecture; The computer marketplace. • Local area networks and e-mail. • The Internet, data security and control.						

Pre-requisite	None	Prohibited module Combination	Computer Literacy modules offered by other faculties and departments.		
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		3	Practicals p.w.		
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Computer Science 211		Module Code			CSC211	
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Computer Science) (3221, 3023), BSc (Mathematics and Statistical Sciences) (3227), BSc (Physical Science) (3233, 3120)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Can implement many algorithms that run correctly on computing machinery. • Can derive and explain the time and space complexity of algorithms. • Is able to tackle software problems from a procedural, object-oriented approach. • Construct models and produce working products from tools that the student has produced ab initio.						
Main Content	• Data structures. • Implementation of algorithms for manipulation. • The time and space complexity of the algorithms. • Efficiency of algorithms. • Correctness of algorithms by induction proofs. • Loop invariants. • Asymptotic bounds for algorithms.						
Pre-requisite	COS101		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			6	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department		Computer Science		
Module Name	Computer Science 212		Module Code		CSC212		
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	20

Programmes in which the module will be offered		BSc (Computer Science) (3221, 3023), BSc (Mathematics and Statistical Sciences) (3227), BSc (Physical Science) (3233, 3120)		Total Learning Time	200 hours
Duration	Semester	Proposed Semester		Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain standard algorithmic methods, algorithm efficiency, algorithm complexity and the ability to apply these. • Explain the nature of computationally intractable problems and the role of heuristics. • Demonstrate improved analytical thinking and problem solving skills. • Explain the interaction between low- and high-level software and hardware and knowledge of the architecture of a typical RISK processor at the conventional machine level. • Explain the limitations and bottlenecks in modern processors and how hardware components and their interconnection affect the performance of a computer.				
Main Content	Algorithms, and Complexity • Recursion, searches and traversal, divide and conquer, greedy algorithms, dynamic programming. • Program correctness; primitive recursive functions, mu-recursive functions, partial functions, Church-Turing thesis, time complexity of algorithms, average case complexity. • Classes P and NP, reducibility, NP-completeness, decidability, certificates. • Probabilistic algorithms. Architecture • System buses. • Internal memory and external storage. • Input/Output. • Instruction sets. • Operands and operations. • Addressing modes and formats. • Processor and register organization.				
Pre-requisite	COS101	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		6	Practicals p.w.		
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Computer Science 311		Module Code			CSC311	
Module Type	Core/ Elective	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered	BSc (Computer Science) (3221, 3023), BSc (Mathematics and			Total Learning Time		300 hours	

		Statistical Sciences) (3227, 3031), BSc (Physical Science) (3233, 3120)		
Duration	Semester	Proposed Semester	First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:			
	In Operating Systems:			
	• Explain the fundamental tasks performed by a modern operating system. • Implement fundamental operating systems tasks and algorithms.			
	Computer Networks:			
Main Content	• Build networks with current network topologies, protocols, operating systems, and applications. • Develop an appreciation of standards and protocols.			
	Introduction to Artificial Intelligence:			
	• Understand an agent-based approach to problem-solving. • Implement and apply uniformed and informed search algorithms. • Implement and apply local search and adversarial search algorithms. • Understand logical agents.			
	Operating Systems:			
Main Content	• History of operating systems. • Operating system concepts and structure. • Emphasis on processes (communication and scheduling). • Basic Input/Output. • Concurrency.			
	Computer Networks:			
	• Communications media. • Network standards and layers. • Communications protocols. • Network architectures. • Client/server and peer-to-peer networks, Network design. • Network operations and operating systems. • Network administration. • Construction and installation of networks.			
	Introduction to Artificial Intelligence:			
Main Content	• Intelligent Agents • Uniformed search strategies • Heuristic functions • Informed search strategies • Local search Algorithms and optimisation algorithms • Adversarial (games) algorithms. • Introduction to logic.			
	Pre-requisite	CSC211 and CSC212	Prohibited module Combination	None
	Co-requisite	None		
	Timetable Requirement per week			
		3	Lectures p.w.	
		2	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Computer Science 311		Module Code			CSC312	
Module Type	Core/ Elective	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Computer Science) (3221, 3023), BSc (Mathematics and Statistical Sciences) (3227), BSc (Physical Science) (3233, 3120)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: In Software Engineering • Interpret and implement principles, processes, methods and tools for quality software development. • Use UML to specify, visualize, construct and document a software system. • Design and implement appropriate human/computer interfaces. In Database Systems • Explain the Relational Model Concepts and Principles. • Represent the Architecture for a Database System. • Design a Relational Database. • Implement a Relational Database. In Machine Learning • Apply linear and logic regression and regularization to data. • Implement artificial and convolutional networks. • Apply reinforcement learning and unsupervised learning methods.						
Main Content	Software Engineering • Definition of software engineering, principles, goals, process, methods and tools. • The evolving role of software. • SE paradigms / process models. • Object-oriented analysis and design. • Introduction to formal specification. • Verification and validation. • Software quality assurance / reliability. • Unified Modeling Language (UML). Database Systems • File systems and databases. • Modeling. • Database design principles. • Database models. • Normalization. • Structured Query Language. Machine Learning • Introduction to machine learning • Linear and Logistic Regression • Artificial Neural networks • Convolutional Neural networks • Reinforcement Learning • Unsupervised learning						

Pre-requisite	CSC211 and CSC212	Prohibited module Combination		Database modules offered by other faculties and departments	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		0	Practicals p.w.		
		2	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Earth Sciences		
Module Name	Earth Materials 111		Module Code			ESC111		
Module Type	Core/ Elective	NQF Level	5	Year Level	1/2	NQF Credit	15	
Programmes in which the module will be offered		BSc (Applied Geology) (3214, 3011), BSc (Biodiversity and Conservation Biology) (3217, 3015), BSc (Environment and Water Science) (3331, 3027), BSc (Physical Science) (3233)				Total Learning Time	150 hours	
Duration	Semester		Proposed Semester			First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Explain basic Earth Materials and their formation. • Identify major rocks and minerals as products associated with these formation processes. • Grasp basic concepts of stratigraphy and Earth dynamics.							
Main Content	• The Earth in the Universe • Structure of the Earth, Isostasy, Plate Tectonics, Paleomagnetism • Rock Cycle, Geologic Time, • Introduction to minerals and rocks, classification of minerals and rocks • Geology of Southern Africa, Earth Dynamics • Historical Geology of Southern Africa • Introduction to structural Geology and mapping techniques							
Pre-requisite	None		Prohibited module Combination			None		
Co-requisite	None							
Timetable Requirement per week			3	Lectures p.w.				
			3	Practicals p.w.				
			1	Tutorials p.w.				
Other teaching modes that does not require time-table								
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA			50%
					FA			50%

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Earth Materials 121		Module Code			ESC121	
Module Type	Core/ Elective	NQF Level	5	Year Level	1/2	NQF Credit	15

Programmes in which the module will be offered		BSc (Applied Geology) (3214, 3011), BSc (Biodiversity and Conservation Biology) (3217, 3015), BSc (Environment and Water Science) (3331, 3027), BSc (Physical Science) (3233)		Total Learning Time	150 hours
Duration	Semester	Proposed Semester		Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate a broad understanding of the origins, nature and interactions of Earth system cycles and processes. • Abstract, synthesise and interrogate theories and concepts related to Earth system cycles. • Source, select and synthesise high quality peer-reviewed academic literature. • Plan and write an academic essay that describes and explains an Earth system cycle or process, and that adheres to academic conventions. • Use and interpret electronic spatial data and hard-copy maps to describe and analyse the regional climatic, geological, hydrological and biophysical setting of a study area.				
Main Content	• Introduction to Earth systems and systems theory • Structure of the atmosphere • Global radiation balance and air circulation • Ocean currents and atmospheric interaction • Climate: global to local • Weathering and mass wasting mechanisms • Weathering-climate links • The hydrological cycle • Flows, fluxes and residence times of water in the Earth system • Controls on biome distribution: global to local				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		1	Practicals p.w.		
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment		CA	50%
				FA	50%

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Environmental and Water Science 211		Module Code			EWP211	
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Applied Geology) (3214, 3011), BSc (Biodiversity and Conservation Biology) (3217, 3015), BSc (Environment and Water Science) (3331, 3027)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	

Main Learning Outcomes	On completion of this module, students should be able to:					
	- Identify, describe hydrological processes; occurring above and on the Earth's surface. - Explain factors accounting for variation in space and time of hydrological processes. - Establish and monitor infrastructure used to measure rainfall, interception and evaporation. - Use point rainfall data to estimate areal rainfall. - Undertake simple water balance analysis of catchments and water bodies. - Identify and describe physical and chemical characteristics of a soil. - Conduct laboratory analyses of soil physical and chemical characteristics, and report on the results. - Develop conceptual models that describe physical and chemical processes of soil formation, and soil erosion. - Develop conceptual models that explain how processes of soil formation and soil erosion work in combination to determine the thickness and physicochemical characteristics of soils					
Main Content	- Formation and measurement of precipitation - Factors influencing interception of precipitation by vegetation and how this process affects the availability of water on the Earth's surface. - Processes involved in the conversion of liquid water to vapour, the return of evaporated water to the atmosphere, and how this affects the availability of water on Earth. - Processes involved in the movement of water above and on the Earth's surface. - Processes relating to the formation of river flows, and factors accounting for the spatial and temporal variation of availability of surface water resources - Physical and biophysical processes that lead to soil development, and which determine variation in soil thickness across landscapes - Physical and chemical properties of soils - Controls on soil production, horizonation and erosion in different physiographic and landscape settings - Soil water and vadose zone hydrological processes - Soil groups of South Africa; classification of soils					
Pre-requisite	ESC111, ESC121		Prohibited module Combination		None	
Co-requisite	None					
Timetable Requirement per week			3	Lectures p.w.		
			2	Practicals p.w.		
			0	Tutorials p.w.		
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%
					FA	50%
Faculty	Natural Sciences		Home Department		Earth Sciences	
Module Name	Environmental and Water Science 221		Module Code		EWP221	
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit
						20

Programmes in which the module will be offered		BSc (Applied Geology) (3214, 3011), BSc (Biodiversity and Conservation Biology) (3217, 3015), BSc (Environment and Water Science) (3331, 3027)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Describe the hydro-geologic system including value, role, key concepts and principles and the hydrologic cycle - Describe and illustrate groundwater occurrences including recharge mechanism and estimations methods - Explain groundwater hydraulics including basic principles of groundwater flow mechanisms - Discuss groundwater chemistry and pollution including water sampling and water quality analysis - Describe and explain groundwater development including drilling and logging of boreholes - Identify and justify groundwater-surface water interaction including examples of groundwater dependence ecosystems - Recognise geohydrology of mining environment including the concept of acid mine drainage (AMD)						
Main Content	- Hydro-geologic system - Groundwater occurrences including recharge mechanism and estimations methods - Groundwater hydraulics including basic principles of groundwater flow mechanisms - Groundwater development i.e drilling and logging of boreholes - Groundwater chemistry and pollution i.e water sampling and water quality analysis - Groundwater-surface water interaction i.e. groundwater dependent ecosystems - Geohydrology of the mining environment i.e. Acid Mine Drainage (AMD)						
Pre-requisite	ESC111, ESC121		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%
Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Environmental and Water Science 222		Module Code			EWP222	
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered	BSc (Environment and Water Science) (3027, 3331)					Total Learning Time	200 hours

Duration	Semester	Proposed Semester	First Semester (Over weekends/ Vacation period)
Main Learning Outcomes	On completion of this module, students should be able to: • Position themselves in the field and record information collected in the field and produce maps at various scales. • Conduct measurements/sampling in the field and in the laboratory to enhance quantitative skills. • Observe, record, describe and report various environmental and water science related features in the field. • Write reports on the conducted field and lab work.		
Main Content	• Develop basic understanding of environmental and water science techniques in the field and laboratory. • Mapping techniques for environmental and water science. • Surface water sampling techniques and data analysis. • Groundwater sampling techniques and data analysis. • Fluvial geomorphology sampling techniques and data analysis. • Sampling techniques for environmental science/ management parameters and data analysis methods for the sampled data. • Reporting data and scientific report writing.		
Pre-requisite	ESC111, ESC121	Prohibited module Combination	None
Co-requisite	EWP211 and EWP221		
Timetable Requirement per week		0	Lectures p.w.
		1	Practicals p.w.
		0	Tutorials p.w.
Other teaching modes that does not require time-table			
50% of the course is conducted in the field outside the University campus			
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA
			100%

Faculty	Natural Sciences			Home Department			Earth Sciences	
Module Name	Geographical Information Systems 311			Module Code			EWP311	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30	
Programmes in which the module will be offered		BSc (Environment and Water Science) (3027, 3331)				Total Learning Time	300 hours	
Duration	Semester		Proposed Semester			First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Describe GIS principles and apply spatial related techniques • Develop, illustrate and apply skills to construct GIS raster and vector maps using environmental and water related data • Demonstrate the use of data queries and spatial data analyses • Develop, demonstrate and practice conceptual understanding of data management, data querying and water related data sets • Describe and explain principles of modelling environmental and water related processes • Develop and apply skills to integrate environmental and water data into computer-based models as well as processing, analysing and evaluating resulting images							
Main Content	• Database management, vector maps and raster maps							

	• Data query and spatial data analysis • Environmental and water related data and GIS • Application of GIS in natural resource studies • Basic statistical data analysis • Correlation and regression analysis • Introduction to cartography: Maps, feature types, map symbolism, spatial relationships and scale concepts				
Pre-requisite	EWP211, EWP221, EWP222	Prohibited module Combination		APG311	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		2	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Environmental and Water Science 312		Module Code			EWP312	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Environment and Water Science) (3027, 3331)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Critically evaluate and apply methods used to analyse a variety of surface water processes and phenomena. • Critically evaluate climate change projections for southern Africa, and their implications for water resources. • Describe the general controls on flow variability in dryland environments. • Interpret metrics of river flow variability, and consider their implications for river ecosystem processes. • Evaluate controls on sediment dispersal through river catchments from global to local scales. • Apply methods used for aquifer characterisation and the evaluation of groundwater hydraulics. • Demonstrate groundwater processes and groundwater-surface water interactions in various hydrogeologic settings using practical examples. • Simulate flow-dynamics of groundwater in three types of boundary conditions using various modelling techniques. • Interpret numerical flow-simulations for groundwater.						
Main Content	• Climatic and hydrological variability, and climate change. • Surface water processes and phenomena driven by precipitation and evapotranspiration. • River flow variability (and methods of analysis). • Hillslope-channel coupling and connectivity. • River sediment delivery. • Aquifer characterisation. • Groundwater processes.						

	• Groundwater hydraulics (and methods of interpretation). • Groundwater modelling (regional and site-specific models for groundwater flow dynamics). • Groundwater-surface water interaction.			
Pre-requisite	EWP211, EWP221, EWP222	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		2	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Environmental and Water Resource Management 321		Module Code			EWP321	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Environment and Water Science) (3027, 3331), BSc (Biodiversity and Conservation Biology) (3217, 3015)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Analyse and apply the concepts and principles of environmental management, impact assessment and water resource management to real-world problems. - Interrogate, synthesise and apply key South African environmental and water policy and legal documents, and their contents. - Explain the purpose and targets of Sustainable Development Goals (SDGs), and develop research and management approaches to addressing the SDGs. - Develop practical interventions for water resources assessment, planning, development, allocation, quality management and protection, costing, monitoring and governance. - Apply the concept of integrated Water Resource Management (IWRM) in the southern African context. - Demonstrate methods, tools, instruments and interventions commonly used in South Africa and internationally to manage water resources and solve environmental problems.						
Main Content	Environmental Management: - Theoretical overview of environmental management principles and practices. - Concepts of sustainable development; Sustainable Development Goals with particular focus on environmental and water-related SDGs. - South African environmental management policy and procedures: National Environmental Management Act, Environmental Impact Assessment Regulations.						

	- Case studies on applications dealing with environmental impacts: instruments, tools and technologies applied to solve environmental problems. Water Resources Management: - Overview of water resources management: global and South African issues, scales of water resources, water safety and security. - Water resources assessment, planning, development, allocation and management interventions. - Water quality factors, impacts and management interventions. - Water costing, monitoring and governance. - South African water policy and regulations: National Water Act and National Water Resource Strategy. - Integrated Water Resource Management (IWRM) concepts. - Case studies on methods, tools, instruments and interventions in water resources management.			
Pre-requisite	EWP211, EWP221	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		2	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Environmental and Water Science 322		Module Code			EWP322	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered	BSc (Environment and Water Science) (3027, 3331)					Total Learning Time	300 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Describe water quality parameters including major processes that govern and modify water quality - Plan and conduct water quality analyses (sampling methods, determining numbers of samples, analyses methods, and selection of chemical species) on the basis of project objectives - Use guidelines to collect meaningful field water quality data through appropriate sampling techniques, including field equipment calibration and sample handling procedures - Develop expertise on field and laboratory data interpretation - Implement data quality assessment and data quality control procedures - Explain the use of isotope hydrology in sampling, and analysing and interpreting water quality data - Compile scientific reports on the results, including research proposals with time plans						
Main Content	- Theoretical background of water quality and field techniques - Design sampling strategy - Setting up field equipment - Field data collection, data analyses including lab work						

	• Quality assurance and quality control reporting • Data interpretation • Report writing, including research proposals				
Pre-requisite	EWP211, EWP221, EWP222	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		2	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Human Biology for Dentistry 105		Module Code			HUB105	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	40
Programme in which the module will be offered		BDS (5101)				Total Learning Time	400 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Understand the importance of cells to operate within the homeostatically controlled internal environment. - Describe the embryonic origins of tissues and the correlations between origin and function of tissue cells. - Understand the physiology of haemostasis and blood types. - Execute basic laboratory investigations and understand the diagnostic value of haematological parameters. - Understand the cellular and biochemical basis immunological mechanisms in the body. - Identify and describe the main anatomical features of structures of the thoracic cavity and relate their structure to specific functions. - Describe the functional histology of, and identify, the structures of the CVS, Respiratory and Renal systems. - Explain the homeostatic mechanisms of the above systems, their neural and endocrine regulation, and the dysfunctions associated with these systems. - Execute basic laboratory investigations.						
Main Content	- Organization of the human body - Principles of homeostatic control - Review of the cell physiology - Method of studying cells - Early embryology - Basic neurology - Connective tissue histology and chemistry - Functional histology of epithelia, cartilage, bone, - Teeth, skin, neural tissue, the lymphatic system and - Muscle - Electrolyte and fluid balance - Blood, haemostasis, blood types, immunology and						

	• Associated abnormalities • Anatomy of the thorax • Mechanics of breathing • Organization of the CVS • The cardiac cycle, Starling's Law and cardiac output • Histology of blood vessels • Haemodynamics • Blood pressure • Control of the CVS • Cardiovascular disease • Structures and histology of the respiratory system • Lung volumes and composition of alveolar air • Transport of O2 and CO2 • Control of breathing • Anatomy and Histology of the kidney • Glomerular filtration • Renal control of body fluids • Acid-base balance			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		6	Lectures p.w.	
		6	Practicals p.w.	
		2	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Human Biology for Pharmacy 113		Module Code			HUB113	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Describe and list the basic structure and general functions of proteins, carbohydrates, lipids, enzymes and nucleic acids. • Define and explain human nutrition and cell metabolism. • Describe and explain the basic anatomy and physiology of the digestive system. • Define, Describe and explain protein synthesis, cell growth and cell division. • Describe and explain the early development of the foetus • Describe the histology of the main tissue types. • Describe, explain and list the gross anatomy and functions of the skeletal and muscle systems, and the characteristics of joints types. • Demonstrate selected laboratory techniques related to the evaluation of tissue types, the muscle and skeletal systems. • Demonstrate appropriate habits and attitudes needed for a career in health sciences and health care promotions.						

Main Content	• The main chemical compounds, nutrition and cellular metabolism • An introduction of the anatomy and physiology of the digestive system. • Essentials of early human development. • Essentials of the histology of tissue types. • Gross anatomy and basic physiology of the skeletal and muscle systems. • Articulations/joints and movement.				
	Pre-requisite	None	Prohibited module Combination		None
	Co-requisite	None			
	Timetable Requirement per week		3	Lectures p.w.	
			1	Practicals p.w.	
1			Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment	CA	50%
				FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Human Biology for Nursing 114		Module Code			HUB114	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Classify and describe the organization of the human body, and the concepts of health, human diseases and disease causing agents. • Describe the various terms used to define the human anatomy. • Define and explain the functions of the skeletal and muscle systems, and the characteristics of different joint types. • Classify and describe the histology and function the main tissue types. • Define and describe the structure-function relationships of components of the generalized animal cells. • Classify and describe the basic structure and general functions of proteins, carbohydrates, lipids, enzymes and nucleic acids. • Explain the concept of human nutrition and cell metabolism. • Define and describe the principles of protein synthesis. • Demonstrate an ability to execute selected laboratory techniques related to the evaluation of the different tissue types, the muscles and bones. • Demonstrate the appropriate skills, habits and attitudes they need to develop for a career in health care promotion.						
Main Content	1. Organization and function from chemical to systemic level a) The scope of human biology and functional organization of the body. b) Homeostasis, human diseases and disease forming organisms c) The main chemical compounds, nutrition and cellular metabolism d) Review of cellular organization and functions of cellular components. e) Essentials of early human development f) Essentials of the histology of tissue types						

	2. Systems of support and movement a) Gross anatomy of the skeletal system b) Gross anatomy of the muscle systems c) Articulations and movement			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		2	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Introduction to the Human Body 117		Module Code			HUB117	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc (Physiotherapy) (8211), BSc (Occupational Therapy) (8111)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the organisation of the human body and the concept of health, human disease and the disease-causing agents. • Describe the gross anatomy and general functions of the skeletal and muscular systems and the characteristics of different joint types. • Describe the components of the locomotor systems. • Explain and apply basic biomechanical concepts. • Explain the anatomy and physiology of movement. • Identify and explain some of the effects of dysfunction of parts of neuromuscular system. • Describe the early development of the foetus. • Explain and classify the histology of the basic tissue types.						
Main Content	Organization and function • The scope of human biology and functional organization of the body. • Essentials of early human development. • Essentials of the histology of the basic tissue types. Systems of support and movement • Gross anatomy of the musculo-skeletal system. • Gross anatomy of nerve supply to muscle groups. • Articulations and movement of the upper limb with special reference to the functions of the hand. • Movements of the lower limb in walking and running. Movements of the trunk.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			3	Practicals p.w.			
			2	Tutorials p.w.			

Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%	
					FA	50%	
Faculty	Natural Sciences		Home Department		Medical Biosciences		
Module Name	Human Biology for Nursing 118		Module Code		HUB118		
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BNurs (8311)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester		First Semester		
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Classify and describe the organization of the human body, and the concepts of health, human diseases and disease causing agents. • Describe the various terms used to define the human anatomy. • Define and explain the functions of the skeletal and muscle systems, and the characteristics of different joint types. • Classify and describe the histology and function the main tissue types. • Define and describe the structure-function relationships of components of the generalized animal cells. • Classify and describe the basic structure and general functions of proteins, carbohydrates, lipids, enzymes and nucleic acids. • Explain the concepts of human nutrition and cell metabolism. • Define and describe the principles of protein synthesis. • Demonstrate an ability to execute selected laboratory techniques related to the evaluation of the different tissue types, the muscle and bones. • Demonstrate the appropriate skills habits and attitudes they need to develop for a career in health care promotion.						
Main Content	Organization and function from chemical to systemic level						
	• The scope of human biology and functional organization of the body • Homeostasis, human diseases and disease forming organisms • The main chemical compounds, nutrition and cellular metabolism • Review of cellular organization and functions of cellular components • Essentials of early human development • Essentials of the histology of tissue types Systems of support and movement • Gross anatomy of the skeletal system • Gross anatomy of the muscle systems • Articulations and movement						
Pre-requisite	None		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%	
					FA	50%	

Faculty	Natural Sciences		Home Department			Medical Biosciences		
Module Name	Human Biology for Pharmacy 123		Module Code			HUB123		
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15	
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours	
Duration	Semester		Proposed Semester			First Semester		
Main Learning Outcomes	On completion of this module, students should be able to:							
	• Describe the organization of human body, the concepts of health, human diseases and disease-causing agents.							
	• Describe and explain the structure-function relationships of components of the generalized body cell, and of example of specialized cells.							
	• Describe, list and explain the functional anatomy of the nervous systems and the major sensory and motor pathways.							
	• Classify, describe and explain the functioning of neurons, the transduction of sensory stimuli, how this information is processed and responded to by the CNS.							
	• Define, list, classify, describe and explain the anatomy and physiology of the various organ systems, viz. the cardiovascular, respiratory, urinary, digestive, endocrine and reproductive systems.							
	• Demonstrate proficiency in selected basic laboratory competencies.							
	• Demonstrate information processing skills, such as searching and evaluating information.							
	• Demonstrate appropriate habits and attitudes needed for a career in health care promotion.							
	Main Content	• The scope of human biology and functional organisation of the body.						
• Homeostasis, human diseases and disease forming organisms.								
• Review of cellular organisation and functions of cellular components.								
• An introduction to the anatomy my and physiology of the nervous, cardiovascular, respiratory, urinary, endocrine and reproductive systems.								
Pre-requisite		None			Prohibited module Combination		None	
Co-requisite		None						
Timetable Requirement per week			3	Lectures p.w.				
			1	Practicals p.w.				
			1	Tutorials p.w.				
Other teaching modes that does not require time-table								
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%	
						FA	50%	

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Human Biology 124		Module Code			HUB124	
Module Type	Core	NQF Level	5	Year Level	2	NQF Credit	15
Programme in which the module will be offered		BNurs (8313)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: Classify, describe and explain the functional anatomy of the nervous system and the major sensory and motor pathways.						

	- Classify, describe and explain the function of neurons, the transduction of sensory stimuli, how this information is processed and responded to by the CNS. - Classify, describe and explain the anatomy and physiology of the various organ systems viz. the cardiovascular, immune, respiratory, urinary, digestive, endocrine and reproductive. - Demonstrate selected basic laboratory competencies. - Demonstrate appropriate habits and attitudes needed for a career in health care promotion. - Demonstrate information processing skills, such as searching and evaluating information.			
Main Content	An introduction to the anatomy and physiology of the cardiovascular, respiratory, urinary, digestive, endocrine, reproductive and nervous system.			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		1	Practicals p.w.	
		2	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Structure and Function of the Human Body 127		Module Code			HUB127	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc (Physiotherapy) (8211), BSc (Occupational Therapy) (8111)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain principles and components of control systems, and examples thereof. • Describe various membrane transport and communication processes of the cell. • Describe functional anatomy of the central nervous system (CNS) and the major sensory and motor pathways. • Explain functioning of neurons, the transduction of sensory stimuli, how this information is processed and responded to by the CNS. • Describe the physiology of the muscular systems, neuromuscular junction, blood and immune system. • Recognize some effects of dysfunction of parts of the neuromuscular system. • Develop information processing skills, such as searching for and evaluating information.						
Main Content	• Principles and components of control systems. • Membranes: general functions, composition, and specific membrane transport processes. • Intercellular signalling: receptors and chemical signals.						

	An understanding of the anatomy and physiology of the central nervous system (CNS), the physiology of the muscular system, blood and immunology.			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		3	Practicals p.w.	
		2	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Introduction to the Human Body 128		Module Code			HUB128	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BNurs (8311)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Define, list, classify, describe and explain the anatomy and physiology of the various organ systems viz. the cardiovascular, immune, respiratory, urinary, digestive, endocrine and reproductive. - Define, list, classify, describe and explain the functional anatomy of the nervous system and the major sensory and motor pathways. - Classify, describe and explain the function of neurons, the transduction of sensory stimuli, how this information is processed and responded to by the CNS. - Demonstrate selected basic laboratory competencies. - Demonstrate appropriate habits and attitudes needed for a career in health care promotion. - Demonstrate information processing skills, such as searching and evaluating information.						
Main Content	An introduction to the anatomy and physiology of the nervous, cardiovascular, respiratory, urinary, digestive, endocrine and reproductive systems.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Human Biology for Dentistry 205		Module Code			HUB205	
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	40
Programme in which the module will be offered		BDS (5101)				Total Learning Time	400 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the role of nutrition in health. • Relate the anatomy of the GIT, and associated structures, to the mechanisms of motility, secretion, digestion and absorption. • Describe the neural and endocrine control of the processes of the digestive system. • Describe the functional anatomy and histology of the major endocrine glands and the reproductive systems • Describe calcium and phosphorous metabolism, its hormonal control and bone metabolism. • Explain the physiology and physical mechanisms that maintain thermal homeostasis. • Describe the menstrual cycle and hormonal context of pregnancy, lactation, contraception and HRT • Describe the development of the head, neck and central nervous system of the fetus. • Describe the anatomy of the head and neck with emphasis on the oral and peri-oral region. • Outline the important functional pathways of the central nervous system. • Explain the structure and function of the central nervous system. • Outline the cranial nerves.						
Main Content	• Describe Body composition nervous system. • List the nutrients and non-nutrients in diet. Describe the: • Anatomy/histology of the GIT. • Mastication and swallowing. • Histology and physiology of salivary glands, stomach, pancreas, liver, biliary system, intestines. • Metabolism. Insulin and glucagon. • Diabetes mellitus. Outline the: • Calcium and bone metabolism. • Hypothalamic and pituitary hormones. • Temperature regulation. • Adrenal glands. • Menstrual cycle • Hormones in pregnancy, lactation, contraception and HRT. • Anatomy male reproductive system. Describe the: • Developmental embryology of the head, neck and central nervous system. • Gross anatomy of the head and neck region. • The cranial nerves.						

	• The functional units of the central nervous system. • Structure and function of sensory pathways. • Structure and function of motor pathways. • The Autonomic system.			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		0	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Human Biology 218		Module Code			HUB218	
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	20
Programme in which the module will be offered		BNurs (8311)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain and differentiate between the various membrane transport and communication processes of the cell. • Explain how genetic information is transferred from DNA to proteins. • Describe the phases and changes cells undergo during their life cycle and the regulation thereof. • Discuss homeostasis with regards to the nervous and endocrine systems. • Describe the anatomical and functional organization of the nervous system. • Explain basic neurophysiology and neural integration. • Describe the components, organization and function of the sensory and motor pathways. • Explain the autonomic nervous system at a basic level. • Explain the basic higher brain functions. • Describe the structure and function of the endocrine system. • Interpret the body systems' structure and physiology and consequences of dysfunction in the various body systems – selected examples. • Demonstrate laboratory competencies related to the investigation of the form and function of the specified systems. • Demonstrate basic competencies in executing investigative experiments, and reporting it in an acceptable scientific style and format						
Main Content	• Membranes: general functions, composition, and specific membrane transport processes • Genes and gene expression • Clinical applications and consequences of cellular dysfunction • Cell cycle, somatic and reproductive cell division • Cellular metabolic processes. • Signalling in the nervous system • Brain, spinal cord, cranial and spinal nerves						

	• General and special senses • Anatomy, physiology and regulation of autonomic nervous system • Higher brain function: memory • Macro- and microscopic structure and coordinated functions of the endocrine system				
Pre-requisite	HUB114 and HUB124; or HUB118 and HUB128	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		1	Practicals p.w.		
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Human Biology 228		Module Code			HUB228	
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	20
Programme in which the module will be offered		BNurs (8311)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Describe the components and organization of the cardiovascular, lymphatic, urinary, digestive and reproductive systems. • Explain the functioning of the cardiovascular, lymphatic, urinary, digestive and reproductive systems. • Explain the metabolism of macromolecules. • Explain the anatomy and physiology of muscle at a basic level. • Discuss muscular systems of the body at a basic level. • Interpret the body systems' structure and physiology and consequences of dysfunction in the various systems – selected examples. • Demonstrate laboratory competencies related to the investigation of the form and function of the specified systems.						
Main Content	• Macro- and microscopic structure and functions of the cardiovascular, respiratory, urinary, digestive and reproductive systems • Topographical (surface) anatomy of these organ systems and vessels • Metabolism of carbohydrates, lipids and proteins • Macro- and microscopic structure and coordinated functions of the reproductive system • Surface anatomy, skeleton, joints, muscles, nerves and movements of the head and neck, upper limb, back, thorax, abdomen, pelvis, perineum and lower limb						
Pre-requisite	HUB114 and HUB124 or HUB118 and HUB128			Prohibited module Combination		None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			1	Tutorials p.w.			

Other teaching modes that does not require time-table									
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%		
						FA	50%		
Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology			
Module Name	Introduction to Science 153		Module Code			ISC153			
Module Type	Core	NQF Level	5	Year Level	1/2	NQF Credit		30	
Programmes in which the module will be offered		BSc (Applied Geology) (3011), BSc (Biodiversity and Conservation Biology) (3015), BSc (Biotechnology) (3007), BSc (Chemical Sciences) (3019), BSc (Computer Science) (3023), BSc (Environment and Water Science) (3027), BSc (Mathematics and Statistical Sciences) (3031), BSc (Medical Bioscience) (3035), BSc (Physical Science) (3120)				Total Learning Time		300 hours	
Duration	Year		Proposed Semester			Both Semesters			
Main Learning Outcomes	On completion of this module, students should be able to: - Engage actively in the processes of reading and writing to learn. - Identify and use appropriately some of the specialist language of your fields of study. - Read a range of texts effectively, purposefully, critically, and with comprehension. - Extract and categorise information to facilitate effective learning strategies. - Summarise and paraphrase texts effectively. - Collect information efficiently for research and writing purposes. - Use computers and other technology to competently demonstrate your learning. - Accurately represent authors' ideas without plagiarising.								
Main Content	- Reading and writing scientific texts - Giving oral presentations. - Improving quantitative literacy, logic, approaches to problem-solving, linking these to other modules Using a computer operating system, email and the Internet, applications software and the integration of these application programmes. - Applying the principles of co-operative learning and teamwork, time-management, diversity, career-planning.								
Pre-requisite	None		Prohibited module Combination			None			
Co-requisite	PHY151 or LFS151; MAM151 or MAM150								
Timetable Requirement per week			5	Lectures p.w.					
			1 x 3	Practicals p.w.					
			0	Tutorials p.w.					
			3	Workshop p.w					

Other teaching modes that does not require time-table								
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%	
Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology		
Module Name	Life Science 151		Module Code			LFS151		
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit		15
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3015), BSc (Biotechnology) (3007), BSc (Environment and Water Science) (3027), BSc (Medical Bioscience) (3035)				Total Learning Time		150 hours
Duration	Year		Proposed Semester			Both Semesters		
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate scientific methodology by recording observations, formulating hypotheses, collecting and critically analysing data and compiling a scientific report. • Link the importance of basic inorganic and organic chemistry to cell organisation. • Explain interactions between major cell organelles, the cell membrane, the role of enzymes in metabolic pathways and protein synthesis. • Identify genetic components related to the inheritance of genetic traits. • Explain the different forms of cell division. • Demonstrate skills in: scientific measurements, operating light microscopes, preparation of slides, scientific drawing and conducting small experiments. • Present information orally in class and in written and typed form using MS Word and MS Excel. • Read, interpret, paraphrase, summarise, compare and synthesize information in the form of literature, figures, graphs and tables. • Demonstrate basic information literacy skills.							
Main Content	• Overview of the process of science and classification of living things. • Cell structure and function of cellular organelles. • Photosynthesis. • Cellular respiration. • DNA replication. • Mitosis, meiosis and control of protein synthesis. • Mendelian genetics. • Modern genetics. • Introduction to biotechnology • Basic principles of evolution.							
Pre-requisite	None		Prohibited module Combination			None		
Co-requisite	None							
Timetable Requirement per week			5	Lectures p.w.				
			1	Practicals p.w.				
			3	Tutorials p.w.				

Other teaching modes that does not require time-table								
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	60%	
						FA	40%	
Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology		
Module Name	Life Science 152		Module Code			LFS152		
Module Type	Core	NQF Level	5	Year Level	2	NQF Credit	15	
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3015), BSc (Biotechnology) (3007), BSc (Chemical Sciences) (3019), BSc (Environment and Water Science) (3027)				Total Learning Time	150 hours	
Duration	Year		Proposed Semester			Both Semesters		
Main Learning Outcomes	On completion of this module, students should be able to: - Describe the nature of science by using as examples the changes made to classification systems to illustrate science as dynamic. - Use the basic rules of classification to classify, describe and identify life forms. - Compare the major characteristics of biological taxa. - Discuss the evolutionary relationships between biological taxa. - Define ecological terminology and explain, with the use of examples, the basic principles of ecology. - Explain the importance of biodiversity in ecological systems. - Identify the value of and threats to biodiversity. - Observe and make scientific drawings of organisms with the aid of microscopes. - Practice dissection skills. - Follow the scientific method to design and conduct a simple experimental data from the laboratory and field. - Organize and interpret simple experimental data from the laboratory and field. - Write scientific abstracts and reports. - Write comparative essays. - Use basic computer programmes such as MS Excel, MS Word and MS Powerpoint. - Present researched material to an audience orally with the use of a Powerpoint presentation and in the form of a scientific poster; and - Apply basic information literacy skills.							
Main Content	- Brief description of the origin of living cells and evolution of higher life forms. - Basic rules of classification, the diversity of life including: Viruses, monera. - Protista, fungi, plants, invertebrates and vertebrates. Biotic and abiotic components of the unifying role of ecology in biology. - Population dynamics, community ecology and ecosystem functioning. - The impact of humans on the environment. The nature and process of science and basic statistics.							

Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week	3	Lectures p.w.		
	1	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Life Science 141		Module Code			LSC141	
Module Type	Core/ Elective	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217), BSc (Biotechnology) (3214), BSc (Chemical Sciences) (3220), BSc (Environment and Water Science) (3331), BSc (Medical Bioscience) (3230), BEd (Mathematics and Natural Sciences) (4516, 4092), BEd (SP and FET) (4405), BDS (5101)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Link the importance of basic inorganic chemistry to cell organisation. • Know the maintenance of life as controlled by the major organic (bio-) molecules. • Explain the interaction between the major cell organelles, the structure and role of cell membranes, the role of enzymes to the various metabolic pathways in cells, the link between protein synthesis and genetic traits, how genetic information can be manipulated in the laboratory. • Identify the various genetic components as related to the inheritance of genetic traits. • Know the different forms of cell division. • Apply practical skills in microscopy. • Assimilate information from various sources. • Interpret and present information in written form.						
Main Content	• Cell structure and organelles and an introduction to processes taking place in them. • DNA replication; DNA control of protein synthesis and thereby biochemical processes, mitosis, meiosis, chromosomes and genes, Mendelian and biochemical genetics, evolution.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			1	Tutorials p.w.			

Other teaching modes that does not require time-table									
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA		60%	
						FA		40%	
Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology			
Module Name	Biodiversity and Ecology 142		Module Code			LSC142			
Module Type	Core/ Elective	NQF Level	5	Year Level	1	NQF Credit		15	
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217), BSc (Biotechnology) (3214), BSc (Chemical Sciences) (3220), BSc (Environment and Water Science) (3331), BEd (Mathematics and Natural Sciences) (4516, 4092), BEd (SP and FET) (4405)				Total Learning Time		150 hours	
Duration	Semester		Proposed Semester			Second Semester			
Main Learning Outcomes	On completion of this module, students should be able to: • Use the basic rules of classification to classify, describe and identify life forms. • Compare the major characteristics of biological taxa. • Discuss the evolutionary relationships between biological taxa. • Define ecological terminology and explain with the use of examples, the basic principles of ecology. • Identify the values of and threats to biodiversity. • Observe and make scientific drawings of organisms with the aid of microscopes. • Practice dissection skills. • Follow the scientific method to design and conduct a simple experiment. • Organize and interpret simple experimental data from the laboratory and field. • Write basic scientific reports.								
Main Content	• Introduction to the history of life and the rules of classification. Diversity of life (bacteria, archaea, eukarya, viruses). Levels of ecology (population, communities, ecosystems, biomes and the biosphere). Awareness of the impact of humans on ecological processes and on biodiversity.								
Pre-requisite	None		Prohibited module Combination			None			
Co-requisite	None								
Timetable Requirement per week			3		Lectures p.w.				
			1		Practicals p.w.				
			1		Tutorials p.w.				
Other teaching modes that does not require time-table									
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA		60%	
						FA		40%	

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Mathematics 115		Module Code			MAM115 (N)	
Module Type	Core	NQF Level	5	Year Level	1/2	NQF Credit	15
Programmes in which the module will be offered		BSc (Applied Geology) (3214, 3011), BSc (Biotechnology) (3211, 3007), BSc (Chemical Sciences) (3019, 3220), BPharm (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Implement mathematical methods and techniques of Differential and Integral calculus in the Life Sciences and Pharmacy.						
Main Content	- Functions and limits - differentiation - curve sketching and applications of differentiation - integration						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Mathematics 116		Module Code			MAM116	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BCom (Accounting) (Extended Curriculum) (1762)			Total Learning Time		100 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Mathematical and computational skills useful in the Economics and Management environment; • Skills to read mathematical text in Economics; • Skills to write mathematical text in Economics; • Skills to interpret correctly literature pertaining to Managerial Sciences that require knowledge of mathematics.						
Main Content	Foundation Content 1. Introduction to the Language of Mathematics • Basic mathematical symbols and signs • Mathematical notations and sets • Bracket pairs in mathematical language 2. Data Handling • Descriptive statistics and measures of central tendency 3. Elementary Algebra • Variables • Exponents, powers and the exponential laws						

• Polynomials versus expressions • Factors and Factorization • Solving quadratic equations • Solving linear inequalities • Rational expressions (i.e. Algebraic fractions) • Solving rational equations Main Content 4. Functions • Linear functions • Quadratic functions • Higher order polynomials • Rational functions • Exponential functions • Logarithmic functions 5. Application of Functions • Simple and Compound interest • Break-even and Market equilibrium • Nominal and Effective rates • Discounting • Annuities: Amortization and Sinking funds • Continuous compounding 6. Special Functions • Compound functions • Combination of functions and composition of functions 7. Limits and Continuity • Introduction to limits • Properties of limits • Continuity 8. Introduction to Differential Calculus • Definition of derivative • Rules of differentiation and rate of change • Derivative of polynomials, rationals, exponential and Logarithmic functions • Marginal analysis • Marginal propensity to consume and save 9. Applications of Differentiation • Increasing and decreasing functions • Relative maxima and minima • Maximization applied to commercial type problems 10. Introduction to Integral Calculus • Definite and indefinite integral • Integration of polynomial, rationals, exponential and logarithmic functions • Integration by substitution • Applications to Managerial Sciences • Consumer and producer surplus			
Pre-requisite	None	Prohibited module Combination	
Co-requisite	None	None	
Timetable Requirement per week		3	Lectures p.w.
		1	Practicals p.w.
		0	Tutorials p.w.

Other teaching modes that does not require time-table									
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA		100%	
Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics			
Module Name	Mathematics 126/127		Module Code			MAM126/127			
Module Type	Core	NQF Level	5	Year Level	1/2	NQF Credit		10	
Programmes in which the module will be offered		BCom (1008), BCom (Accounting) (1021, 1753), BCom (Financial Accounting) (1175)				Total Learning Time		100 hours	
Duration	Semester		Proposed Semester			S1 - MAM126 S2 - MAM127			
Main Learning Outcomes	On completion of this module, students should be able to: • Mathematical and computational skills useful in the Economics and Management environment. • Skills to read mathematical text in Economics • Skills to write mathematical text in Economics • Skills to interpret correctly literature pertaining to Managerial Sciences that require knowledge of mathematics.								
Main Content	Mathematics of Finance 1. Simple and Compound interest • Definitions and Applications • Nominal and Effective rates • Continuous compounding • Discounting 2. Simple and Complex Annuities • Present and Future Values • Simple and Compound depreciation and amortization • Sinking funds with deduction 3. Introduction to Differential Calculus • Functional notation; functions and limits • Graphs of polynomials, exponential functions • Break-even and Market equilibrium • Definition of derivative • Rules of differentiation for polynomial, exponential and Logarithmic functions 4. Applications of Differentiation • Increasing and decreasing functions • Relative maxima and minima • Maximization applied to commercial type problems 5. Introduction to Integral Calculus • Definite and indefinite integral • Integration of polynomial, exponential and logarithmic functions • Integration by substitution and parts • Applications to Managerial Sciences • Consumers' and suppliers' surplus • Business Application								

Pre-requisite	QSC131 or QSC132 or QSA131	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		0	Practicals p.w.		
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Introduction to Mathematics 150		Module Code			MAM150	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3015), BSc (Biotechnology) (3007), BSc (Environment and Water Science) (3027), BSc (Medical Bioscience) (3035)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Implement mathematical methods and techniques of Differential and Integral calculus in the Life Sciences and Pharmacy						
Main Content	• An introduction to the language of Mathematics including bracket pairs in Mathematics, the use of brackets in expressions and formulae. Equations versus inequalities. Absolute value. • Functions- • Simple polynomial-, power-, and rational functions, their graphs and their sum, difference, product and quotient. The composite function. • Trigonometric functions. • Exponential and logarithmic functions-infusing the exponential and logarithmic laws and surds. Exponential growth and decay. • Log-log and semi-log plots. • Limits and continuity of simple functions. The natural logarithm. • Rates of change, definitions of derivative, rules for finding derivative-including logarithmic differentiation. Higher order derivatives. • Application of derivatives –curve sketching, maxima and minima, applications of maxima and minima –infusing volume and surface area, Newton’s Method. • Implicit differentiation. • Integration, the anti-derivative and indefinite integral, including standard integrals, method of substitution and integration by parts. • The definite integral, including areas under curves and areas between curves. • Improper integrals. • Numerical integration-the Trapezoidal Rule and Simpson’s Rule.						

Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		1	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Mathematics 151		Module Code			MAM151	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc (Applied Geology) (3011), BSc (Chemical Sciences) (3019), BSc (Computer Science) (3023), BSc (Mathematics and Statistical Sciences) (3031), BSc (Physical Science) (3120)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Have a firm understanding of how to use the basic properties of functions (composition, transformation, combinations, and graphs in different contexts) • Use mathematical induction to prove a statement involving a positive integer. • Understand and use the binomial theorem in various applications. • Use the concepts of limits, continuity and derivative with an emphasis on meanings in different mathematical contexts. • Use rates of change, relates rates and differentiation correctly in solving real-world problems.						
Main Content	• Sets, real numbers, Cartesian coordinates, intro to proximity analysis. • Function, mapping diagram, limit, algebra of limit • Mathematical induction • Main types of functions that occur in calculus; transcendental functions [constant, power-law, polynomial, rational, radical, trigonometric, inverse trigonometric, exponential, hyperbolic trigonometric, logarithmic, combinations and compositions] • Binomial theorem and Pascal's triangle • Limits, continuity and the derivative of a function; existence theorems [mean value theorems for derivative, intermediate value theorem, extreme value theorem, Rolle's theorem] • Rules of differentiation rules and implicit differentiation. • Application of derivatives [curve sketching, polynomial approximations, indeterminate forms, l'Hopital's rule, local and global extrema, critical points, non-differentiability, related rates]. • Additional course-material will be covered, as compared with the corresponding mainstream module.						

Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		4	Lectures p.w.		
		1	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		70%
			FA		30%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Mathematics 152		Module Code			MAM152	
Module Type	Core	NQF Level	5	Year Level	2	NQF Credit	15
Programmes in which the module will be offered		BSc (Applied Geology) (3011), BSc (Chemical Sciences) (3019), BSc (Computer Science) (3023), BSc (Mathematics and Statistical Sciences) (3031), BSc (Physical Science) (3120)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Use different methods to determine the definite integral as a limit of Riemann sums • Use the basic methods of integration correctly to solve problems • Have working knowledge of the definite integral, the fundamental theorem of calculus, the mean value theorem for integrals, and techniques of integration • Use basic methods for solving matrix equations • Use vectors and coordinate systems to solve problems in 3-dimensional space • Use the basic properties of numbers including complex numbers in different contexts • Have solid grasp of first order differential equations, direction fields, initial value problem and the integral equation.						
Main Content	• Anti-derivative and indefinite integral • Riemann sum and definite integral • Fundamental theorem of calculus • Properties of definite integral, mean value theorem for integrals • Differentials, techniques of integration [polynomial, rational, exponential, logarithmic, trigonometric, inverse trigonometric function, substitution, inverse substitution, trigonometric substitution, partial fraction, integration by parts, reduction formulae, complete the square, recursive integral, linear quadratic irreducible, factor theorem] • Improper integrals • Application of integrals [areas under curves, volumes, arc length and average value of a function, differentials] • Vectors, dot product, cross product and equations of lines and planes in 3-dimensional space Row-reduced echelon forms and Gauss-Jordan elimination Solutions of systems of linear equations using determinants, Cramer's rule, and inverses of matrices						

	- Complex numbers [field properties, Argand diagram, roots of unity, de Moivre theorem] - First order ordinary differential equations [variables separable, homogeneous, and linear]; direction fields, initial value problem, integral equation.			
Pre-requisite	MAM151	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		4	Lectures p.w.	
		1	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	70%
			FA	30%

Faculty	Natural Sciences		Home Department				Mathematics and Applied Mathematics	
Module Name	Differential and Integral Calculus and Linear Algebra 105			Module Code			MAT105	
Module Type	Core/ Elective	NQF Level	5	Year Level	1	NQF Credit	30	
Programmes in which the module will be offered		BSc (Mathematics and Statistical Sciences) (3227), BSc (Chemical Sciences) (3220), BSc (Physical Science) (3233), BSc (Computer Science) (3221)				Total Learning Time	300 hours	
Duration	Year		Proposed Semester			Both Semesters		
Main Learning Outcomes	On completion of this module, students should be able to: • Have a firm understanding of how to use the basic properties of functions (composition, transformation, combinations, and graphs in different contexts} • Use mathematical induction to prove a statement involving a positive integer • Use the basic properties of numbers including complex numbers in different contexts • Use the concepts of limits, continuity and derivative with an emphasis on meanings in different mathematical contexts • Use rates of change, relates rates and differentiation correctly in solving real-world problems • Use different methods to determine the definite integral as a limit of Riemann sums • Use the basic methods of integration correctly to solve problems • Have working knowledge of the definite integral, the fundamental theorem of calculus, the mean value theorem for integrals, and techniques of integration • Use basic methods for solving matrix equations • Use vectors and coordinate systems to solve problems in 3-dimensional space • Use the basic properties of numbers including complex numbers in different contexts							

	• Have solid grasp of first order differential equations, direction fields, initial value problem and the integral equation.				
Main Content	Part One (Differential Calculus) • Sets, real numbers, Cartesian coordinates, introduction to proximity analysis • Function, mapping diagram, limit, algebra of limit. • Mathematical induction • Main types of functions that occur in calculus; elementary functions of calculus, transcendental functions [constant, power-law, polynomial, rational, radical, trigonometric, inverse trigonometric, exponential, hyperbolic trigonometric, logarithmic, combinations and compositions] • Binomial theorem and Pascal's triangle • Limits, continuity and the derivative of a function; existence theorems [mean value theorems for derivative, intermediate value theorem, extreme value theorem, Rolle's theorem] • Rules of differentiation and implicit differentiation Application of derivatives [curve sketching, polynomial approximations, indeterminate forms, l'Hopital's rule, local and global extrema, critical points, non-differentiability, related rates). Part Two (Integral Calculus and Linear Algebra): • Anti-derivative and indefinite integral • Riemann sum and definite integral • Fundamental theorem of calculus • Properties of definite integral [mean value theorem for integrals] • Differentials, techniques of integration [polynomial, rational, exponential, logarithmic, trigonometric, inverse trigonometric, substitution, inverse substitution, trigonometric substitution, partial fraction, integration by parts, reduction formulae, complete the square, recursive integral, linear quadratic irreducible, factor theorem) • Improper integral • Application of integrals [areas under curves, volumes, arc length and average value of a function, differentials] • Vectors, dot product, cross product and equations of lines and planes in 3-dimensional space • Row-reduced echelon forms and Gauss-Jordan elimination • Solutions of systems of linear equations using determinants, Cramer's rule, and inverses of matrices • Complex numbers [field properties, Argand diagram, roots of unity, de Moivre's theorem] • First order ordinary differential equations [variables separable, homogeneous, and linear]; direction fields, initial value problem, integral equation.				
	Pre-requisite	None		Prohibited module Combination	None
	Co-requisite	None			
	Timetable Requirement per week			4	Lectures p.w.
				1	Practicals p.w.
				0	Tutorials p.w.
	Other teaching modes that does not require time-table				
	Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	70%
				FA	30%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Mathematics 211		Module Code			MAT211	
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Mathematics and Statistical Sciences) (3227, 3031), BSc (Chemical Sciences) (3220, 3019), BSc (Physical Science) (3233, 3120), BSc (Computer Science) (3221, 3023)				Total Learning Time	200 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: Advanced Calculus • Know how to use the Principle of Mathematical Induction to prove statements. • Understand infinite sequences and series; be familiar with different tests for convergence of sequences and series. • Perform power series expansion of functions and find its interval and radius of convergence. • Use functions of several variables, surfaces and coordinate transformations, contours, level surfaces and traces. • Determine tangent planes and linear approximations, partial differentiation, implicit differentiation, limits, continuity and differentiability of functions of several variables. • Use Lagrange multipliers, maxima and minima, Second derivative test for functions of several variables, chain rule for functions of several variables. • Perform double and triple integrals, iterated integrals, surface integrals, the Jacobian, change of variables for double and triple integrals. • Perform basic operations in Vector Algebra. • Determine vector function, tangent vector, acceleration, TNB-gframe and osculating sphere on a curve. • Evaluate and interpret line integrals. • Understand Independence of Path. • Be familiar with the properties of vector fields, as well as grad, div and curl differential operators. • Use Green's Theorem to evaluate line integrals and double integrals. • Apply Stokes' Theorem to vector fields. • Understand Surface Independence. • Use the Divergence Theorem to evaluate surface integrals, volume integrals and the flux of a vector field across a surface. • Solve integral problems in mechanics.						
Main Content	Part One: • Limits, continuity and differentiability of functions of several variables [in-depth theoretical approach, combined with application, using visualisation as a technique in the Cartesian plane for different coordinate systems, mapping diagram technique] • Quadric surfaces, polar coordinates and parametric equations, spherical and cylindrical coordinates [Klein bottle, Mobius strip, embedding in R4]						

	- Partial derivative; chain rule, maxima and minima of functions of two and three variables, Lagrange Multipliers [Clairaut's theorem, consequences exposure to PDE application problems] - Multiple Integrals, surface integrals, multiple integrals in polar, spherical and cylindrical coordinates [double and triple Riemann sums, iterated integrals, Fubini's theorem, change order of integration in 2 and 3 dimensions, regions of integration, Jacobians, surface area] - Sequences and series [epsilon-delta definition, applications; infinite series, alternating series, power series, Taylor and Maclaurin series, convergence tests, absolute convergence, conditional convergence, examples]. Part Two: - Vector Algebra and vector function of curves and surfaces - Tangent-Normal-Binormal frame and osculating sphere - Vector fields - Line integrals and independence of path and Green's theorem - Grad, div and curl [geometrical interpretation, 2- and 3-dimensional Laplace operator] - Oriented surfaces, connected regions [Klein bottle, Mobius strip, embedding in R⁴] - Surface integrals and flux, Stokes' theorem and divergence theorem [applications of divergence theorem to gravity, electromagnetism, fluid flow and heat].			
Pre-requisite	MAT105 or (MAT103 and MAT104) or (MAM151 and MAM152)	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		2	Lectures p.w.	
		1	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Practicals - Every alternate week				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	70%
			FA	30%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Mathematics 212		Module Code			MAT212	
Module Type	Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Mathematics and Statistical Sciences) (3227, 3031), BSc (Chemical Sciences) (3220, 3019), BSc (Physical Science) (3233, 3120), BSc (Computer Science) (3221, 3023)				Total Learning Time	200 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Formulate differential equations from given physical situations. • Solve linear ordinary differential equations. • Interpret the solutions in the given physical context.						

	• Solve linear systems of differential equations. • Use Newton's method to solve systems of nonlinear equations. • Use different methods for polynomial interpolation. • Use different methods for numerical differentiation and numerical integration with error estimates.				
Main Content	Differential Equations: • First order ordinary differential equations (general and particular solutions, velocity and acceleration, direction fields and solution curves, existence and uniqueness of solutions; linear equations, homogeneous equations, and applications) • Second order linear differential equations • Introduction to systems of differential equations (first order systems and their applications; the eigenvalue method for solving homogeneous linear systems) • Phase-plane analysis (population models, acceleration and velocity models) • Introduction to series solution of ordinary and partial differential equations • Some self-study/familiarity with the method of Laplace transforms for solution of ODEs is recommended. Numerical Methods: • Methods for solving nonlinear equations • Iterative methods solving linear systems of equations • Interpolation and polynomial approximation • Numerical differentiation • Numerical integration • Numerical methods for ordinary differential equations and their systems.				
Pre-requisite	MAT105 or (MAT103 and MAT104) or (MAM151 and MAM152)	Prohibited module Combination		None	
Co-requisite	MAT211				
Timetable Requirement per week		2	Lectures p.w.		
		1	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Mathematics 221		Module Code			MAT221	
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered	BSc (Mathematics and Statistical Sciences) (3227, 3031), BSc (Chemical Sciences) (3220, 3019),					Total Learning Time	200 hours

		BSc (Physical Science) (3233, 3120), BSc (Computer Science) (3221, 3023)			
Duration	Year	Proposed Semester		Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Comprehend basic concepts associated with vectors, linear equations, matrices, linear transformations and eigenvectors and eigenvalues. • Analyse arguments in the above contexts. • Construct counter-examples in the above contexts. • Structure logically valid arguments in the above contexts • Comprehend basic concepts associated with sets, relations, functions, groups, rings, and fields. • Analyse arguments in the above contexts. • Construct counter-examples in the above contexts. • Structure logically valid arguments in the above contexts.				
Main Content	Part One: • Mathematical induction; well- ordering principle; proof by induction; proof by deduction; counter-example; proof by contradiction • Set theory • Vectors in R^n: vector addition and scalar multiplication, dot product, norm and distance in R^n • Systems of linear equations, solutions of systems of linear equations. • Matrix operations [matrix addition and scalar multiplication, matrix multiplication, transpose of a matrix] • Matrices and systems of linear equations [echelon form, elementary row operations, square matrices, algebra of square matrices, invertible matrices) • Vector spaces and subspaces [example, linear combinations, linear spans, row and column space of a matrix] • Basis and dimension [linear dependence and independence, subspaces, rank of a matrix, applications to linear equations] • Linear transformations [kernel and image of a linear transformation, singular and non-singular transformations) • Matrices and linear transformations, change of basis • Properties of determinants and applications to linear equations • Eigenvalues and eigenvectors [diagonalisation, characteristic polynomial, Cayley-Hamilton theorem, minimal polynomial, characteristic and minimum polynomials of linear transformations]. Part Two: • Integers and equivalence relations, well ordering principle, division algorithm theorem • Greatest common divisor, relatively prime integers, modular arithmetic, equivalence relations • Properties of mappings and permutations, properties of binary operations • Groupoids, identities in groupoids, commutative groupoids, associative groupoids, groupoids that satisfy cancellation properties • Groups [examples such as integers, solutions of equations in C: complex roots of unity, general linear group in R, orthogonal group in R, Z_n] elementary properties of groups, subgroups • Fundamental theorem of cyclic groups • The symmetric group, cycle notation, odd and even permutations				

	• Introduction to rings and fields [examples, such as polynomial rings and fields] • Finite fields			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	MAT211			
Timetable Requirement per week		2	Lectures p.w.	
		1	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Practicals - Every alternate week				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	70%
			FA	30%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Discrete Mathematics 232		Module Code			MAT232	
Module Type	Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Mathematics and Statistical Sciences) (3227, 3031), BSc (Physical Science) (3233, 3120)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Solve problems in elementary set theory and logic. • Use combinatorial techniques to solve various enumerative combinatorial problems. • Formulate and solve counting problems in algebra and graph theory.						
Main Content	• Introduction to axiomatic set theory and logic. • Enumerative combinatorics. • Introductory graph theory.						
Pre-requisite	APM112 and (MAT105 or MAT103)		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department		Mathematics and Applied Mathematics		
Module Name	Mathematics 311		Module Code		MAT311		
Module Type	Core/ Elective	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered	BSc (Mathematics and Statistical Sciences) (3227, 3031), BSc (Physical Sciences) (3233, 3120),				Total Learning Time	300 hours	

		BSc (Computer Science) (3221, 3023)		
Duration	Year	Proposed Semester	Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the notions of open subset, boundedness and finiteness, connectedness and compactness in Euclidean space. • Explain the notions of continuity and uniform continuity of functions between subspaces of Euclidean space, and in relation to sequences, compactness and connectedness. • Use properties of complex numbers and complex functions in problem solving. • Use the properties of analytic functions to solve problems. • Use contour integrals in various situations. • Use Taylor and Laurent series to solve problems • Use Residue theory to solve problems.			
Main Content	Part 1 • Real number system, irrational number, infimum, supremum, completeness axiom, Archimedean principle, density of rational numbers • Function and inverse function, images and inverse image, image sets and preimage sets; open sets, closed sets; countable, and uncountable sets • Metric spaces [metric on $\mathbb{R}^n$, discrete metric, sup metric on function spaces] • Boundedness, cluster point, squeeze theorem, limit, limit properties with application in $\mathbb{R}^n$, discrete metric • Nested intervals, compactness, Bolzano-Weierstrass theorem, Heine-Borel theorem • Cauchy sequences, completeness • Continuity, uniform continuity, uniform Cauchy criterion, uniform convergence, fixed point theorems • Connectedness, interior, closure, boundary of a set • Preservation of compactness and connectedness. Part 2: • Complex numbers and properties, the complex plane, sets of points, neighbourhoods, applications • Elementary complex functions, complex mappings, parametric curves, multiple-valued functions, reciprocal function • Limits and continuity [polynomial functions and rational functions] branches, applications • Differentiability, entire functions, analyticity, Cauchy- Riemann equations, harmonic functions, orthogonal families of level curves, gradient fields, applications • Complex exponential and complex logarithmic functions, complex powers, complex trigonometric functions, complex hyperbolic functions, exponential mapping, logarithmic mapping, trigonometric mapping • Contour integral and properties [complex integrals, Cauchy's theorem, Cauchy-Goursat theorem, fundamental theorem for contour integrals, Cauchy's integral formula, Liouville's theorem, fundamental theorem of algebra, applications] • Taylor and Laurent series [complex sequences and series, test for divergence, absolute and conditional convergence, tests for			

	convergence, Taylor's theorem, Maclaurin series, Laurent's Theorem, zeros and poles] Residue theory [residue at a simple pole, residue theorem, evaluation of improper integrals, Cauchy principal value, indented contours, summing infinite series, and applications].				
Pre-requisite	MAT211	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		2	Lectures p.w.		
		1	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Practicals – every alternate week					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment		CA	70%
				FA	30%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Mathematics 312		Module Code			MAT312	
Module Type	Core/ Elective	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Mathematics and Statistical Sciences) (3227, 3031), BSc (Physical Sciences) (3233, 3120), BSc (Computer Science) (3221, 3023)				Total Learning Time	300 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: - Analyse and simulate mathematical models for single and interacting population systems - Perform stability analysis, find equilibrium solutions, and phase portraits - Implement predator-prey, competitive, and cooperative models - Know the fundamentals of epidemiology in various real- world contexts - Formulate and solve optimal control problems using analytical and computational methods.						
Main Content	- Single and interacting population models (equilibrium solutions, stability analysis, direction fields and phase portraits) - Predator-prey models - Competitive and cooperative species models - Epidemic models - Dynamic optimisation in discrete and continuous time (applications to environmental management) - Introduction to optimal control and its applications to epidemiology - Numerical methods to solve differential equations and their systems arising from the topics mentioned above.						
Pre-requisite	MAT211 and (MAT212 or MAT221)		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table								
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%	
Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics		
Module Name	Mathematics 321		Module Code			MAT321		
Module Type	Core/ Elective	NQF Level	7	Year Level	3/4	NQF Credit	30	
Programmes in which the module will be offered		BSc (Mathematics and Statistical Sciences) (3227, 3031), BSc (Physical Science) (3233, 3120), BSc (Computer Science) (3221, 3023)				Total Learning Time	300 hours	
Duration	Year		Proposed Semester			Both Semesters		
Main Learning Outcomes	On completion of this module, students should be able to: • Use their knowledge of the basic theory of groups to provide examples and counter-examples of various concepts. • Carry out proofs of mathematical statements • Use new knowledge in unfamiliar but similar situations • Generalize concepts from Groups to Rings • Use different methods to test for irreducibility of polynomials. • Use the theory to construct finite fields • Use their knowledge of field extensions to prove the impossibility of certain geometric constructions using ruler and compass.							
Main Content	Part One • Mappings and operations • Introduction to group, inverse, identity, permutation, symmetry group, subgroup, dihedral group, groups of finite order, groups of infinite order, Abelian group, non-Abelian groups, Cayley tables • Relation, equivalence relation, congruence, division algorithm, integer modulo n, greatest common divisor, Euclidean algorithm, fundamental theorem of arithmetic, Fermat's Little Theorem • Group properties, generator, order of an element, direct product, cosets, Lagrange's theorem, properties of group order • Homomorphism, isomorphism, finite and infinite cyclic group, isomorphism of cyclic group, subgroup of cyclic group, Cayley's theorem, normal subgroup, kernel, range • Quotient group, fundamental homomorphism theorem, consequences. Part Two • Rings, integral domains, subrings, fields, isomorphisms, characteristic • Ordered integral domains, the integers, field of quotients, field of rational numbers • Ordered fields, field of real numbers, field of complex numbers, complex roots of unity • Polynomial, division algorithm, polynomial factorisation, unique factorisation domain • Ring homomorphism, ideal, quotient ring, quotient ring of $F[X]$. factorisation and ideal							

	• Field extension: adjoining root, finite field; polynomial equation: polynomial root, rational root, conjugate, introduction to Galois theory Impossible geometric constructions.			
Pre-requisite	MAT221	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week	2	Lectures p.w.		
	1	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Practicals – every alternate week				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	70%
			FA	30%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Mathematics 322		Module Code			MAT322	
Module Type	Core/ Elective	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Mathematics and Statistical Sciences) (3227, 3031), BSc (Physical Science) (3233, 3120), BSc (Computer Science) (3221, 3023)				Total Learning Time	300 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Implement the basic tools of the geometric Brownian motion of stock prices. • Use their knowledge of elementary cases of pricing via arbitrage of options to solve problems. • Use their knowledge of vanilla and exotic options and their valuation to solve problems. • Use the Black-Scholes formula to solve option pricing problems. • Use the n-period binomial model method to solve problems. • Use Monte Carlo simulation approach to solve option pricing problems.						
Main Content	• Elementary probability theory • Normal random variables • Geometric Brownian motion • Present value analysis • Introduction to derivatives • Pricing contract via arbitrage, The arbitrage theorem • The Black-Scholes formula • Vanilla options • Multi-period binomial model method • Option valuations by expected utility • Exotic options • Portfolio optimization • Autoregressive models and mean reversion • Simulations-random walk, Monte Carlo methods						
Pre-requisite	MAT211 and (MAT212 OR MAT221)		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week		2		Lectures p.w.			
		1		Practicals p.w.			
		0		Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%
Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Introduction to the Human Body 111		Module Code			MBS111	
Module Type	Core	NQF Level	5	Year Level	1/2	NQF Credit	15
Programmes in which the module will be offered		BSc (Medical Bioscience) (3230, 3035), BSc (Dietetics) (8231), BSc (Sport and Exercise Science) (8051), BSc (Complementary Health Sciences) (8104)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completing of this module, students should be able to: • Explain the organization of the human body, and the concepts of homeostasis, health, human diseases and disease causing agents. • Describe the early development of the fetus. • Identify and discuss the histology of the main tissue types. • Explain the gross anatomy and general functions of the skeletal and muscle systems, and the characteristics of joints types. • Demonstrate competencies in selected laboratory techniques related to the evaluation of tissue types, the muscle and skeletal systems.						
Main Content	Organisation and function from chemical to systemic level: • The scope of human biology and functional organisation of the body. • Homeostasis, human disease, disease causing agents. • Essentials of early human development. • Essentials of the histology of tissue types Systems of support and movement: • Gross anatomy of the skeletal system • Gross anatomy of the muscular system • Articulations and movement Application of knowledge in writing a literature review on a chosen topic and a powerpoint presentation.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3		Lectures p.w.		
			3		Practicals p.w.		
			1		Tutorials p.w.		
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Introduction to the Human Body 121		Module Code			MBS121	
Module Type	Core	NQF Level	5	Year Level	1/2	NQF Credit	15
Programmes in which the module will be offered		BSc (Medical Bioscience) (3230,3035), BSc (Dietetics) (8231), BSc (Sport and Exercise Science) (8051), BSc (Complementary Health Sciences) (8104)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completing of this module, students should be able to: • Demonstrate a good knowledge of the function of neurons, the transduction of sensory stimuli, and how this information is processed and transduced by the CNS. • Describe the functional anatomy of the nervous system and discuss the major sensory and motor pathways. • Explain the basic anatomy and physiology of selected organ systems, for example, blood, immunity and the lymphatic system, structure and function of muscles, digestive system, and the vital role of the cardiovascular system in maintaining homeostasis. • Demonstrate selected basic laboratory competencies. • Illustrate competency in experimentally-based data handling basic statistical analyses. • Demonstrate information processing skills such as searching for and evaluating information						
Main Content	• Overview of the neuron and the major neural pathways of the CNS: • The macro- and microscopic characteristics of blood and the lymphatic system, muscles, digestive system and cardiovascular system. • The control mechanisms that help regulate and integrate the diverse functions and component parts of the systems referred to above.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Microbiology for Nurses 123		Module Code			MBS123	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BNurs (8312)			Total Learning Time		100 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Understand the history and philosophy of medical microbiology and the microorganisms of medical importance.						

	- Understand the host-parasite relationships, thus explaining why some microbes live in symbiosis with the host and others cause infection. - Apply the basic principles of immunity, infection control and its application in clinical medicine			
Main Content	- The science of medical microbiology (brief history of the science, importance of microbes and their relationship with infectious diseases, the germ theory of disease) - Cell structure and taxonomy (structure of Eukaryotic and Prokaryotic cells; Microbial classification and reproduction) - In vitro and in vivo control of microbial growth. - Microbial ecology - Epidemiology - Health care-associated infections - Laboratory diagnosis of infectious diseases - Pathogens of infectious diseases (difference between primary and secondary pathogens; pathogenesis of infectious diseases; mechanisms by which specific pathogens escape the host immune response) - Infectious diseases of different body systems (bacterial, viral, fungal and parasitic infections of the different body systems; notifiable infections; treatment and prevention of infectious disease)			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		2	Lectures p.w.	
		0	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Microbiology for Nurses 130		Module Code			MBS130	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BNurs (8313)				Total Learning Time	100 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: - Understand the history and philosophy of medical microbiology and the microorganisms of medical importance. - Understand the host-parasite relationships, thus explaining why some microbes live in symbiosis with the host and others cause infection. - Apply the basic principles of immunity and infection control and understand its application in clinical medicine.						
Main Content	- The science of medical microbiology (brief history of the science, importance of microbes and their relationship with infectious diseases, the germ theory of disease) - Cell structure and taxonomy (structure of Eukaryotic and Prokaryotic cells; Microbial classification and reproduction) - In vitro and in vivo control of microbial growth.						

	• Microbial ecology • Epidemiology • Health care-associated infections • Laboratory diagnosis of infectious diseases • Pathogens of infectious diseases (difference between primary and secondary pathogens; pathogenesis of infectious diseases; mechanisms by which specific pathogens escape the host immune response) • Infectious diseases of different body systems (bacterial, viral, fungal and parasitic infections of the different body systems; notifiable infections; treatment and prevention of infectious disease)			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		0	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Structure and Function of the Human Body 214		Module Code			MBS214	
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	15
Programme in which the module will be offered		BSc (Occupational Therapy) (8111)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the macro-and microscopic structure of the specified systems. • Discuss the mechanisms whereby the specified organs systems, individually and collectively, contribute to body homeostasis. • Describe how functions of these systems are regulated and coordinated. • Describe the functional organisation of the nervous system and respiratory system. • Relate the consequences of dysfunction in the various body systems- selected examples. • Demonstrate laboratory competencies related to the investigation of the structure and function of the specified systems. • Demonstrate basic competencies in executing investigate experiments, and report it in an acceptable scientific style and format.						
Main Content	• Organisation and function of the central and peripheral nervous systems. • Macro-and microscopic structure and coordinated functions of the respiratory system.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week	3	Lectures p.w.		
	1	Practicals p.w.		
	1	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Structure and Function of the Human Body 217		Module Code			MBS217	
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	20
Programmes in which the module will be offered		BSc (Physiotherapy) (8211), BSc (Occupational Therapy) (8111)			Total Learning Time		200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the macro-microscopic structure of the specified systems. • Explain the mechanisms whereby the specified organs systems, individually and collectively, contribute to body homeostatis. • Describe how the functions of these systems are regulated and coordinated. • Describe the Functional organisation of the nervous system. • Relate the consequences of dysfunction in the various body systems – selected examples. • Demonstrate laboratory competencies related to the investigation of the structure and function of specified systems. • Demonstrate basic competencies in executing investigative experiments, and report it in an acceptable scientific style and format.						
Main Content	• Macro- and microscopic structure and coordinated functions of the respiratory system						
Pre-requisite	HUB117 and HUB127		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Structure and Function of the Human Body 218		Module Code			MBS218	
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	20
Programmes in which the module will be offered	BSc (Physiotherapy) (8211), BSc (Occupational Therapy) (8111)					Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	

Main Learning Outcomes	On completion of this module, students should be able to:					
	• Explain the macro-microscopic structure of the specified systems. • Explain the mechanisms whereby the specified organs systems, individually and collectively, contribute to body homeostasis. • Describe how the functions of these systems are regulated and coordinated. • Relate the consequences of dysfunction in the various body systems – selected examples. • Demonstrate laboratory competencies related to the investigation of the structure and function of specified systems. • Demonstrate basic competencies in executing investigative experiments, and report it in an acceptable scientific style and format.					
	Main Content	• Embryology macro-and microscopic structure and coordinated functions of the cardiovascular and urinary systems, including acid base regulation. • Embryology, macro- and microscopic structure and coordinated functions of the endocrine and reproductive systems.				
		Pre-requisite	HUB117 and HUB127		Prohibited module Combination	None
		Co-requisite	None			
		Timetable Requirement per week				
				3	Lectures p.w.	
1				Practicals p.w.		
1	Tutorials p.w.					
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%
					FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Introduction to the Human Body 231		Module Code			MBS231	
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Medical Bioscience) (3230, 3035), BSc (Sport and Exercise Science) (8051), BSc (Dietetics) (8281)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the functional organisation of the nervous system • Explain the physiology of special senses. • Explain the control of movement. • Explain some of the effects of loss of function of parts of the neuromuscular system • Explain the macro- and microscopic structure of the specified systems • Demonstrate knowledge of how the functions of these systems are regulated and coordinated • Explain the consequences of dysfunction in the various body systems – selected examples • Carry out experimental work • Write an experimental report in an acceptable scientific style						

Main Content	• Organisation and function the of central and peripheral nervous systems • Macro- and microscopic structure and coordinated functions of the endocrine system. • Basic concepts of thermoregulation.				
	Pre-requisite	LSC141 or MBS111 and MBS121	Prohibited module Combination		None
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		1	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		60%
			FA		40%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Introduction to the Human Body 232		Module Code			MBS232	
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Medical Bioscience) (3230, 3035), BSc (Natural Medicine) (8344), BSc (Chinese Medicine) (8106), BSc (Sport and Exercise Science) (8051), BSc (Dietetics) (8281)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Have a detailed knowledge of the macro- and microscopic structure of the specified systems. • Have a detailed knowledge of the mechanisms whereby the specified organs systems, individually and collectively, contribute to body homeostasis. • Explain of how the functions of these systems are regulated and coordinated. • Explain intercellular signalling: receptors and chemical signals (local, neurological and hormonal). • Explain the consequences of dysfunction in the various body systems – selected examples. • Master laboratory competencies related to the investigation of the form and function of the specified systems. • Master basic competencies in executing investigative experiments, and reporting it in an acceptable scientific style and format. • Explain the metabolism of carbohydrates, lipids and proteins.						
Main Content	• Macro- and microscopic structure and coordinated functions of the endocrine, reproductive, urinary and digestive systems. • Cellular metabolic processes.						
Pre-requisite	LSC141 or MBS111; and MBS121			Prohibited module Combination		None	
Co-requisite	None						

Timetable Requirement per week	3	Lectures p.w.		
	2	Practicals p.w.		
	1	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Anatomical and Molecular Systems Biology 331		Module Code			MBS331	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Medical Bioscience) (3230, 3035), BSc (Sport and Exercise Science) (8051)			Total Learning Time		300 hours
Duration	Semester	Proposed Semester			First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Explain and evaluate advanced cellular and molecular control mechanisms and the molecular basis of selected diseases, e.g. cancer, cystic fibrosis, Alzheimer's and malaria. • Explain communication in the nervous system. • Explain macro - and micro-anatomy of the male and female reproductive health including fertility/infertility, family planning and STD's. • Demonstrate competencies in basic laboratory principles (laboratory mathematics, buffer systems, pH, preparation of solutions, pipetting techniques, use of weighing instruments, etc.) and scientific research methodology (literature search, description of data with the aid of statistical methods). • Demonstrate competency in basic information management. • Find, summarise and interpret scientific literature. • Quantify and interpret scientific data.						
Main Content	Control mechanisms: • Normal and abnormal cell growth with special reference to program cell death (apoptosis) cancer • Cell communication and signal transduction pathways and their relation to disease states. • Application of neuro regulators with receptors and mediation of function through these receptors. Applied Reproduction: • Macro- and micro-anatomy of female and male reproductive systems with special reference to humans. • Advanced comparative aspects of gamete production, transport, function and control of gestation and parturition. • Fertility and infertility. • Artificial control of reproduction and reproductive health, including family planning and contraception. • Pathophysiology of the reproductive system						

	Research methodology 1: • Selection of and planning of a research project; Budget, Funding and proposal writing and Protocol writing • Execution of research: Various hands on laboratory techniques; Generation and capturing of data • Data collection and analysis: Methods in collection of different categories of scientific data; Quantification and interpretation of scientific data • Literature research and interpretation: Overview of all methods of literature retrieval and organization. • Application of knowledge in writing a literature review on a chosen topic and a PowerPoint presentation			
Pre-requisite	MBS231 and MBS232	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		5	Lectures p.w.	
		3	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences				Home Department		Medical Biosciences	
Module Name	Impact of Lifestyle on Cardiopulmonary Systems and Metabolism 332				Module Code		MBS332	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30	
Programmes in which the module will be offered		BSc (Medical Bioscience) (3230, 3035), BSc (Sport and Exercise Science) (8051)				Total Learning Time	300 hours	
Duration	Semester			Proposed Semester		Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Explain and evaluate the physiological basis of the lung development and selected cardiovascular and lung diseases. • Explain the principles that underpin selected techniques used in heart and lung research. Students should also be able to interpret the data. • Explain and evaluate the nutrition and metabolism in maintenance of health and the consequences of obesity and anorexia nervosa. • Explain and evaluate the role of exercise in health maintenance. • Explain and evaluate the role of exercise in maintaining energy balance and reducing/preventing obesity. • Describe changes to muscle histology during exercise. • The student should be competent in the planning, execution and presentation of the research project.							
Main Content	Anatomical, Molecular and Systems Biology Lifestyle and health: • The physiological basis of: elected cardiac, lung and vascular diseases. • Role of lifestyle on development and health • Histopathology/pathophysiology of the respiratory system with reference to emphysema, AIDS and tuberculosis • Muscle adaptations to endurance and resistance training.							

	• Muscle structure adaptations to a decreased physical activity. • Controlling food intake and factors that control hunger and satiety with regards to energy expenditure and its effects on obesity. Research Methodology - Project management. • Application of skills gained in the first semester • Research methodology lectures requantification and interpretation of scientific data. Data collection and analysis • Methods in collection of different categories of scientific data • Quantification and interpretation of scientific data Execution of research • Conduct laboratory research • Generation of data Writing of a scientific report on the research project. • Application of knowledge in writing a report on the research project and presentation of a seminar			
Pre-requisite	MBS231, MBS232 and MBS331	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		3	Practicals p.w.	
		2	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Medical Microbiology 251		Module Code			MIC251	
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Medical Bioscience) (3230, 3035), BSc (Complementary Health Sciences) (8104, 8106)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the history and philosophy of medical microbiology and the microorganisms of medical importance. • Grasp host-parasite relationships, thus explaining why some microbes live in symbiosis with the host and others cause infection. • Describe the basic principles of immunity • Understand infection control and its application in clinical medicine						
Main Content	The history and philosophy of medical microbiology; Microbes as parasites (micro- and macroparasites, intracellular and extracellular life, systems of classification); Eukaryotes (disease-causing fungi, protozoa, helminthes, arthropods); Prokaryotes (structure, nutrition, growth and survival, gene expression); Host-parasite interactions (characteristics and evolution of parasitism, the infectious process, parasite and host adaptations, basic immunity); Hospital acquired infections (predisposing factors, types and dynamics); Infection control (use of antimicrobials, sterilisation,						

	disinfection and decontamination procedures, laboratory safety rules, protective clothing, adequate waste disposal, diminishing reservoirs, social and environmental factors, health education, vector control, immunisation).			
Pre-requisite	LSC141	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		2	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Medical Microbiology 252		Module Code			MIC252	
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered	BSc (Medical Bioscience) (3230, 3035), BSc (Complementary Health Sciences) (8104, 8106)					Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Classify and explain normal human microflora and discuss the role they play in disease prevention. - Explain the scope of immunology. - Classify and explain the pathogenic mechanisms of medically important parasites - Classify and explain the features of surface and systemic infections and discuss the human host responds to these infections.						
Main Content	- Immunology: Normal flora of the human body; characteristics and evolution of parasitism; parasite and host adaptations; cells and organs of the immune systems; innate; alternative complement pathway; - Major Histocompatibility complex; humoral immunity; classic complement pathway; cell-mediated immunity; immunoregulatory mechanisms; pathologic consequences of the immune response. - The host community: Classes of host populations; Co-evaluation of host and parasite; herd immunity; cycles of disease. - Emergence factors for infectious disease: human population shifts and behaviour; microbial adaptations and change; breakdown of public health measures; emerging infectious diseases; factors favouring global eradication of an infectious disease. - The infectious process: An overview of bacterial, viral fungal and protozoal pathogenesis; features of surface and systemic infections; mechanisms of spread through the body; Genetic determinants of spread and replication. - Microbial pathogenic mechanisms: The role of bacterial biofilms; parasite survival; interactions of Human cancer viruses with their hosts.						
Pre-requisite	MIC251		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week	3		Lectures p.w.				
	2		Practicals p.w.				
	0		Tutorials p.w.				
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%	
					FA	50%	
Faculty	Natural Sciences		Home Department		Medical Biosciences		
Module Name	Medical Microbiology 353		Module Code		MIC353		
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Medical Bioscience) (3230, 3035)			Total Learning Time	300 hours	
Duration	Semester		Proposed Semester		First Semester		
Main Learning Outcomes	On completion of this module, students should be able to:						
	- Identify the signs and symptoms of system-based bacterial, viral and obligatory intracellular infections. - Explain the aetiology, pathogenesis and control of these microorganisms. - Explain and evaluate the replication cycles of viruses causing infections in humans. - Explain the collection and transport of biological specimens for laboratory diagnosis. - Explain the laboratory diagnostic and research methodology for the detection and characterisation of these prokaryotic, viral and other intracellular pathogens. - Explain and evaluate the molecular biology techniques used in the detection and characterisation of these prokaryotic, viral and obligatory intracellular pathogens.						
Main Content	- The language of clinical microbiology - Transmission, aetiology, pathogenesis, treatment and prevention of bacterial and viral system-based infections - Detection and characterisation of the bacterial and viral pathogens in clinical specimens - Diagnostic and research methodology including molecular biology and molecular epidemiology						
Pre-requisite	MIC251 and MIC252		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week	3		Lectures p.w.				
	2		Practicals p.w.				
	1		Tutorials p.w.				
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%	
					FA	50%	

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Infections caused by Prokaryotic Parasites 354		Module Code			MIC354	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Medical Bioscience) (3230, 3035)			Total Learning Time	300 hours	
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Explain and evaluate the value of basic entomology.						
	• Design the characteristics and life cycle of fungal, helminthic and protozoan parasites.						
	• Argue the aetiology, pathogenesis and control of systems-based parasitic infections.						
Main Content	• Design proper collection and transport systems of biological diagnosis and characterisation of medically important parasites.						
	• The principles of parasitology; the role of arthropods as vectors for human infections						
	• The characteristics of the major groups of medically important fungi, helminthes and protozoa and the diseases they cause						
	• Laboratory detection and characterization of eukaryotic pathogens						
Pre-requisite	MIC251 and MIC252		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Medical Microbiology for Dentistry 355		Module Code			MIC355	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	20
Programmes in which the module will be offered		BSc (Medical Bioscience) (3230, 3035)				Total Learning Time	200 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: - Describe the causative agent, reservoir, mode of transmission, signs and symptoms, pathogenesis, treatment and basic laboratory diagnosis of the major oral infections and infectious diseases of the body systems. - Apply antimicrobial stewardship and infection control in the clinical environment.						
Main Content	The main course content includes: - Basic immunology including the ecosystems of the oral cavity and other organ systems Bacterial, viral, fungal and protozoal causes of: - Infections of the body's surfaces and skeletal system - Infections of the respiratory tract						

	• Oral endogenous infections and their effect on distant body sites (e.g. Cardiovascular, pregnancy, alzheimers etc); • Salivary gland infections and cervicofacial actinomycosis • Infections of the digestive system and food intoxication • Infections of the genito-urinary tract, the cardiovascular and lymphatic systems and the central nervous system • Common childhood infections and fever of unknown origin • Antimicrobial stewardship, sterilization and infection control.				
Pre-requisite	BDP220	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		2.5	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%	
			FA	50%	

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmaceutics 231		Module Code			PCE231	
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students, should be able to: • Explain the principles of Good Manufacturing Practice and the relevant legislation affecting manufacturing processes. • Identify the various dosage forms and the appropriate use of reference material. • Describe the various properties of chemicals to identify and select those which will result in quality pharmaceutical products. • Know how to change particle sizes through the appropriate use of machinery and how this changes physicochemical properties. • Identify, analyse and solve pharmaceutical problems using calculations. • Be capable of the extemporaneous preparation and appropriate labelling of various dosage forms.						
Main Content	• The language of Pharmaceutics: • Introduction • Reference usage • Manufacturing legislation • Physicochemical properties- • Liquid state: • Solutions and solubility • Surface and interfacial phenomena • Solid state: • Polymorphism • Unit processes: • Communion • Compounding						

Pre-requisite	MAM115, PHC123, PHY112	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		1	Practicals p.w.		
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences			Home Department		School of Pharmacy	
Module Name	Pharmaceutics 232			Module Code		PCE232	
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the dynamic and relevance of disperse systems and how various physicochemical properties of these can be manipulated to achieve stable systems as pharmaceutical products. • Explain the rheological properties of liquids and semi solids and how these can be altered for improved stability of products where these are used. • Explain the concepts of particle size analysis, mixing and granulation. • Be able to identify, analyse and solve pharmaceutical problems using calculations. • Be capable of the extemporaneous preparation and appropriate packaging of various dosage forms.						
Main Content	• Disperse systems: • Colloidal • Suspensions • Emulsions • Rheology and Viscosity • Particle size analysis • Unit processes: • Mixing • Granulation and formulation for tableting • Compounding						
Pre-requisite	MAM115, PHC123, PHY112			Prohibited module Combination		None	
Co-requisite	None						
Timetable Requirement per week			3		Lectures p.w.		
			1		Practicals p.w.		
			1		Tutorials p.w.		
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)			Method of Student Assessment		CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmaceutics 331		Module Code			PCE331	
Module Type	Core	NQF Level	7	Year Level	3	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain formulation techniques and the technology required in the production of pharmaceuticals. • Describe Good Manufacturing Practices and their application in the production of quality pharmaceutical products. • Describe the medicines registration process. • Plan and execute the production of laboratory scale samples of various dosage forms.						
Main Content	• Introduction to pharmaceutical technology • Tablets and tableting • Filtration and extractions • Heat transfer, evaporation and drying • Distillation • Medicine registration • Principles of Good Manufacturing Practices						
Pre-requisite	PCE231 and PCE232		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences			Home Department			School of Pharmacy		
Module Name	Pharmaceutics 333			Module Code			PCE333		
Module Type	Core	NQF Level	7	Year Level	3	NQF Credit	15		
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours		
Duration	Semester		Proposed Semester			First Semester			
Main Learning Outcomes	On completion of this module, students should be able to: • Describe the requirements for sterilization and sterility testing. • Explain anti-microbial limitation and testing: preservatives, antiseptic and disinfectants. • Describe pyrogen free formulations. • Discuss the processes involved in antibiotic manufacture. • Implement the appropriate use of surgical dressings. • Evaluate medical waste disposal and the impact on environmental health care. • Plan and execute the preparation of extemporaneous sterile products. • Apply kinetics in the determination of expiry dates.								
Main Content	• Sterilisation and Sterility testing • Anti-microbial limitation or destruction- preservatives antiseptics and disinfectants								

	• Sources of microbial contamination • Selection and use of antimicrobials • Pharmaceutical application of microbiological techniques • Pyrogens • Antibiotic manufacture • Biological assays • Kinetics • Surgical dressings • Waste disposal and environmental health care • Practical work				
Pre-requisite	PHC123, PCE231 and PCE232	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		1	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences		Home Department			School of Pharmacy		
Module Name	Pharmaceutics 401		Module Code			PCE401		
Module Type	Core	NQF Level	8	Year Level	4	NQF Credit	15	
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours	
Duration	Year		Proposed Semester			Both Semesters		
Main Learning Outcomes	On completion of this module, students should be able to: • Describe advances in drug delivery systems and areas of specialisation. • Apply such methods to formulate and manufacture pharmaceuticals. • Critically evaluate the value of such developments for drug delivery.							
Main Content	• Recent developments in pre-formulation, formulation, and manufacture from currently published pharmaceuticals literature.							
Pre-requisite	PCE331 and PCE333		Prohibited module Combination			PPR404, PCH402 and PHC403		
Co-requisite	None							
Timetable Requirement per week			1	Lectures p.w.				
			0	Practicals p.w.				
			1	Tutorials p.w.				
Other teaching modes that does not require time-table								
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%	
						FA	50%	

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacy 431		Module Code			PCE431	
Module Type	Core	NQF Level	8	Year Level	4	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	

Main Learning Outcomes	On completion of this module, students should be able to: • Apply the principles of pre-formulation and formulation to manufacture and package quality, safe and effective drug delivery systems for standard, modified and novel dosage forms and apply this information to solve delivery problems.				
Main Content	• Pharmaceutical Preformulation • Drug delivery systems: • Immediate, modified and novel dosage forms • Packages and packaging • Veterinary pharmaceutical delivery systems				
Pre-requisite	PCE331 and PCE333	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		0	Practicals p.w.		
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%	
			FA	50%	

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmaceutical Chemistry 232		Module Code			PCH232	
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Describe the therapeutic or medicinal value of plants and methods of identifying and evaluating medicinal plants, or crude drugs derived from such plants. • Demonstrate acquired skills in performing the isolation, chemical and physical analysis and quality assurance of phytochemicals. • Explain the influence of the three-dimensional properties and behaviour of molecules and drugs on their biological activity.						
Main Content	• Plants of pharmaceutical importance: Evaluation and quality assurance of herbal medicines, selected chromatographic techniques and phytochemicals. • Stereochemistry: The study of the three-dimensional structure of molecules and drugs. • Practical work: Assay and analysis of phytochemicals. Computer-based stereochemical practicals.						
Pre-requisite	CHE114, CHE124 and PCH233 or a minimum of 40% final mark in PCH233			Prohibited module Combination		None	
Co-requisite	None						
Timetable Requirement per week				3	Lectures p.w.		
				1	Practicals p.w.		
				0	Tutorials p.w.		

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences		Home Department	School of Pharmacy	
Module Name	Pharmaceutical Chemistry 233		Module Code	PCH233	
Module Type	Core	NQF Level	6	Year Level	2
Programme in which the module will be offered		BPharm (3305)		NQF Credit	15
				Total Learning Time	150 hours
Duration	Semester	Proposed Semester		First Semester	

Main Learning Outcomes	On completion of this module, students should be able to: - Describe the theoretical and practical principles involved in the analysis and quality control of drugs. - Apply chemical and physicochemical methods as well as instrumental methods of analysis of pharmaceutical substances. - Interpret spectra obtained from spectroscopic techniques to deduce and/or identify the chemical structure of organic compounds. - Apply the principles of quality assurance in the registration and control of medicines. - Perform validation of experimental methods and their results.				
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Main Content	- Theory and principles of quality control. - Methods of quality control, including: Complexometric titrations and Spectrophotometric techniques (Ultra violet-, Infra-red- and Atomic spectroscopy). - Advanced analytical methods, including: Gas chromatography; High-performance liquid chromatography; Nuclear magnetic resonance spectroscopy and Mass spectroscopy - Practical work: Laboratory techniques; methods of analysis.				
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Pre-requisite	CHE114 and CHE124	Prohibited module Combination	None		
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		1	Practicals p.w.		
		1	Tutorials p.w.		

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences		Home Department	School of Pharmacy	
Module Name	Pharmaceutical Chemistry 312		Module Code	PCH312	
Module Type	Core	NQF Level	7	Year Level	3
Programme in which the module will be offered		BPharm (3305)		NQF Credit	15
				Total Learning Time	150 hours
Duration	Semester	Proposed Semester		First Semester	

Main Learning Outcomes	On completion of this module, students should be able to:				
	• Be able to explain drugs as organic chemicals whose biological activities derive from their chemical structures and physico-chemical properties. • Have an understanding of the chemistry of certain drugs, which includes their discovery, design, synthesis, structure activity relationships and molecular mechanism of action.				
Main Content	• Drug discovery and design of molecules that may exert an effect on the function and survival of living cells. • Organic chemistry of drug receptor-, drug enzyme interactions and metabolic reactions. • Prodrugs and drug delivery systems. • Synthesis, design, molecular mechanism of action, physico-chemical properties and structure-activity relationships of selected drugs, including: • Anesthetics; Sedative-hypnotics; Antiseizure agents; Antipsychotic agents; Opioid Analgesics; Antiparkinsonian agents and Radiopharmaceuticals. Practical work: • Synthesis and characterisation of selected drugs. Case studies and tutorials				
Pre-requisite	PCH232, PCH233 and BTN216		Prohibited module Combination	None	
Co-requisite	None				
Timetable Requirement per week			3	Lectures p.w.	
			1	Practicals p.w.	
			0	Tutorials p.w.	
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment	CA	50%
				FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmaceutical Chemistry 322		Module Code			PCH322	
Module Type	Core	NQF Level	7	Year Level	3	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the chemistry of certain drugs, which includes their synthesis, structure activity relationships, molecular mechanism of action and design. • Describe drugs as organic chemicals whose biological activities derive from their chemical structures and physicochemical properties. • Apply the main medicinal chemistry principles to commonly encountered drugs.						
Main Content	• Synthesis, design, physicochemical properties, structure-activity relationships and molecular mechanism of action of selected drugs,						

	including: Antihistamines; Steroidal drugs and Chemotherapeutic agents. • Case studies and tutorials			
Pre-requisite	PCH232, PCH233 and PCH312 or a minimum of 40% final mark in PCH312	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		1	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmaceutical Chemistry 402		Module Code			PCH402	
Module Type	Elective	NQF Level	8	Year Level	4	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Have a comprehensive understanding of the process of drug discovery and development from the identification of novel drug targets to the introduction of new drugs into clinical practice. • Be able to identify and validate novel and existing drug targets. • Be able to explain advances in the use of computational, combinatorial, microwave and green chemistry. • Be able to apply advanced drug design principles in order to design and synthesise novel therapeutically active compounds/drugs and elucidate the respective structures.						
Main Content	• Advanced Drug Design principles including: • New approaches to drug discovery. • Drug target identification, validation and assay development. • Advanced QSAR and ADME prediction models • Computer-aided drug design. • Drug synthesis and reactions. • Advanced structural elucidation utilizing sophisticated analytical techniques, including: NMR; MS; IR and HPLC. • Combinational-, microwave- and green chemistry.						
Pre-requisite	PCH312 and PCH322		Prohibited module		PCE401, PPR404 and		
Co-requisite	None		Combination		PHC403		
Timetable Requirement per week			1	Lectures p.w.			
			0	Practicals p.w.			
			1	Tutorials p.w.			

Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%
Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacology for Nurses 210		Module Code			PCL210	
Module Type	Core	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BNurs (8312, 8313)				Total Learning Time	200 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Discuss the role of the nurse in the management of drugs • Use pharmacological terms correctly in nursing and midwifery practice • Explain the mechanism of action of drugs, identify adverse effects, contraindications and interactions of drugs in the general use of medication in nursing and midwifery practice • Apply pharmacological knowledge correctly in the practice of nursing and midwifery						
Main Content	• General terms used in pharmacology • Pharmacy Regulation and the role of the nurse in the management of drugs • Qualitative pharmacokinetics • Basic pharmacodynamics • Aspects of pharmacotherapy (including adverse drug reactions, drug-drug & drug-herbal medicine interactions, and toxicity) • Autonomic nervous system • Antimicrobial chemotherapy including antiretrovirals • Drugs for respiratory tract infections • Drugs for cardiovascular diseases • Antidiabetics • Hormones and hormone antagonists: Contraceptives • Drugs for blood diseases • Drugs for gastrointestinal tract disease • NSAIDs and Analgesics • Opioids and Anaesthetic agents • Psychiatric/CNS drugs • Poisoning and emergencies • Drugs used in maternal and reproductive healthcare • Review of the pharmacotherapy of selected disease states •						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Dental Pharmacology 305		Module Code			PCL305	
Module Type	Core	NQF Level	7	Year Level	3	NQF Credit	20
Programme in which the module will be offered		BDS (5101)				Total Learning Time	200 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Apply the basic principles underpinning the action of drugs to rational pharmacotherapy • Discuss the mechanism of action, side effects and potential drug interactions of drugs used in the treatment of dental conditions and common medical conditions. • Discuss the mechanism of action, side effects and potential drug interactions of drugs used in selected common disease states. • Apply rational pharmacotherapy to the treatment of dental conditions and common medical conditions.						
Main Content	• Basic principles of drug action • Local anaesthetics • Drugs acting on the respiratory system • Antimicrobial chemotherapy • Cancer chemotherapy • Drugs and the immune system • Drugs acting on the cardiovascular system • Drugs acting on the Endocrine system • Drugs acting on the Central Nervous System • Analgesic and anti-inflammatory agents • The P-drug concept and basic principles of pharmacotherapy						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			4	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacology for Dietetics 315		Module Code			PCL315	
Module Type	Core	NQF Level	7	Year Level	3	NQF Credit	10
Programme in which the module will be offered		BSc (Dietetics) (8281)				Total Learning Time	100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the basic principles of pharmacology and medicine classification. • Describe the principles associated with medicine administration. • Explain and apply basic pharmacokinetic and pharmacodynamic principles.						

	• Explain basic pharmacotherapeutic principles. • Apply pharmacotherapeutic approaches recommended for gastro-intestinal tract disorders, endocrine system, and nutrition related diseases.			
Main Content	• Concepts and principles of pharmacology • Routes and processes of medicine administration • Medicine absorption, distribution, metabolism and excretion • Receptors and mechanism of action of medicines • Factors influencing variability of medicine responses • Adverse reactions to medicines • Medicines used in the treatment of gastro-intestinal disorders, liver, bladder, endocrine system, nutrition-related diseases • Important food-drug interactions • Pharmacotherapy of diseases where nutrition plays a role			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	DIE316, DIE326			
Timetable Requirement per week		2	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacology and Nursing Practice: Dispensing 410		Module Code			PCL410	
Module Type	Core	NQF Level	8	Year Level	4/5	NQF Credit	10
Programmes in which the module will be offered		BNurs (8312, 8313)				Total Learning Time	100 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Apply pharmacological care principles in the management of a patient's medicine and health needs within a dispensing setting. • Apply pharmacological principles related to the ethics and rational prescribing within a dispensing setting. • Apply the principles of good pharmacy practice in the care, control and dispensing of medicines.						
Main Content	• Good Pharmacy Practice • Dispensing practices • Medicine management, supply and storage • Medicine history and reconciliation, including self-medication • Identifying and reporting adverse drug reactions • Anti-microbial stewardship • Relevant medicine legislation • Medicine recalls • Medicine waste disposal • Creating medicine labels • Latin abbreviations						

	• Role of the dispensing nurse • Pharmacy ethics			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		6	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacy 116		Module Code			PHA116	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programme in which the module will be offered	BPharm (3305)					Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Identify and apply learning styles and various teaching and learning tools and concepts. • Understand and apply concepts related to information literacy • Describe the roles and specific skills sets of pharmacists across different pharmacy practice settings • Understand and apply concepts related to interprofessional learning and interprofessional healthcare practice • Explain and describe the roles played by all members of an interdisciplinary healthcare team. • Explain and apply professionalism within the current legislative and good practice framework for pharmacy in South Africa • Describe the legislative framework for Pharmacy Practice in South Africa • Describe and interpret the structure of the South African healthcare system. • Explain Primary Health Care (PHC) principles and practices and describe the main features of the PHC approach • Describe and understand Pharmacy as a profession • Explain the relevance of basic sciences in pharmacy. • Explain, interpret and apply medical terminology Effectively communicate and evaluate information from various media findings in the form of word documents, spreadsheets, and/or slide presentations.						
Main Content	• Introduction to the Pharmacy profession in South Africa • Primary Health Care • Interdisciplinary Healthcare practice • Interprofessional learning and interprofessional healthcare practice • Roles and responsibilities of the pharmacist in South Africa • Pharmacy-related legislation: Pharmacy Act 53 of 1974; Medicines and Related Substances Act 101 of 1965, and related legislation • Medical and Pharmaceutical terminology						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week		4		Lectures p.w.					
		0		Practicals p.w.					
		1		Tutorials p.w.					
Other teaching modes that does not require time-table									
Assessment Module Type		Continuous Assessment (CA)		Method of Student Assessment		CA		100%	
Faculty		Natural Sciences		Home Department		School of Pharmacy			
Module Name		Pharmacology for Nursing 204		Module Code		PHA204			
Module Type		Core	NQF Level	6	Year Level	2	NQF Credit		20
Programmes in which the module will be offered			BNurs (8311), BSc (Complementary Health Sciences) (8104)				Total Learning Time		200 hours
Duration		Year		Proposed Semester		Both Semesters			
Main Learning Outcomes		On completion of this module, students should be able to:							
		• Explain the basic principles of the action and uses of drugs • Discuss all aspects of pharmacotherapeutics for the treatment of selected disease states. • Apply the knowledge of pharmacology in the practice of nursing							
Main Content		• General terms used in Pharmacology • Basic pharmacodynamics • Qualitative pharmacokinetics • Adverse effects of drugs • Pharmacy Regulation and the role of the nurse in the management of drugs • Drugs affecting the autonomic, somatic and sensory nervous systems • Drugs affecting the cardiovascular system • Analgesics and anti-inflammatory drugs • Antihistamines • Drugs affecting the respiratory system • Hormones and hormone antagonists: including antidiabetics • Drugs affecting the digestive tract • Antimicrobial and other anti-infective drugs: including antiretrovirals and antituberculosis drugs • Drugs affecting the central nervous system: including antiepileptics • Drugs that affect the haemopoetic system							
		Pre-requisite		None		Prohibited module Combination		None	
Co-requisite		None							
Timetable Requirement per week				4		Lectures p.w.			
				0		Practicals p.w.			
				1		Tutorials p.w.			
Other teaching modes that does not require time-table									
Assessment Module Type		Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%	
						FA		50%	

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacy 408		Module Code			PHA408	
Module Type	Core	NQF Level	8	Year Level	4	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Have a broad understanding of research methods and the principles involved in the research. • Be able to plan and conduct a research project. • Be able to write up the results of a research project and perform an oral presentation on the research project. • Have an understanding of the need for and the appropriate procedures involved in all research.						
Main Content	• Research methods • Introduction • Structure of a research proposal • Aspects of basic research • Research project						
Pre-requisite	PCE331, PCE333, PCH312, PCH322, PHC313, PHC323, PPR314, PPR324		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week			1	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment		CA		100%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacy 426		Module Code			PHA426	
Module Type	Core	NQF Level	8	Year Level	4	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Describe what pharmaceutical biotechnology entails and evaluate its impact on disease management. • Describe the manufacturing process of a biopharmaceutical including in-process control. • Discuss the process of generating monoclonal antibodies. • Understand the concepts of antisense therapy, gene therapy and stem cell therapy. • Apply biopharmaceutics principles in the development of drug delivery systems.						
Main Content	• Principles of pharmaceutical biotechnology, including: • Recombinant DNA Technology and Therapeutic Agents						

	• Polymerase Chain Reaction • Gene Therapy • Antisense Therapeutic Agents • Monoclonal Antibodies and Therapeutic Agents • Protein and Peptide Drugs • Principles of biopharmaceuticals, including: • Concepts and definitions of biopharmaceutics. • GIT physiology and drug absorption: Physiological factors influencing oral drug absorption, transit of medicines in the GIT and barriers to absorption. • Bioavailability: physicochemical, physiological, drug and dosage form factors. • Biopharmaceutical property assessment. • Biopharmaceutics Classification System (BCS). • Dosage regimens. • Bioavailability considerations of modified release and targeted delivery.				
Pre-requisite	PCH312 and PCH322, PCE331 and PCE333	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		0	Practicals p.w.		
		2	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacy 427		Module Code			PHA427	
Module Type	Elective	NQF Level	8	Year Level	4	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Monitor and evaluate experiential learning programmes using social accountability criteria. • Analyse organizational structure and culture of pharmacy role-players. • Synthesize the data and societal influences to design a practical intervention. • Implement a quality improvement intervention. • Manage change in a diverse environment. • Reflect critically on change implementation in relation to personal and professional transformation.						
Main Content	Social accountability • Active citizenship and social justice • Principles, values and framework for monitoring and evaluation Change management • Health systems strengthening • Fundamentals of quality improvement • Organizational structure and culture • Advocacy						

	• Negotiation and leadership Communication • Behavior change • Marketing / pitching ideas • Dealing with difference Critical reflection • Critical theory and transformative theory • Understanding and managing ambiguity			
Pre-requisite	PPR314 and PPR324	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week	1	Lectures p.w.		
	0	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacology and Clinical Pharmacy 123		Module Code			PHC123	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programme in which the module will be offered	BPharm (3305)					Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Describe the different types of microorganisms and their classification which are of relevance to pharmacy. - Describe bacterial resistance and basic mechanisms of these. - Describe hospital acquired infections based on predisposing factors, different types and the dynamics of it. - Describe basic immunity for the use vaccines and other methods. - Have an understanding of relevant terms in Pharmacology. - Have an understanding of the various pharmacodynamics concepts and be able to apply them. - Define the basic terms used in pathology. - Understand the basic principles of pathology. - Describe the pathophysiology and clinical manifestations of the following conditions: inflammation, benign and malignant tumours, circulatory disorders, granulomatous diseases.						
Main Content	- Medical Microbiology - The science of microbiology with emphasis on the identification of major groups - Basic morphology and physiology of bacteria, viruses and fungi - Basic bacterial genetics to understand resistance - Host defences against infections. - Immunological basis of vaccination - Basic principles of pathology: Inflammation; Cell injury; Environmental and nutritional pathology; Haemodynamics; Circulatory disorders; Granulomatous diseases; Growth disorders; Neoplasia.						

	Introduction to Pharmacology • Overview • Drug nomenclature • Pharmacodynamics • Definitions • Drug-receptor concepts • Pharmacokinetics • Definitions • ADME • Drug administration and dosage forms • Introduction to drug receptor action within the Autonomic Nervous System				
Pre-requisite	HUB123	Prohibited module Combination		None	
Co-requisite	HUB113				
Timetable Requirement per week		3	Lectures p.w.		
		1	Practicals p.w.		
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacology and Clinical Pharmacy 213		Module Code			PHC213	
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Apply aspects of qualitative pharmacokinetics. • Explain the theory and relevance of drug interactions, adverse drug reactions including drug allergy and toxicity. • Classify and discuss drugs affecting the peripheral nervous system. • Discuss the chemical mediators of inflammation and how drugs affect them. • Explain fundamental concepts in pharmacotherapy and clinical pharmacy. • Discuss the concepts of pharmaceutical care, eliciting a patient history, rational drug use, record-keeping and laboratory data. • Construct a patient history profile and interpret laboratory data. • Apply the theoretical knowledge learnt to computer simulation practicals. • Assess the socio-economic and psychosocial factors that impact on community health. • Demonstrate competence in clinic-based activities in the provision of preventive health care for priority diseases. • Demonstrate competence in physical assessment and clinical skills.						
Main Content	Qualitative Pharmacokinetics Concepts of pharmacotherapy Drugs affecting peripheral nervous system. Drugs affecting the respiratory system						

	Inflammation and Autacoids. Concepts in Clinical Pharmacy • Pharmaceutical care • History-taking • Rational drug use (concept and process) • Record-keeping • Laboratory data Dermatology I • Dermatitis • Urticaria • Drug eruptions • Pruritis • Acne • Fungal skin infections Practicals • Computer Simulation Pharmacology Practical Sessions • Clinical skills (ENT, blood glucose and screening, interpreting laboratory findings, introduction to therapeutic work-up notes) Experiential learning • Community Health profiling: demographics, disease burden in vulnerable communities, history-taking, non-drug management of endemic diseases, psychosocial aspects of care, screening for communicable and non-communicable diseases and health promotion.						
Pre-requisite	PHC123 and HUB113 and HUB123 and PHA116		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%	
					FA	50%	
Faculty	Natural Sciences		Home Department		School of Pharmacy		
Module Name	Pharmacology and Clinical Pharmacy 223		Module Code		PHC223		
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)			Total Learning Time	150 hours	
Duration	Semester		Proposed Semester		Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Explain the pharmacology of drugs used to treat disorders related to the immune system and gastro intestinal tract. • Apply pharmacotherapeutic concepts to various disease states. • Apply the knowledge acquired in the clinical case studies using a step-wise pharmaceutical care approach.						

	• Demonstrate skills in history-taking, clinical reasoning, decision making and communication through participation in case study laboratories and experiential learning. • Apply the knowledge acquired in the clinical case study laboratories. • Demonstrate skills in assessing maternal and child health care needs in underserved population groups.					
Main Content	Immunopharmacology • Analgesics and anti-inflammatories • Drugs used in gout ENT conditions (allergic rhinitis, otitis media /externa) • Drugs affecting the gastro-intestinal system • Oral cavity (ulcers, gingivitis, herpes) • Pregnancy and infant care • Immunisation and childhood illnesses • Family planning • Urinogenital system diseases • HIV and TB • Vitamins • Case Study Laboratories: • PBLs for Primary health care • Experiential learning: • Service learning in Primary Health care clinics: Maternal and child health.					
Pre-requisite	HUB113, HUB123, PHC123, PHA116, or a final mark of more than or equal to 40% for PHC213	Prohibited module Combination		None		
Co-requisite	None					
Timetable Requirement per week		3	Lectures p.w.			
		1	Practicals p.w.			
		1	Tutorials p.w.			
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%	
			FA		50%	
Faculty	Natural Sciences		Home Department		School of Pharmacy	
Module Name	Pharmacology and Clinical Pharmacy 313		Module Code		PHC313	
Module Type	Core	NQF Level	7	Year Level	3	
Programme in which the module will be offered		BPharm (3305)			NQF Credit	15
					Total Learning Time	150 hours
Duration	Semester		Proposed Semester		First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the etiology, pathophysiology, differential diagnosis, pharmacological treatment, non-pharmacological treatment, prevention, guidelines for referral and evidence-based treatment guidelines of various disease states.					

	• Discuss clinical applications of drugs in the treatment of various disease states and the possible adverse effects and drug interactions. • Apply the knowledge acquired in the clinical case laboratories.						
Main Content	Cardiovascular and renal drugs • Dyslipidaemia • Drugs affecting the blood (haematopoietic system, including anticoagulants) • Hypertension, Ischaemic heart diseases and stroke • Cardiac arrhythmias • Congestive cardiac failure Chemotherapeutic agents • Antimicrobial agents • Antifungal agents • Antiviral agents • Antiparasitic agents • Anticancer agents Case study laboratories: PBL Case study laboratories: PBL • Cardiovascular system • Infectious diseases						
	Pre-requisite	PHC213 and PHC223		Prohibited module Combination	None		
	Co-requisite	None					
	Timetable Requirement per week			3	Lectures p.w.		
				1	Practicals p.w.		
		0	Tutorials p.w.				
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment	CA	50%		
				FA	50%		
Faculty	Natural Sciences		Home Department		School of Pharmacy		
Module Name	Pharmacology and Clinical Pharmacy 323		Module Code		PHC323		
Module Type	Core	NQF Level	7	Year Level	3	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester		Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Explain concepts related to quantitative pharmacokinetics and their relevance in therapeutic drug monitoring. • Discuss various disease states, the clinical applications of drugs in the treatment of these diseases and the possible adverse effects and drug interactions with particular emphasis on evidence-based guidelines. • Apply the knowledge acquired in the clinical case laboratories. • Demonstrate pharmaceutical and clinical skills in the health services.						
Main Content	Drugs acting on endocrine system • Hormones of the pituitary and thyroid glands • Pancreatic hormones and anti-diabetic drugs • Adrenalcorticosteroids and adrenocortical antagonists • The gonadal hormones and inhibitors						

	- Agents that affect bone mineral homeostasis CNS pharmacology: - General and Local anaesthetics - Anxiolytic and hypnotic drugs - Anti-depressant drugs - Neuroleptic drugs (anti-psychotic agents) - Drugs used in Parkinson's disease - Anticonvulsants Quantitative pharmacokinetics Therapeutic Drug Monitoring Clinical pharmacy concepts: - Antimicrobial stewardship - Evidence-based medicine - Medicine use evaluations - Case study laboratories					
Pre-requisite	PHC213 and PHC223 and PHC313 or a final mark of more than or equal to 40% for PHC313		Prohibited module Combination		None	
Co-requisite	None					
Timetable Requirement per week			3	Lectures p.w.		
			1	Practicals p.w.		
			0	Tutorials p.w.		
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment	CA	50%	
				FA	50%	
Faculty	Natural Sciences		Home Department	School of Pharmacy		
Module Name	Pharmacology and Clinical Pharmacy 403		Module Code	PHC403		
Module Type	Elective	NQF Level	8	Year Level	4	
Programme in which the module will be offered		BPharm (3305)			NQF Credit	15
					Total Learning Time	150 hours
Duration	Year		Proposed Semester		Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: - Have a broad understanding of principles involved in pre-clinical and clinical trials including pharmaco-epidemiology as post marketing surveillance - Be able to describe the content of a clinical trial protocol. - Have a good understanding of the South African Good Clinical Practice. - Have an understanding of the relevance and importance of statistics to clinical research.					
Main Content	Preclinical trials: - Types of preclinical studies - Pharmacokinetic studies - Pharmacodynamic studies - Animal studies: toxicity and safety					

	Clinical trials • Introduction to Good Clinical Practice: ethical, legal and regulatory issues in clinical research • Phases of clinical trials • Types of clinical trials • Pharmaco-epidemiology • Introduction • Post marketing surveillance • Information and data management in clinical research			
Pre-requisite	PHC313 and PHC323	Prohibited module Combination	PCE401, PCH402 and PPR404	
Co-requisite	None			
Timetable Requirement per week		1	Lectures p.w.	
		0	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	40%
			FA	60%

Faculty	Natural Sciences			Home Department			School of Pharmacy		
Module Name	Pharmacology and Clinical Pharmacy 405			Module Code			PHC405		
Module Type	Core	NQF Level	8	Year Level	4	NQF Credit		30	
Programme in which the module will be offered		BPharm (3305)				Total Learning Time		300 hours	
Duration	Semester		Proposed Semester			Second Semester			
Main Learning Outcomes	On completion of this module, students should be able to: • Undertake patient consultations at health care facilities, establish a therapeutic relationship to identify their medicine-related concerns. • Review patient medication history. • Evaluate evidence-based drug treatment options with identified patient/disease factors. • Balance benefit and risks of drug therapy options. • Document and communicate identified therapeutic discrepancies and justify recommended therapeutic options with training supervisors. • Identify the social determinants of health that impact on patient treatment adherence.								
Main Content	• Bedside teaching and learning: consult with patient, obtain medication history, record and interpret clinical and laboratory data, compile evidence base to assess drug treatment options. • Chronic care (clinics/ clubs): Address patient concerns on chronic medicine use, identify social issues impacting on treatment adherence, promote health and wellness								
Pre-requisite	PCE331, PCE333, PCH312, PCH322, PHC313, PHC323, PPR314, PPR324					Prohibited module Combination		None	
Co-requisite	None								
Timetable Requirement per week			0	Lectures p.w.					
			0	Practicals p.w.					
			1	Tutorials p.w.					

Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment		CA		100%
Faculty	Natural Sciences		Home Department		School of Pharmacy		
Module Name	Pharmacology and Clinical Pharmacy 423		Module Code		PHC423		
Module Type	Core	NQF Level	8	Year Level	4	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester		First Semester		
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Assess and compare adverse events at population level as well as individual patient level.						
	• Analyze and validate pharmacovigilance systems.						
	• Analyze and evaluate pharmacovigilance methods.						
	• Highlight and motivate the importance of spontaneous reporting of adverse drug reactions (ADRs) and the key role players.						
	• Describe the use of the various epidemiological measures of association and effect in pharmacoepidemiology.						
	• Apply the various measures of association and effect to the design and evaluation of pharmacoepidemiology studies.						
	• Evaluate and assess literature on pharmacoepidemiology studies.						
	• Design evidence based interventions to solve and identified pharmacoepidemiology problems.						
	• Evaluate the quality and validity of evidence based medicine.						
Main Content	• Formulate a clinical question.						
	• Gather evidence to answer a clinical question and apply it to the care of a given patient.						
	• Historical and legal background to pharmacovigilance.						
	• Pharmacological bases for adverse events.						
	• Basic concepts and terminology used in pharmacovigilance.						
	• Adverse drug reporting (ADR) and its effect on adherence.						
	• Types of medication errors and sources of adverse drug reporting in the health system.						
	• Spontaneous and cohort event monitoring mechanisms.						
	• Classification system of medication errors.						
	• ADR –relationship/causality assessment, harmful effects and interventions.						
Pre-requisite	PHC313, PHC323, PPR314, PPR324		Prohibited module Combination		None		
	None						
	Timetable Requirement per week		3		Lectures p.w.		
			0		Practicals p.w.		
			1		Tutorials p.w.		

Other teaching modes that does not require time-table								
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%	
Faculty	Natural Sciences		Home Department			School of Pharmacy		
Module Name	Global Pharmacy 424		Module Code			PHC424		
Module Type	Elective	NQF Level	8	Year Level	4	NQF Credit	15	
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours	
Duration	Semester		Proposed Semester			First Semester		
Main Learning Outcomes	On completion of this module, students should be able to:							
	• Demonstrate cultural competence via communication skills and development of a culture-specific pharmacy knowledge base.							
	• Analyze the management of HIV/AIDS in a resource-limited environment and compare the effectiveness of therapeutic regimens to those used in the United States.							
	• Develop an appropriate treatment algorithm for tuberculosis and apply it during patient care activities.							
	• Design a patient work-up system for collection of patient-specific data needed for developing daily therapeutic plans.							
	• Formulate a strategy for effective verbal and non-verbal communication with patients, preceptors, and healthcare personnel.							
	• Establish a working relationship with hospital and clinic personnel as well as personnel from University of the Western Cape.							
	• Show a sense of confidence that allows for active participation with the medical team in a variety of settings.							
	• Show critical thinking and problem-solving skills to optimize patient therapy using a rational evidence-based approach to decision making.							
	• Review the differences in medication selection and availability differences between South Africa and the United States.							
Main Content	• Show competency in the transition of knowledge learned throughout the curriculum to a patient-oriented practice setting.							
	• Describe the South African healthcare system and compare it to the U.S. system;							
	• Review common disease states (e.g., hypertension, heart failure, diabetes) and their treatment encountered across all patient populations.							
	• Provide drug information in-services to hospital personnel, pharmacy staff and present patient cases reviewing both American and South African treatment guidelines.							
	Using bedside teaching as a method, the main content is as follows:							
	• Burden of disease in South Africa, with a focus on HIV and TB.							
	• Health care systems in South Africa, with a focus on the public health care system.							
	• Legal framework for pharmacy services in South Africa.							
	• Standard Treatment Guidelines and the Essential Medicines List for South Africa.							
	• Pharmaceutical care and the principles thereof for direct-patient-care.							
• Patient education on drug and disease management.								

Pre-requisite	Determined by home university	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		0	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
No Timetable Requirements.					
Onsite Learning Only.					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacology and Clinical Pharmacy 425		Module Code			PHC425	
Module Type	Elective	NQF Level	8	Year Level	4	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Discuss the principles and importance of rational medicine use within health systems. • Apply quantitative methods to investigate medicine use. • Describe and apply the principles of evidence based medicine as part of the decision making process and formulary management. • Discuss the role and function of Pharmaceutical and Therapeutics Committees (PTCs).						
Main Content	Course Content: The course content will be based on the following: • Importance of rational medicines use and problems associated with irrational medicine use; • Identification of medicines use problems using quantitative methods such as, prescribing indicator studies, application of defined daily dose (DDD), ABC, and analyses; • Develop, implement and evaluate the results of a Medicines Use Evaluation (MUE); • Application of the Essential Medicines Concept and the development of Standard Treatment Guidelines (STGs) and Essential Medicines Lists (EMLs) using evidence-based decision making principles; • The role of the pharmacist in the Pharmaceutical Therapeutic Committees (PTCs); • Development of a plan for a defined medicine use problem to quantify, investigate and intervene, with evaluation.						
Pre-requisite	PHC323, PPR324		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table							
No Timetable Requirements. Onsite Learning Only.							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%
Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacology and Clinical Pharmacy 426		Module Code			PHC426	
Module Type	Elective	NQF Level	8	Year Level	4	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Compare the physiology of paediatric and adult patients. • Evaluate the impact of paediatric physiology on the pharmacokinetics and pharmacodynamics of medications. • Educate paediatric patients and caregivers regarding safe medicine use. • Evaluate safe medication use in paediatric patient care and the challenges with paediatric formulations. • Promote the importance of childhood immunisations and vaccine preventable diseases. • Demonstrate competence in the assessment of pharmacotherapy of common medical diseases/problems in paediatric patients. • Evaluate the pharmacotherapeutic management of common childhood diseases/conditions. • Apply pharmaceutical care principles to paediatric patients. • Apply evidence-based medicine principles to paediatric patients. • Review Standard Treatment Guidelines/evidence-based guidelines and pharmacotherapeutic options related to paediatric disorders. • Perform paediatric calculations commonly encountered in practice. • Critically evaluate the role of the pharmacist in paediatric patient care.						
Main Content	The course content will be based on the following: • Safe paediatric medicine use. • Common paediatric conditions/diseases. • Paediatric care plan development. • Evidence-based medicine in paediatrics. • Paediatric specific calculations. • The role of the pharmacist in paediatric medicine and patient care. • Content description of common paediatric conditions: • Disease overview • Epidemiology • Etiology • Pathophysiology • Clinical Presentation • Diagnosis • Laboratory and diagnostic investigations • Prognosis • Pharmacologic and non-pharmacologic treatment • Pharmacoeconomic considerations						

	• Evaluation of therapeutic outcomes (effectiveness and safety)			
Pre-requisite	PHC313, PHC323	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		2	Lectures p.w.	
		0	Practicals p.w.	
		2	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 106		Module Code			PHY106	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc (Physiotherapy) (8211)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • A knowledge and understanding of basic ideas relating to elementary mechanics. • Understand the role of elementary mechanics in everyday life. • A knowledge and understanding of basic ideas relating to electricity and magnetism. • Understand natural phenomena and various technologies by applying the basic concepts of electricity and magnetism. • Work in a laboratory environment and record, represent and interpret the data and deliver a public presentation.						
Main Content	• Linear and non-linear motion including acceleration, free fall and projectile motion. • Momentum, impulse and collisions. • Work, power, potential and kinetic energy, conservation of energy and applications • Rotational inertia, torque, center of mass and angular momentum. • Electrostatics and current electricity including lightning, electric shielding, the Van de Graaff Generator, electrical circuits and safety. • Magnetism including permanent magnets, electromagnets and applications. • Electromagnetic Induction, Faraday's law, generators and transformers.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3		Lectures p.w.		
			3		Practicals p.w.		
			1		Tutorials p.w.		
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		60%
					FA		40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 111		Module Code			PHY111	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc (Physical Science) (3233), BSc (Mathematics and Statistical Sciences) (3277), BSc (Chemical Sciences) (3220), BSc (Computer Science) (3221)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completing of this module, students should be able to: • Have knowledge and understanding of introductory mechanics of kinematics, dynamics and applications. • Be able to apply mechanics theory in everyday life situations. • Be able to work in a laboratory environment and record, represent and interpret data.						
Main Content	• Introduction to vectors • Kinematics: Motion in one dimension and in a plane • Dynamics: Newton's laws of motion and their applications • Application of Newton's laws: Translational equilibrium, circular motion, gravitational forces and fields, satellite motion. • Work and energy • Impulse and momentum • Rotational Equilibrium						
Pre-requisite	None		Prohibited module Combination			PHY113, PHY118, PHY116	
Co-requisite	MAT103 or MAT105						
Timetable Requirement per week			3	Lectures p.w.			
			3	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	60%
						FA	40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 112		Module Code			PHY112	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completing of this module, students should be able to: • Have an ability to sensibly discuss in conceptual terms fundamentals of introductory mechanics, elasticity, hydrostatics, X-rays and electricity. • Have an ability to solve both qualitative and quantitative problems in relation to everyday life with special reference to the context of pharmacy. • Have an ability to work in a physics laboratory environment that draws upon fundamentals in recording, representing and interpreting data.						
Main Content	• An introductory appreciation of science as a way of knowing. • An introduction to vectors: representation, adding, subtracting, scalar product, vector product						

	- Equilibrium of a rigid body: Newton's Laws, first condition of equilibrium, friction, second condition of equilibrium (moment of a force / torque); Energy (Work-Energy theorem; machines and efficiency) - Elasticity: stress, strain and Young's modulus. - Hydrostatics: pressure in a fluid, buoyancy, flotation, Pascal's principle, Archimedes' principle, Bernoulli's principle. - Atomic physics: Atoms, molecules, compounds and mixtures; atomic structure; models of the atom and atomic spectra; quantisation of energy levels and Louis de Broglie matter waves. - Radioactivity: Radiation and effects of radiation on humans; radioactive isotopes and carbon dating. - Electricity: Coulomb's law, electric field, electrical potential energy, capacitors and dielectrics, simple circuits. - Relevant applications to pharmacy will be discussed.			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		3	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 113		Module Code			PHY113	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BDS (5101)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Have an ability to sensibly discuss in conceptual terms fundamentals of introductory mechanics, elasticity, hydrostatics, X-rays and electricity. • Have an ability to solve both qualitative and quantitative problems in relation to everyday life with special reference to the context of dentistry. • Have an ability to work in a physics laboratory environment that draws upon fundamentals in recording, representing and interpreting data.						
Main Content	• An introductory appreciation of science as a way of knowing • An introduction to vectors: representation, adding, subtracting, scalar product, vector product. • Equilibrium of a rigid body: Newton's Laws, first condition of equilibrium, friction, second condition of equilibrium (moment of a force / torque). • Elasticity: stress, strain and Young's modulus. • Hydrostatics: pressure in a fluid, buoyancy, flotation, Pascal's principle, Archimedes' principle, Bernoulli's principle. • X-rays: description, creation, and safety. • Electricity: Coulomb's law, electric field, electrical potential energy, capacitors and dielectrics, simple circuits. • Relevant application to dentistry and pharmacy will be discussed						
Pre-requisite	None		Prohibited module Combination			PHY111, PHY116	
Co-requisite	None						

Timetable Requirement per week	3	Lectures p.w.		
	3	Practicals p.w.		
	1	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences			Home Department			Physics and Astronomy		
Module Name	Physics 115			Module Code			PHY115		
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15		
Programme in which the module will be offered		BNurs (8313)				Total Learning Time	150 hours		
Duration	Year		Proposed Semester			Both Semesters			
Main Learning Outcomes	On completion of this module, students should be able to: • Explain an understanding of the nature of science • Describe an understanding of the social, ethical and environmental dimensions of science • Solve problems using modern physics and heat principles • Solve problems by applying introductory kinematics and dynamics principles. • Solve problems by applying mechanics theory to everyday life situations • Work in a laboratory environment and record, present and interpret data • Access science texts and communicate in a variety of forms: laboratory report, essay, poster presentation								
Main Content	• The Language of Physics: Units, Measurements and basic mathematical calculations • Understand the scientific method • Electrostatics-Electric field and electrostatic force between point charges • Electricity-Ohm's law applicable to basic electric circuits • Waves (light and sound) • The Atom nature of matter • Kinematics molecular theory-Explanation of the state of matter • Heat and temperature • Kinematics: Motion in one dimension and in a plane • Dynamics: Newton's law of motion and their applications, Work Energy and Power • Basic operational principles behind selected nursing instruments work (e.g. Thermometer, blood pressure instruments, etc.)								
Pre-requisite	None		Prohibited module Combination			PHY113, PHY118, PHY111			
Co-requisite	None								
Timetable Requirement per week		3							Lectures p.w.
		0	Practicals p.w.						
		2	Tutorials p.w.						
Other teaching modes that does not require time-table									
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		60%		
					FA		40%		

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 116		Module Code			PHY116	
Module Type	Core/ Elective	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc (Applied Geology) (3214), BSc (Biotechnology) (3211), BSc (Computer Science) (3211), BSc (Environment and Water Science) (3331), BSc (Mathematics and Statistical Sciences) (3227)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Understand and apply the scientific approach to problem solving. • Understand various technologies by applying the basic ideas of modern physics and thermodynamics. • A knowledge and understanding of basic ideas relating to elementary mechanics. • Understanding the role of elementary mechanics in everyday life. • Be able to work in a laboratory environment and record, represent and interpret data.						
Main Content	• Ideas and terminology of science as well as the impact of science on our world using a conceptual exploration platform, focusing on the atomic nature of matter. • The nature of the atom, Atomic spectra, the Bohr model of the atom and basic concepts of quantum mechanics. • Types of radiation, radioactive decay, decay schemes and c-14 dating. • Nuclear Fission, nuclear reactors and fusion. • Concepts of temperature and heat, specific heat and thermal expansion. • Heat transfer by conduction, convection and radiation and applications. • Linear and non-linear motion including acceleration, free fall and projectile motion. • Newton's three laws of motion. • Momentum, impulse and collisions. • Work, power, potential and kinetic energy, conservation of energy and applications • Rotational inertia, torque, center of mass and angular momentum.						
Pre-requisite	None		Prohibited module Combination			PHY113, PHY118	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			3	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	60%
						FA	40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 118		Module Code			PHY118	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc (Physiotherapy) (8211)				Total Learning Time	150 hours

Duration	Semester	Proposed Semester	First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Understand introductory basic physics principles as for non-physicists. • Establish a foundation from which to view and understand nature and everyday situations. • Show a connected framework of knowledge and thought that can lead to a better understanding of political and social issues that depend on scientific data.			
	Main Content • The Nature of Matter and Radioactivity • The Language of Physics; Measurements; Forces • Work, Energy, Power and Machines • Pressure; States of Matter; Gases • Heat; Electrostatics and Electricity • Electromagnetic Radiation • Light and Sound			
Pre-requisite	None	Prohibited module Combination	PHY111, PHY116	
Co-requisite	None			
Timetable Requirement per week	3		Lectures p.w.	
	0		Practicals p.w.	
	1		Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 121		Module Code			PHY121	
Module Type	Core/ Elective	NQF Level	5	Year Level	1/2	NQF Credit	15
Programmes in which the module will be offered		BSc (Physical Science) (3233, 3120), BSc (Applied Geology) (3214, 3011), BSc (Chemical Sciences) (3220, 3019), BSc (Computer Science) (3221, 3023), BSc (Mathematics and Statistical Sciences) (3227, 3031)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate knowledge and understanding of introductory vibrations and waves theory and applications in sound • Show knowledge and understanding of geometrical optics and applications of it in everyday life • Be able to work in a laboratory environment and record, represent and interpret data.						
Main Content	• Simple harmonic motion: Energetics of SHM, simple pendulum, damped oscillations, forced oscillations and resonance • Mechanical waves: Basic wave properties, sinusoidal travelling waves, speed of a wave, energy in wave motion, reflection of waves, interference, standing waves and resonance • Sound: sound waves, intensity and intensity levels, beats, Doppler effect						

	• Geometrical optics: ray optics, reflection and refraction, image formation by mirrors and lenses • Electrostatics: point electrical charges, electric field, electric flux, Gauss' law • Electric potential, Electric currents and Resistance, DC circuits • Capacitors and dielectrics. • Magnetic field and forces, motion of charged particle in a B-field • Electromagnetic induction, inductance, AC circuits				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	MAT105 or MAT103				
Timetable Requirement per week		3	Lectures p.w.		
		3	Practicals p.w.		
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment		CA	60%
				FA	40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 126		Module Code			PHY126	
Module Type	Core/ Elective	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc (Applied Geology) (3214), BSc (Biotechnology) (3211), BSc (Computer Science) (3211), BSc (Environment and Water Science) (3331), BSc (Mathematics and Statistical Sciences) (3227)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Have knowledge and understanding of basic ideas relating to wave phenomena, electricity and magnetism. - Be able to understanding natural phenomena and various technologies by applying the basic concepts of vibrations, waves, electricity and magnetism. - Be able to work in a laboratory environment and record, represent and interpret data and deliver a public presentation.						
Main Content	- Description of wave motion, standing waves, Doppler effect, radar gun and shock waves - The origin and nature of sound, reflection and refraction of sound, forced vibrations, natural frequency, resonance, interference, pitch and loudness. Ultrasound applications and compact discs - Electromagnetic waves transparent and opaque media, selective absorption and transmission, natural phenomena (e.g. why the sky is blue) - Reflection, refraction, diffraction, interference and polarization - Electrostatics and current electricity including lightning, electric shielding, the Van de Graaf Generator, electrical circuits and safety. - Magnetism including permanent magnets, electromagnets and applications.						

	• Electromagnetic Induction, Faraday's law, generators and transformers.			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week				
	3	Lectures p.w.		
	3	Practicals p.w.		
	1	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 151		Module Code			PHY151	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc (Applied Geology) (3011), BSc (Chemical Sciences) (3019), BSc (Computer Science) (3023), BSc (Mathematics and Statistical Sciences) (3031), BSc (Physical Science) (3120)			Total Learning Time		150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate an understanding of the nature of science. • Demonstrate an understanding of the social, ethical and environmental dimensions of science. • Apply modern physics and heat principles to solve problems. • Apply introductory kinematics and dynamics principles to solve problems. • Apply mechanics theory to everyday life situations. • Work in a laboratory environment and record, represent and interpret data. • Access science texts and communicate in a variety of forms: laboratory report, essay, poster presentation.						
Main Content	• The nature of the atom, Atomic spectra, the Bohr model of the atom and basic concepts of quantum mechanics. • Types of radiation, radioactive decay, decay schemes and C-14 dating. • Nuclear fission, nuclear reactors and fusion. • Concepts of temperature and heat, specific heat and thermal expansion. • Heat transfer by conduction, convection and radiation and applications • Introduction to vectors and mathematics skills required for mechanics • Kinematics: Motion in one dimension and in a plane • Dynamics: Newton's laws of motion and their applications • Application of Newton's laws: Translational equilibrium, circular motion, gravitational forces and fields, satellite motion. • Work and energy • Impulse and momentum • Rotational Equilibrium						
Pre-requisite	None		Prohibited module Combination			PHY113, PHY118, PHY 111	
Co-requisite	MAM151						

Timetable Requirement per week		3		Lectures p.w.			
		3		Practicals p.w.			
		1		Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		60%
					FA		40%
Faculty	Natural Sciences		Home Department		Physics and Astronomy		
Module Name	Physics 152		Module Code		PHY152		
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc (Applied Geology) (3011), BSc (Chemical Sciences) (3019), BSc (Computer Science) (3023), BSc (Mathematics and Statistical Sciences) (3031), BSc (Physical Science) (3120)				Total Learning Time	150 hours
Duration	Year		Proposed Semester		Both Semesters		
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Apply the physics principles of vibrations and waves to solve problems. • Apply the physics principles of vibrations and waves to everyday life situations e.g. Light and sound. • Apply the physics principles of introductory electricity and magnetism to solve problems. • Apply the physics principles of introductory electricity and magnetism to everyday life situations. • Work in a laboratory environment and record, represent and interpret data. • Access science texts and communicate in a variety of forms: laboratory report, essay, poster presentation. • Apply basic statistics to measurement and uncertainty in data.						
Main Content	• Mathematics skills required for waves, electricity and magnetism • Simple harmonic motion: Energetics of SHM, simple pendulum, damped oscillations, forced oscillations and resonance • Mechanical waves: Basic wave properties, sinusoidal travelling waves, speed of a wave, energy in wave motion, reflection of waves, interference, standing waves and resonance • Sound: sound waves, intensity and intensity levels, beats, Doppler effect • Electrostatics: point electrical charges, electric field, electric flux, Gauss' law • Electric potential, Electric currents and Resistance, DC circuits • Capacitors and dielectrics. • Magnetic field and forces, motion of charged particle in a B-field • Electromagnetic induction, inductance, AC circuits						
	Pre-requisite	None		Prohibited module Combination		PHY121	
Co-requisite	None						
Timetable Requirement per week		3		Lectures p.w.			
		3		Practicals p.w.			
		1		Tutorials p.w.			

Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	60%
						FA	40%
Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 212		Module Code			PHY212	
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Physical Science) (3233, 3120), BSc (Chemical Sciences) (3220, 3019), BSc (Computer Science) (3221, 3023), BSc (Mathematics and Statistical Sciences) (3227, 3031)				Total Learning Time	200 hours
Duration	Semester	Proposed Semester			First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Use crucial concepts to solve problems in modern physics, interference, diffraction, and thermodynamics. • Demonstrate practical laboratory skills. • Implement simple numerical computational algorithms to solve physics problems on the computer. • Demonstrate writing and presentation skills. • Explain the progression and nature of science in relation to modern physics.						
Main Content	• Temperature and ideal gases; heat flow and First Law of Thermodynamics; Second Law of Thermodynamics, heat engine, entropy. • Wave physics, including interference and diffraction. • Basic modern physics, including relativity, wave-particle duality, photo-electric effect, blackbody radiation, line spectra, foundations of modern quantum mechanics. • Basic computational physics. • Laboratory experiments.						
Pre-requisite	PHY111 and PHY121 or PHY151 and PHY152 MAT105 or MAM151 or MAT103		Prohibited module Combination			None	
Co-requisite	MAT211 or MAT104 or MAM152						
Timetable Requirement per week			3	Lectures p.w.			
			4	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	60%
						FA	40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 217		Module Code			PHY217	
Module Type	Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Physical Science) (3233, 3120), BSc (Chemical Sciences) (3220, 3019), BSc (Computer Science) (3221, 3023), BSc (Mathematics and Statistical Sciences) (3227, 3031), BSc (Environment and Water Science) (3331)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain basic ideas in Astronomy to a wide audience.						
Main Content	• The scientific method, technology and the development of astronomical ideas; the night sky; space travel and the solar system; stellar evolution; galaxies; evidence for the Big Bang and cosmology; Current research questions and SA contributions to Astronomy; Communicating basic astronomical ideas: tourism, the media and science education.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			4	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 218		Module Code			PHY218	
Module Type	Core	NQF Level	5	Year Level	2	NQF Credit	15
Programme in which the module will be offered		BSc (Occupational Therapy) (8111)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Understand introductory basic physics principles as for non-physicists. • Establish a foundation from which to view and understand nature and everyday situations. • Show a connected framework of knowledge and thought that can lead to a better understanding of political and social issues that depend on scientific data.						
Main Content	• The Nature of Matter and Radioactivity • The Language of Physics; Measurements; Forces • Work, Energy, Power and Machines • Pressure; States of Matter; Gases • Heat; Electrostatics and Electricity • Electromagnetic Radiation • Light and Sound						

Pre-requisite	None	Prohibited module Combination	PHY111, PHY116	
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		0	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 222		Module Code			PHY222	
Module Type	Core/ Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Physical Science) (3233, 3120), BSc (Chemical Sciences) (3220, 3019), BSc (Computer Science) (3221, 3023), BSc (Mathematics and Statistical Sciences) (3227, 3031)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Solve problems in Classical Mechanics and Electrodynamics. • Utilize mathematical and computer skills in problem solving. • Interpret and represent the basic subject matter, with emphasis on the unity of electric and magnetic phenomena. • Do experimental work related to topics in Classical Mechanics and Electrodynamics. • Write and interpret practical reports.						
Main Content	• Newton's equation of motion, Conservation of energy and conservative forces. Velocity dependent non-conservative damping forces. Conservative systems in 3-D, Central force motion. Dynamics of many-body systems, Dynamics of rigid bodies. Non-inertial reference frames. • Electrostatics: The electrostatic Field, Divergence and Curl of electrostatic fields, Electric Potential, work and energy in electrostatics, Conductors. • Magnostatics: The Lorenz Force, Biot-Savart Law, Divergence and Curl of magnostatic Fields, Vector Potential. • Electrodynamics: The emf, Faraday's Law, Maxwell's Equations. • Electromagnetic waves: The Wave equation.						
Pre-requisite	(PHY111 and PHY121) or (PHY151 and PHY152) and (MAT105 or MAT103 or MAM152)			Prohibited module Combination		None	
Co-requisite	None						
Timetable Requirement per week				3	Lectures p.w.		
				3	Practicals p.w.		
				1	Tutorials p.w.		

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	60% 40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy		
Module Name	Physics 227		Module Code			PHY227		
Module Type	Elective	NQF Level	6	Year Level	2/3	NQF Credit	20	
Programmes in which the module will be offered		BSc (Physical Science) (3233, 3120), BSc (Chemical Sciences) (3220, 3019), BSc (Computer Science) (3221, 3023), BSc (Mathematics and Statistical Sciences) (3227, 3031), BSc (Environment and Water Science) (3331, 3027)				Total Learning Time	200 hours	
Duration	Semester		Proposed Semester			Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Describe the environmental issues involved in energy generation • Be knowledgeable on environmental physics and applications. • Present on issues related to the environment. • Perform laboratory work on environmental physics • Explain how energy is generated • Explain and represent environmental issues around energy supply • Do experimental work related to energy issues.							
Main Content	• Introduction to Environmental Science, Environmental Physics, Environment and technology • Traffic and transport, Environment and Industry, Atmospheric Physics, Water in the Environment. • The Energy Future, Electricity Generation, Fossil (coal, liquid, gas), Nuclear, Hydrogen economy, variety hydro-energy. • Experimental laboratory and computational applications							
Pre-requisite	(PHY116 and PHY126) or PHY111 or PHY151		Prohibited module Combination			None - except from timetable clash groups		
Co-requisite	None							
Timetable Requirement per week			3	Lectures p.w.				
			0	Practicals p.w.				
			2	Tutorials p.w.				

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	60% 40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 312		Module Code			PHY312	
Module Type	Core	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Physical Science) (3233, 3120), BSc (Computer Science) (3221, 3023)				Total Learning Time	300 hours

Duration	Semester	Proposed Semester	First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Solve the Schrödinger equation for simple potentials and the Hydrogen atom. • Use the Mathematical structure of quantum mechanics to describe quantum systems. • Calculate properties of nuclei using nuclear models. • Be able to interpret scientific data. • Write scientific reports. • Perform simple nuclear experiments.			
Main Content	Quantum Physics: • Mathematical structure of quantum mechanics: Operators, eigen-states. Schrödinger equation. Free particle, wave packet. One-dimensional problems: infinite square well, harmonic oscillator. Angular Momentum, orbital and spin quantum numbers. Hydrogen atom, quantum numbers and electron clouds. Building of the Periodic Table of elements Nuclear Physics: • Bulk nuclear properties, nucleon structure, nuclear structure models, nuclear decay and excitation, nuclear reactions, applied nuclear physics, social impact of nuclear physics, particle physics, unification theories, and structure of Universe. Laboratory component: • This component provides students with the practical skills to execute, record and analyse data in especially nuclear and quantum physics experiments. Scientific report writing and presentation skills are developed further. Non-linear least squares, distributions in physics and other elements of data analysis are studied.			
Pre-requisite	PHY212 and PHY222 and MAT211 and MAT221	Prohibited module Combination	None - except from timetable clash groups	
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		4	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences			Home Department			Physics and Astronomy	
Module Name	Physics 314			Module Code			PHY314	
Module Type	Core/ Elective	NQF Level	7	Year Level	3/4	NQF Credit	30	
Programmes in which the module will be offered		BSc (Physical Science) (3233, 3120), BSc (Computer Science) (3221, 3023)				Total Learning Time	300 hours	
Duration	Semester		Proposed Semester			First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Solve the Schrödinger equation for simple potentials and the Hydrogen atom.							

	• Use the Mathematical structure of quantum mechanics to describe quantum systems. • Apply Quantum physics for the determination of properties of Physics systems. • Be able to interpret scientific data. • Write scientific reports. • Perform simple nuclear experiments in a controlled laboratory environment.			
Main Content	• Mathematical structure of quantum mechanics: • Operators and basic perturbation theory, eigen-states. • Schrödinger equation. • Free particle, wave packet and Heisenberg's uncertainty principle. • One-dimensional problems: infinite square well, harmonic oscillator. • Angular Momentum, orbital and spin quantum numbers. • Hydrogen atom, quantum numbers and electron clouds. • Building of the Periodic Table of elements • Laboratory component: • This component provides students with the practical skills to execute, record and analyse data in quantum physics experiments. Scientific report writing and presentation skills are developed further. Non-linear least squares, distributions in physics and other elements of data analysis are studied.			
Pre-requisite	PHY212 and PHY222 and MAT211 and MAT221	Prohibited module Combination		None - except from timetable clash groups
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		1	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 317		Module Code			PHY317	
Module Type	Elective	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Physical Science) (3233,3120), BSc (Computer Science) (3221, 3023)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the concept of energy sources. • Discuss and explain renewable energy sources. • Do standard evaluations for applications of renewable energy sources. • Know about energy issues in SA. • Discuss international issues on energy supply and demand. • Explain the practical applications of renewable energy sources. • Do standard PV systems sizing						
Main Content	• Renewable fuels and electricity generation, Design of solar cells, Standard silicon solar cell technology incl. Crystalline, Polycrystalline,						

	Thin film amorphous PV systems and design, Wind energy, Wave and Tidal energy. • Overview of energy in SA, energy technology in SA. Fossil, nuclear and renewable energy sources. International energy supply and demand. Future scenarios. • Practical: applications of renewable energies in the laboratories and in practice, Site visits of industrial applications			
Pre-requisite	PHY227	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		4	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 322		Module Code			PHY322	
Module Type	Core/ Elective	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Physical Science) (3233, 3120), BSc (Computer Science) (3221, 3023)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Integrate and find coherence from the experimental results using the Solid state theory • Explain thermal and electronic properties of crystals • Demonstrate how the theory should be used to solve problems • Solve the problems numerically using computer programs						
Main Content	• Crystal structures, X-ray diffraction, Defects in crystals: vacancies, impurities, dislocations and grain boundaries. Boltzmann, Maxwell and Planck distributions. Phonons in crystals, Debye theory of the lattice specific heat. Electrons in metals: conduction electrons, electrical conductivity, motion in a magnetic field. Free electron model: Fermi-Dirac statistics, Fermi energy. Electrons in semiconductors: energy bands, intrinsic and doped semiconductors, mobility, Hall effect, p-n junctions. • Practical component provides students with the skills to acquire and analyse data in solid state physics experiments, to apply the computational methods and to hone the report writing and presentation skills.						
Pre-requisite	PHY212 and PHY222 and MAT211 and MAT221			Prohibited module Combination		None - except from timetable clash groups	
Co-requisite	None						
Timetable Requirement per week				3	Lectures p.w.		
				4	Practicals p.w.		
				1	Tutorials p.w.		

Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	60%
						FA	40%
Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 324		Module Code			PHY324	
Module Type	Core/ Elective	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Physical Science) (3233, 3120), BSc (Computer Science) (3221, 3023)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Apply the kinetic theory of gases to describe the properties of matter on the basis of thermodynamics • Apply the methods of statistical mechanics to explain the probable behaviour of a physical system • Devise the relations between statistical and thermodynamic quantities from theoretical considerations • Propose and use appropriate theoretical concepts in solving general thermodynamics problems. • Analyze the theory of statistical mechanics in order to describe Solid State Physics phenomena from an ensemble standpoint. • Devise and integrate Thermal and Solid-State Physics practical experiments with computational methods in the practical laboratory.						
Main Content	• Temperature, reversible processes and work, The First Law of Thermodynamics, The Second Law of Thermodynamics, Entropy and the Third Law of Thermodynamics • Probability and statistics, microscopic view of Entropy, Maxwell–Boltzmann statistics, ideal gases, Fermi-Dirac (FD) statistics • Classification and structure of solids, classical theory of the heat capacity of lattices, the quantized linear oscillator, Einstein's theory on the heat capacity of a solid, Debye's theory on the heat capacity of a solid, electronic heat capacity, • The free electron gas in a metal, electrical conduction in solids, density of states of intrinsic and doped semiconductors, energy bands of semiconductors • Practical component provides students with the skills to acquire and analyse data in thermal and solid-state physics experiments, to apply the computational methods and to hone scientific report writing and presentation skills.						
Pre-requisite	PHY212 and PHY222 and MAT211 and MAT221		Prohibited module Combination			None - except from timetable clash groups	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			1	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	60% 40%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 327		Module Code			PHY327	
Module Type	Elective	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered	BSc (Physical Science) (3233, 3120), BSc (Computer Science) (3221, 3023)				Total Learning Time	300 hours	
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the physics and mathematics of important ideas in astronomy. • Operate a telescope. • Analyse data from telescopes. • Use computational techniques to model data and simulate astronomical objects. • Explain advanced astronomical ideas to a wide audience.						
Main Content	Analysing astronomical spectra and photometry; Concepts in general relativity applied to cosmology and black holes; Astrophysical fluids; Nucleosynthesis; Technology and science of SA's astronomical facilities; Communicating advanced astronomical ideas: tourism, the media and science education						
Pre-requisite	(MAT105 or MAT103 or MAM152) and (PHY217 or PHY212)			Prohibited module Combination		None - except from timetable clash groups	
Co-requisite	None						
Timetable Requirement per week				3	Lectures p.w.		
				4	Practicals p.w.		
				1	Tutorials p.w.		

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacy Practice 214		Module Code			PPR214	
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	10
Programme in which the module will be offered		BPharmacy (3305)				Total Learning Time	100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Describe and apply the philosophies of pharmaceutical care, Universal Health Coverage and Public Health. • Demonstrate the ability to communicate professionally. • Describe and analyse the legal, professional and policy framework within which the pharmacy profession operates.						

	- Describe and analyse the link between the South African Constitution, human rights and health. - Describe the functions of SAHPRA. - Describe the purpose and scope of the Pharmacy and Medicines Act. - Differentiate between the scopes of practice of personnel authorised to work in pharmacy settings. - Describe and apply the principles of the SAPC code of conduct. - Describe the standards available for various sectors of pharmacists and the importance of adhering to these for the pharmacy profession. - Apply selected Pharmacy Act regulations. - Critically evaluate topical health and pharmacy-related topics.			
Main Content	- Philosophies of pharmacy practice - Relevant policy and legislation - Standards and procedures in practice			
Pre-requisite	PHA116 and PHC123	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		1	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacy Practice 224		Module Code			PPR224	
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Describe, apply and analyse the legislation governing the sale and pricing of medicines. - Describe and apply inventory control and medicine supply management principles and practice. - Demonstrate the ability to develop Standard operating procedures. - Describe and apply the phases of dispensing. - Demonstrate the ability to use computer dispensing packages.						
Main Content	The Medicines and Related Substances Act 101 Of 1965 - Control of the sale of medicines (Section 22A) - Validity of prescriptions - Label, PILs, package insert - Medicine pricing and rationale for pricing - Introduction to medicine classification (Schedules) - Generic substitution - Medicine supply management principles - Inventory management and stock control. - Medicine storage - Procurement of medicine - Selection of medicine - Standard Operating Procedures						

	• Dispensing practice according to GPP • Dispensing calculations				
Pre-requisite	PCE231, PHC213 and PPR214 or a minimum of 40% final mark in each of these level 2 modules	Prohibited module Combination		None	
Co-requisite	PHC223				
Timetable Requirement per week		4	Lectures p.w.		
		1	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment		CA	50%
				FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacy Practice 314		Module Code			PPR314	
Module Type	Core	NQF Level	7	Year Level	3	NQF Credit	15
Programme in which the module will be offered	BPharm (3305)					Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Describe, interpret and apply the fee structure related to services for which pharmacists may levy professional fees. - Describe, interpret and apply the legislation related to the advertising and marketing of medicines and pharmaceutical services. - Critically evaluate medicine advertisements. - Describe and analyze various communication styles and identify how to adopt these in specific patient or inter- professional interactions. - Demonstrate the ability to interact with individual patients and recognize specific counselling needs relative to medicine adherence. - Describe and analyze factors affecting health behaviours. - Analyze and advise on the general principles relating to responsible self-medication. - Differentiate between ethics, morality and ethical theories and principles and apply these concepts to professional situations. - Demonstrate the ability to reflect, plan, implement and evaluate experiential learning in SLIP and other practice sites and record these in an ePortfolio. - Demonstrate an understanding of the professional obligation of pharmacists in communities. - Apply professional expertise and skill to initiate pharmaceutical and therapeutic care in line with community needs.						
Main Content	<u>Pharmacy Act No 53 of 1974</u> - Services for which pharmacists may levy a fee - Medicines and Related <u>Substances Act 101 of 1965</u> - Advertising and marketing of medicines and services <u>Patient-centred pharmacy practice</u> - Communication - Patient adherence						

	• Patient counselling • Pharmaceutical and therapeutic care based on community needs • Psychosocial aspects of pharmacy practice • Pharmacy ethics • Healthcare Ethics: concepts and theories <u>Service Learning in Pharmacy (SLIP)</u>			
Pre-requisite	PPR214 and PPR224	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week	4	Lectures p.w.		
	0	Practicals p.w.		
	1	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacy Practice 324		Module Code			PPR324	
Module Type	Core	NQF Level	7	Year Level	3	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain interpret and apply human resources management theories and legislation in a pharmacy practice context. • Describe marketing strategies and - o Explain the significance of marketing in modern pharmacy practice settings. - o Apply transactional and relationship marketing. - o Apply relationship marketing as a personal and professional marketing opportunity. • Plan a pharmacy marketing research project; • Analyse and interpret the SAPC disciplinary procedure for registered persons who deviate from the code of conduct. • Interpret the SAPC requirements for licensing of premises. • Describe and explain the main components of a dossier for registration of medicines. • Describe and apply the legislative requirements for medicine registration. • Explain and apply the concept of section 21 medicines and how to obtain access to such medicines. • Demonstrate an understanding of the professional role of pharmacists in community health care. • Describe, initiate, apply and evaluate strategies to promote the profession through the provision of values-based services. • Demonstrate the ability to apply therapeutic and pharmaceutical care at community level. • Demonstrate the ability to reflect, plan, implement and evaluate experiential learning in SLIP and other practice sites and record these in an ePortfolio. • Apply basic financial management principles in a business enterprise.						

Main Content	• Human Resources Management • Labour Relations Act 66 of 1965 • Marketing and business practice • Pharmacy Act No 53 of 1974 - o SAPC role: conduct and disciplinary procedures - o Licensing of premises • Medicines and Related substances Act 101 of 1965 - o Registration of medicines - o Dossiers • Section 21 items • Community-centred pharmacist-initiated therapy • Financial accounting • Service Learning in Pharmacy				
	Pre-requisite	PPR214 and PPR224, PPR314 or a minimum of 40% final mark in PPR314	Prohibited module Combination		None
	Co-requisite	None			
	Timetable Requirement per week		4	Lectures p.w.	
			0	Practicals p.w.	
1			Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacy Practice 404		Module Code			PPR404	
Module Type	Elective	NQF Level	8	Year Level	4	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Understand the role of pharmacoeconomics in pharmaceutical therapy, the pharmaceutical industry and health outcomes research. • Define and understand the terminology associated with pharmacoeconomics and health outcomes research. • Identify and apply the main economic principles used in pharmacoeconomics. • Cost-benefit analysis. • Cost-minimisation analysis. • Cost-utility analysis. • Cost-effectiveness analysis. • Apply the different pharmacoeconomic principles to practical case scenarios. • Identify and evaluate how the various perspectives influence pharmacoeconomic analysis and the decision making process. • Evaluate and analyse Cost-of-illness studies and understand their financial implications. • Understand the role and application of modelling in pharmacoeconomic research.						

	- Understand the role and application of pharmacoeconomics in clinical trials and early drug development. - Evaluate pharmacoeconomic literature and health outcomes research critically.			
Main Content	- Pharmacoeconomic theories and principles, including - Pharmacoeconomic analyses - Pharmacoeconomic modelling - Relevant pharmacoeconomic and health outcomes literature - Applications of the above in clinical trials and drug development			
Pre-requisite	PPR314 and PPR324	Prohibited module Combination	PCE401, PCH402 and PHC403	
Co-requisite	None			
Timetable Requirement per week	1	Lectures p.w.		
	0	Practicals p.w.		
	1	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacy Practice 414		Module Code			PPR414	
Module Type	Core	NQF Level	8	Year Level	4	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Interpret miscellaneous pharmacy related legislation • Identify and interpret any amendments to legislation applicable to pharmacy practice; • Interpret aspects of the Medical Schemes Act and Managed Healthcare and its role in Pharmacy in South Africa; • Discuss the benefits and challenges of the intergration of complementary and alternative medicines into traditional pharmacy practice; • Critically evaluate queries through the use of various information sources; • Identify drugs of abuse and discuss patterns of drug abuse, as well as the role of the pharmacist; • Evaluate the role of the pharmacist in issues relating to reproductive health; • Apply financial management principles and ratios in a business enterprise; • Demonstrate the ability to reflect, plan, implement and evaluate experiential and formal learning in the context of SAPC competency standards and record these in an ePortfolio; • Plan for future practice based on foreseeable trends.						
Main Content	• Miscellaneous pharmacy related legislation - o Medical Schemes Act (MSA): managed health care (MHC) - o Amended legislation • Selected Acts and regulations impacting pharmacy practice in South Africa.						

	• Disposal/ destruction of medicines • Integration of Complementary and alternative medicines into pharmacy practice • Medicines Information • Substance abuse • Reproductive health - o Infertility - o Cervical cytology - o Menopause - o Common gynaecological complaints - o Contraception (other than methods in the EML) • Financial management • Continuing professional development and a professional portfolio of practice • Scenario-based future planning			
Pre-requisite	PPR314 and PPR324	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		4	Lectures p.w.	
		0	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacy Practice 434		Module Code			PPR434	
Module Type	Elective	NQF Level	8	Year Level	4	NQF Credit	15
Programme in which the module will be offered		BPharm (3305)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Identify and analyse key policy, legislative and regulatory frameworks applicable to medicines; • Explain the importance of product selection, planning and procurement in managing medicines; • Evaluate key principles applicable to the storage and distribution of medicines; • Identify and analyse Inventory Management Systems for medicines; • Evaluate the use of Logistics Management Information Systems in managing medicines supply; • Advocate for stakeholders to adopt Rational Medicines Use principles in medicines supply management and • Critically evaluate financial management mechanisms applicable to managing medicines.						
Main Content	• Policy, Laws and Regulations applicable to medicines; • Analysis of product selection, forecasting/quantification and supply planning for medicines; • Key principles of procurement and quality assurance; • Storage and distribution of medicines; • Use of Inventory Management Systems for managing medicines;						

	- Logistics Management Information Systems for managing medicines and - Rational Medicines Use and key financing mechanisms for managing medicines.			
Pre-requisite	PPR314 and PPR324	Prohibited module Combination	PCE401, PCH402, PHC403 and PHC425	
Co-requisite	None			
Timetable Requirement per week				
		1	Lectures p.w.	
		0	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Physiology 319		Module Code			PSE319	
Module Type	Core	NQF Level	7	Year Level	3	NQF Credit	5
Programme in which the module will be offered		BSc (Dietetics) (8281)				Total Learning Time	50 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Discuss the basics of oncology, and how diet influences the development of cancers. • Describe the basic structure and function of the GIT. Discuss the physiology of gastrointestinal disorders and explain the nutritional implications and adjustments. • Discuss the importance of the placenta in nutrition. • Discuss the implications of renal disturbances on nutritional requirements. • Explain the importance of laboratory testing. • Give normal values for selected parameters.						
Main Content	• Topic based discussions to supplement therapeutic nutrition. • Topics include • Basics of oncology, renal • Diseases, physiology of the placenta, • Gastrointestinal disorders, small bowel syndrome, etc.						
Pre-requisite	MBS231 and MBS232 (MBS21, MBS212, MBS221, MBS222)		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Physiology 329		Module Code			PSE329	
Module Type	Core	NQF Level	7	Year Level	3	NQF Credit	5
Programme in which the module will be offered		BSc (Dietetics) (8281)				Total Learning Time	50 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Discuss the physiology of						
	• CV diseases such as hypertension an atherosclerosis						
	• Explain how diet influence the development of CV diseases						
	• Discuss the physiology of the new born						
Main Content	• Explain the nutritional need during pregnancy, in the new born and in the premature infant						
	• Discuss the physiology of AIDS						
	Topic based discussions to supplement therapeutic nutrition.						
	Topics include, cardiovascular diseases, AIDS, premature development, etc						
Pre-requisite	MBS231 and MBS232 (MBS211, MBS212, MBS221, MBS222)		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Quantitative Skills for Commerce 131/132		Module Code			QSC131/132	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BCom (1008), BCom Law (7211), BCom (Financial Accounting) (1175)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			S1 - QSC131 S2 - QSC132	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate the ability to solve problems with Commercial applications involving: • Ratio, rates and simple proportion • Discounts and markdowns, breakeven analysis • Time value of money • Straight-line and parabolic graphs with particular reference to revenue/cost/profit supply/demand • Exponential and logarithmic curves • Drawing and interpreting statistical charts and calculating mean, median, mode						

Main Content	• Ratio and proportions, rates, approximations • Pricing a product, profits, breakeven analysis • Simple interest, compound interest, annuities • Straight line graphs and parabolas with link to finance and economics e.g. supply/demand • Exponential and logarithmic equations and graphs with link to finance and economics • Descriptive statistics and measures of central tendency.				
	Pre-requisite	None	Prohibited module Combination	None	
	Co-requisite	QLC141/142			
	Timetable Requirement per week		3	Lectures p.w.	
			0	Practicals p.w.	
1			Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Quantitative Skills for Commerce 141 (Maths Group)		Module Code			QSC141	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BCom (Extended Curriculum) (1761)			Total Learning Time		150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate an understanding of the core basic mathematical skills. • Understand number systems and basic operations with proficiency of ordinary and decimal fractions and percentages. • simplify algebraic expressions and solve basic algebraic equations, including simultaneous equations. • Understand direct and inverse proportionality. • Draw linear and quadratic graphs using the table and algebraic methods. • Demonstrate an understanding of descriptive statistics (measures of central tendency and dispersion) • Understand and solve problems related to time value of money: interest, savings accounts, annuities, loans, amortizations and depreciation. • Be proficient in solving basic mathematical problems relating to Commerce and Economic applications, involving: • Ratios, rates, simple proportion, discounts, mark-ups, mark-downs and break-even analysis. • Straight-line and Parabolic graphs with particular reference to revenue/ cost/ profit/ supply/ demand. • Demonstrate proficiency with the application of exponential and logarithmic curves.						

Main Content	Main Content - Ratio and proportions, rates, approximations. - Pricing a product, profits, breakeven analysis - Simple interest, compound interest, lump sum, annuities, amortization - Straight line graphs and parabolas with links to finance and economics e.g. supply/demand. - Exponential and logarithmic equations and graphs with link to finance and economics - Descriptive statistics and measures of central tendency. Foundational Support: Practiced during extra tutorial - Problem Solving Skills. - Number Systems: Integers, Rational Numbers etc. - Basic Mathematical Operations and Calculator Skills. - Fractions, Decimal fractions and Percentages. - Powers, Indices and roots. - Measurement, Shape and space - Basic Algebra: Variables, expressions, equations, Linear and Quadratic equations. - Product and Factors. - Graphs: Ordered Pairs. Linear and Quadratic. Additional Supporting initiative - Last lecture of the week to be used to recap week's content and to administer unannounced spot tests that will be marked by peers. 2nd tutorial of the week will be facilitated by one of 3 student groups whose rotating leader will take turns to prepare and explain a 5 minute presentation on given topic, or concept. - Healthy competition will be encouraged. Assessment changes - MCQ to be applied to 20% of all formative assessments, except gateways.							
	Pre-requisite		None		Prohibited module Combination		None	
	Co-requisite		None					
	Timetable Requirement per week			4	Lectures p.w.			
				0	Practicals p.w.			
				1	Tutorials p.w.			
	Other teaching modes that does not require time-table							
	Assessment Module Type		Continuous Assessment (CA)		Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Quantitative Skills for Commerce 142 (Maths Literacy Group)		Module Code			QSC142	
Module Type	Core	NQF Level	5	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BCom (Extended Curriculum) (1761)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	

Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate an understanding of the core basic mathematical skills. • Understand number systems and basic operations with proficiency of ordinary and decimal fractions and percentages. • simplify algebraic expressions and solve basic algebraic equations, including simultaneous equations. • Understand direct and inverse proportionality. • Draw linear and quadratic graphs using the table and algebraic method. • Demonstrate an understanding of descriptive statistics (measures of central tendency and dispersion) • Understand and solve problems in relation to time value of money: interest, savings accounts, annuities, loans, depreciation and amortization. • Be proficient in solving other basic mathematical problems arising from Commerce and Economic applications, involving: ratios, rates, simple proportion, discounts and markdowns, break-even analysis. • Straight-line and Parabolic graphs with particular reference to revenue/ cost/ profit, supply/demand. • Demonstrate proficiency with the application of exponential and logarithmic curves.				
Main Content	Main Content • Ratio and proportions, rates, approximations. • Pricing a product, profits, breakeven analysis • Simple interest, compound interest, annuities • Straight line graphs and parabolas with link to finance and economics e.g. supply/demand. • Exponential and logarithmic equations and graphs with link to finance and economics • Descriptive statistics and measures of central tendency. Foundation Content • Problem Solving and mathematical thinking skills • Number Systems: Integers, Rational Numbers. • Basic Mathematical Operations and Calculator Skills. • Fractions, Decimal fractions and Percentages. • Powers, Indices, roots and logarithms. • Measurement, Shape and space • Basic Algebra: Variables, expressions, equations, Linear and Quadratic equations. • Product and Factors. • Graphs: Ordered Pairs. Linear and Quadratic • Measure of central tendency and dispersion				
Pre-requisite	None		Prohibited module Combination		None
Co-requisite	None				
Timetable Requirement per week			4	Lectures p.w.	
			0	Practicals p.w.	
			2	Tutorials p.w.	
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment	CA	100%

Faculty	Natural Sciences			Home Department			Physics and Astronomy		
Module Name	Radiation Physics 220			Module Code			RAP220		
Module Type	Core	NQF Level	6	Year Level	2	NQF Credit		5	
Programme in which the module will be offered		BDS (5101)				Total Learning Time		50 hours	
Duration	Semester		Proposed Semester			Second Semester			
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the interaction of radiation with matter. • Describe the instrumentation used to produce x-rays. • Discuss the factors affecting the quality of x-rays. • Discuss the factors affecting the quality of x-ray images. • Explain the biological effects and measurement of radiation.								
Main Content	• Structure of matter: the atom, atomic energy levels, electromagnetic radiation, production of x-rays • The x-ray tube: the anode, cathode, transformers, voltage rectification, basic x-ray circuit • Physics of x-ray production: Bremsstrahlung, characteristic x-rays, x-ray energy spectrum, operating characteristics • Interaction of radiation with matter: ionisation, photoelectric effect, Compton scattering, pair production • Production of x-ray images: image formation and contrast • Factors affecting the quality of x-ray images: radiographic contrast, scattered radiation and contrast, radiographic receptors • Measurement of absorbed dose: absorbed dose, dose measurements • Radiation protection: patient exposure and protection, personnel exposure and protection								
Pre-requisite	None		Prohibited module Combination			None			
Co-requisite	None								
Timetable Requirement per week			3	Lectures p.w.					
			0	Practicals p.w.					
			1	Tutorials p.w.					
Other teaching modes that does not require time-table									
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA		40%	
						FA		60%	

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 111		Module Code			STA111	
Module Type	Core/ Elective	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217), BSc (Computer Science) (3221), BSc (Environment and Water Science) (3331), BSc (Mathematics and Statistical Sciences) (3227)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	

Main Learning Outcomes	On completion of this module, students should be able to: • Recognize the importance of statistics in both private and public sectors. • Summarize data into summary measures (e.g. mean and standard deviation). • Do simple statistical analysis. • Use a computer to analyse data.			
Main Content	• Descriptive statistics • Simple linear regression analysis • Probability • Sampling distributions • Inferential Statistics • Chi-Square • Anova			
Pre-requisite	None	Prohibited module Combination		STA125, STA141, STA142, STA151, BUS131, BUS132
Co-requisite	None			
Timetable Requirement per week		4	Lectures p.w.	
		5	Practicals p.w.	
		3	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 121		Module Code			STA121	
Module Type	Core/ Elective	NQF Level	5	Year Level	1/2	NQF Credit	15
Programmes in which the module will be offered		BSc (Mathematics and Statistical Sciences) (3227, 3031)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Obtain basic knowledge on questionnaire design and planning of surveys. • Analyse survey data. • Obtain informed understanding of Official Statistics. • Obtain fundamental knowledge on different data collection types.						
Main Content	Quantitative methods: • Questionnaire design and analysis • Planning of surveys • Introduction of Official Statistics • Censuses • Vital registration (civil registration) • Surveys (general household surveys, family surveys, demographic and health surveys)						
Pre-requisite	None					None	

Co-requisite	Knowledge of STA111/ STA125/ STA151 or BUS131/ BUS132 or STA141/ STA142 or equivalent	Prohibited module Combination		
Timetable Requirement per week		0	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 125		Module Code			STA125	
Module Type	Core/ Elective	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc (Biodiversity and Conservation Biology) (3217), BSc (Biotechnology) (3211), BSc (Chemical Sciences) (3200), BSc (Computer Science) (3221), BSc (Environment and Water Science) (3331), BSc (Mathematics and Statistical Sciences) (3220)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Recognize the importance of statistics in both private and public sectors. • Summarise data into summary measures (e.g. mean and standard deviation). • Do simple statistical analysis. • Use a computer to analyse data						
Main Content	• Descriptive statistics • Simple linear regression analysis • Probability • Sampling distributions • Inferential Statistics • Chi-square • Anova						
Pre-requisite	None		Prohibited module Combination			STA111, STA141, STA142, STA151, BUS131, BUS132	
Co-requisite	None						
Timetable Requirement per week			4	Lectures p.w.			
			5	Practicals p.w.			
			3	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 141/142		Module Code			STA141/142	
Module Type	Elective	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BA (2101)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			S1 - STA141 S2 - STA142 (BA students can only select this option)	
Main Learning Outcomes	On completion of this module, students should be able to: - Understand the importance of statistics in both private and public sectors. - Summarise basic data into summary measures (e.g. mean and standard deviation). - Prepare simple statistical analysis. - Use a computer to analyse data.						
Main Content	- Descriptive statistics - Simple linear regression analysis - Probability - Sampling distributions - Inferential Statistics - Chi-square - Anova						
Pre-requisite	QLC141, QLC142 FOR EMS Students or an Equivalent for Arts Students			Prohibited module Combination		STA111, STA125, STA141, STA151, BUS131, BUS132	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	60%
						FA	40%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 151		Module Code			STA151	
Module Type	Core/ Elective	NQF Level	5	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc (Applied Geology) (3011), BSc (Computer Science) (3023), BSc (Environment and Water Science) (3027), BSc (Medical Bioscience) (3035)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	

Main Learning Outcomes	On completion of this module, students should be able to: • Recognise the importance of statistics in both the private and public sectors. • Summarise the data into a few summary measures (e.g. Mean and standard deviation). • Do simple statistical analysis. • Use a computer to analyse the data and present data graphically. • Science communication; scientific investigation. • Probabilistic reasoning skills. • Introduction to sampling techniques.			
Main Content	Introductory Statistics • Descriptive statistics; Regression analysis, • Introductory probability; Confidence intervals • Hypothesis testing; Statistical computer literacy • Presentation and computer skills directed toward probabilistic reasoning; Non-parametric statistics • Sampling techniques			
Pre-requisite	None	Prohibited module Combination	STA111, STA125, STA141, STA142, BUS131, BUS132	
Co-requisite	None			
Timetable Requirement per week	3		Lectures p.w.	
	3		Practicals p.w.	
	1		Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 211		Module Code			STA211	
Module Type	Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Computer Science) (3221, 3023), BSc (Mathematics and Statistical Sciences) (3227, 3031)			Total Learning Time		200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Understand probability theory. • Apply discreet and continuous probability distributions. • Apply moments and moment generation functions. • Understand sampling and sampling distributions. • Develop statistical computer literacy skills.						
Main Content	Distribution theory: • Definition of statistical terms • Probability theory • Discrete and continuous probability distributions • Moments and moment generating functions • Sampling and sampling distributions • Manipulating and summarising data with reports and graphs						

Pre-requisite	(MAT105 or (MAM151 + MAM152)) and (STA111 or STA125 or STA151)	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		4	Lectures p.w.		
		3	Practicals p.w.		
		2	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 221		Module Code			STA221	
Module Type	Elective	NQF Level	6	Year Level	2/3	NQF Credit	20
Programmes in which the module will be offered		BSc (Computer Science) (3221, 3023), BSc (Mathematics and Statistical Sciences) (3227, 3031)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Perform statistical inference with the use of estimation and hypothesis tests. • Explore linear models in more detail. • Perform analysis of variance and categorical data analysis.						
Main Content	• Inference and Regression • Central Limit Theorem • Transformations; • Point and interval estimation; • Hypothesis tests • Regression analyses using matrices • ANOVA • Categorical data analyses						
Pre-requisite	STA211 (at least qualified to write the exam for STA211)		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department		Statistics and Population Studies		
Module Name	Statistics 331		Module Code		STA331		
Module Type	Elective	NQF Level	7	Year Level	2/3	NQF Credit	30

Programmes in which the module will be offered		BSc (Mathematics and Statistical Sciences) (3227, 3031)		Total Learning Time		300 hours	
Duration	Semester		Proposed Semester		First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Describe and explain multivariate distributions (discrete and continuous) • Demonstrate advanced inference • Explain statistical programming techniques						
Main Content	• Multivariate distribution Theory • Multivariate distributions (discrete and continuous) • Limit theories • Advanced estimation and hypothesis testing • Statistical programming techniques						
Pre-requisite	STA211 and STA221		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%	
					FA	50%	

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 332		Module Code			STA332	
Module Type	Elective	NQF Level	7	Year Level	3/4	NQF Credit	30
Programmes in which the module will be offered		BSc (Mathematics and Statistical Sciences) (3227, 3031), BSc (Computer Science) (3221, 3023)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Execute advanced linear models for estimation. • Perform advanced analysis of variance and multivariate regression. • Design and execute data analysis using statistical package; • Write reports summarising and interpreting results.						
Main Content	Advanced Inference and linear models • Advanced linear models • Analysis of variance and advanced regression analysis. • Understanding and interpretation of computer output						
Pre-requisite	STA211 and STA221		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3		Lectures p.w.		
			1		Practicals p.w.		
			1		Tutorials p.w.		
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies		
Module Name	Statistics for Commerce 101		Module Code			STC101		
Module Type	Core	NQF Level	5	Year Level	2	NQF Credit	15	
Programmes in which the module will be offered		BCom (Extended Curriculum) (1761), BCom Accounting (Extended Curriculum) (1762)				Total Learning Time	150 hours	
Duration	Year		Proposed Semester			Both Semesters		
Main Learning Outcomes	On completion of this module, students should be able to: - Analyse data sets by applying appropriate visualisation techniques to show trends and patterns inherent a data series. - Demonstrate an understanding of statistical theory regarding probability, probability distribution and sampling distributions. - Demonstrate understanding of frequency-based distribution. - Apply frequency-based probability statistics to draw conclusions from samples.							
Main Content	[A] Foundational - Population and samples - Descriptive versus inferential statistics - Basic concepts (Parameter, sample data, statistic) - Graphing distributions - Types of distributions [B] Descriptive statistics with commercial applications - Descriptive statistics: tabular and graphical representation. - Descriptive statistics: numerical measures. - Index numbers. - Analyse and visualise data to solve commerce problems [C] Statistical theory - Introduction to probability. - Discrete and continuous distributions. - Sampling and sampling distribution. - Chi-Square Test. [D] Inferential statistics with commerce applications - Interval estimation - Hypothesis testing - Statistical inference about means and proportions with two populations. - Inferences about population variances. - Test of goodness of fit and independence. - ANOVA. - Simple regression and correlation analyses. - Statistical inference to solve commerce problems - Time series							
Pre-requisite	QSC141/QSC142/MA M116		Prohibited module Combination			STA111/ STA125/ STA142/ STA151/ BUS132		
Co-requisite	None							
Timetable Requirement per week			4		Lectures p.w.			
			0		Practicals p.w.			
			1		Tutorials p.w.			

Other teaching modes that does not require time-table				
Contact with lecturer / tutor: 100 hrs				
Self study: 50hrs				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

POSTGRADUATE MODULE DESCRIPTORS

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Research Project 709		Module Code			APG709	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	30
Programme in which the module will be offered	BSc Hons (Applied Geology) (3710)					Total Learning Time	300 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Do extensive literature research on a particular research topic. • Perform independent to semi-independent field or laboratory-based research or experiments. • Interpret chemical, geological, structural or other types of data. • Write a comprehensive report based on measurements and observations obtained and the interpretation thereof. • Present the results and interpretation orally.						
Main Content	• Research topic literature review • Research project proposal • Geological setting • Field and/or laboratory work • Obtaining analytical results • Interpretation of data • Report writing • Oral presentation of report						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Advanced Geology 721		Module Code			APG721	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	30
Programme in which the module will be offered	BSc Hons (Applied Geology) (3710)					Total Learning Time	300 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Explain and describe the geological evolution of the African continent as a whole or parts thereof either with regards to time periods or regions of Africa. • Explain the tectono-magmatic and tectono-metamorphic evolution of portions of Africa. • Explain the formation of the principal mineral and hydrocarbon deposits of Africa.						

	- Independently map a large area in terms of lithologies and geological structures and produce a detailed and comprehensive geological map. - Analyse the geological information obtained during mapping and interpret the geological processes and history of the area in terms of formation of the rock types as well as the deformation history of the area and place the total into a larger geotectonic framework. - Describe and explain microscale deformation and structural features in rocks and how these develop. - Relate large- and microscale structural scale features in rocks to the overall geotectonic and structural evolution and history of the rocks of an area.					
Main Content	- A review of the Geological Time Scale. - The geology and tectonic evolution of the Kalahari, Congo-Tanzania and West African Cratons, as well as tectogenetic cycles and Mobile Belts in Africa. - Overall anorogenic and orogenic magmatism and tectonic evolution of Africa. - Archaean and Proterozoic Metallogeny (Gold, Tin and Copper). - The development of sedimentary Basins in Africa. - The development of precious and base metal and hydrocarbon potentials. - Structural Geology in the geotectonic context. - Stress and its origin in the Earth's crust - Deformation mechanisms leading to strain, as found in compressional, extensional and strike-slip regimes. - Application of microtectonics to assist in the unravelling of the geological history of an area. - Microtectonics and their role in determining the geodynamic and structural evolution of an area - Semi-independent geological field mapping with data collection - Interpretation of the lithological and structural data gathered - Analysis and interpretation of petrographic thin sections and chemical analyses - Complete geological interpretation of the mapped field area based on their own data					
Pre-requisite	None		Prohibited module Combination		None	
Co-requisite	None					
Timetable Requirement per week			10	Lectures p.w.		
			1	Practicals p.w.		
			0	Tutorials p.w.		
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%
					FA	50%
Faculty	Natural Sciences		Home Department		Earth Sciences	
Module Name	Applied Geochemistry 722		Module Code		APG722	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit
Programme in which the module will be offered		BSc Hons (Applied Geology) (3710)			20	
					Total Learning Time	
					200 hours	

Duration	Semester	Proposed Semester				Second Semester
Main Learning Outcomes	On completion of this module, students should be able to: Understand the principles of low temperature geochemical modeling as it relates to: - Weathering and associated products. - Element speciation in aqueous phase modeling and the influence of redox processes. - Factors controlling mineral stability in aquaeous phase. - Adsorption processes. Demonstrate an understanding of the principles underlying: - Sample preparation for geochemical analysis; sample digestion, and partial/ selective extraction techniques. - Quality control and quality assurance. - The use of GFAAS, XRF and ICP-MS in geochemical analysis. Demonstrate an understanding of the principles underlying: - The selection and design of various geochemical techniques. - Execution of geochemical exploration programme. - Evaluation and interpretation of geochemical data.					
	Main Content - Sample preparation for geochemical analysis; sample digestion, and partial/ selective extraction techniques - Quality control and quality assurance - The use of GFAAS, XRF and ICP-MS in geochemical analysis. Low temperature geochemistry covering: - Chemical weathering and associated products - Aqueous geochemistry: speciation modeling and redox processes - Weathering and mineral stability diagrams. - Adsorption processes Geochemical prospecting techniques - Lithogeochemistry - Stream sediment geochemistry - Regolith geochemistry - Biogeochemistry - Evaluation of geochemical data					
Pre-requisite	None		Prohibited module Combination		None	
Co-requisite	None					
Timetable Requirement per week			3	Lectures p.w.		
			3	Practicals p.w.		
			0	Tutorials p.w.		
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Economic Geology and Exploration Techniques 723		Module Code			APG723	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Applied Geology) (3710)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	

Main Learning Outcomes	On completion of this module, students should be able to:											
	• Understand the various techniques used to explore for, and discover an economic ore deposit. • Have an understanding of various geological processes that cause the formation of economic mineral deposits and which may be used to discover a mineral deposit. • Understand the role of the geologist in an exploration program. • Have an understanding of the various geological and geochemical techniques used during an exploration program. • Have an understanding of the principles of application of geophysics to petroleum and mineral exploration, which include the following methods: Electrical, Magnetic, Density, Seismic, Potential Field, Electromagnetic, Nuclear, Logging Techniques and Tools. • Will demonstrate an understanding of the principles underlying the selection and design of various geophysical, geochemical and drilling techniques, the execution of an exploration program, and the evaluation and interpretation of geophysical, geochemical and drilling data. • Be able to start assessing various types of geological data to evaluate the viability of a mineralized area. • Have an introductory understanding of mineral economics and the forces behind mineral exploration. • Have an understanding of ore reserve calculations.											
	Main Content											
	Pre-requisite	APG321				Prohibited module Combination			None			
	Co-requisite	None										
	Timetable Requirement per week					10		Lectures p.w.				
						12		Practicals p.w.				
10						Tutorials p.w.						
Other teaching modes that does not require time-table												
Assessment Module Type		Continuous and Final Assessment (CFA)				Method of Student Assessment			CA		50%	
									FA		50%	
Faculty		Natural Sciences				Home Department			Earth Sciences			
Module Name		Geology 724				Module Code			APG724			
Module Type		Core		NQF Level		8	Year Level		1	NQF Credit		20
Programme in which the module will be offered				BSc Hons (Applied Geology) (3710)						Total Learning Time		200 hours

Duration	Semester	Proposed Semester	First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Understand how hydrocarbon is formed and accumulated • Calculate the maturity of source rocks • Conceptualise and explain different trap mechanisms • Determine all reservoir properties such as porosity and permeability • Calibrate logs/cores • Estimate the volume of hydrocarbon present in reservoirs			
Main Content	• Hydrocarbon accumulation • Sedimentary environments • Basins and sequence stratigraphy • Reservoir Diagenesis • Determination of Source Rock Maturity • Hydrocarbon traps • Log Analysis • Core Analysis • Calibration of Log/Core • Volume estimates • Reserves			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		5	Lectures p.w.	
		2	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Research Project 701		Module Code			ASP701	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	30
Programme in which the module will be offered	BSc Hons (Astrophysics) (3793)					Total Learning Time	300 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Conduct a research project in astrophysics • Prepare a presentations for communicating research findings. • Prepare a report on the research topic • Be able to program in Python • Work in the Linux environment						
Main Content	• This module involves a research project done in collaboration with an astrophysics researcher as supervisor. • Students will need to write a report and give a presentation on their findings.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table								
No Timetable Requirements.								
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA		100%

Faculty	Natural Sciences		Home Department		Physics and Astronomy			
Module Name	Topics in Physics 702		Module Code		ASP702			
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10	
Programme in which the module will be offered		BSc Hons (Astrophysics) (3793)			Total Learning Time		100 hours	
Duration	Semester		Proposed Semester			First Semester		
Main Learning Outcomes	On completion of this module, students should be able to:							
	Electromagnetic theory: • Apply the general ideas of EM theory in various fields of physics. • Apply Maxwell's equations in its application to astronomy. • Calculate the energy involved in electromagnetic radiation. Radiative Transfer • Calculate energy levels in simple atoms • Calculate transitions in atoms relevant to stars							
Main Content	• Vectors and Tensors • Maxwell's Equations • Electromagnetic Waves • Radiative Processes							
Pre-requisite	None		Prohibited module Combination			None		
Co-requisite	None							
Timetable Requirement per week			3	Lectures p.w.				
			0	Practicals p.w.				
			1	Tutorials p.w.				
Other teaching modes that does not require time-table								

Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%
					FA	50%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	General Astrophysics 703		Module Code			ASP703	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Astrophysics) (3793)			Total Learning Time		100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Calculate the properties of stars. • Calculate the light output of stars. • Calculate the equilibrium conditions in stars. • Explain the different types of galaxies. • Explain galaxy evolution. • Explain the distance scales used in the cosmos. • Calculate the expansion of the universe.						

Main Content	• Introduction to Multi-wavelength astronomy, focused on the research strengths of South Africa, including Stellar Astronomy, Galaxies, Cosmology			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		0	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Computational Methods in Astronomy 704		Module Code			ASP704	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Astrophysics) (3793)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Write programmes to solve astrophysical problems. • Fit data using the least square method. • Solve differential equations numerically. • Find periodicity in data.						
Main Content	• Scientific Programming as it pertains to Astrophysics and Space Science.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			1	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Astrophysical Physics Topics 706		Module Code			ASP706	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Astrophysics) (3793)			Total Learning Time		100 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Calculate atomic properties. • Calculate the statistical distributions related to stars and galaxies. • Describe and calculate the hydrodynamical fluid flow in astrophysics.						

Main Content	• Atomic Physics • Statistical Mechanics • Astrophysical Fluid Dynamics			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		0	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Geospace 707		Module Code			ASP707	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Astrophysics) (3793)			Total Learning Time		100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain what is meant by and the characteristics of Geospace • Calculate the physical parameters relevant to geospace • Describe plasma systems						
Main Content	• Environment, Dynamics and structure of Geospace • Physics of the region of space from 50 km above the earth to the sun's magnetopause • Introduction to plasma physics						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	General Relativity 708		Module Code			ASP708	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Astrophysics) (3793)			Total Learning Time		100 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Apply Special Relativity. • Use Curved Manifolds and Spacetime. • Use Einstein's Field Equations. • Calculate Gravitational waves. • Explain the phenomenon of Black holes.						

Main Content	Einstein's theory of gravity known as General Relativity				
	• Special Relativity. • Tensors. • Tensors in SR. • Curved Manifolds and Spacetime. • Einstein's Field Equations. • Gravitational waves. • Black holes.				
	Pre-requisite	None	Prohibited module Combination		None
	Co-requisite	None			
	Timetable Requirement per week		3	Lectures p.w.	
			0	Practicals p.w.	
			1	Tutorials p.w.	
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%	
			FA	50%	

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Observational Techniques 709		Module Code			ASP709	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Astrophysics) (3793)			Total Learning Time		100 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Calculate stellar positions • Explain how telescopes work and operate • Explain the detection methods used with modern telescopes.						
Main Content	• Basics Observational Techniques • Telescopes • Detectors • Photometry • Virtual Observatory • Near Infrared Techniques • Polarimetry • Radio telescopes • Radio interferometry • Aperture synthesis • Astronomical Data Reduction • SALT						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Cosmology 710		Module Code			ASP710	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Astrophysics) (3793)				Total Learning Time	100 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Calculate geometric aspects of the universe. • Explain the measurement, origin and structure of the cosmic wave background. • Describe the large scale structure of the universe.						
Main Content	• Geometry • Observational parameters • Cosmic Wave Background • Early Universe • Large scale structures • Baryon acoustic oscillations and dark energy						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences			Home Department			Physics and Astronomy		
Module Name	Space Science Topics 711			Module Code			ASP711		
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	20		
Programme in which the module will be offered		BSc Hons (Astrophysics) (3793)				Total Learning Time	200 hours		
Duration	Semester		Proposed Semester			Second Semester			
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the important aspects in Space weather • Calculate the wave behaviour in the solar wind and the magnetosphere • Explain the dynamics of solar flares								
Main Content	• Space Technology • Space Weather • Plasma Physics • Fluid Dynamics • Solar Physics • Space waves								
Pre-requisite	None			Prohibited module Combination			None		
Co-requisite	None								
Timetable Requirement per week				3	Lectures p.w.				
				0	Practicals p.w.				
				1	Tutorials p.w.				

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Advanced Topics in Astrophysics 712		Module Code			ASP712	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Astrophysics) (3793)			Total Learning Time		200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Describe issues around galaxy formation. - Explain the evolution of galaxies.						
Main Content	- This module will contain topics of research interest to the researchers who are available locally as well as visitors. - The content will depend on the research expertise of these researchers, but galaxy formation and evolutions will be emphasized.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			1	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences			Home Department			Physics and Astronomy	
Module Name	Electrodynamics 713			Module Code			ASP713	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit		10
Programme in which the module will be offered		BSc Hons (Astrophysics) (3793)				Total Learning Time		100 hours
Duration	Year		Proposed Semester			Both Semesters		
Main Learning Outcomes	On completion of this module, students should be able to: • Do calculations using Vectors and Tensors. • Apply Maxwell's equations for simple systems to find fields. • Calculate the properties of electromagnetic waves, electromagnetic energy and momentum. • Obtain the paths for charged particles in fields. • Calculate the radiative processes that take place in antennas.							
Main Content	• Vectors and Tensors • Maxwell's Equations • Electromagnetic Waves • Relativity and Electromagnetism • Electromagnetic energy and momentum • Single particle motion							

	• Radiative Processes • Antennas			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week	3		Lectures p.w.	
	0		Practicals p.w.	
	1		Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Research Project 700		Module Code			BCB700	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	40
Programme in which the module will be offered		BSc Hons (Biodiversity and Conservation Biology) (3731)				Total Learning Time	400 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Execute research under supervision.						
	• Formulate research question/s.						
	• Review the appropriate literature and contextualise a study.						
	• Collect field and/or lab data.						
	• Statistically analyse data.						
	• Summarise data in tabular/graphic format.						
	• Interpret data in the context of the literature and the research question/s.						
Main Content	• Communicate research findings in oral and written formats.						
	• Subject specific, depending upon the topics offered by staff in any one year.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No timetable requirements			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department		Biodiversity and Conservation Biology		
Module Name	Systematics and Evolution 731		Module Code		BCB731		
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Biodiversity and Conservation Biology) (3731)			Total Learning Time		200 hours
Duration	Semester	Proposed Semester			Second Semester		

Main Learning Outcomes	On completion of this module, students should be able to:			
	- Find taxonomic information by selecting appropriate taxonomic treatments or revisions for a particular group of organisms. - Understand species concepts related to angiosperm and animal diversity. - Understand the links between zoological and botanical nomenclature, and the conservation of those organisms. - Demonstrate competence in basic laboratory techniques including DNA extraction, amplification, and sequencing. - Perform and present appropriate data analyses.			
Main Content	- Diversity and evolution of Cape Flora - Diversity of plant and animal families in South Africa - Conservation of taxa with special emphasis on Red Data Species - Modern species concepts and classification - Botanical and Zoological nomenclature - Electronic keys and descriptive databases - Plant and animal classification - Sources of taxonomic information - Evolutionary and ecological processes maintaining species diversity - Herbarium/museum curating techniques and Flora writing - Molecular systematic techniques and analyses			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		2	Lectures p.w.	
		1	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Biodiversity Information Management 736		Module Code			BCB736	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Biodiversity and Conservation Biology) (3731)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Analyse spatial information and derive patterns. - Spatially model information for the management of natural resources and prediction of landscape processes. - Compile a self-executing installation CD/DVD of spatial data with a viewer to explore the data, complete with compiled meta-data and help facilities. - Evaluate the requirements for developing a biodiversity/taxonomic data base.						

Main Content	• Geographical Information Systems (GIS) and remote-sensing; • Mapping using a Global Positioning System (GPS) and analysis of this data; • Digitizing using online resources; • Geostatistics and spatial interpolations for modeling point data; • Use of spatial data to develop species distribution data and to define meta-populations and identify species with conservation-critical distributions; • General Internet mapping; • Classification of satellite imagery (supervised, unsupervised and object-based approaches); • Development principles of a biodiversity/taxonomic database.				
	Pre-requisite	None	Prohibited module Combination		None
	Co-requisite	None			
	Timetable Requirement per week		4	Lectures p.w.	
			4	Practicals p.w.	
			0	Tutorials p.w.	
	Other teaching modes that does not require time-table				
	Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Marine Biology 738		Module Code			BCB738	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered	BSc Hons (Biodiversity and Conservation Biology) (3731)					Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Read and understand the relevant literature. • Design and execute simple marine biological field studies. • Analyse simple field data. • Write a professional research report on a marine biological topic.						
Main Content	The course is a blend of practical field work, structured discussion classes and research in the field of pelagic and benthic ecology. Topics include: • Introductory physical and chemical oceanography; • The role that the physical environment plays in shaping pelagic organisms (from individuals to communities); • The natural and not so natural factors affecting the diversity and functioning of pelagic ecosystems; and the role of biological interactions in structuring the communities of rocky intertidal habitats. The emphasis throughout, will be on Southern Africa.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			1	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Herpetology 739		Module Code			BCB739	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Biodiversity and Conservation Biology) (3731)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Converse competently regarding herpetological diversity, evolution, ecology, and conservation. • Contribute novel virtual museum data to an appropriate national database. • Gather appropriate herpetological data in the field as part of a group. • Analyse, present and report herpetological research.						
Main Content	• Evolution of tetrapods. • Systematics of reptiles and amphibians. • Natural history and ecology of reptiles and amphibians. • Conservation of southern African and global herpetofaunal populations. • Current herpetological science trends. • Field survey, data collection, data analysis, and professional research writing.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Research Method 742		Module Code			BCB742	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Biodiversity and Conservation Biology) (3731)			Total Learning Time		200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Plan and execute a research project. • Interpret research data and research literature. • Write-up research material for submission as a scientific paper. • Communicate research findings orally and in a poster format.						
Main Content	• Introduction to sampling design, research techniques, instruments, and laboratory safety.						

	• Developing a project proposal and planning field research. • Presentation of data as tables and figures. • Reading and writing skills and interpreting scientific data. • Preparing and presenting a scientific oral and poster presentation. • Preparation of scientific research for publication.			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		4	Lectures p.w.	
		4	Practicals p.w.	
		4	Tutorials p.w.	
Other teaching modes that does not require time-table				
Block teaching within the Department.				
No timetable requirements				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Quantitative Ecology 743		Module Code			BCB743	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Biodiversity and Conservation Biology) (3731)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Understand the concepts of alpha, beta and gamma diversity. • Know and understand the current hypotheses that attempt to explain species assembly processes in space and time (e.g. Neutral and niche mechanisms). • Be able to collect ecological data at the appropriate scale, which would lend themselves to a quantitative analysis of points 1 and 2, above. • Be able to use the R software and associated packages to undertake the analyses required in point 3, above. • Be able to interpret the outcomes of the above analyses and use it to quantitatively characterise points 1 and 2, above. • Be able to communicate the findings by written and oral means.						
Main Content	• Core theoretical framework: ecological hypotheses underlying the processes of species assembly in space and time, including neutral and niche-based mechanisms, and historical events; overview of the currently known and understood distributional patterns of major groups of organisms at global, regional and local scales; consideration of sampling designs aimed at capturing these patterns and drivers so as to arrive at a processed based understanding of species assembly. • Field-activities: data collection aimed at a quantitative test of the relevant hypotheses, above. • Statistical/software competence: the management and analysis of ecological data; reproducible and collaborative research; the use of R as a tool for the analysis multivariate ecological data; multivariate techniques such as nMDS, PCA, RDA and cluster analysis; graphical data summaries and visualisations.						

Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	BCB744			
Timetable Requirement per week		2	Lectures p.w.	
		2	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Block teaching within the Department. No timetable requirements				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Biodiversity and Conservation Biology	
Module Name	Biostatistics 744		Module Code			BCB744	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Biodiversity and Conservation Biology) (3731)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Design basic experiments. • Manage data. • Use appropriate proprietary and open-source software to analyse data.						
Main Content	• Training and skills development with appropriate analytical software including SPSS and R. • Data management.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			4	Lectures p.w.			
			4	Practicals p.w.			
			4	Tutorials p.w.			
Other teaching modes that does not require time-table							
Block teaching within the Department. No timetable requirements							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Molecular Biology and Bioinformatics Practical 701		Module Code			BTY701	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Biotechnology) (3707)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Practical and technical experience. • Laboratory safety knowhow. • Scientific presentation skills. • Scientific writing skills.						

Main Content	• Molecular basis of viral vector technologies • Types of viral vectors • Viral vectors for vaccine delivery • Viral vectors for gene therapy • Molecular biology practical component				
	Pre-requisite	None	Prohibited module Combination	None	
	Co-requisite	None			
	Timetable Requirement per week	5	Lectures p.w.		
		5	Practicals p.w.		
0		Tutorials p.w.			
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%	
			FA	40%	

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Literature Review 702		Module Code			BTY702	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Biotechnology) (3707)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Basic scientific writing skills. • Scientific sentence construction. • How to perform a literature survey. • Literature preparation for research project.						
Main Content	• Literature review construction • Literature survey • Preparation of scientific introductions • Academic writing						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No Timetable Requirements			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Research Project 703		Module Code			BTY703	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	40
Programme in which the module will be offered	BSc Hons (Biotechnology) (3707)					Total Learning Time	400 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Preparation of research proposal. • Conducting laboratory experiments. • Scientific presentation skills.						

	- Statistical data analysis. - Constructing a mini-thesis.			
Main Content	- Biotechnology research project - Scientific methodology - Scientific writing - Research article preparation - Data analysis - Laboratory training - Health and safety training - Chemical safety training			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week No Timetable Requirements		0	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA 100%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Advanced Plant Biotechnology 704		Module Code			BTY704	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Biotechnology) (3707)			Total Learning Time	100 hours	
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Plant transformation and regeneration. - The mechanisms and regulation of plant gene expression. - Plant stress signaling.						
Main Content	- Transformation and regeneration of plants - Expression systems and gene suppression - Disease and pest (fungi, virus, bacteria, insect) resistance - Molecular tools for plant breeding - Crop modification and improvement						
Pre-requisite	60% Pass Average Geology Third Year		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
No Timetable Requirements.							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment		FA		100%

Faculty	Natural Sciences		Home Department		Biotechnology	
Module Name	Genomics 709		Module Code		BTY709	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit 10

Programme in which the module will be offered		BSc Hons (Biotechnology) (3707)		Total Learning Time	100 hours
Duration	Year	Proposed Semester		Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Fundamental understanding of genes and chromosomes. • Features of the genome. • Mapping of the genome. • Human genetic diseases.				
Main Content	• DNA and chromosome structure, organization and function • Genes in pedigrees • Physical and genetic mapping • Mutations and instability of DNA • Selected genome projects • Selected genetic diseases				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week No Timetable Requirements		0	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Apoptosis 710		Module Code			BTY710	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Biotechnology) (3707)				Total Learning Time	100 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • A sound knowledge of cell growth control. • Interpret scientific literature regarding the cell cycle and the role of cell signaling in the control aspects of the cell cycle. • Know about different types of cell death, including apoptosis and the induction of apoptosis by various mechanisms. • Gain knowledge on the role of apoptosis in tumourigenesis.						
Main Content	The Cell Cycle • Checkpoints in the cell cycle • Cyclin-dependent kinases • Inhibitors of cyclin-dependent kinase Cell Death Pathways • Necrosis, Apoptosis, atrophy and terminal differentiation, genetic control of programmed cell death Apoptosis and disease						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week	2	Lectures p.w.	Lectures for 4 weeks only	
	0	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Advanced Food Microbiology 711		Module Code			BTY711	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Biotechnology) (3707)			Total Learning Time		100 hours
Duration	Term		Proposed Semester			Term 3 or Term 4	
Main Learning Outcomes	On completion of this module, students should be able to: • A detailed understanding of the molecular basis of detection of pathogens in complex food samples. • Familiarity with recent review and research literature in the subject area. • Information on applications of HACCP and other food safety management tools. • Microbial control and quality. • Problem solving and sanitation.						
Main Content	• Diversity of food borne microorganisms • Adaptations to food environments • Food borne pathogens • Management of food safety						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No Timetable Requirements			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Forensic DNA Analysis 714		Module Code			BTY714	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programmes in which the module will be offered		BSc Hons (Biotechnology) (3707), BSc Hons (Medical Bioscience) (3721)				Total Learning Time	100 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Understand the scientific basis of DNA typing. • Have detailed knowledge of forensic DNA typing systems.						

	- Understand the procedures used in forensic DNA analysis: DNA isolation, determining quantity & quality of DNA, RFLP analysis, PCR amplification, analysis of PCR products, automated analysis systems. - Interpret DNA typing results when there are many complicating factors: multiple contributors, DNA degradation, extraneous substances, and system-specific interpretational issues. - Assess the strength of forensic evidence: determination of genetic concordance, evaluation of results, frequency estimation calculations, population substructure, and likelihood ratios.				
Main Content	Introduction: - What is forensics - The criminal justice system Working with the dead: - Time of death, decomposition and fire, events at death. - Forensic archaeology and excavation of human remains - Osteobiography Working with the living: - The nature of physical evidence - The scientific basis of DNA typing - An overview of forensic DNA typing systems - Procedures for forensic DNA analysis - Interpretation of DNA typing results. - Assessing the strength of forensic evidence				
	Pre-requisite	None	Prohibited module Combination	None	
	Co-requisite	None			
	Timetable Requirement per week		2		Lectures p.w.
			0		Practicals p.w.
0			Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%	
			FA	50%	

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Special Topics 717		Module Code			BTY717	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Biotechnology) (3707)				Total Learning Time	100 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: Understand this specialist study of a topic within Biotechnology, not covered in other BTY Honours modules.						
Main Content	- A specialist study of a topic within Biotechnology, not covered in other BTY Honours modules. - Course structure variable, to suit the size of group, and topic covered.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Proteomics 718		Module Code			BTY718	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Biotechnology) (3707)			Total Learning Time		100 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Understanding of the various aspects of Proteomics such as experimental design, sample preparations, quantification, separation, comparative analysis, identification etc. • Also learning Proteomics as a broad research tool and its various applications in molecular biology research particularly in plant biotechnology and medical research.						
Main Content	Proteomics Phase 1 (Week 1) • Proteomics definition, background and principle • Protein sample preparation for proteomics studies • Protein quantification methods • Protein separation methods Proteomics Phase 2 (Week 2) • Protein staining approaches (gel based) • Sample comparative analysis (gel based) • Protein staining approaches (non-gel based) • Sample comparative analysis (non-gel based) Proteomics Phase 3 (Week 3) • Protein Identification methods (mass spectrometry) Proteomics Phase 4 (Week 4) • Proteomics applications						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.	Lectures for 4 weeks only		
			0	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Biofuels 719		Module Code			BTY719	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Biotechnology) (3707)			Total Learning Time		100 hours
Duration	Year		Proposed Semester			Both Semesters	

Main Learning Outcomes	On completion of this module, students, should be able to: • An understanding of the types of biofuels available and currently in development. • Knowledge of the various aspects of the biofuels value chain and how each step requires improvement. • An appreciation of the current state of the art in first and second generation biofuels. • Knowledge of how metabolic engineering and pretreatment methods are being used to improve biofuel yields and decrease costs of biofuel production.			
Main Content	• Types of biofuels including first generation and second generation biofuels • Analysis of the entire biofuels value chain from feedstock to fuel and energy • Biodiesel, biogas and bioethanol: current state of the art • The prospect of algae in fuel production • Biomass pretreatment and hydrolysis • Improvement of state of the art technologies: metabolic engineering to achieve consolidated bioprocessing • Next generation technologies and drop-in fuels			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		1	Lectures p.w.	
		0	Practicals p.w.	
		3	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Molecular Mechanisms of Cancer 721		Module Code			BTY721	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Biotechnology) (3707)				Total Learning Time	100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • The basic manifestations of cancer. • The development of the field of cancer research during the 20th century. • The roles of oncogenes and tumour suppressors in its regulation.						
Main Content	• The Nature of Cancer • Tumour Viruses • Cellular Oncogenes • Growth Factors and their Receptors • Cytoplasmic Signalling Circuitry • Tumour Suppressor Genes • pRb and Control of the Cell Cycle • p53 and Apoptosis						
Pre-requisite	None					None	

Co-requisite	None	Prohibited module Combination		
Timetable Requirement per week		2	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Biotechnology	
Module Name	Next Generation Sequencing 722		Module Code			BTY722	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Biotechnology) (3707)				Total Learning Time	100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • An understanding of the various sequencing technologies available. • Been exposed to novel applications of sequencing technology in the biotechnology industry.						
Main Content	• Overview of sequencing technology – Sanger, 454, Illumina and Ion Torrent • Sequencing in genomics and metagenomics. • Applications in transcriptomics. • Applications in viromics for studying gene delivery systems. • Future applications in metabolomics.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			0	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department			Chemistry	
Module Name	Chemistry 701		Module Code			CHM701	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Chemistry) (3734)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Apply the methods of mass spectroscopy (MS), nuclear magnetic resonance (NMR) spectroscopy, graphite furnace atomic absorption spectrometry (GF-AAS) and inductively coupled plasma atomic emission spectrometry (ICP-AES) to collect, analyse and interpret analytical data, for either qualitative and/or quantitative purposes.						

	• Apply the principles of mass spectroscopy (MS), and nuclear magnetic resonance (NMR) spectroscopy to elucidate structural aspects of simple organic and inorganic molecules e.g. alcohols, aldehydes, ketones and esters. • Evaluate the technical and operational principles of MS instrumentation in order to derive plausible combinations to analyse simple organic and inorganic molecules. • Apply the principles of graphite furnace atomic absorption (GF-AAS) and inductively coupled plasma atomic emission spectrometry (ICP-AES) to elucidate compositional aspects of simple organic and inorganic molecules. • Perform qualitative investigations in order to determine the elemental composition of real samples. • Conduct research using the library, the web and other sources of information. • Reference sources of information correctly. • Use the internet and computer-based word-processing, spreadsheet, and presentation software to complete tasks. • Recognise the relationship of chemistry to society, technology and the environment. • Present a clear, well-structured oral presentation and well-structured research report. • Work productively in co-operative learning groups. • Demonstrate life-long learning capabilities and an appreciation of chemistry as discipline in a wider context.			
Main Content	• Advanced mass spectroscopy: Ion sources, analyser and detector types, and spectral interpretation. • Advanced nuclear magnetic resonance (NMR) spectroscopy: ¹H-NMR, ¹³C-NMR, ¹H correlation spectroscopy (COSY), hetero-nuclear correlation spectroscopy (HETCOR) and the nuclear Overhauser effect (NOE). • Advanced atomic spectrometry: High sensitivity sampling techniques, graphite furnace atomic absorption spectrometry (GF-AAS) and inductively coupled plasma atomic emission spectrometry (ICP-AES).			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department		Chemistry
Module Name	Chemistry 702		Module Code		CHM702
Module Type	Core	NQF Level	8	Year Level	1
Programme in which the module will be offered		BSc Hons (Chemistry) (3734)		NQF Credit	15
				Total Learning Time	150 hours
Duration	Semester		Proposed Semester		First Semester

Main Learning Outcomes	On completion of this module, students should be able to:						
	• Apply accepted rules to classify terpenoids, carbohydrates and alkaloids into subgroups, in order to better understand their chemistry. • Develop synthetic strategies for the laboratory preparation of terpenoids, carbohydrates and alkaloids, based on chemical reactivity and stereochemical orientation. • Apply the principles of predictive reaction pathways in order to identify plausible products derived from the terpenoids, carbohydrates and alkaloids. • Apply the fundamental principles of molecular orbital theory and orbital symmetry for the confirmation and identification of the products of selected organic reactions. • Develop strategies for advanced synthetic methods based on functional group inter conversion and one-group and two-group disconnections. • Conduct research using the library, the web and other sources of information. • Reference sources of information correctly. • Use the internet and computer-based word-processing, spreadsheet, and presentation software to complete tasks. • Recognise the relationship of chemistry to society, technology and the environment. • Present a clear, well-structured oral presentation and well-structured research report • Work productively in co-operative learning groups. • Demonstrate life-long learning capabilities and an appreciation of chemistry as discipline in a wider context.						
	Main Content						
	• Terpenes- classification, synthesis and reactions • Steroids-conformational aspects and reactivity • Carbohydrates- classification, structural representation, reactions, functions, nomenclature, structure • Determination Orbital symmetry- Pericyclic reactions • Molecular Orbital Theory • Thermal and Photochemical reactions • Advanced synthetic methods- The disconnection approach, Functional group interconversion						
	Pre-requisite	None		Prohibited module Combination		None	
	Co-requisite	None					
	Timetable Requirement per week			3	Lectures p.w.		
				0	Practicals p.w.		
				0	Tutorials p.w.		
	Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%	
					FA	50%	
Faculty	Natural Sciences		Home Department		Chemistry		
Module Name	Chemistry 703		Module Code		CHM703		
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Chemistry) (3734)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester		Second Semester		

Main Learning Outcomes	On completion of this module, students should be able to:				
	• Apply chemical trends of the main group elements to explain the chemistry and reactivity of inorganic rings, cages and chains. • Apply the principles of crystal field theory in order to identify bonding and orientation in simple inorganic molecules. • Apply the principles of molecular orbital theory in order to identify bonding and orientation in metal-ligand coordination complexes. • Apply the trends and principles of spectrochemical properties in order to characterise transition metal complexes. • Justify the composition of simple bio-inorganic compounds based on the fundamental principles of chemical reactivity and properties. • Classify, and propose pathways for the synthesis of organometallic compounds of the types: T-M Alkyls, T-M Alkenes, T-M Carbenes, and T-M Carbynes. • Apply the principles of transition metal compound chemistry to selected industrial homogeneous catalysis processes. • Conduct research using the library, the web and other sources of information. • Reference sources of information correctly. • Use the internet and computer-based word-processing, spreadsheet, and presentation software to complete tasks. • Recognise the relationship of chemistry to society, technology and the environment. • Present a clear, well-structured oral presentation and well-structured research report • Work productively in co-operative learning groups. • Demonstrate life-long learning capabilities and an appreciation of chemistry as discipline in a wider context.				
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Programme in which the module will be offered		BSc Hons (Chemistry) (3734)	Total Learning Time	150 hours
Duration	Semester	Proposed Semester	First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:			
	• Explain postulates, derive expressions for, and calculate rate constants and thermodynamic parameters based on the theories of simple homogeneous gas phase reactions: collision and transition state (activated complex) theories. • Derive expressions and calculate kinetic parameters for unimolecular reactions. • Apply the principles, and laws governing different types of photochemical reactions in order to solve theoretical problems. • Distinguish between the different types of adsorption isotherms. • Apply the principles and mathematical underpinnings of the electrochemistry of kinetics and mitigation measures to analysis of real samples. • Relate the concept of founding quantum mechanical theory on postulates. • Demonstrate the mathematical competency to solve a variety of problems involving mathematical operators. • Construct a quantum mechanical operator from the classical expression for the corresponding operator. • Use the properties of the electronic states of the hydrogen ion to construct approximate wave functions and electron configurations for homonuclear diatomic molecules. • Apply general properties of molecular orbitals and criteria for formation of good bonding orbitals, to predict the qualitative properties of electronic states of heteronuclear diatomic molecules. • Apply the principles of rotational and vibrational spectra as well as the origin of selection rules to the interpretation of spectra. • Apply the principles of Raman Spectroscopy in order to predict which normal modes of a polyatomic molecule will be Raman active. • Apply the principles of electron spin resonance to spectral analysis and data interpretation. • Solve problems involving probabilities of molecular states using the statistical mechanical distribution. • Calculate values of molecular partition functions for atoms, diatomic molecules, and polyatomic molecules. • Calculate values of thermodynamic functions of dilute gases from molecular partition functions • Conduct research using the library, the web and other sources of information. • Reference sources of information correctly. • Use the internet and computer-based word-processing, spreadsheet, and presentation software to complete tasks. • Recognise the relationship of chemistry to society, technology and the environment. • Present a clear, well-structured oral presentation and well-structured research report			

	• Work productively in co-operative learning groups. • Demonstrate life-long learning capabilities and an appreciation of chemistry as discipline in a wider context.			
Main Content	• Advanced theories of chemical reactivity: unimolecular reactions; adsorption isotherms; homogeneous, heterogeneous and enzyme catalysis; photochemical reactions; electrode kinetics; and fast reactions. • Theories, principles and procedures for solving quantum mechanical problems for simple systems: variation and perturbation theorems, Born-Oppenheimer approximations; valence bond and molecular orbital theories. • Symmetry and group theory. Atomic structures of hydrogen-like and helium-like atoms; angular momentum. Molecular orbital and valence bond theories applied to the structures of the hydrogen molecule ion, the hydrogen molecule and other polyatomic molecules. • Theories and applications of rotational and vibrational spectroscopy. Canonical and molecular partition functions and their applications in the statistical thermodynamics of entropy, internal energy, enthalpy, free energy and heat capacity.			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences			Home Department			Chemistry	
Module Name	Chemistry 705			Module Code			CHM705	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15	
Programme in which the module will be offered		BSc Hons (Chemistry) (3734)				Total Learning Time	150 hours	
Duration	Semester		Proposed Semester			First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Apply the principles of homogenous and heterogeneous catalysis to the evaluation of chemical processes at work in real life examples from the petroleum Industry. • Apply the principles and classification of electro catalysis to the differentiation of catalytic processes in chemical technologies. • Identify special features and advantages that may enhance the efficiency of electrochemical processes. • Apply the definition of oxygen reduction and hydrogen evolution reactions to the delineation of electrochemical windows in various electrolytes. • Identify the basic principles of process design. • Evaluate the advantages and disadvantages of adsorption and chemisorption in process design and model these as Temkin adsorption isotherms.							

	• Characterize and evaluate the electrochemical processes at work in fuel cells and batteries in terms of structural properties, surface spectroscopy and surface morphology using cyclic voltammetry chronoamperometry, AC electrochemical techniques, AFM, HRSEM and HRTEM. • Rationalise the technical, economic and social considerations of the hydrogen economy and hydrogen storage in terms of physical and chemical processes, costs of catalysts and electrode materials, best practice for storage and characterisation of materials and processes. • Conduct research using the library, the web and other sources of information. • Reference sources of information correctly. • Use the internet and computer-based word-processing, spreadsheet, and presentation software to complete tasks. • Recognise the relationship of chemistry to society, technology and the environment. • Present a clear, well-structured oral presentation and well-structured research report • Work productively in co-operative learning groups. • Demonstrate life-long learning capabilities and an appreciation of chemistry as discipline in a wider context.					
Main Content	• Application of homogenous and heterogenous catalysis to the petroleum Industry. • Characterisation of different types and applications of electrocatalysis in chemical technology. • Electrochemical Processes: Special features and advantages. • Electrochemical reactions based on oxygen reduction and hydrogen evolution. • Basic principles of process design. • Principles of adsorption and chemisorption and the Temkin adsorption isotherm. • Introduction to characterization techniques for studying the electrochemical processes, structural properties, surface spectroscopy and surface morphology of fuel cells and batteries. • Technical, economic and social considerations of the hydrogen economy and hydrogen storage.					
Pre-requisite	None		Prohibited module Combination		None	
Co-requisite	None					
Timetable Requirement per week			3	Lectures p.w.		
			0	Practicals p.w.		
			0	Tutorials p.w.		
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%
					FA	50%
Faculty	Natural Sciences		Home Department		Chemistry	
Module Name	Chemistry 707		Module Code		CHM707	
Module Type	Core	NOF Level	8	Year Level	1	NOF Credit
						30

Programme in which the module will be offered		BSc Hons (Chemistry) (3734)			Total Learning Time		300 hours		
Duration		Semester		Proposed Semester			Second Semester		
Main Learning Outcomes		On completion of this module, students should be able to:							
		• Apply the principles, reasoning and mathematical competencies developed to date to the design, synthesis, characterisation, qualification and quantification of materials and methods relevant to the research focus of the mini-project selected.							
		• Propose a rationale, research question(s), methodology, resource requirements, research plan, preliminary reading list and timeline for the mini-project, in clear, written English and using the terminology, style and format customary to the field of chemistry research.							
		• Work productively and (where applicable) co-operatively to collect and critically evaluate experimental data, with recognition and sensitivity for the impact of the research on society, technology and the environment, following a clear literature precedent.							
		• Inform the research process using literature resources from the library, the web and other sources, referencing these correctly, in the format that is customary in the field.							
Main Content		• Present clear, well-structured oral and written presentations using current technologies and software applications.							
		• Pose questions to extend the research beyond the scope of the mini-project, under guidance of a mentor.							
		• Introduction to the research focus of the department by means of post graduate brochure, departmental research open days or other suitable and innovative ways.							
		• Consideration and selection of research mini-project from a compilation of contemporary offerings prepared by the department each year in a printed format.							
		• Oral and written presentation of a research proposal developed under the guidance of the principal researcher(s) and their research team.							
Pre-requisite		None		Prohibited module Combination			None		
Co-requisite		None							
Timetable Requirement per week				0	Lectures p.w.				
				0	Practicals p.w.				
				0	Tutorials p.w.				
Other teaching modes that does not require time-table									
Assessment Module Type		Final Assessment (FA)		Method of Student Assessment			FA		100%
Faculty		Natural Sciences		Home Department			Chemistry		
Module Name		Chemistry 708		Module Code			CHM708		
Module Type		Core	NQF Level	8	Year Level	1	NQF Credit		15

Programme in which the module will be offered		BSc Hons (Chemistry) (3734)	Total Learning Time	150 hours
Duration	Semester	Proposed Semester	Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate an understanding of advanced chemical concepts and their (potential) applications in industry within a specified set of advanced topics • Demonstrate an ability to write a mini-review article on a given advanced topic • Conduct chemistry research using the library, relevant search engines and other sources of information. • Reference sources of information in accordance with standard practice in advanced chemistry. • Recognise the relationship of chemistry to society, technology and the environment, as it relates to the specified set of advanced topics. • Present a clear, well-structured oral presentation and well-structured research report • Work productively in co-operative learning groups.			
Main Content	A selection of advanced topics in the following specialist disciplines: Biosensors: Introduction to biosensors Design of biosensors for medical and environmental applications Catalysis: Polymerisation reactions Hydrogenation Biocatalysis Electrochemistry: Fuel cell technology Corrosion chemistry Battery development technology Environmental/Green Chemistry: Atom economy Waster prevention Renewable materials Materials Science: Materials processing and manufacturing Biomaterials Superconductors Nanoscience: Functional nanomaterials Nanomedicine Molecular engineering Natural Products Chemistry: Biosynthesis of terpenoids Formation of polyisoprene chains Cyclization and modifying reactions Organometallic Chemistry: Oxidation reactions Bioinorganic complexes			

	Green complexes				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		2	Practicals p.w.		
		4	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Research Project Computer Science 716		Module Code			CMP716	
Module Type	Core	NQF Level	8	Year Level	1/2	NQF Credit	30
Programme in which the module will be offered	PGDip (Computer and Media Application) (1627)				Total Learning Time	300 hours	
Duration	Semester		Proposed Semester			First Semester of Second (or subsequent) Year of enrollment	
Main Learning Outcomes	On completion of this module, students should be able to: - Engage with clients to define and scope a business need/challenge and/or opportunity. - Utilise theoretical knowledge acquired in terms of data structuring, management, statistical procedures and mining to propose data analytics solutions to the identified business needs/challenges. - Select the appropriate business intelligence and data analytical processes and methodologies to effectively explore and address a business need in the area of Computer Science - Apply the selected process and methodologies according to sound project management principles within the appropriate time line. - Interpret, and present the outcome of the advanced data analytics and business intelligence project within the context of a client's business reality.						
Main Content	- Application of theoretical knowledge of data mining and data analytics in the context of industry-based problem solving through the following areas of content: - Problem identification and scoping in business terms in the area of Computer Science - Data structuring, data management, and statistical manipulation applied to solve an identified problem. - Application of methodology, data analytics, and business intelligence process using a structured project format. - Communication and multi-stakeholder engagement to enable data analytics and business intelligence led outcome to an identified problem. - Presentation of project outcomes (analytics and intelligence) in the client's business context.						

	• Business communication • Research reporting.			
Pre-requisite	BIA715 (Introduction to Data Mining) and 45 further credits made up of any combination of the following modules: BIA710, BIA711, BIA712, BIA713, BIA714	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		4	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment	FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Simulation 710		Module Code			COF710	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Statistical Science) (3737), BSc Hons (Computational Finance) (3739)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Examine a basic approach to statistical simulation and develop basic Matlab skills.						
Main Content	- Introduction to Matlab; - Generating random numbers from different distributions; - Cellular automata; - Systems of linear equations; - Modelling time series; - Simulating queues; - Bootstrapping.						
Pre-requisite	STA331; STA332; MAT211; MAT221; Matlab programming or equivalent		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week			1	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		
					FA		
				50%			
				50%			

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Computational Finance 711		Module Code			COF711	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Statistical Science) (3737), BSc Hons (Computational Finance) (3739)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Apply fundamentals of time domain methods and develop elementary analysis skills.						
Main Content	• Classical time series methods; strong and weak stationarity, and non-stationarity; ARMA models – acf, pacf, stationarity and invertibility; evaluation of residuals; ARIMA models; seasonality in ARIMA models; estimation and prediction; spectral density functions; periodogram; example analyses using appropriate software						
Pre-requisite	STA211; STA221; MAT211; MAT221 or equivalent		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			1	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Computational Finance 712		Module Code			COF712	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Statistical Science) (3737), BSc Hons (Computational Finance) (3739)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Apply the laws of large numbers: • The Weak Law of Large Numbers. • The Central Limit Theorem. • Generate random observations from specified distributions, e.g. The inverse transform method, acceptance-rejection method.						
Main Content	• Review concepts in Probability. • Random variables, distributions, cumulative distribution functions, densities. Generating moments of functions: • Cumulants, probabilities, factorial moments.						

	• Basic inequalities, e.g. Tchebychev inequality. Approximate methods, the Delta method. Introduction to Stochastic Processes. The Poisson process. Branching processes. Introduction to Markov Chains.			
Pre-requisite	STA331 or equivalent	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		2	Practicals p.w.	
		4	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment	FA	100%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Stochastic Calculus for Finance 714		Module Code			COF714	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Mathematical Science) (3736), BSc Hons (Computational Finance) (3739)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Have a working knowledge of Ito calculus including stochastic control.						
Main Content	- Probability spaces - Random walk and Brownian motion - Discrete and continuous martingales - Stopping times - Quadratic variation - Ito integral and basic formulae - Stochastic differential equations - Black-Scholes theory - Stochastic control: formulae and applications						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2		Lectures p.w.		
			1		Practicals p.w.		
			0		Tutorials p.w.		
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		60%
					FA		40%

Faculty	Natural Sciences		Home Department		Mathematics and Applied Mathematics		
Module Name	Financial Engineering 716		Module Code		COF716		
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15

Programmes in which the module will be offered		BSc Hons (Mathematical Science) (3736), BSc Hons (Computational Finance) (3739)		Total Learning Time	150 hours
Duration	Semester	Proposed Semester		Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Have basic knowledge about modeling and evaluation of financial derivatives.				
Main Content	- Forward, Futures and Options Contracts, - Futures Markets and the Use of Futures for Hedging, - Forward and Futures Prices, - Properties of Stock Option Prices (Put-Call Parity), - Valuation of Stock Options, - Ito's Lemma and its Applications, - Analytical Approach of Black-Scholes to Price Stock Options, - Greeks, - Numerical Approaches of Pricing Stock Options (Binomial Trees, Monte Carlo Simulation, Finite Difference Methods) - The Role of Martingales and Measures in Derivatives.				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		2	Lectures p.w.		
		1	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment		CA	50%
				FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Computational Finance 726		Module Code			COF726	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Computational Finance) (3739), BSc Hons (Statistical Science) (3737)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Summarise knowledge and analyse risk management problems.						
Main Content	Risk measurement, analytical methods of risk management, value-at-risk methodologies, risk modelling.						
Pre-requisite	STA331 or equivalent		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			4	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Advanced Software Engineering 726		Module Code			COS726	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Computer Science) (3735)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Create a Formal Mathematical Specification using Z. • Demonstrate the use of box structure specification for analysis, design and verification. • Demonstrate domain engineering and component-based development. • Analyse and blend conventional principles, concepts and methods with Object-Oriented and component-based SE. • Create high quality Web applications. • Restructure (reconstruct/re-engineer) software to improve quality. • Demonstrate the use automated tools in support of the SE process. • Explain and be aware of the latest developments within SE.						
Main Content	• Formal Methods; • Z Specification Language • Cleanroom Software Engineering (SE); • Component-Based SE; • Client/Server SE; • Web Engineering; • Reengineering; • Computer-Aided SE; • Recent publications (e.g. IEEE, ACM, etc) concerning the latest trends and developments with SE						
Pre-requisite	CSC312		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Project Analysis and Design 731		Module Code			COS731	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Computer Science) (3735)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Understand and implement the Analysis and Design activities of the Software Development Life-cycle. • Analyze User Requirements (e.g. UML/OOA/HLD). • Write a Project Proposal. • Write a User Requirements Document.						

	• Write a Requirements Analysis Document. • Produce a Project Plan. • Write a User Interface Specification for a GUI, API, etc. • Produce a design (e.g. UML/OOD/LLD). • Produce a prototype. • Write a Project Report on Analysis and Design. • Present the analysis, design and prototype to an audience.				
Main Content	• Requirements Engineering • Analysis modeling and documentation (e.g. UML/OOA) • Object-Oriented Analysis • Design Engineering • Design Concepts (e.g. UML/OOD) • Design Models and Documentation • User Interface Specification and Design • Prototyping • Project Presentation				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		1	Lectures p.w.		
		5	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Project Implementation and Testing 732		Module Code			COS732	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered	BSc Hons (Computer Science) (3735)					Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Understand and implement the Implementation and Testing activities of the Software Development Life-cycle. • Implement a project from design documentation (produce code). • Refine design documentation. • Devise evaluation criteria and test cases. • Test a program against a test-suite. • Write a Users Guide. • Write a Project Report on Implementation and Testing. • Present/Demonstrate the final program to an audience.						
Main Content	• Code Generation • Compilation • Testing Strategies • Testing Tactics • Debugging • Software Program • Software User Guide • Software Program						

	• Project Presentation			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		1	Lectures p.w.	
		5	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences			Home Department			Computer Science		
Module Name	ICT4D			Module Code			COS734		
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit		15	
Programme in which the module will be offered		BSc Hons (Computer Science) (3735)				Total Learning Time		150 hours	
Duration	Semester		Proposed Semester			Second Semester			
Main Learning Outcomes	On completion of this module, students should be able to: • Critically evaluate development and how technology fits in with development. • Describe the interactions between technologists, social scientists and communities by examining multiple case studies. • Understand multi-disciplinary and trans-sector ict4d research, research methods, evaluation and practice. • Engage with seminal and topical ict4d research outputs. • Partake in independent and group outputs through practicals, papers, presentations and workshops. • Demonstrate awareness of ethics, community engagement and challenges to sustainability particular to ict4d contexts.								
Main Content	The main course content is divided into three parts: Part 1: What is ICT4D? • Definitions of development, e.g. Techno-centric and social-centric; • Theory of development, e.g. Sen and capabilities theory; and also theory of technology, e.g. Amplification theory. • Ethics and intent Part 2: Domains • Community networks • Education • Disability • Health • Governance Part 3: Critique • Challenges, e.g. Barriers to access, ict4d failures • Gender and ict • Decolonisation and post-colonial thinking								
Pre-requisite	None			Prohibited module Combination			None		
Co-requisite	None								
Timetable Requirement per week				1		Lectures p.w.			
				4		Practicals p.w.			
				0		Tutorials p.w.			

Other teaching modes that does not require time-table							
iKamva discussions							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%
Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	An Introduction to the Internet of Things 735		Module Code			COS735	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Computer Science) (3735)			Total Learning Time	150 hours	
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Discuss the history, current state and evolution of the Internet-of-Things • Understand and describe the structure of the Internet-of-Things as well as different instantiations and implementations of the IoT architecture. • Identify and describe the impact of the IoT on Big data generation and applications. • Critically evaluate and describe the role and impact of Cloud/Fog Computing technologies in the IoT. • Analyze and describe key applications and use-cases of the Internet-of-Things. Implement a real-world IoT use-case focusing on an urban or rural IoT application by clearly analysing, specifying and developing the main building blocks and layers of the IoT architecture of the use-case.						
Main Content	After an overview of the IoT, the course content is centered around a four layer architecture that includes						
	i) IoT device layer ii) IoT networking layer iii) IoT adaptation layer and iv) IoT application layer. The main topics to be covered include • Overview of the Internet-of-Things (IoT) including ○ IoT evolution ○ IoT structure and architecture ○ IoT and Big Data ○ IoT and Cloud Computing • IoT sensing layer – Sensor, Actuator, /RFID, GPS technologies • IoT networking layer including IoT communication standards and protocols. • IoT adaptation layer including IoT-based fog/cloud storage and analytics. • IoT application layer with use-cases in smart cities and smart villages						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			0	Lectures p.w.		Block	
			0	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment		CA	100%	
Faculty	Natural Sciences		Home Department		Computer Science		
Module Name	Big Data Engineering 736		Module Code		COS736		
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Computer Science) (3735)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester		First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the history, current state and evolution of Big Data as well as its relation to the Interne-of-Things (IoT) and cloud computing technologies. • Evaluate the difference between data engineering and data science and how they complement each other. • Describe the main characteristics of Big Data including its volume, velocity, variety, veracity and value. • Identify and describe the impact of the IoT on Big data generation and applications. • Understand and describe the role and impact of Cloud/Fog Computing technologies in Data engineering. • Critically evaluate Big Data architectures and describe some of its ecosystems and data engineering tools. • Implement a real-world Big Data use-case focusing on an urban or rural application by clearly analysing, specifying and developing the main building blocks and layers of the Big Data architecture of the use-case and developing tools for its implementation.						
Main Content	The main topics to be covered include • Big Data Engineering vs Big Data Science • Big Data Landscape including ◦ Big Data sources ◦ Big Data characteristics ◦ Big Data Architectures • Big Data batch processing including Hadoop/MapReduce/Mahout • Big Data Stream Processing including Spark/Scala/MLIB • Big Data Ecosystems including tools and platforms • Big Data File Systems including HDFS • Big Data Applications and Use Cases.						
Pre-requisite	None		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment		CA	100%	

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Computer Science 737		Module Code			COS737	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered	BSc Hons (Computer Science) (3735), BSc Hons (Statistical Science) (3737)				Total Learning Time	150 hours	
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain and design an agent-based approach to problem-solving. • Implement and apply key Artificial Intelligence algorithms and techniques. • Discuss and apply logical agents and first-order logic.						
Main Content	• Constraint Satisfaction Problems • Logic Agents • First-order logic • Knowledge Representation • Swarm Intelligence • Fuzzy Logic • Reinforcement Learning						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Cybersecurity 738		Module Code			COS738	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Computer Science) (3735)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Describe and report on the history of cybersecurity, the current state cybersecurity and its evolution.						
	• Distinguish, evaluate and compare the different the algorithms, standards and protocols of cryptology.						
	• Critically evaluate various information security risks, discuss and demonstrate possible counter measures.						
	• Understand and describe the different aspect and concepts of digital forensics, and identify possible solutions.						
Main Content	• Discuss and distinguish the different legislation that governs cybersecurity in South Africa.						
	The main course content is divided into the following parts:						
	• Cryptography and Cryptology						
	• Algorithms, Standards and Protocols						
	• Digital Forensics						
• Information Security Risks							

	• Legislation and Governance • Cybersecurity Awareness • Malware (or Intrusion Detection) • Social Engineering • Ethics			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week	0	Lectures p.w.		
	0	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	4IR Special Topics 739		Module Code			COS739	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Computer Science) (3735)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First or Second Semester (depending on demand and staffing resources)	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate the in depth subject knowledge and critical skills pertaining to a topic in Computer Science that is not covered within our current offerings and is an extension to an existing Computer Science NQF level 8 module.						
Main Content	Special topic content as determined by the lecturer, visiting academic or industrial partner in consultation with the HoD.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.		Block	
			2	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Computer Science 740		Module Code			COS740	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered	BSc Hons (Computer Science) (3735)				Total Learning Time	150 hours	
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Describe and report on the history of cloud computing, the current state and emerging trends.						

	- Understand and describe the different aspects and concepts of cloud computing and its place in 4IR. - Distinguish, evaluate and compare the different cloud computing technologies, service and deployment models. - Critically evaluate and discuss the various cloud architectures and protocols. - Discuss and distinguish cloud applications and the provisioning of services within clouds. - Design and implement a project based on a cloud architecture, associated technologies, services and deployment models.			
Main Content	The main course content is divided into the following parts: - Deployment models - Technologies related to cloud computing - Cloud computing service models - Cloud computing Architectures - Cloud technologies - Cloud protocols - Emerging trends - Applications and service provisioning in clouds			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		0	Lectures p.w.	Block
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA
				100%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Computer Science 741		Module Code			COS741	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered	BSc Hons (Computer Science) (3735)				Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Evaluate and compare the importance of applied artificial intelligence given real-world scenarios. • Discuss the model considerations of different AI models, how to interpret their results, and the advantages and disadvantages of each model. • Use a rapid development environment to apply different machine learning and artificial intelligence models to a given case study, analyse the results and critically reflect on the effectiveness, impact and implications of the developed solutions. • Develop and demonstrate a practical AI solution that solves a real life data-intensive challenge by applying relevant AI models and following the steps of the AI development methodology. • Develop an AI model to solve a real-life data-intensive problem, implement the solution following a structured AI development						

	methodology, and critically evaluate its effectiveness in addressing the problem.							
Main Content	• Introduction to a business-driven AI development methodology (e.g. CRISP-DM), that includes problem understanding and data preparation. • Using a low-code/no-code environment and case studies to practically implement a basket of models. Case studies emphasize data requirements as well as model considerations, advantages and disadvantages and the impact on results. • The basket of models includes: ◦ Classification Models • Correlation Methods • Association Rules • Clustering ◦ Predictive Categorization • Linear Discriminant Analysis • k-Nearest Neighbors • Naïve Bayes ◦ Machine Learning Models • Regression (Linear and Logistic) • Decision Trees • Neural Networks ◦ Text Mining ◦ Generative AI (such as Large Language Models (LLMs)) • The evaluation and deployment of AI solutions (including aspects such as cross-validation and model optimization) • AI and Data Ethics							
Pre-requisite	CSC311, CSC312		Prohibited module Combination		STA701, STA737, STA738, STA739, STA760			
Co-requisite	None							
Timetable Requirement per week			0	Lectures p.w.	Block teaching			
			0	Practicals p.w.				
			0	Tutorials p.w.				
Other teaching modes that does not require time-table								
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment		CA	100%		
Faculty	Natural Sciences		Home Department		Computer Science			
Module Name	Computer Science 742		Module Code		COS742			
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15	
Programme in which the module will be offered		BSc Hons (Computer Science) (3735)			Total Learning Time		150 hours	
Duration	Semester		Proposed Semester		First Semester			
Main Learning Outcomes	• On completion of this module, students should be able to: • Critically reflect on and critique the history and current state of Immersive Technologies. • Investigate, evaluate and discuss Immersion and Stereoscopy techniques and the role of 3D vision in interactive immersive applications.							

	• Critically analyse, reflect on, and apply the interaction concepts of immersive systems, incorporating the latest perspectives and prospects in the immersive technology field. • Systematically apply their knowledge of 2D and 3D asset creation using appropriate tools and techniques. • Analyse and implement the transfer of created assets into a cross-platform software development suite. • Explain and implement the proper content development workflow. • Develop, create and deploy a basic interactive application within a software development suite.				
Main Content	• Introduction to the evolution of immersive hardware and software solutions. • Introduction to 2D and 3D asset development software applications. • Practical creation of 2D and 3D assets for use in interactive applications. • Good practice in creating assets for immersive systems. • Basic import, export, and optimisation of 2D and 3D assets. • Developing basic interactive applications. • Introduction and use of other forms of multimedia within created applications.				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		1	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Computer Science	
Module Name	Computer Science 743		Module Code			COS743	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Computer Science) (3735)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Be conversant with and implement code in the C# programming language. • Integrate C# code into a cross-platform application development suite. • Create interactive applications within a cross-platform application development suite. • Create basic mobile applications for mobile devices. • Evaluate, compare and implement different AR frameworks to create AR applications. • Implement 360-degree environments for immersive application development. • Create hand-tracking-enabled applications for optical sensor devices. • Develop augmented reality (AR) applications for mobile devices.						

	• Develop virtual reality (VR) applications for Head Mounted Display (HMD) devices. • Create integrative immersive applications.			
Main Content	• Introduction to C# programming language. • Programming in an application development suite. • Introduction to mobile application development. • Introduction to AR frameworks. • Introduction to AR application development. • Advanced mobile application development. • Introduction to VR application development. • Immersive application development.			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	Block
		1	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA
				100%

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Environmental Assessment and Management 701		Module Code			EWP701	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Environment and Water Science) (3780)			Total Learning Time		200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Apply the principles of environmental management and impact assessment. • Comply with, and evaluate environmental management and water laws, Environmental Impact Assessment (EIA) procedures and relevant institutions. • Coordinate EIA projects and write EIA areports as practitioner. • Compile specialist studies on water resources assessment, planning, allocation, management, protection and monitoring. • Review EIA reports and specialist studies as regulatory authority. • Prepare and implement environmental management plans.						
Main Content	• Introduction and principles of environmental management and impact assessment. • Overview key aspects of environmental management law, impact assessment regulations and water laws, and relevant institutions. • Overview of EIA process and procedures. • Methods and tools in environmental management and EIA (Environmental Risk Assessment, Strategic Environmental Assessment) with specific reference to water resources assessment, planning, allocation, management, protection and monitoring. • EIA reports and license application documents.						

	- Guidelines of specialist studies (biodiversity, socio-economic, visual and aesthetic, heritage) with particular focus on water aspects and hydrogeology. - Preparation and implementation of environmental management plans. - Case study of EIA with particular focus on impacts on water resources.			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week Taught in-house. No formal timetable requirement		0	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Advanced Geographical Information Systems 702		Module Code			EWP702	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Environment and Water Science) (3780)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate understanding and knowledge on the use of various GIS softwares (ArcGIS, IWLIS) and tools (programming in Rand Google Earth Engine) to view and manipulate spatial data. • Execute complex raster and vector GIS analysis in a GIS environment. • Understand the structure, advantages and limitations of raster datasets. • Engage with advanced forms of spatial analysis such as interpolation, network analysis, and various forms of spatial modelling. • Construct and manipulate vector and raster data sets. • Explore and solve spatial problems using GIS techniques and technology. • Apply specific GIS tools for generalisation and multi-criteria analysis. • Engage in the creation, management and delivery of spatial data in an online GIS environment. • Interpret and present the results of spatial data analysis. • Plan and execute a complex applied GIS project that would require students to demonstrate an understanding of spatial data acquisition, data quality checks, accuracies, and database development complexities, error management, as well as overall project execution.						
Main Content	• Structure of GIS softwares- ArcGIS, ILWIS. • Basic programming • GIS data (attribute and spatial data) • Spatial data acquisition and data analytics. • Spatial data query • Geocoding • Spatial data analysis • Multi-criteria/ overlay analysis						

	• Terrain analysis • GIS data management • Coordinate systems • Data editing and topology • Geodatabase • Metadata						
Pre-requisite	None		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%	
					FA	50%	
Faculty	Natural Sciences		Home Department		Earth Sciences		
Module Name	Groundwater Hydrology 703		Module Code		EWP703		
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Environment and Water Science) (3780)			Total Learning Time		200 hours
Duration	Semester		Proposed Semester		First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • Apply geophysical techniques to investigate groundwater problems. • Apply Darcy's law, groundwater flow equations; and pumping tests to investigate groundwater flow dynamics, groundwater flow dynamics and groundwater- surface water Interactions, taking cognisance of relevant assumptions and simplifications. • Develop conceptual, analytical and numerical models for catchment-scale modeling of groundwater resources. • Select and apply appropriate geochemical tools to establish the quality of water in the subsurface. • Apply principles and concepts of contaminant hydrogeology and mass transport processes in groundwater regimes including techniques/methods for site investigation for groundwater remediation and transport of contaminants. • Critically evaluate groundwater governance and institutional arrangements including frameworks for groundwater management.						
Main Content	• Groundwater geophysics: Techniques • Groundwater hydraulics/ flow dynamics • Groundwater modeling for with case studies • Geochemical processes of groundwater • Contaminant hydrogeology and mass transport processes for remediation • Estimation of groundwater recharge and surface and groundwater interaction • Theoretical measurements and application of isotopic techniques in groundwater • Groundwater management and governance • Field hydrogeology: Practical work activities						

Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week	2	Lectures p.w.		
	2	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences			Home Department			Earth Sciences		
Module Name	Surface Water Hydrology 704			Module Code			EWP704		
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20		
Programme in which the module will be offered		BSc Hons (Environment and Water Science) (3780)				Total Learning Time	200 hours		
Duration	Semester		Proposed Semester			Second Semester			
Main Learning Outcomes	On completion of this module, students should be able to: • Measure and monitor hydrological parameters using state-of-the art instruments and infrastructure. • Analyse hydrological data to understand hydrological processes in catchments and river channels. • Develop simple models of catchment runoff processes, and execute model calibration and validation procedures. • Apply hydrological models to investigate and resolve water resource management problems.								
Main Content	• Analysis of spatial and temporal variations of precipitation • Infiltration and soil water movement • Methods for measuring and estimating evapotranspiration rates • Runoff formation in different environmental settings • River flow measurement • Hydrograph analysis • Flooding and flood frequency analysis • Hydrological modelling • Introduction to the use of earth observation/remote sensing for hydrological assessments • Hydrological effects of land use changes • Surface water quality and water pollution • Reservoir yield estimation								
Pre-requisite	None		Prohibited module Combination			None			
Co-requisite	None								
Timetable Requirement per week			2	Lectures p.w.					
			2	Practicals p.w.					
			0	Tutorials p.w.					
Other teaching modes that does not require time-table									
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA		50%	
						FA		50%	

Faculty	Natural Sciences			Home Department			Earth Sciences		
Module Name	Research Project or Internship 705			Module Code			EWP705		
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	30		
Programme in which the module will be offered	BSc Hons (Environment and Water Science) (3780)					Total Learning Time	300 hours		
Duration	Year		Proposed Semester			Both Semesters			
Main Learning Outcomes	On completion of this module, students should be able to: EITHER: - Conduct research of limited scope on an Environmental and Water Science related project of his/her choice or recommended by his/her supervisor within the fields of expertise in the Department and compile a scientific report on the research. OR - Develop a full research proposal on a suitable topic that may enable him/her to continue directly with a Master's study on completion of the Honours program. OR - In the case of employed students or students on an internship, compile a report on a project they may be working on. - This report should be of equivalent status to the Honours project.								
Main Content	- Conducting a thorough literature survey - Identifying a research problem - Formulating a research methodology - Data collection - Data analysis - Reporting on the results in a report of 5000 – 7500 words								
Pre-requisite	None			Prohibited module Combination			None		
Co-requisite	None								
Timetable Requirement per week No Timetable Requirements				0	Lectures p.w.				
				0	Practicals p.w.				
				0	Tutorials p.w.				
Other teaching modes that does not require time-table									
Assessment Module Type	Continuous and Final Assessment (CFA)			Method of Student Assessment			CA	20%	
							FA	80%	

Faculty	Natural Sciences		Home Department			Earth Sciences	
Module Name	Anthropocene Rivers 707		Module Code			EWP707	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Environment and Water Science) (3780)			Total Learning Time		200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Evaluate physiographic controls on river and wetland characteristics and processes; • Classify river and wetland environments in southern Africa; • Use simple morphodynamic models to investigate and visualise river and wetland dynamics; • Diagnose, evaluate and monitor river and wetland ecosystem integrity;						

	- Quantify ecosystem service provision in river and wetland environments, and; - Design river and wetland restoration interventions that identify and address the causes of ecosystem degradation.				
Main Content	- Principles of fluvial geomorphology - River and wetland classification - Geomorphology and biogeomorphology of wetlands in drylands - Assessment of river and wetland ecological integrity - Assessment of river and wetland ecosystem services - River and wetland degradation and restoration				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week Taught in-house. No formal timetable requirement.		1	Lectures p.w.		
		1	Practicals p.w.		
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Ordinary Differential Equations 707		Module Code			MAM707	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Mathematical Science) (3736)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Solve ordinary differential equations analytically - Interpret the solutions of these ordinary differential equations.						
Main Content	- First Order Differential Equations: Direction fields, linear differential equations, orthogonal trajectories of curves, existence and uniqueness theorems. - Linear differential equations of second and higher order: Existence and uniqueness theory. - Solution by undetermined coefficients, solution by variation of parameters. - Systems of Differential equations: Phase plane, critical points, stability, Liapunov functions and Qualitative methods. - Series solution of differential equations and special functions: Power series methods, Frobenius method, Sturm-Liouville problems. Laplace transforms.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2		Lectures p.w.		
			0		Practicals p.w.		
			1		Tutorials p.w.		

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Number Theory 709		Module Code			MAM709	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered	BSc Hons (Mathematical Science) (3736)					Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Have a firm foundation in elementary number theory.						
Main Content	- Divisibility and Primes, - Modular Arithmetic, - Fundamental theorems of Modular Arithmetic, - Cryptography, Primality Testing, Applications, - Primitive Roots, Quadratic Congruences, Diophantine Equations, Quadratic Reciprocity Law, Elliptic Curves.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2		Lectures p.w.		
			1		Practicals p.w.		
			0		Tutorials p.w.		

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Graph Theory 710		Module Code			MAM710	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Mathematical Science) (3736)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First or Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Have a firm foundation in graph theory						
Main Content	- Definitions of Graphs - Connectivity - Characterization of Trees - Eulerian and Hamiltonian graphs - Planar graphs - Matching - Colorings						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week	2	Lectures p.w.		
	1	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Partial Differential Equations 717		Module Code			MAM717	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Mathematical Science) (3736)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Solve the PDEs analytically.						
Main Content	- First order PDE (Linear equations of the first order, Quasi-linear equations, Non-linear first order PDE); - Second order PDE (One-dimensional wave equation, vibrations of strings: Riemann's method, method of separation of variables. Laplace equation; - BVPS, Maximum and Minimum principles; Cauchy, Dirichlet and Neumann problems, Harnack's theorem, Green's function. Wave equation, Heat equation, Duhamel's principle. Kelvin's inversion theorem). Elements of distribution theory: weak solutions, Sobolev spaces.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Measure and Integration 727		Module Code			MAM727	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Mathematical Science) (3736)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Have a firm foundation in elementary measure theory.						

Main Content	• Measures, The Lebesgue Measure, Measurable Functions, Integration, Product Measures, Fubini's Theorem, The Radon-Nikodym Theorem, Differentiation, LP spaces.				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		2	Lectures p.w.		
		1	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment		CA	50%
				FA	50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Computational Linear Algebra 728		Module Code			MAM728	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Mathematical Science) (3736), BSc Hons (Computational Finance) (3739), BSc Hons (Computer Science) (3735)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First or Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - A strong foundation in computational methods in Numerical Linear Algebra. - Proficiency in the use of MATLAB for solving problems in Numerical Methods.						
Main Content	- INTRODUCTION TO MATLAB - MATRIX FACTORIZATIONS: LU-factorization, QR-factorization, Cholesky factorization. - MATRIX AND VECTOR NORMS: Properties of the $\ \cdot\ _1$, $\ \cdot\ _2$ and $\ \cdot\ _\infty$ norms of vectors in $\mathbb{R}^n$ Properties of the $\ \cdot\ _1$, $\ \cdot\ _2$, $\ \cdot\ _\infty$ and $\ \cdot\ _F$ norms of an $m \times n$ matrix. - ERROR ANALYSIS OF LINEAR SYSTEMS: The Condition Number of a Matrix. Ill-Conditioned systems. Perturbation Analysis of Linear Systems. - CURVE FITTING AND APPROXIMATION OF FUNCTIONS: Least-squares approximations of over-determined and under-determined systems. Least-squares approximation of continuous functions; The Hilbert Matrix. Piece-wise polynomial interpolation - ITERATIVE METHODS FOR LINEAR SYSTEMS: The Jacobi and Gauss-Seidel Methods. The SOR method. Applications to the solution of linear systems with banded coefficient matrices. - THE SINGULAR VALUE DECOMPOSITION (SVD) The singular value decomposition of an $m \times n$ matrix. The Moore-Penrose Inverse. Rank k approximation of a matrix.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week	2	Lectures p.w.		
	1	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	60%
			FA	40%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Mathematical Physics 731		Module Code			MAM731	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Mathematical Science) (3736)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • A good understanding of mathematics used in physics • An introduction to differential geometry, relativity, and relativistic quantum mechanics • Perform mathematical analysis and computations in differential geometry, relativity, and relativistic quantum mechanics • Skills in various aspects of mathematical physics and apply them to solve problems in differential geometry, relativity, and relativistic quantum mechanics.						
Main Content	• Vector and tensor calculus • Special functions and differential equations of physics Introduction to: • Differential geometry • Hamiltonian mechanics • Lie groups and Lie algebras • Relativistic quantum mechanics • Special Relativity and General Relativity Black hole spacetimes: • Schwarzschild, Kerr, and Kerr-Newman black holes						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2		Lectures p.w.		
			0		Practicals p.w.		
			1		Tutorials p.w.		
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	70%
						FA	30%

Faculty	Natural Sciences		Home Department	Mathematics and Applied Mathematics	
Module Name	Category Theory and Order Theory 732		Module Code	MAM732	
Module Type	Elective	NQF Level	8	Year Level	1
				NQF Credit	15

Programme in which the module will be offered		BSc Hons (Mathematical Science) (3736)		Total Learning Time		150 hours	
Duration		Semester		Proposed Semester		Second Semester	
Main Learning Outcomes		On completion of this module, students should be able to: • Identify different ordered structures and know their defining properties. • Use the point-free methods of frames and locales to extend those of general topology. • Express common mathematical constructions and properties in the language of categories, functors and natural transformations. • Use the methods and notation of order theory and category theory in mathematical proofs.					
Main Content		• Partially ordered sets, semi-lattices and lattices • Ideals and filters • Completeness • Monotone maps and Galois connections • Frames and locales • Categories and functors • Natural transformations • Adjoint functors • Reflective subcategories					
Pre-requisite		None		Prohibited module Combination		None	
Co-requisite		None					
Timetable Requirement per week				2		Lectures p.w.	
				1		Practicals p.w.	
				0		Tutorials p.w.	
Other teaching modes that does not require time-table							
Assessment Module Type		Continuous Assessment (CA)		Method of Student Assessment		CA	
						100%	

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Topology 737		Module Code			MAM737	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Mathematical Science) (3736)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Have a firm foundation in general topology						
Main Content	• Set Theory and Metric Spaces • Topological Spaces • Product • Subspaces and Quotients • Convergence • Separation and Countability • Compactness • Connectedness • The Fundamental Group.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week	2	Lectures p.w.
	1	Practicals p.w.
	0	Tutorials p.w.

Other teaching modes that does not require time-table

Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%
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Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics		
Module Name	Functions of a Complex Variable 738		Module Code			MAM738		
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15	
Programme in which the module will be offered		BSc Hons (Mathematical Science) (3736)				Total Learning Time	150 hours	
Duration	Year		Proposed Semester			Both Semesters		
Main Learning Outcomes	On completion of this module, students should be able to: • Have a Firm Foundation in Complex Function Theory							
Main Content	• Algebra and Geometric representation of complex numbers • Analytic functions • Elementary topology • Conformal mapping and Riemann Mapping Theorem • Linear transformations • Cauchy's Theorem and Integral Formula • Residue theory • Taylor and Laurent series • Harmonic functions • Elliptic functions and Weierstrass theory • Analytic continuation, Riemann surfaces and Picard's Theorem.							
	Pre-requisite		None			Prohibited module Combination		None
	Co-requisite		None					
	Timetable Requirement per week			2		Lectures p.w.		
				1		Practicals p.w.		
				0		Tutorials p.w.		

Other teaching modes that does not require time-table

Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Group Theory 739		Module Code			MAM739	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Mathematical Science) (3736)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Be well-prepared for more advanced studies in group theory.						

	• Apply their knowledge to give a complete classification of groups of a given order. • Apply their knowledge in areas where group theory is required.			
Main Content	• Isomorphic Theorems, Zassenhaus' lemma, Automorphism groups, Group Action, Sylow theorem, Solvable groups, Permutation Groups, Free Abelian Groups			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		2	Lectures p.w.	
		1	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Coding Theory 747		Module Code			MAM747	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Mathematical Science) (3736), BSc Hons (Computer Science) (3735)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Understand how mathematics, in particular, linear algebra, number theory, groups, finite fields, is applied in real life. • Understand how the error-detection and error-correction capabilities of codes are used in the process of data storage and data transmission between sources. • Understand the inter-relationship between the parameters of a code and its limitations. • Understand how the properties of selected known codes such as Reed-Solomon, BCH, Reed-Muller are used in practical situations.						
Main Content	• Error-detection, error-correction, encoding, decoding, dual codes, finite fields, BCH, Reed-Solomons, reed-Muller, burst errors, erasures, interleaving						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Project 748		Module Code			MAM748	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	30
Programmes in which the module will be offered		BSc Hons (Mathematical Science) (3736), BSc Hons (Computational Finance) (3739)				Total Learning Time	300 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Do independent research on a chosen topic. • Read relevant literature. • Write a report which displays clarity of understanding, logical reasoning, the correct use of language (including subject vocabulary).						
Main Content	• Students choose a given topic in consultation with the lecturer.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Rings and Modules 757		Module Code			MAM757	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Mathematical Science) (3736)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First or Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Well prepared for more advanced studies in rings and modules. • Well equipped to understand the intricate relationship between modules and their underlying rings. • Able to apply their knowledge in areas such as representation theory.						
Main Content	• Jacobson radical, Artinian rings, Schur's lemma, Projective modules, Free modules, Semi-simple rings, Wedderburn Thoeorem, Tensor products, Prime ideals and Nullstellensatz, Krüll's intersection Theorem, Localisation						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Functional Analysis 767		Module Code			MAM767	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Mathematical Science) (3736)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Have a firm foundation in functional analysis.						
Main Content	• Normed linear spaces and Banach spaces • bounded linear operators • Inner product spaces and Hilbert spaces • Baire category theorem • Open mapping theorem • Closed graph theorem • Dual spaces, reflexive spaces • Hahn-Banach theorem • Banach-Steinhaus theorem • Riesz representation theorem						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences		Home Department			Mathematics and Applied Mathematics	
Module Name	Numerical Analysis 777		Module Code			MAM777	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Mathematical Science) (3736)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First or Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Find numerical approximations to systems of linear equations, using both direct and iterative methods. • Find numerical approximation for eigen-values and eigenvectors. • Find polynomial approximation of functions, with error estimate. • Find numerical solutions to boundary-value problems for ODEs.						

	Find numerical solutions to parabolic and hyperbolic partial differential equations.			
Main Content	Perturbation Analysis of linear systems, matrix factorisation, Error estimates and Iterative Refinements, QR-algorithm, Hermite Interpolation, cubic spline interpolation, Gaussian quadrature, Least squares approximations, Finite-difference method for linear and non-linear problems, Rayleigh Ritz method, Parabolic and hyperbolic PDE's.			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		2	Lectures p.w.	
		1	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department				Medical Biosciences	
Module Name	Specialist Module in Epidemiology 701		Module Code				MBS701	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10	
Programme in which the module will be offered		BSc Hons (Medical Bioscience) (3721)				Total Learning Time	100 hours	
Duration	Year		Proposed Semester				Both Semesters	
Main Learning Outcomes	On completion of the module students should be able to: - Understand the criteria commonly used to evaluate causal relationships. - Evaluate the quality and comparability of data. - Understand the major study designs. - Define exposure variables, outcome variables, extraneous variables and measures of their frequency. - Understand and calculate commonly used health measures, such as relative risk, attributable risk, and odds ratio; select appropriate methods for estimating such measures. - Define appropriate comparison groups for epidemiologic studies. - Interpret descriptive and inferential statistics resulting from data analysis and draw relevant conclusions. - Apply the concepts of confounding and bias to describe variables; describe appropriate methods for addressing each. - Critique the study design and quantitative methods used in published literature and appropriately interpret the findings. - Identify key sources of epidemiologic data. - Formulate and apply epidemiologic methodology to identify a specific research problem, develop a hypothesis, and design a study to investigate the issue. - Understand key issues in Research Ethics.							
Main Content	- Basics of Epidemiology and Research Ethics for biomedical sciences. - Critique and presentation of original article. - Presenting the scientific report.							
Pre-requisite	None		Prohibited module Combination				None	
Co-requisite	None							

Timetable Requirement per week		4		Lectures p.w.			
		0		Practicals p.w.			
		0		Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%
Faculty	Natural Sciences		Home Department		Medical Biosciences		
Module Name	Advanced Technique Studies 702		Module Code		MBS702		
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered	BSc Hons (Medical Bioscience) (3721)				Total Learning Time	200 hours	
Duration	Year		Proposed Semester		Both Semesters		
Main Learning Outcomes	On completion of this module, students should be able to: The students will develop competence in various hands-on laboratory techniques, information management and generation of data						
Main Content	- The student will have to make a selection of a minimum number of departmentally approved techniques to constitute this module. - The selection of techniques will be determined by the availability of lecturers and by the specific techniques required by the student's research project. - Examples of techniques that may be available: - Basic laboratory calculations - Laboratory etiquette and protocols - General lab behaviour - Working in a group - General lab rules - Project management, etc. - Microscopy - Enzyme linked immunosorbent assays (ELISA's) - Bioinformatics Electrophysiological techniques - Molecular biology techniquesPolymerase Chain Reaction (PCR) - Gel electrophoresis (SDS-PAGE) - Western blots - Dot blots - Chromatography - Other useful assays - Bradford assay - Cytotoxicity assay						
Pre-requisite	None		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week	10		Lectures p.w.				
	0		Practicals p.w.				
	0		Tutorials p.w.				
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Research Project 704		Module Code			MBS704	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	50
Programme in which the module will be offered	BSc Hons (Medical Bioscience) (3721)			Total Learning Time		500 hours	
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: The student will demonstrate competency in the planning, execution and presentation of a departmentally approved research project.						
Main Content	- Selection of and planning of a research project - Budget - Protocol writing - Execution of research - Generation of data - Project management - Quantification and interpretation of scientific data - Writing of a scientific report on the research project - Presenting project (e.g. PowerPoint) – public lecture						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No time table requirement			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Specialist Module in Anatomical Science 705		Module Code			MBS705	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Medical Bioscience) (3721)			Total Learning Time		100 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate a sound knowledge of theoretical aspect regarding morphological, histological and molecular techniques currently used for research in basic science. • Demonstrate competence in various hands-on laboratory techniques and dissection techniques. • Perform laboratory calculations, analyse data, and interpret scientific literature. • Write and present a scientific report with proper referencing. • Demonstrate competence in planning, execution and presentation of a dissection assignment.						
Main Content	• The module provides an introduction to molecular techniques used in scientific research, • In addition, it offers basic training on writing and publishing scientific reports.						

	• Comprehensive lectures and dissection sessions will be introduced to enable graduates to specialise as anatomists.			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week No time table requirement		3	Lectures p.w.	
		3	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Microbiology of Female Morbidity 707		Module Code			MBS707	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered	BSc Hons (Medical Bioscience) (3721)					Total Learning Time	100 hours
Duration	Term		Proposed Semester			Fourth Term	
Main Learning Outcomes	On completion of this module, students should be able to: - Explain how women's anatomy predisposes them to certain infections. - Identify the role of women as childbearers, caregivers, users and managers of the local environment. - Explain and evaluate how the above roles impact on women's susceptibility to disease.						
Main Content	- Latrogenic causes of female morbidity and mortality (hospital acquired infections, side effects from drug/antibiotic over-use, malpractice). - Environmental and occupational diseases (diseases due to air pollution and pesticides, health risks from migrant labour, waterborne infections) - The microbial aetiology of female genito-urinary tract infections (cystitis, toxic shock syndrome, sexually transmissible diseases, septic abortion. - Infections transmitted from mother to infant (those causing life-threatening pre-term and/or low weight births - Infections that challenge the fetus causing birth defects. - Effects of hormone shifts on women's oral health (oral manifestations during menstruation; oral contraceptives and pregnancy; estrogen supplements during menopause).						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			3	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
No time table requirement							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Microbiology of Female Morbidity 709		Module Code			MBS709	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Medical Bioscience) (3721)			Total Learning Time		100 hours
Duration	Term		Proposed Semester			First Term	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate a sound knowledge of cell growth control. • Interpret scientific literature regarding the cell cycle and the role of cell signalling in the control aspects of the cell cycle. • Compare and contrast different types of cell death, including apoptosis and autophagy and the induction of apoptosis ny various mechanisms. • Appraise the role of tumorigenesis.						
Main Content	• The Cell Cycle • Checkpoints in the cell cycle • Cyclin-dependent kinases • Inhibitors of cyclin-dependent kinase • Cell Death • Necrosis, Apoptosis, atrophy and terminal • Differentiation, genetic control of programmed cell death • Apoptosis and disease						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No time table requirement			10	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences			Home Department			Medical Biosciences	
Module Name	Foetal Origins of Adult Disease 712			Module Code			MBS712	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10	
Programme in which the module will be offered		BSc Hons (Medical Bioscience) (3721)				Total Learning Time	100 hours	
Duration	Semester		Proposed Semester			First or Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to: • The learner will understand the importance of normal development during foetal life and the consequences of in utero compromise (e.g. due to lifestyle) on the future health of the individual							
Main Content	• The “foetal origins hypothesis”. • The role of the placenta in foetal programming: Maternal nutrition and foetal programming. Placental weight and long-term outcomes. Foetal adaptations and developmental plasticity. The placenta and foetal cardiovascular adaptations. The placenta and foetal body composition. The placenta and foetal endocrinology and metabolism. Maternal influences on the placenta							

	• Placental insufficiency and inheritance vs. maternal malnutrition. Impact of life style e.g. alcohol abuse, smoking, socio-economic factors.				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		0	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Advanced Studies in Reproduction and Health 713		Module Code			MBS713	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Medical Bioscience) (3721)			Total Learning Time		100 hours
Duration	Term		Proposed Semester			Third or fourth Term	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate a sound knowledge of basic reproductive anatomy and physiology. • Interpret the concepts of reproductive health. • Demonstrate, in terms of infertility, the role of reproductive habits and disease. • Appraise the assisted reproductive strategies and implications there off. • Recognize the importance of various strategies in terms of family planning and future directions.						
Main Content	• Structural and functional considerations of the male and the female reproductive systems • The concept of 'Reproductive Health' • The problem of infertility in females and males • Infertility and disease/HIV/endometriosis • Assisted reproduction: Advances and problems • Family planning and contraception						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No time table requirement			5	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department		Medical Biosciences		
Module Name	Applied Immunology 714		Module Code		MBS714		
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10

Programme in which the module will be offered		BSc Hons (Medical Bioscience) (3721)		Total Learning Time	100 hours	
Duration	Semester	Proposed Semester		First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: - Describe and compare the different signalling pathways involved in immune responses - Explain and discuss the role of the immune system in fighting intracellular and extracellular pathogens and cancers - Use their knowledge of the impacts of pharmaceuticals, pollutants, natural products, and viral proteins on immune defences. - Design and analyse experiments to monitor the activation of immune pathways by different stimulants.					
Main Content	- The Immune Pathways Phagocytosis: - Extracellular Pathogen associated molecular pattern receptor (PAMP) pathways. - Intracellular pathogen recognition receptors - Interferon pathways - Complement pathways - Inflammation - Humoral Immunity - Cell Mediated Immunity - Immune defences against: - Extracellular pathogens: specifically, bacteria - Intracellular pathogens: specifically, viruses - Cancers - The effects of immune pathway modulation on immune defences. - Hands-on laboratory experience in immunology techniques such as antibody production, ELISA, Western Blotting, lymphocyte stimulation assays, and reporter assays to measure immune signaling.					
Pre-requisite	None	Prohibited module Combination		None		
Co-requisite	None					
Timetable Requirement per week		5	Lectures p.w.			
		0	Practicals p.w.			
		0	Tutorials p.w.			
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%
					FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Applied Exercise Science 715		Module Code			MBS715	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered		BSc Hons (Medical Bioscience) (3721)				Total Learning Time	100 hours
Duration	Semester		Proposed Semester			First or Second Semester	
Main Learning Outcomes	On completion of this module, students will be able to: • Explain the physiological response to exercise. • Explain physiological adaptations occurring from various types of training.						

	• Explain intrinsic and extrinsic factors affecting the above. • Explain the different exercise testing regimes.			
Main Content	• Physiological response to exercise at sea level. • Physiological adaptations to strength and endurance training and its regulation (including diet and doping). • Physiological response and adaptations to different environments (warm, cold, deep sea, altitude and spaceflight). • Nature vs. Nurture - what makes an elite athlete? • Exercise tests			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		10	Lectures p.w.	Presented in a block over 4 weeks
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Nutrition in Health and Disease 716		Module Code			MBS716	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	10
Programme in which the module will be offered	BSc Hons (Medical Bioscience) (3721)					Total Learning Time	100 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Evaluate how vital macro- and micronutrients drive metabolic pathways and physiological function important for human health - Critically evaluate and predict the risk factors, causes, and health consequences of inadequate and excessive nutrient intake - Analyse the relationship between nutrient intake and poor health including the mechanism(s) of action - Explore and critique the role of nutrition in addressing diseases on a population, group or individual scale.						
Main Content	- Role of macro-and micronutrients in homeostasis and human health - Role of nutrient intake and nutritional status, at different life stages, in the development of diseases (e.g. Diabetes, Cardiovascular disease, Hypertension) - Risk factors, causes, and health consequences of inadequate and excessive nutrient intake - Role of nutrition in approaches to prevent and or treat common diseases or health conditions - Nutrient-related topics from current scientific literature						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No timetable requirement			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%
Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 709		Module Code			PHY709	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	30
Programme in which the module will be offered		BSc Hons (Physical Science) (3714)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Be exposed to the world of telecommunication. • Be given a chance to use their physics skills in that field. • Be able to understand and apply the knowledge gained in all areas of signals and telecommunications. Experimental laboratory: students will • Be introduced to a research environment. • Be able to run a specific experiment or measurement linked to one of the running research programmes in the department. • Be able to use advanced equipment, to run an unknown measurement or analysis, to connect the information with published data and present it in a research report.						
Main Content	Communication and electronics: • Detectors and Electronics: A detailed discussion of the mathematics of signals, including Fourier transform methods • An introduction to the field of data transmission • A discussion of computers, computer networks etc. • Data – its coding compression and transmission • A discussion of the Internet, Broadband networks • Some practical work related to the above Experimental laboratory: • One selected experiment in advance experimental analysis from the following fields: • Experimental Nuclear Physics, • Physics Education, • Materials Science, • Semiconductors, • Solid State Physics, • Computational Physics, • Photovoltaics, • Superconductors • Electron Microscopy						
	Pre-requisite	None		Prohibited module Combination			None
Co-requisite	None						
Timetable Requirement per week			0	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table								
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%	
Faculty	Natural Sciences		Home Department			Physics and Astronomy		
Module Name	Physics 720		Module Code			PHY720		
Module Type	Core/ Elective	NQF Level	8	Year Level	1	NQF Credit	30	
Programmes in which the module will be offered		BSc Hons (Physical Science) (3714), BSc Hons (Computer Science) (3735)				Total Learning Time	300 hours	
Duration	Semester		Proposed Semester			First Semester		
Main Learning Outcomes	On completion of this module, students should be able to: Quantum Physics: • Apply the general ideas of Quantum Mechanics in various fields of modern physics and chemistry. Mathematical Methods: • Apply the required advanced mathematical methods to solve physical problems. Electromagnetism: • Apply the general ideas of EM theory in various fields of physics. Computational physics: • Write and develop computer codes for calculation of physical data analysis, and generate computer simulations of physical models.							
Main Content	Quantum Physics: • Foundations of Classical Mechanics, Lagrange and Hamilton formalism. • Schrödinger equation. • Quantum amplitudes and state vectors. • Perturbation theory. • Particle scattering. • Angular momentum, spin. Math Methods: A selection of mathematical functions applied in physics such as: • Fourier transforms, matrices, • Legendre functions, Bessel functions, Eigen functions, • Green's functions, • Calculus of Variations, Tensor analysis. • Binomial, Poisson and Gaussian distributions. Electromagnetism: • Electromagnetism: Potential formulations of electrodynamics • Gauge transformations • EM waves in conducting and non-conducting media • Electromagnetic radiation. Computation Physics: • Spreadsheet physics. • Numerical analysis using Fortran 77, • ordinary differential equation, • boundary values and eigenvalue problems, • Gaussian quadrature.							

	- matrix operations, Monte Carlo methods - Numerical analysis using packages such as Mathematica				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		1	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	25%	
			FA	75%	

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 721		Module Code			PHY721	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered		BSc Hons (Physical Science) (3714)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Solve qualitative and quantitative problems associated with accelerators. • Solve and fix problems relate to accelerators and accelerator applications. • Explain the various types of accelerators and how each of them operates.						
Main Content	• Overview of Accelerators • Maxwell equations • AC optics • Beam Diagnostics • Linear Accelerators • Beam Optics • Ion Sources • RF Systems • Vacuum System • Injection • Extraction • Application of Accelerators						
	Pre-requisite	None	Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			4	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department		Physics and Astronomy		
Module Name	Physics 722		Module Code		PHY722		
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20

Programme in which the module will be offered		BSc Hons (Physical Science) (3714)		Total Learning Time	200 hours	
Duration	Semester		Proposed Semester		First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the relationship between the generalised principles of Thermodynamics and the detailed postulates of Statistical Mechanics. • Implement advanced quantum concepts and approximations to complex systems.					
Main Content	• Thermodynamic probability • Equation of state for a perfect gas • Legendre Transforms • Bose-Einstein and Fermi-Dirac distributions • Two Electron Systems • Scattering • WKB and Born Approximations • Transition Probabilities • X-ray Emission and Absorption Spectra					
Pre-requisite	None		Prohibited module Combination		None	
Co-requisite	None					
Timetable Requirement per week			3	Lectures p.w.		
			1	Practicals p.w.		
			0	Tutorials p.w.		
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%
					FA	50%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 723		Module Code			PHY723	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	30
Programme in which the module will be offered	BSc Hons (Physical Science) (3714)				Total Learning Time		300 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	This is an introduction to the theoretical and experimental aspects of nuclear physics. On completion of this module, students should be able to: • Understand and apply the nuclear physics knowledge gained in nuclear physics application areas, such as research, isotope manufacturing or medical applications. • Apply ethical issues related to nuclear physics. • Do some nuclear experiments. • Apply the theoretical knowledge of nuclear physics.						
Main Content	• Introduction to Nuclear Physics: Nuclear Models: Shell model, Collective Model and experimental methods • Scattering theory • Experimental research at the National Accelerator Centre and elsewhere • Applied nuclear physics Radiation protection, Medical applications of isotopes • The use of radioactivity Nuclear issues in society • Selected topics in bulk nuclear properties. nucleon structure						

	• The nuclear force and two-nucleon system • Nuclear structure models • Nuclear decay and excitation, nuclear reactions • Applied nuclear physics • Social impacts of nuclear physics. • Applied Nuclear Physics: Reactor Physics • Advance Topics in a selected field (eg: environmental radioactivity, Radon, etc)			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	PHY720			
Timetable Requirement per week		3	Lectures p.w.	
		1	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences			Home Department		Physics and Astronomy	
Module Name	Physics 724			Module Code		PHY724	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered	BSc Hons (Physical Science) (3714)					Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Solid State Physics: • Apply the basics of modern theory of crystals. • Apply experimental methods to study crystals and know the important electronic properties of metals and semiconductors. • Do literature searches in related subjects and to know the main research tools in the field. Statistical Physics: • Be able to apply the basics of statistical mechanics. • Know various applications in kinetic theory of gases, phonon and electron systems in solids. • Be able to apply the basis of thermal electromagnetic radiation. • Be able to explain the general ideas of entropy, statistical distribution functions. • Be able to use the principles in various research fields. Semiconductor Physics: • Be able to apply the basic concepts of semiconductor theory and electronic characteristics. • Be able to apply the physics concepts of amorphous material and amorphous semiconductors. • Be able to understand the electronic processes occurring in semiconductor material and its relation to the characteristics of semiconductors.						
Main Content	Solid State Physics: • Crystal structures, Bravais lattices, elements of crystal symmetry • Reciprocal lattice and X-ray diffraction • Phonons and inelastic neutron scattering						

	• Electron energy bands in metals and semiconductors. • Free-electron model. Statistical Physics: • Statistical mechanics: • Combinatorics, distribution function, partition function, entropy. • Equipartition of energy. • Ideal gas. • Paramagnetics. • Phonons. Photons and black-body radiation. • Planck formula. • Fermi-Dirac statistics of conducting electrons. Semiconductor Physics: • Crystalline and amorphous semiconductors, • doping, band theory. • Electronic transport, mobility, resistivity. • Recombination. • Poisson equation, Ohm's law, continuity equation. • P-N Junctions. • Application: photovoltaic solar cell			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		1	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	40%
			FA	60%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 725		Module Code			PHY725	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programme in which the module will be offered	BSc Hons (Physical Science) (3714)					Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Materials Science: • Apply advanced concepts in Materials Science and engineering. • Apply the theoretical understanding to industrial applications.						
Main Content	Material Science: • Diffusion mechanisms. • Phase diagrams. • Mechanical properties of solids • Metals • Polymers and composites • Nanmaterials • Advance topics (eg Magnetic materials, superconductors, photovoltaics, etc)						
Pre-requisite	PHY724		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week	3	Lectures p.w.		
	0	Practicals p.w.		
	1	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	40%
			FA	60%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics 726		Module Code			PHY726	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	30
Programmes in which the module will be offered		BSc Hons (Physical Science) (3714), BSc Hons (Computer Science) (3735)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	Photon based techniques: - Obtain and analyse spectra from Rutherford backscattering spectroscopy. - Obtain and analyse spectra from Elastic Recoil Detection Analysis. - Obtain and analyse spectra from PIXIE spectroscopy. Photon based: - Explain the optical and electronic reaction of materials when exposed to electromagnetic waves in the range from ultra violet to infrared. - Know what material characteristics are determined from these reactions. - Understand the processes and theory of absorption, reflection and transmission of EM waves in dielectric materials. - Obtain and analyse X-ray diffraction spectra. - Obtain and analyse Raman and Infrared spectra. Electronmicroscopy: - Know the theoretical concepts of electron diffraction and electron interaction with matter. - Apply it to the use of transmission and scanning electron microscopy. - Use the scanning and transmission electron microscopes. - Extract all the information contained in the images and spectra obtained using these instruments. Mechanical / electrical techniques: - Set up the equipment and obtain and analyse the results from STM, AFM, 4-point probes, Hall effect and profilometry.						
Main Content	Ion Based Techniques: - Rutherford Back Scattering spectroscopy RBS - Elastic recoil detection analysis ERDA - PIXIE						
	Photon based techniques (XRD, Raman, Infrared, UV Vis, laser techniques): - Lattice vibrations, Specific heat theory. - Debye model phonons, lattice waves, dispersion. - Semiconductor p-n junction and diode equation.						

	- Macroscopic optical properties, Lorentz oscillator. - Absorption, dispersion and application dielectric theory. - Infrared spectrometry Electron based: Electron Microscopy: - Electron interaction with matter, - Basics of SEM, detectors, - Resolution, magnification, depth of focus, depth of field. - Basics of EDS analysis, detectors, qualitative and quantitative analysis. - Basics of TEM, electron diffraction, imaging. - Sample preparation techniques. Mechanical/Electrical based - Scanning Tunneling Microscopy STM - Atomic Force Microscopy AFM - Hall effect measurements - Profilometry			
Pre-requisite	PHY724	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		1	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	40%
			FA	60%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Studies 703		Module Code			POP703	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Population Studies) (3738)			Total Learning Time		150 hours
Duration	Term		Proposed Semester			First Term	
Main Learning Outcomes	On completion of this module, students should be able to: - Appreciate the extent to which official statistics dominate the lives of citizens on a daily basis. - Describe the processes of data collection, processing, analysis, and dissemination by Statistics South Africa. - Learn some important calculations techniques used by Statistics South Africa that are not taught in other courses in the Department of Statistics.						
Main Content	History of Official Statistics; Why should governments gather statistics; The future of statistics; Census 1996 and Census 2001; Vital statistics in South Africa; Labor Force and General Household Surveys in South Africa; Economic and Industrial Surveys; Population projections; Index numbers; Development and poverty.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Studies 706			Module Code		POP706	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Population Studies) (3738), BSc Hons (Statistical Science) (3737)				Total Learning Time	150 hours
Duration	Term		Proposed Semester			First Term	
Main Learning Outcomes	On completion of this module, students should be able to: • Interpret and explain surveys and census. • Draw questionnaires. • Conduct a demographic survey.						
Main Content	• Data collection processes • How to conduct a survey • Design of questionnaire • Practical case						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Studies 707		Module Code			POP707	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Population Studies) (3738), BSc Hons (Statistical Science) (3737)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Calculate a rate, Calculate proportion and ratios, Draw a “Lexis’Diagram”. • Analyse mortality. • Analyse any demographic event. • Read and construct any demographic event tables. • Know how to prepare and analyse data. • Know how to evaluate quality of secondary data. • Know how to respond to any demographic event. • Prepare policy statements on any demographic issue.						

Main Content	• Introduction to demography; Calculation of rate, ratios and proportions, Lexis' Diagram • Demographic analysis of mortality • Direct and indirect methods of analysis • Deep analysis of any demographic events • Transversal and longitudinal analysis • Advanced demographic analysis of mortality • Analysis of families and parity ratios; Analysis of fertility and fecundity • Overview of migrations issues					
	Pre-requisite	None	Prohibited module Combination		None	
	Co-requisite	None				
	Timetable Requirement per week		3	Lectures p.w.		
			0	Practicals p.w.		
			0	Tutorials p.w.		
	Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%		
			FA	50%		

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Studies 708			Module Code		POP708	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Population Studies) (3738)				Total Learning Time	150 hours
Duration	Term		Proposed Semester			Fourth Term	
Main Learning Outcomes	On completion of this module, students should be able to: • Know and interpret how to analyse gender issues. • Know how to analyse contraception's questions. • Know how to analyse maternal and infant issues. • Know how to analyse reproductive health data. • How to provide policies on those questions.						
Main Content	• Direct analysis of gender equity • Direct and indirect analysis of maternal and child mortality and morbidity • Direct analysis of contraception • Critical analysis of data on reproductive health • Use of available data (surveys) • Use of demographic and health surveys (DHS)						
Pre-requisite	POP707 or equivalent knowledge			Prohibited module Combination		None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)			Method of Student Assessment		CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Studies 709		Module Code			POP709	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Population Studies) (3738)			Total Learning Time		150 hours
Duration	Term		Proposed Semester			Fourth Term	
Main Learning Outcomes	On completion of this module, students should be able to: - Examine linkages between demographic and social issues from an analytical perspective. - Up-dated examination of contemporary debates on the relationships between population and employment, education, elderly and social exclusion variables. - In-depth knowledge of theoretical perspectives on the social dimensions of demographic variables such as fertility, mortality and migration.						
Main Content	- Background and importance of social demography - Structural approach to social issues from a demographic perspective (age structures and education, employment, elderly) - The demography of socially excluded - Analytical approaches to social demography (persistence of high fertility, health and demographic transition, migration and urbanisation)						
Pre-requisite	POP707 or equivalent knowledge		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3		Lectures p.w.		
			0		Practicals p.w.		
			0		Tutorials p.w.		
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Studies 710		Module Code			POP710	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Population Studies) (3738), BSc Hons (Statistical Science) (3737)			Total Learning Time		150 hours
Duration	Term		Proposed Semester			Third Term	
Main Learning Outcomes	On completion of this module, students should be able to: • Use mathematics to analyse demographic events from discrete to continuous forms. • Propose models of integrating population variables into development planning. • Use population models for prospective demographic analysis.						
Main Content	• Instantaneous mortality function. Expression for expectancy of life. Analytical expression for mortality functions. Nuptiality function. • Fertility functions. Selected applications. Instantaneous crude birth and death rates.						

	• Population dynamics functions (exponential, stable, Malthusian, stationary) and applications. Population modeling and development planning. Applications to sectoral development.			
Pre-requisite	POP707 and some mathematical background in differential and integral calculus.	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Research Project (Population Studies) 761		Module Code			POP761	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	30
Programme in which the module will be offered		BSc Hons (Population Studies) (3738)			Total Learning Time		300 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: - Develop a short research proposal from a research question in the field of Population Studies or Demography. - Conduct a literature review on current research trends in the field of study. - Perform the relevant data analyses required to address the research question at the Honours level. - Report the research outcomes in a format suitable for publication including critical evaluations of the data's significance and contextual relevance.						
Main Content	- Development of a research proposal under the guidance of a research supervisor(s); - Exposure to research in the field of Population Studies or Demography including the completion of a research project report under the guidance of the supervisor(s).						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No Timetable requirements. Ad hoc workshops by arrangement.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 501		Module Code			STA501	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Population Studies) (3738), BSc Hons (Medical Bioscience) (3721)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: At the end of the module be able to critically apply data analysis techniques.						
Main Content	- Exploratory Data Analysis and Graphical Methods, - Probability Theory, - Statistical Models and the Normal Distribution, - Sampling Distributions, - Hypothesis Testing, - Categorical Data Analysis, - Analysis of Variance, - Correlation and Regression Analysis.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			1	Lectures p.w.			
			1	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
No Timetable requirements. Ad hoc workshops by arrangement.							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 701		Module Code			STA701	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Statistical Science) (3737)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Analyse a multivariate data set using an appropriate statistical technique required by the research hypothesis and execute various techniques to check for the assumption of normality.						
Main Content	Data summaries and graphical techniques, analysis and covariance structures, inferences about multivariate and linear models, multivariate normality, discrimination and classification						
Pre-requisite	STA331; STA332; (STA739) or equivalent; knowledge of SAS			Prohibited module Combination		None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment	FA	100%

Faculty	Natural Sciences		Home Department			Statistics	
Module Name	Statistics 702		Module Code			STA702	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Computational Finance) (3739)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Formulate research questions and a problem statement, conduct • a broad scan literature review, design a research methodology • produce a working proposal for research • implement a rudimentary research project • Conduct a quantitative analysis with SPSS • Communicate their research findings in a scientific manner						
Main Content	• Conceptual foundations of research, Theory in quantitative research, problem statement and hypotheses formulation, research design and measurement, validity and reliability of the research instruments, methods of observations, data processing and analysis, interpretive framework, research report writing up, ethics in scientific research, introductory qualitative research, SPSS applications to quantitative data analysis						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	STA501						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment	CA 50% FA 50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 705		Module Code			STA705	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Statistical Science) (3737), BSc Hons (Computational Finance) (3739)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Apply knowledge of statistical inference.						
Main Content	• Consistency of estimators; method of moments and maximum likelihood estimation; the delta method; confidence intervals; power and relative efficiency; likelihood ratio tests						
Pre-requisite	STA331; STA332; MAT211: or equivalent			Prohibited module Combination		None	

Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		2	Practicals p.w.	
		4	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Experimental Design 710		Module Code			STA710	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Statistical Science) (3737)			Total Learning Time		150 hours
Duration	Term		Proposed Semester			First Term	
Main Learning Outcomes	On completion of this module, students should be able to: • Apply knowledge of principles in experimental design. Acquisition of advanced skills in statistical analysis of data from experiments (chemistry, physics, industry, environment, biochemistry, medical science, pharmacy, education). Examination of different techniques of experimental design such as completely randomised design, randomised complete block design, latin square design, split-plot design, means separation. Exposure to practical examples drawn from different disciplines where such techniques are widely applied. Use of Chi-square in experimental data. Use of analysis of variance in assessing the robustness and fitness in regression analysis.						
Main Content	• Introduction and principles in experimental design • General considerations one one-way and two-way analysis of variance • The completely randomised design • The randomised complete block design • Mean separation methods • Latin square design; Split plot design • Incomplete block designs						
Pre-requisite	STA331; STA332 or equivalent		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Biostatistics 733		Module Code			STA733	
Module Type	Core/ Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Population Studies) (3738), BSc Hons (Statistical Science) (3737)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Explain various terms used in Bio statistical analyses and apply various statistical techniques applicable to medical related data.						
Main Content	Study design in medical related research; exploratory medical research including risk ratio's, odds ratio's, mortality and morbidity rates; Inferential medical research including non-parametric analyses, survival analysis and meta-analysis; regression analysis; and the analyses of a Bio statistical dataset using a statistical package						
Pre-requisite	STA331; STA332 or equivalent		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 736		Module Code			STA736	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered	BSc Hons (Statistical Science) (3737)					Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Apply introductory knowledge on the development of statistical genetics. Practical application and interpretation of linkage analysis, variance components methods and association analysis.						
Main Content	• Introduction to relevant aspects of human genetics; Mendelian inheritance, • Linkage analysis, • Association analysis of dichotomous and quantitative traits for case-control and family based study designs, • Introduction to software for human genetical analyses.						
Pre-requisite	STA701, STA733 and COF712		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week	3	Lectures p.w.		
	0	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistical Modelling 737		Module Code			STA737	
Module Type	Core/ Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Statistical Science) (3737)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Understand concepts of statistical modelling. • Compare variable selection techniques. • Apply and test linear and non-linear modeling. • Evaluate and select classification methods. • Understand sampling methods. • Assess statistical models.						
Main Content	• Prepare data for modelling (missing data imputation, transformations, resampling, parsing) • Explore variable selection techniques • Data modelling and assessment						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			1	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 738		Module Code			STA738	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered		BSc Hons (Statistical Science) (3737)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First or Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Prepare various data structures for analysis on various cloud enabled platforms:						

	• Apply contemporary statistical analytics and statistical learning techniques using cloud-based platforms; • Apply specialized programming skills needed for cloud analytics; • Use a variety of analytical software packages in the cloud environment.				
Main Content	• Data preparation for analytics in the big data environment will be taught. • Supervised and unsupervised statistical learning techniques will be used in a cloud-based environment. • Various cloud-based platforms will be explored to prepare students for statistical analysis in the big data environment. • Skills will be developed for managing and scheduling analysis processes in the cloud environment.				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		2	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment		FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 739		Module Code			STA739	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	15
Programme in which the module will be offered	BSc Hons (Statistical Science) (3737)					Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Apply and understand matrix algebra for multivariate analysis; • Characterise multivariate data; • Perform and interpret principal component analysis; • Calculate and interpret discriminant analysis results; • Apply the taught theory to data problems using SAS and/or RStudio.						
Main Content	Topics in matrix algebra such as vector and matrix theory, linear transformations and systems of linear transformations will be introduced to understand multivariate analysis with specific emphasis on principal component analysis and discriminant analysis.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 760		Module Code			STA760	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Statistical Science) (3737), BSc Hons (Computer Science) (3735)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Design and execute various Data Mining techniques, from extraction of huge data repositories to modeling of the data. Techniques covered would include: • Decision tree analyses, clustering, visual data analyses, regression and classification.						
Main Content	• Data management and processing; • Statistical inference techniques explored, Classification, association and clustering, post-processing of discovered structures.						
Pre-requisite	STA331; STA332 or equivalent		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment		FA		100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Honors Project 761		Module Code			STA761	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	30
Programmes in which the module will be offered	BSc Hons (Statistical Science) (3737), BSc Hons (Computational Finance) (3739)					Total Learning Time	300 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Make a small contribution with limited scope to knowledge in the field of Statistics, Population Studies and Demography or Computational Finance and its interaction with another field or discipline, following upon a research question with the potential to make such a satisfactory contribution. • Carry out and report on this research in a format suitable for publication. • Concomitant outcomes will normally include personal skills development supporting the required research and writing activities under the guidance of the supervisor(s).						
Main Content	• The primary task is to develop the research proposal towards its envisaged goal(s), conducting the research and preparing the research						

	project under the guidance of the supervisor(s), before submitting it for assessment; Research proposals may derive from any area of Statistics, Population Studies and Demography or Computational Finance in interaction with another field or discipline, and may include substantial historical and/or comparative studies			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		0	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment	FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Studies 790		Module Code			STA790	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		BSc Hons (Population Studies) (3738), BSc Hons (Statistical Science) (3737)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Use the appropriate projective methods for the estimation of future growth of population by age, urban/rural population, households, labour force and school enrolments. • Incorporate past demographic trends in the elaboration of projective outputs. • Relate the projective methods learned in class with those applied in the world reports produced by international organizations such as UNFPA, UN, WHO. • Use software to produce projective computations.						
Main Content	• Scope and data for population projections, projections by sex and age, projection of urban and rural population, projection of labour force, projection of households for local planning, projection of school level enrolments and projection by level of education, selected topics, computerized-assisted applications.						
Pre-requisite	POP707 or equivalent only		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%	
					FA	50%	

Faculty	Natural Sciences		Home Department			Institute for Water Studies, Department of Earth Sciences	
Module Name	IWRM in Practice 731		Module Code			WAT731	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programmes in which the module will be offered		PGDip (IWRM) (3880)			Total Learning Time		200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Conduct an original review and critique of the application of IWRM in a water resources management project. • Evaluate the extent to which the principles of IWRM have been applied in practice (provide a well-reasoned reflective write-up). • Integrate an understanding of various aspects of water resources management in evaluating management practice in the water sector. • Write a well-reasoned, clear, coherent and professionally formatted report on the application of IWRM in practice.						
Main Content	• The National Water Strategy, Catchment Management Agencies and Management Plans. • Tools in IWRM (Resource-Directed Measures, water conservation and demand management strategies, communication, public participation, water pricing, water economics). • Water demand and use: Water use license applications. • Culture and organisations: Values, norms, practices, perceptions • Stakeholder engagement perspective of organisations. • Power and decision making in organisations: roles and responsibilities. • Leadership challenges in complex organisations.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No lecture venue required			0	Lectures p.w.		Block teaching by the Department	
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Institute for Water Studies, Department of Earth Sciences	
Module Name	Water Resources Management 732		Module Code			WAT732	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programmes in which the module will be offered		PGDip (IWRM) (3880)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	- Describe how the water resources system works (processes, principles, interactions, functions, components, approaches) - Explain water resources planning, implementation and information						

	- Assess various methods for protecting water resources - Explain and conduct water resources assessment and monitoring - Evaluate principles and approaches for water demand and use - Evaluate water resources allocation, authorisation and licensing - Assess water management institutions and water governance				
Main Content	- Water resources system including hydrological cycle - Water resources information, planning, and implementation - Water resources protection, water policy implementation - Water resources assessment and monitoring - Water demand and use in various sectors - Water resources allocation, authorization and licensing - Water management institutions and water governance				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week No lecture venue required	0	Lectures p.w.	Block teaching by the Department		
	0	Practicals p.w.			
	0	Tutorials p.w.			
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment		CA	50%
				FA	50%

Faculty	Natural Sciences		Home Department			Institute for Water Studies, Department of Earth Sciences	
Module Name	Spatial Analysis for IWRM 733		Module Code			WAT733	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programmes in which the module will be offered		PGDip (IWRM) (3880)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Conduct a fully-integrated spatial analysis to support water management in a selected study area, and to use the spatial analysis products to describe and illustrate the physiographic setting of the area, and to identify and characterise spatial aspects of water resources management. • Apply GIS and remote sensing concepts and analysis tools to investigate spatial aspects of surface water processes, groundwater processes, geomorphology, and land cover / use. • Develop GIS-based models of Earth surface processes and spatial decision-making processes.						
Main Content	• Introductory GIS and remote sensing tutorials, on data capture, management and storage, preparation, analysis, and display. • Terrain analysis for analytical hydrology / geomorphology models. • Remote sensing; spectral indices for vegetation, surface water, groundwater, and fire mapping. • Analytical modelling of physical processes. • Modelling the environmental decision making process (suitability mapping).						

Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week No lecture venue required	0	Lectures p.w.	Block teaching by the Department		
	0	Practicals p.w.			
	0	Tutorials p.w.			
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences			Home Department		Institute for Water Studies, Department of Earth Sciences	
Module Name	Environmental Management and Water Regulations 734			Module Code		WAT734	
Module Type	Core	NQF Level	8	Year Level	1	NQF Credit	20
Programmes in which the module will be offered		PGDip (IWRM) (3880)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain management tools for managing natural resources. • Assess management practices for natural resources using appropriate techniques/approaches. • Describe environmental governance including frameworks, policies, regulations and institutions. • Evaluate various methods for prevention and remediation of environmental problems. • Explain the relations between water and the constitution. • Explain the role of legal instruments in protecting and using water. • Evaluate water management strategies and enforcement mechanisms using legal tools including criminal and civil regulations.						
Main Content	• Environmental management tools and management practices. • Environmental impact assessment and environmental governance. • Methods for preventing and remediating water pollution. • Water and the constitution for water protection and use. • Water management strategies and enforcement mechanisms.						
Pre-requisite	None			Prohibited module Combination		None	
Co-requisite	None						
Timetable Requirement per week No lecture venue required			0	Lectures p.w.		Block teaching by the Department	
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)			Method of Student Assessment		CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Institute for Water Studies, Department of Earth Sciences	
Module Name	Water Quality Management 735		Module Code			WAT735	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	20
Programmes in which the module will be offered		PGDip (IWRM) (3880)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Present a comprehensive conceptual understanding of water quality, including the major processes responsible for governing and modifying water quality. • Plan and conduct water quality analyses based on project objectives, and implement data quality assessment and data quality control procedures. • Apply expertise in field and laboratory data interpretation to quantify the water quality of surface and groundwater. • Develop a scientific report evaluating and interpreting the results of water quality analyses.						
Main Content	• Theoretical background for surface and groundwater quality. • Chemical, physical and biological processes responsible for water quality variation. • Water quality classification. • Designing sampling strategies, setting up field equipment, field data collection, and analyses. • QA and QC Reporting, data interpretation, and report writing.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Institute for Water Studies, Department of Earth Sciences	
Module Name	Aquatic Ecosystems Management 736		Module Code			WAT736	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	20
Programmes in which the module will be offered		PGDip (IWRM) (3880)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Distinguish between and classify different types of river and wetland ecosystem. • Describe and evaluate fundamental hydrogeomorphological controls on the dynamics of rivers and wetlands.						

	Undertake basic assessments of ecological integrity and ecosystem service provision for a river or wetland in South Africa, and develop a management plan based on these assessments.						
Main Content	- Ecosystem classification systems theory, and classification systems for South African river and wetland types. - Applied fluvial geomorphology; river and wetland dynamics in dryland environments, the hydrological and geomorphological basis of river and wetland ecological integrity. - Bioindicators of ecological integrity. - Ecological integrity assessment approaches for rivers and wetlands. - Ecosystem services in river and wetland environments; description, measurement and management.						
Pre-requisite	None	Prohibited module Combination		None			
Co-requisite	None						
Timetable Requirement per week		0	Lectures p.w.				
		0	Practicals p.w.				
		0	Tutorials p.w.				
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%	
					FA	50%	
Faculty	Natural Sciences		Home Department		Institute for Water Studies, Department of Earth Sciences		
Module Name	Environmental Education and Water Security 737		Module Code		WAT737		
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	20
Programmes in which the module will be offered		PGDip (IWRM) (3880)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester		Second Semester		
Main Learning Outcomes	On completion of this module, students should be able to: - Explain the philosophical thinking, principles, assumptions, and practices of environmental education and water education in an academic setup and in a community setup. - Demonstrate application of the philosophical thinking, principles, and practices of environmental education in a water education and awareness project. - Explain the link between water and security, water and health, and water and development and water and environment. - Conduct a critical review of the processes and context of international, national and local agendas for achieving the UN Sustainable Development Goals using IWRM principles. - Evaluate various mechanisms for ensuring peaceful management and utilization of transboundary water resources for development. - Apply principles of SMART and SWOT analysis in designing, implementing, motoring, evaluating, and managing water projects. - Develop a proposal for a 'water and development', 'water in development' or 'water for development' project.						

Main Content	• Environmental and water education and community [Awareness and participation] • Water and development • Water for health and water for food • Water and Sustainable Development Goals • Peaceful management and utilization of transboundary water resources for the development • Research proposals and projects in the water sector using IWRM principles and assumptions at various scales • Philosophy, principles and practice of environmental education. • Environmental education and the community: public participation, sustainable living. • Water security, human security and development. • Water in development including water and gender. • Water for health and water for food. • Rural water and water for cities: Water access related issues.				
	Pre-requisite	None		Prohibited module Combination	None
	Co-requisite	None			
	Timetable Requirement per week No formal timetable requirement. Taught in-house.			0	Lectures p.w.
				0	Practicals p.w.
				0	Tutorials p.w.
	Other teaching modes that does not require time-table				
	Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
				FA	50%

Faculty	Natural Sciences		Home Department			Institute for Water Studies, Department of Earth Sciences	
Module Name	Water Demand Management for Sustainable Development 738		Module Code			WAT738	
Module Type	Elective	NQF Level	8	Year Level	1	NQF Credit	20
Programmes in which the module will be offered		PGDip (IWRM) (3880)				Total Learning Time	200 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Engage in water demand management initiatives at municipal or provincial level using IWRM principles. • Identify and define inter-relationships between socio-political processes, technical processes, and hydrological processes in water demand management. • Conduct a basic assessment of the feasibility of water demand management strategies, interventions, options, innovations, and technologies in different settings (urban, rural, agricultural, and mine water demand management) and select appropriate methods/. tools/measures for different settings. • Apply legal and regulatory frameworks to address challenges in the water demand management sector.						

	Understand principles, concepts indicators, and targets of Sustainable Goals including techniques/methods to measure progress toward achieving such goals.					
Main Content	- Water conservation and water demand management: concept and principles, assumption, theoretical underpinning, and application in different contexts/settings - Urban and rural water demand management - Agriculture and mine water demand managementUtilization/allocation/economics - Roles and measures and planning consideration of water demand management - Programs and projects of WDM - Policy and institutional context of WDM - Legal and regulatory frameworks of WDM - Sustainable Development Goals					
	Pre-requisite	None	Prohibited module Combination		None	
	Co-requisite	None				
	Timetable Requirement per week		0	Lectures p.w.		
	No formal timetable requirement. Taught in-house.		0	Practicals p.w.		
		0	Tutorials p.w.			
Other teaching modes that does not require time-table						
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA	50%
					FA	50%

Faculty	Natural Sciences		Home Department		Mathematics - AIMS		
Module Name	Mini-Thesis (Research Project) 803		Module Code		AIM803		
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	60
Programme in which the module will be offered		MSc (Mathematical Science) (3849)			Total Learning Time	600 hours	
Duration	Semester		Proposed Semester		First Semester		
Main Learning Outcomes	On completion of this module, students should be able to:						
	- Produce a scientific report under the supervision of an expert supervisor in the mathematical/ physical sciences. - Critically review and interpret scientific or mathematical material. - Demonstrate the ability to perform experiments - on a computer or otherwise - and report the results.						
Main Content	Mini-thesis (Research Project)						
Pre-requisite	None		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week		0		Lectures p.w.			
No Timetable Requirements.		0		Practicals p.w.			
		0		Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment		FA	100%	

Faculty	Natural Sciences		Home Department			Mathematics - AIMS	
Module Name	Communication Skills, Computing and LaTeX 811		Module Code			AIM811	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	7
Programme in which the module will be offered		MSc (Mathematical Science) (3849)			Total Learning Time		70 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate the ability to communicate their work, both verbally and through writing. • Be able to write and present objectively. • Understand and know the importance of effective science report writing and referencing. • Understand the aims computing facilities; good security practice; acceptable use policies. • Understand ubuntu desktop and google apps; netiquette. • Work efficiently learning latex, texmaker, and graphics programs for typesetting scientific reports, command line and software development skills. • Understand professional and ethical issues in the scientific environment.						
Main Content	• This course introduces the students to the essentials of English communication, writing and presentation in a scientific environment, and also to the AIMS computing facilities and packages. • The course covers an introduction to LaTeX using texmaker, followed by working through the book: http://en.wikibooks.org/wiki/LaTeX, and associated documents such as the American Mathematical Society. Time is spent on scientific writing and presentation. General queries about the Ubuntu desktop, Google Apps, and using various science software, interpreters, or compilers via a simple text editor and terminal, a language specific interpreter, and optionally a full front-end for some languages (notebook for SAGE, rstudio for R) are dealt with. • The course also covers how to install Ubuntu and maintain one's own computer with Free Software scientific applications added, in a similar way the AIMS desktops are managed and updated. It explores the Ubuntu ecosystem, release schedules, security, software sources. If time allows: installing from source in various programming languages. We cover using cloud services for backups and syncing preferences, plus recovering a lost password. We end with a brief look at possible futures of technology.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No Timetable Requirements.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Mathematics - AIMS	
Module Name	Mathematical Problem Solving 812		Module Code			AIM812	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	7
Programme in which the module will be offered		MSc (Mathematical Science) (3849)			Total Learning Time		70 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Understand and gain practical experience of different approaches to problem solving and research.						
Main Content	The consideration of a variety of elementary, but challenging problems in different branches of pure mathematics. Investigations, comparisons of different methods of attack, literature searches, solutions and generalisations of the problems will arise in discussions in class.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No Timetable Requirements.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
No Timetable Requirements							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Mathematics - AIMS	
Module Name	Problem Solving in Physics 813		Module Code			AIM813	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	7
Programme in which the module will be offered		MSc (Mathematical Science) (3849)			Total Learning Time		70 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Understand that Physics describes the real world using the language of mathematics. - Gain practical experience of different approaches to problem solving and research.						
Main Content	Problem solving techniques in a variety of physics areas. This includes using different reference frames, estimating orders of magnitude, dimensional analysis, and numerical approaches will be used. Examples will be taken from physics of moving objects electrodynamics, gravity, movement of molecules in gases and elementary particle physics.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No Timetable Requirements.			0		Lectures p.w.		
			0		Practicals p.w.		
			0		Tutorials p.w.		

Other teaching modes that does not require time-table									
Assessment Module Type	Continuous Assessment (CA)			Method of Student Assessment			CA	100%	
Faculty	Natural Sciences			Home Department			Mathematics - AIMS		
Module Name	Scientific Software Development in Python 814			Module Code			AIM814		
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	7		
Programme in which the module will be offered		MSc (Mathematical Science) (3849)				Total Learning Time	70 hours		
Duration	Semester		Proposed Semester			First Semester			
Main Learning Outcomes	On completion of this module, students should be able to: Demonstrate the ability to craft simple computational models.								
Main Content	The students will be presented the fundamentals of imperative programming, that is, sequential, conditional and iterative flow structures. This will allow students to solve simple problems by means of the computer. Modularity will be introduced as the tool that will allow students to tackle larger problems by decomposing them into smaller ones. Once this basis is well established, several higher level concepts will be introduced such as the use of files, strings, and an overview of the scipy library.								
Pre-requisite	None			Prohibited module Combination			None		
Co-requisite	None								
Timetable Requirement per week No Timetable Requirements.				0	Lectures p.w.				
				0	Practicals p.w.				
				0	Tutorials p.w.				
Other teaching modes that does not require time-table									
Assessment Module Type	Continuous Assessment (CA)			Method of Student Assessment			CA	100%	

Faculty	Natural Sciences			Home Department			Mathematics - AIMS		
Module Name	Experimental Mathematics with SAGE 815			Module Code			AIM815		
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit		7	
Programme in which the module will be offered		MSc (Mathematical Science) (3849)				Total Learning Time		70 hours	
Duration	Semester		Proposed Semester			First Semester			
Main Learning Outcomes	On completion of this module, students should be able to: Understand a short contextual history of computing have gained an introduction to the sage interface and programming skills needed for working in the scientific environment.								
Main Content	- Emphasis is placed on how one would explore and analyse mathematical problems involving computation and the tools and methods needed for doing this quickly. - Problems of various kinds from analysis, number theory and linear algebra are investigated.								

Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week No Timetable Requirements.		0	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Mathematics - AIMS	
Module Name	Selected Topics in the Mathematical Sciences 816		Module Code			AIM816	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	7
Programme in which the module will be offered		MSc (Mathematical Science) (3849)			Total Learning Time		70 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Gain an understanding of a mathematical/ statistical area which builds on and complements the problem solving and computational courses. • This course is designed to lead on to the advanced topics in mathematical sciences.						
Main Content	• Areas covered include concepts in analysis, geometry, discrete mathematics and statistics						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No Timetable Requirements.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Mathematics - AIMS	
Module Name	Professional Development and Entrepreneurism 817		Module Code			AIM817	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	7
Programme in which the module will be offered		MSc (Mathematical Science) (3849)			Total Learning Time		70 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Be introduced to selected employability, business entrepreneurship and presentation skills.						
Main Content	These skills include a selection from: Employability: CV writing, business communications, presentations, interviewing, career search strategies, leadership and negotiation.						

	Business skills: - Building basic business skills such as financial analysis, management and leadership, problem analysis, with on-going focus on communication and presentation skills. Entrepreneurial skills: - Developing project management, business planning, budgeting, marketing and presentation skills based on a relevant country/development issue.				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week No Timetable Requirements.		0	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Mathematics - AIMS	
Module Name	Advanced Topics in Mathematics Science A 818		Module Code			AIM818	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	77
Programme in which the module will be offered		MSc (Mathematical Science) (3849)			Total Learning Time		770 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate an independent, inquiring approach to contemporary mathematics. • Demonstrate the ability to embark on advanced graduate work in a cutting edge area of mathematics. • Understand the steps in and evaluate the suitability of modelling physical systems. • Demonstrate the ability to employ abstract structures and forms of reasoning to understand novel mathematical behaviour.						
Main Content	• Selected topics from: Probability Theory and Statistics, Differential equations, Number Theory, Topology, Geometry, Algorithms, Applied Complex Analysis, Computer Algebra, Numerical Analysis, Algebraic Graph Theory.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No Timetable Requirements.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Mathematics - AIMS	
Module Name	Advanced Topics in Mathematics Science B 819		Module Code			AIM819	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	77
Programme in which the module will be offered		MSc (Mathematical Science) (3849)			Total Learning Time		770 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Understand the steps in an independent, inquiring approach to contemporary Physics. - Demonstrate the ability to embark on advanced graduate work in a cutting edge area of Physics. - Understand the steps in and evaluate the suitability of modelling physical systems. - Demonstrate the ability to apply mathematical methods.						
Main Content	Selected topics from: Electromagnetism, Relativity, Quantum mechanics, Quantum computing, Quantum field theory, Non-linear waves, Topology and Geometry, Robotics, Statistical inference, Complex systems, Modelling discrete and continuous systems						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No Timetable Requirements.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department			Mathematics - AIMS	
Module Name	Advanced Topics in the Mathematical Sciences C 820		Module Code			AIM820	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	77
Programme in which the module will be offered		MSc (Mathematical Science) (3849)			Total Learning Time		770 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate an independent, inquiring approach to Applied and Computational, Mathematics and its applications. • Show evidence of a proficiency in constructing mathematical models. • Demonstrate the ability to embark on advanced graduate work in a cutting edge area of Applied and Computational Mathematics and its applications. • Demonstrate the ability to employ appropriate computational methods to understand mathematical behaviour.						
Main Content	Selected topics from: • Probability Theory and Statistics, Algorithms, Wave equations, Applied Complex Analysis, Computer Algebra, Mathematical modelling of						

	natural systems, Signal processing, Numerical Analysis, Industrial Mathematics, Statistical Mechanics				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week No Timetable Requirements.		0	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Mathematics - AIMS	
Module Name	Advanced Topics in the Mathematical Sciences D 821		Module Code			AIM821	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	77
Programme in which the module will be offered		MSc (Mathematical Science) (3849)				Total Learning Time	770 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate an independent, inquiring approach to Statistics, Mathematics and its applications. • Show evidence of a proficiency in constructing models of, and analysing, discrete data. • Demonstrate the ability to embark on advanced graduate work in a cutting-edge area of Statistics, Mathematics and its applications. • Demonstrate the ability to employ appropriate computational techniques.						
Main Content	Selected topics from: • Probability Theory and Statistics, Differential Equations, Algorithms, Applied Complex Analysis, Mathematical modelling of natural systems, Signal Processing, Numerical Analysis, Industrial Mathematics, Statistical Mechanics						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No Timetable Requirements.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department		Mathematics - AIMS	
Module Name	Advanced Topics in the Mathematical Sciences E 822		Module Code		AIM822	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit 77

Programme in which the module will be offered		MSc (Mathematical Science) (3849)		Total Learning Time	770 hours
Duration	Semester	Proposed Semester		First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Demonstrate an independent, inquiring approach to Computer Science, Mathematics and its applications. • Show evidence of a proficiency in designing and implementing algorithms for the efficient solution of computational problems. • Demonstrate the ability to embark on advanced graduate work in a cutting edge area of Statistics, Mathematics and its applications. • Demonstrate the ability to apply mathematical methods.				
Main Content	Selected topics from: • Probability Theory and Statistics, Number Theory, Geometry, Algorithms, Applied Complex Analysis, Computer Algebra, Numerical Analysis, Algebraic Graph Theory, Robotics				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week No Timetable Requirements.		0	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Computational Finance 817		Module Code			COF817	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		MSc (Statistical Science) (3838), MSc (Computational Finance) (3093)			Total Learning Time		150 hours
Duration	Semester	Proposed Semester			First Semester		
Main Learning Outcomes	Students should be able to critically develop the ability to devise statistical simulation algorithms and Matlab skills						
Main Content	Introduction to Matlab; generating random numbers from different distributions; cellular automata; systems of linear equations; modelling time series; simulating queues; bootstrapping.						
Pre-requisite	STA211; STA221; MAT211; MAT221		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			1	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Computational Finance 818		Module Code			COF818	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MSc (Computational Finance) (3093)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Summarise and critically evaluate the international monetary system, foreign exchange markets, measurement of risk and return, foreign currency risk management.						
Main Content	International monetary system, foreign exchange markets, measurement of risk and return, foreign currency risk management						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			4	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Computational Finance 821		Module Code			COF821	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MSc (Computational Finance) (3093)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Develop software for financial applications; • Present data structures and algorithms; • Critically evaluate results.						
Main Content	• Programs for financial application; • Data structures and algorithms. • Project work.						
Pre-requisite	STA331 or equivalent		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			4	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Computational Finance 826		Module Code			COF826	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		MPhil (Population Studies) (3921), MSc (Statistical Science) (3838), MSc (Computational Finance) (3093)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Apply confident use of statistical inference techniques						
Main Content	• Consistency of estimators; method of moments and maximum likelihood estimation; the delta method; confidence intervals; power and relative efficiency; likelihood ratio tests.						
Pre-requisite	STA331; STA332; MAT211; MAT221 or equivalent		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			4	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Computational Finance 827		Module Code			COF827	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		MSc (Computational Finance) (3093), MSc (Statistical Science) (3838)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Critically summarise knowledge and analyse risk management problems.						
Main Content	Risk measurement, analytical methods of risk management, value-at-risk methodologies, risk modelling.						
Pre-requisite	STA331 or equivalent		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			4	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Computational Finance 828		Module Code			COF828	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		MSc (Statistical Science) (3838), MSc (Computational Finance) (3093)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Critically apply the laws of large numbers: the Weak Law of Large Numbers, the Central Limit Theorem. - Generate random observations from specified distributions, e.g. the inverse transform method, acceptance-rejection method.						
Main Content	- Review advanced concepts in Probability. Random variables, distributions, cumulative distribution functions, densities. Generating moments of functions: cumulants, probabilities, factorial moments. - Basic inequalities, e.g. Tchebychev inequality. Approximate methods, the Delta method. Introduction to Stochastic Processes. The Poisson process. Branching processes. Introduction to Markov Chains.						
Pre-requisite	STA331 or equivalent		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			4	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Computational Finance 829		Module Code			COF829	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		MSc (Statistical Science) (3838), MSc (Computational Finance) (3093)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Critically apply a broad overview of time series methods and develop advanced analysis skills.						
Main Content	- Classical time series methods; strong and weak stationarity, and non-stationarity; ARMA models – acf, pacf, stationarity and invertibility; evaluation of residuals; ARIMA models; seasonality in ARIMA models; estimation and prediction; - Spectral density functions; periodogram; Example analyses using SPSS						

Pre-requisite	STA211; STA221; MAT211; MAT221 or equivalent		Prohibited module Combination	None	
Co-requisite	None				
Timetable Requirement per week	3	Lectures p.w.			
	2	Practicals p.w.			
	4	Tutorials p.w.			
Other teaching modes that does not require time-table					
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment	FA	100%

Faculty	Natural Sciences		Home Department			Statistics	
Module Name	Computational Finance 832		Module Code			COF832	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MSc (Computational Finance) (3093)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Critically evaluate standard approaches to statistical arbitrage including market neutral strategies such as pair trading, value-based or contrarian methods, momentum-based strategies, cointegration-based trading, and technical analysis.						
Main Content	Standard approaches to statistical arbitrage including marked neural strategies such as pairs trading, value-based and contrarian methods, momentum-based strategies, cointegration-based trading, and technical analysis.						
Pre-requisite	MAN781; MAN756 or equivalent.		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			4	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics	
Module Name	Computational Finance 835		Module Code			COF835	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MSc (Computational Finance) (3093)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Critically evaluate structural and intensity based or reduced form approach to the value of assets, pricing models.						
Main Content	Structural and intensity based or reduced form approach to the value of assets, pricing models						

Pre-requisite	STA331 or equivalent	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		2	Practicals p.w.		
		4	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment		FA	100%

Faculty	Natural Sciences		Home Department			Statistics	
Module Name	Computational Finance 836		Module Code			COF836	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MSc (Computational Finance) (3093)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Critically: Apply regression analysis and linear models to equity portfolio management: Simple and multiple linear regression, model selection, residual analysis, diagnostics, detection of multi-collinearity, nonstandard conditions and transformations, principal components and factor analysis.						
Main Content	Regression analysis and linear models with application to equity portfolio management: simple and multiple linear regression, model selection, residual analysis, diagnostics, detection of multi-collinearity, nonstandard conditions and transformations, principal components and factor analysis						
Pre-requisite	STA331 or equivalent		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			2	Practicals p.w.			
			4	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Physics and Astronomy/ Chemistry/ Medical Biosciences/ Biotechnology/ School of Pharmacy	
Module Name	Nanoscience Research Project 803/804		Module Code			NSS/ NSB/ NSP 803/804	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	100
Programme in which the module will be offered		MSc (Nanoscience) (3089)				Total Learning Time	1000 hours
Duration	Year		Proposed Semester			Both Semesters	

Main Learning Outcomes	On completion of this module, students should be able to: • Design, perform and report independent research work on nanobiomedical science, nanochemistry or nanophysics. • Use specialised nanoscience equipment to perform specific research work. • Use new or standard nanoscience techniques and procedures in performing the research work.				
Main Content	• Research project topics in biomedical sciences, chemistry or physics				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week No Timetable Requirements.		0	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment		FA	100%

Faculty	Natural Sciences		Home Department			Chemistry	
Module Name	Central Concepts in Nanoscience 809		Module Code			NSS809	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	4
Programme in which the module will be offered		MSc (Nanoscience) (3089)			Total Learning Time		40 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the structural basis of nanoscience. • Classify nanomaterials. • Identify journals and other sources of information on nanomaterials. • Discuss applications of nanomaterials in chemistry, physics and nanomedicine and the basis of such applications.						
Main Content	Introduction: • What is nanoscience? History, nature and development of nanoscience. Properties of nanomaterials. Types of nanomaterials - nanotubes, nanorod, nanoparticles, etc. Different fields of nanoscience. Interdisciplinary aspects. Practical matters: • Sources of information and nanoscience literature. Principles and techniques of nanoscience - analytical approaches, synthetic approaches, research methodology, experimental protocols. Applications to nanoscience: • Devices and nanotechnology, South African landscape, current devices and possibilities. Mathematics for nanoscience: • Modelling and computation. Nanoscience and society: • Ethics, safety, toxicology, regulation and standards. This will include environmental protection, public awareness, government policy and monitoring, networks.						

	Fundamentals of bionanotechnology: - Physiological, cellular, biological properties of nanobiosystems; nanoscience and systems for in-vivo and in-vitro procedures. Nanomedicine: - Techniques for nanomedicines which will include drug delivery systems, medical devices, regenerative medicine, biosensors and diagnostic systems; biological, physical and chemical characterization of nanosystems; in-vitro and in-vivo applications of nanobiosensor devices.				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		2	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%	
			FA	50%	

Faculty	Natural Sciences		Home Department			Chemistry	
Module Name	Management for Nanoscientists 810		Module Code			NSS810	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	40
Programme in which the module will be offered		MSc (Nanoscience) (3089)			Total Learning Time		40 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain entrepreneurial process and its components. • Explain the fundamental concepts, characteristics and criteria for successful small business development and entrepreneurial process. • Discuss fundamental principles and procedures of successful project management. • Discuss the requirements for managing new product development and innovation ventures.						
Main Content	Entrepreneurship • Business concepts and models: services, innovative products and business growth. • Recognizing, assessing and testing opportunity. • The entrepreneurial management process. • Identification, protection and management of Intellectual property. • Patents Small Business Management • Principles and fundamental concepts of small business development and management; including pricing, credit, personal selling, advertising, sales promotion and product marketing and human resources management. • Small business and e-commerce: planning, financing, marketing plan, operational plan and factors affecting the growth of small business. Project management • Project characteristics.						

	• Project life cycle: initiation, planning, execution and closeout. • Managing new product development. • Principles of project management: application of knowledge; standards and regulations; understanding the project environment; management skills; interpersonal skills; influence; leadership; motivation; negotiation; and problem solving skills. • Managing innovation. • Networks for projects.				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		2	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences		Home Department			Biotechnology/ School of Pharmacy/ Medical Biosciences	
Module Name	Advanced Nanobiomedical Science 811		Module Code			NSS811	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	48
Programme in which the module will be offered		MSc (Nanoscience) (3089)				Total Learning Time	480 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • The purpose is to provide students with courses in fundamental nano-biomedical sciences. • It consists of two major topics: Fundamentals of bionanotechnology and the study of nanomedicine.						
Main Content	This module includes an in depth study of: Fundamentals of bionanotechnology • Nanoparticles (NP) • Synthesis, assembling and encapsulation of NP; isolation/ purification and characterization of NP • Biomedical applications of nanomaterial devices • Diagnostics (colorimetric, fluorometric, biosensors, surface Plasmon resonance), Therapeutics, Imaging • Physiological response to nanomaterials • Systematic and cellular immunological responses to different nanomaterials • Development and linking the biological component to the nanomaterials Nanomedicine • Disease Molecular origin of diseases, complexity of diseases (primary vs secondary manifestations), Disease diagnostics based on molecular changes Using nanomaterials for diagnostics						

Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		7	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences		Home Department			Biotechnology/ Medical Biosciences	
Module Name	Experimental Techniques in Nanobiomedical Sciences 812		Module Code			NSS812	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	16
Programme in which the module will be offered		MSc (Nanoscience) (3089)				Total Learning Time	160 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Perform experiments on the most recent developments in cloning, protein expression, purification, identification and quantification. • Discuss the principles and applications of immunoassays. • Perform experiments on flow cytometry: from basic concepts to multicolour analysis. • Discuss the basic principles of PCR amplification, quantification and latest development in molecular biology. • Perform experiments and analyse results on the most recent developments in different imaging techniques. • Discuss the principles and relative merits of a range of techniques for the production of nanostructures including mono- and multilayers and encapsulation techniques. • Explain the principles and relevance of nanoparticle bio-junction. After attending the course be able to identify the proper analysis tool for a particular problem.						
Main Content	• The purpose of the course is to provide students with live demonstrations on research equipment used for nanoresearch. It consists of theoretical and practical exposure to: • Theoretical background and practical exposure to the most recent developments in protein expression, purification, identification and quantification; different imaging techniques; molecular biology, genomics; immunoassays.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			3	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences		Home Department			Medical Biosciences	
Module Name	Foundations of Nanobiomedical Sciences for Non-Biologists 813		Module Code			NSS813	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	4
Programme in which the module will be offered		MSc (Nanoscience) (3089)				Total Learning Time	40 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Describe the interacting forces between molecules in a biological system. - Describe the basic structure and function of each macromolecule. - Describe the components of the cell, metabolic reactions of organelles such as respiration, interaction of molecules and transport mechanisms. - Describe the principles of the basic assays used in the biological sciences.						
Main Content	- Biological chemistry: Bonds, acids and bases, chemical reactions, enzymes - Macromolecules: Carbohydrates, lipids, proteins, nucleic acids - Cells and house-keeping functions: Cell structure and metabolic processes						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA FA	50% 50%

Faculty	Natural Sciences		Home Department			Chemistry	
Module Name	Advanced Nanochemistry 821		Module Code			NSS821	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	48
Programme in which the module will be offered		MSc (Nanoscience) (3089)			Total Learning Time		480 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Discuss the synthesis and characterisation techniques suitable for producing organic and inorganic nanomaterials. • Use simple models (e.g. particles in a box, tight binding, molecular orbitals) to describe the electronic structure of molecular and solid state nanosystems.						

	• Use simple models and examples to describe how the electronic structure of nanosystems is influenced by electron-electron interactions (charge, spin) and coupling to the vibrations. • Explain electronic conduction through nanosystems and identify different regimes (Ballistic, Coulomb Blockade etc).				
Main Content	Advanced Nanochemistry Synthesis • Synthetic methods: Electrosynthesis, chemical, thermal and microwave synthesis. Thin Film Deposition Methods. Physical synthetic methods including carbon arc discharge, laser ablation, thermal chemical vapor deposition (CVD), catalytic synthesis and plasma synthesis. • Properties of nanomaterials: Chemical, electrochemical, spectroscopic, microscopic, mechanical, electrical and optical properties of materials • Synthetic nanomaterials: Ceramics, glasses, polymers, fullerenes, nanotubes, graphenes, carbon nanotubes, metal oxides and catalysts (PGMs etc), nanocrystals, nanocomposites, nano-alloys, quantum dots, zeolites, MOFs and dendrimers. Group Theory for Chemists • Symmetry groups, vibrational analysis, orbital Analysis and molecular spectroscopy. Advanced Characterisation Techniques • Spectroscopy, FTIR, Raman, liquid and SS-NMR, UV - VIS, XPS, XRD, XRF. • Microscopy: TEM, SEM, AFM. • Physical Techniques: Hall Effect, TGA, BET, Contact Angle, fluorescence, etc. • Electroanalysis and electrochemical technology: Voltammetry, scanning electrochemical microscopy, electrochemiluminescence, spectroelectrochemistry and electrochemical impedance spectroscopy. Applications of nanomaterials • Energy Devices (Fuel cells, ion batteries, Catalysis Supercapacitors, Photovoltaic cells). • Health (Sensors and biosensors; biomaterials; drug delivery; food preservation, spoilage and contaminants). • Catalysis (Reaction dynamics; adsorption isotherms, thermodynamics; homogeneous and heterogeneous catalysis) • Environmental Analysis (Mining and minerals; water and pollution) • Beneficiation (Materials and manufacturing; chemical processes).				
	Pre-requisite	CHE701 and CHE704	Prohibited module Combination	None	
	Co-requisite	None			
	Timetable Requirement per week		7	Lectures p.w.	
			0	Practicals p.w.	
			1	Tutorials p.w.	
	Other teaching modes that does not require time-table				
	Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
				FA	50%

Faculty	Natural Sciences		Home Department			Chemistry	
Module Name	Experimental Techniques in Nanochemistry 822		Module Code			NSS822	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	16
Programme in which the module will be offered		MSc (Nanoscience) (3089)				Total Learning Time	160 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Synthesise, characterise and apply nanomaterials in sensor technology, development of biomaterials, drug delivery and in food preservation and food quality determination. • Use advanced characterisation techniques and instruments to study nanomaterials.						
Main Content	• Chemical, electrochemical or physical synthesis of specific nanomaterials and catalysts and their characterisation for applications in energy devices, sensors and catalysis. • The practicals should involve the use of spectroscopic (FTIR, Raman, liquid and SS-NMR, UV – VIS, XPS, XRD, XRF), microscopic (TEM, SEM, AFM) and physical techniques (Hall Effect, TGA, BET, Contact Angle, fluorescence, etc) in the analysis of nanomaterials.						
Pre-requisite	CHE701 and CHE705		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			0	Lectures p.w.			
			3	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			Chemistry	
Module Name	Foundations of Nanochemistry for Non-Chemists 823		Module Code			NSS823	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	4
Programme in which the module will be offered		MSc (Nanoscience) (3089)			Total Learning Time		40 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: - Classify organic and inorganic nanomaterials and identify their various structural aggregations. - Explain the structural, bonding, physical and chemical properties of classes of nanomaterials. - Discuss the use of spectroscopic and microscopic techniques in structure and property elucidation.						
Main Content	- Nanomaterials: ceramics, glasses, polymers, fullerenes, graphene, carbon nanotubes, polymeric and inorganic nanostructures, metal oxides, nano-powders, nanocomposites, nano-alloys and quantum dots. - Structural properties in nanochemistry: chemical crystallography (introduction to bonding, crystal structures and properties), reactivities						

	of nanostructured materials, physical and chemical properties, processing nanostructured materials. Structural tools in nanochemistry: spectroscopy (FTIR, NMR, UV-Vis), microscopy (TEM, SEM, AFM) and physical methods (XRD, XPS, Auger).				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		2	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%	
			FA	50%	

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Advanced Nanophysics 831		Module Code			NSS831	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	48
Programme in which the module will be offered		MSc (Nanoscience) (3089)				Total Learning Time	40 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain basic physical phenomena on the nanoscale, basic principles, uses and limitations of a wide variety of materials, theory of characterisation techniques applicable to the elucidation of the structure, chemistry and properties of nanostructured materials. • Identify the pertinent parameters amenable to characterisation, governing a general description of nanostructured materials and devices. • Discuss the principles and relative merits of a range of techniques for the production of nanostructures including ultra-thin films and multilayers. • Discuss the standard application of spectroscopic, microscopic and chemical characterization tools in literature, and after attending the course the student should be able to identify the proper analysis tool for a particular problem.						
Main Content	The purpose is to provide students with information about fundamental nanophysics with regard to the study of the phenomena and concepts induced by the extreme smallness of size structures. It consists of two major topics: Fundamental nanophysics and nanoscientific methodologies. This module includes an in depth study of: • Quantum mechanics • The fundamentals of nanophysics • The quantum nature and construction of atoms, molecules and nanoparticles • Structural properties in nanophysics • Crystallography of nanostructured materials, physical and chemical properties of nanoparticles and interfaces, processing of nanostructured materials.						

	- Nanomaterials for nanoscience and nanotechnology - Different nanomaterials (ceramics, semiconductors, glasses, polymers, fullerenes, graphene, carbon nanotubes, polymeric and inorganic nanostructures, metal oxides, nano-powders, nanocomposites, nano-alloys and quantum dots). - Synthesis, nano-structures and nano-devices - Nano-analytical characterisation methods (theory) Analysis tools in nanophysics: spectroscopy (FTIR, UV-Vis, Raman), structural and analytical (TEM, EDS, EELS, SEM, WDS)				
Pre-requisite	PHY724, PHY725, PHY726	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		6	Lectures p.w.		
		1	Practicals p.w.		
		1	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA	50%	

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Experimental Techniques in Nanophysics 832		Module Code			NSS832	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	16
Programme in which the module will be offered		MSc (Nanoscience) (3089)				Total Learning Time	160 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Discuss the basic principles, uses and limitations of a wide variety of characterisation techniques applicable to the elucidation of the structure and properties of nanostructured materials. • Identify the pertinent parameters, amenable to characterisation, governing a general description of nanostructured materials and devices. • Explain the principles and relative merits of a range of techniques for the production of nanostructures including ultra-thin films and multilayers. • Describe the standard application of spectroscopic, microscopic and physical characterisation tools in practice. After attending the course be able to identify the proper analysis tool for a particular problem.						
Main Content	The purpose of the course is to provide students with live demonstrations on research equipment used for nanoresearch. It consists of theoretical and practical exposure to: • Advanced high level synthesis • Advanced high level analytical techniques • Nano-analytical characterisation methods (experimental) • Analysis tools in nanophysics: spectroscopy (FTIR, UV-Vis, Raman), structural and analytical (TEM, EDS, EELS, SEM, WDS, EBSD, AFM, STM, XRD, XPS, Auger).						

Pre-requisite	PHY724, PHY725, PHY726	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		0	Lectures p.w.		
		3	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Foundations of Nanophysics for Non-Physicists 833		Module Code			NSS833	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	4
Programme in which the module will be offered		MSc (Nanoscience) (3089)			Total Learning Time	40 hours	
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Classify inorganic nanomaterials identifying their various structural aggregations and applications. - Explain the structural, bonding, physical and chemical properties of different classes of nanomaterials. - Discuss the use of spectroscopic, microscopic and analytical techniques in structure and property elucidation.						
Main Content	- Quantum physics: atoms, molecules and nanoparticles - Nanomaterials: ceramics, semiconductors, glasses, polymers, fullerenes, graphene, carbon nanotubes, polymeric and inorganic nanostructures, metal oxides, nano-powders, nanocomposites, nano-alloys and quantum dots. - Structural properties in nanophysics: crystallography (introduction to bonding, crystal structures and properties), reactivities of nanostructured materials, physical and chemical properties of nanoparticles and interfaces, processing of nanostructured materials. - Analysis tools in nanophysics: spectroscopy (FTIR, NMR, UV-Vis), structural and analytical (TEM, EDS, EELS, SEM, WDS, EBSD, AFM, XRD, XPS, Auger).						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			2	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Mini Thesis 803/804		Module Code			PAR803/804	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	60
Programme in which the module will be offered	MSc (Pharmacy Administration and Policy Regulation) (3859)					Total Learning Time	600 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	Applied Research Project Research gives scientists, practitioners and policymakers essential knowledge to use in making decisions and thus it is an essential activity which encourages them to contribute to this process and practice it in their professional lives through the research activity that form a core component of the MSc. in Pharmacy Administration and Pharmacy Policy On completion of this module, students should be able to:						
	• Comprehend the principle steps in pharmaceutical research and development based on incomplete and often contradictory data and objectives.						
	• Evaluate the suitability of research methodologies for the purpose of undertaking research in pharmaceutical medicine.						
	• Demonstrate expertise in formulating, conducting and reporting an independent and ethical research project.						
	• Critically review and interpret the literature relating to drug research and development.						
	• Demonstrate the ability to formulate and employ effective and ethical leadership tools in Regulatory Science by completing a substantial body of research.						
	• Demonstrate the ability to analyse and evaluate data, information and/or surveyed opinions gathered (depending on the nature of the research project) and discuss findings at a level appropriate to a Level 9 research project.						
Main Content	Applied Research Project / Mini-Thesis • The general aims of the applied research project are to develop conceptual and academic rigour in applied research, and to develop participants' ability in the identification, analysis and presentation of independent applied research through appropriate inferences and recommendations in one of the key subject areas within pharmaceutical medicine. • Participants will conduct research and report through a written minithesis. The modules will encourage participants to contribute towards this process and practice it in their professional lives through the research and other activities that form a core component of the MSc. in Regulatory Sciences.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No Timetable Requirements.			0		Lectures p.w.		
			0		Practicals p.w.		
			0		Tutorials p.w.		
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment		FA		100%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Ethics for Pharmacists 811		Module Code			PAR811	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	10
Programme in which the module will be offered		MSc (Pharmacy Administration and Policy Regulation) (3859)			Total Learning Time		100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students will be able to: - Evaluate the ethical suitability of research methodologies for the purpose of undertaking research in pharmaceutical medicine especially with respect to vulnerable groups. - Critically review the ethical issues involved in the undertaking of clinical research. - Recognise the subjectivity in ethical issues which impact on the research of human and non-human subjects and construct reasonable arguments for their own or others opinion. - Demonstrate the ability to formulate effective and ethical leadership tools in Pharmaceutical Medicine.						
Main Content	- Current ethical issues that affect the pharmaceutical and health industries. - Ethical Issues in medical research and their benefit to society. - Humanitarian issues and Ethics						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week Online Course. No Timetable requirements.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Common Technical Document 812		Module Code			PAR812	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	10
Programme in which the module will be offered	MSc (Pharmacy Administration and Policy Regulation) (3859)				Total Learning Time		100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Write up a CTD with a view to submitting it to a regulatory authority for medicines' registration and regulation. • Evaluate and compare the elements that are specific to the pharmaceutical industry of South Africa with those in organisations such as ICH. • Assist applicants in the preparation of their dossiers which are being developed for different types of medicines including New Chemical Entities and Generics. • Contribute to discussions of various interest groups (such as ICH members) on changes to the current CTD.						

Main Content	• A CTD is composed of various elements which are arranged in modules. Which address matters such as Administrative and prescribing Information; quality, Good Manufacturing Practice, various summaries, Non-clinical and clinical study reports			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week Online Course. No Timetable requirements.	0	Lectures p.w.		
	0	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Regulatory Sciences 813		Module Code			PAR813	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	10
Programme in which the module will be offered	MSc (Pharmacy Administration and Policy Regulation) (3859)					Total Learning Time	100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: - Interpret the regulation of Generic medicines and Biosimilars. - Contribute to the further development of the above-mentioned regulations. - Provide valuable assistance to those applicants aiming to market these two groups of medicines.						
Main Content	- The study of the continual advancement and development of the legislation, regulations and science - especially of the Biosimilars - The investigation of aspects of quality, safety and efficacy of these medicinal products and the effects of these factors on the access by patients of these agents and finished products						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week Online Course. No Timetable requirements.			0		Lectures p.w.		
			0		Practicals p.w.		
			0		Tutorials p.w.		
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department		School of Pharmacy		
Module Name	The Regulation of Complementary and Traditional Medicines 814		Module Code		PAR814		
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	10
Programme in which the module will be offered	MSc (Pharmacy Administration and Policy Regulation) (3859)				Total Learning Time	100 hours	

Duration	Semester	Proposed Semester	First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:			
	• Understand complementary and traditional medicine. • Understand the regulatory framework for such medicines globally. • Critically examine international practices and what are the most important therapeutic, quality and safety issues to consider when regulating them. • Understand the legal status of complementary and traditional medicines around the world. • Analyze different approaches and evaluate possible best practices in this area. • Critically examine herbal medicine, vitamins and minerals, nutritional supplements, homoeopathy, aromatherapy, African, Ayurvedic, Chinese, African and other traditional medicines.			
Main Content	• The critical study of complementary and traditional medicines including their definition and use in their respectively healing systems. • The module will also explore the acceptable use of these medicines in today's cultures especially where they are used with evidence-based medicines of Western origin. • The module will also equip students to explore and to critically analyse the national and international legislation of these medicines			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		0	Lectures p.w.	
Online Course. No Timetable requirements.		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Regulatory Sciences and Medical Devices 815		Module Code			PAR815	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	10
Programme in which the module will be offered		MSc (Pharmacy Administration and Policy Regulation) (3859)				Total Learning Time	100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Understand and contextualise major issues affecting the development of medical devices. • The regulatory control of medical devices, with special reference to the EMA CE-Mark and other key international legislation. • Review the key standards, conformity assessments, clinical issues, safety and risk assessment as well as medical devices containing drugs and the latest developments in diagnostics.						
Main Content	• Major issues impacting medical devices and their regulatory control both national and internationally with special attention given to those medical devices containing drugs e.g. drug-eluting stents						

Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week Online Course. No Timetable requirements.		0	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacovigilance 816		Module Code			PAR816	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	10
Programme in which the module will be offered		MSc (Pharmacy Administration and Policy Regulation) (3859)			Total Learning Time		100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Appreciate the effect on patient safety and a product's licence by the different types of adverse events/reactions. • Identify the key safety reporting requirements, according to the type of adverse event/reaction (in clinical research or in commercial products). • Construct and evaluate benefit/risk assessments throughout the lifecycle of a medicine, with reference where applicable to issues around pre- and post-market authorisation.						
Main Content	• Pharmacovigilance deals with drugs and their safety. The topics include the prediction and analysis of the actions of drugs upon the body. Students will be taught to report signals, the different types of adverse drug events and lastly, they are taught how to construct and evaluate benefit/risk assessments. • This module will give participants a working knowledge of all aspects of drug safety monitoring and risk management and will enable them to predict and analyse drug interactions, and recognise the implications for prescribing and product licence. The participant will also be familiarized with pharmacovigilance best practice which includes monitoring strategies and action plans and be fully cognisant of regulatory reporting requirements.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week Online Course. No Timetable requirements.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Regulatory Affairs 817		Module Code			PAR817	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	10
Programme in which the module will be offered		MSc (Pharmacy Administration and Policy Regulation) (3859)			Total Learning Time		100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Evaluate the regulatory issues affecting the pharmaceutical industry. • Compare the regulation of medicines in different global markets. • Develop and critically appraise product-related information to ensure adherence to ethical and legal provisions. • Compare and evaluate the different roles of Regulatory and Medical Affairs departments in order to propose consistent strategies involving the marketing of pharmaceuticals, medical information, the role of a pharmaceutical company's country medical department and licensing agreements.						
Main Content	• Regulatory Affairs covers the key regulatory and medical affairs issues including Good Manufacturing Practice (cGMP) affecting the national and international Industry today, from an overview of the major drug regulators and the requirements during finished medicinal product development to medical information and regulations including prescribing issues to life cycle management issues.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week Online Course. No Timetable requirements.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Research Methods for Pharmacists 818		Module Code			PAR818	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	10
Programme in which the module will be offered		MSc (Pharmacy Administration and Policy Regulation) (3859)			Total Learning Time		100 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Understand the principle steps in pharmaceutical research and development based on incomplete and often contradictory data and objectives. • Evaluate the suitability of research methodologies for the purpose of undertaking research in pharmaceutical medicine. • Demonstrate expertise in formulating, conducting and reporting an independent and ethical research project. • Critically review and interpret the literature relating to drug research and development.						

	• Demonstrate the ability to formulate and employ effective and ethical leadership tools in Regulatory. • Science by completing a substantial body of research. • Demonstrate the ability to analyse and evaluate data, information and/or surveyed opinions gathered (depending on the nature of the research project) and discuss findings at a level appropriate to a Level 9 research project.			
Main Content	• The study of conceptual and academic rigour in applied research, and then to develop participants' ability in the identification, analysis and presentation of independent applied research, through appropriate inferences and recommendations in one of the key subject areas (modules) within pharmaceutical medicine.			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week Online Course. No Timetable requirements.	0	Lectures p.w.		
	0	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Online Course. No Timetable requirements.				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Leadership and Negotiation 819		Module Code			PAR819	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	10
Programme in which the module will be offered		MSc (Pharmacy Administration and Policy Regulation) (3859)				Total Learning Time	100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	Leadership and Negotiation On completion of this module, students should be able to: • Demonstrate the ability to judge, select and/or formulate effective leadership tools. • Demonstrate mastery of a set of practical leadership skills. • Review, evaluate and propose solutions to scenarios typical for leaders in industry and arrive at possible solutions working in a team environment. • Recognise the subjectivity in ethical and leadership issues and construct reasonable arguments for one's opinion or recommendations. [A limited number of places, 5 are also available on the onsite version of this course which takes place over five days at Harvard University. Additional travel, accommodation and other expenses are born by students who wish to attend the onsite course.]						
	Main Content • Teaching and learning a set of practical leadership skills • Leadership simulations. • Analysis of case studies and discussion of articles on leadership, participants apply the lessons learned to ongoing, real-world cases. • Theory and practice of negotiation in the context of organisational leadership and management.						

	• Development of a framework for diagnosing and intervening in organisational systems to generate adaptive work.				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week Online Course. No Timetable requirements.		0	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA		50%
			FA		50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Clinical Trials and its Statistics 820		Module Code			PAR820	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	20
Programme in which the module will be offered		MSc (Pharmacy Administration and Policy Regulation) (3859)			Total Learning Time		200 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Statistics and Clinical Trial Management - Critically appraise the principle steps in the clinical trial phases that can often be based on incomplete and sometimes contradictory data and objectives. - Evaluate the suitability of research methodologies for the purpose of undertaking clinical trials as adaptive as possible. - Critically review the issues (including legal, ethical, clinical and/or commercial) involved when undertaking clinical trials. - Demonstrate the ability to plan and develop clinical trials in accordance with legislative requirements and Good Clinical Practice (GCP) procedures. - Critically interpret and manage statistical data used in clinical development and discriminate between relevant and non-relevant data and be able to justify such decisions.						
Main Content	- The Clinical Trial Management module covers the practical management of the development of a medicinal product from Phase I to IV using the latest flexible and adaptive research techniques to optimise efficiency in the development process. It also teaches participants how to critically review and evaluate statistical data and to leverage the information and knowledge created to contribute to the competitive advantage of their organisation. - It is a practical course on how to plan and manage clinical trials cognizant of the pharmacoeconomic conditions and leadership challenges the participant will need to address when leading clinical development programs. - Practical exercises are presented on how to plan and manage clinical trials cognizant of the pharmacoeconomic conditions and leadership challenges the participant will need to address when leading clinical development programs.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week	0	Lectures p.w.		
	0	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Health Economics 821		Module Code			PAR821	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	10
Programme in which the module will be offered		MSc (Pharmacy Administration and Policy Regulation) (3859)			Total Learning Time		100 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Evaluate the commercial and pharmaco-economic issues pertaining to drug development. • Explain the principles of health economics and discuss their application in the development of medicines. • Evaluate the principles of economics and the methodology behind the statistical and/or data analysis tools used for health economics.						
Main Content	• Economics affecting the individual and linked areas of the pharmaceutical industry with a special focus on the important stage of “drug to market” and • Impact on healthcare on the economy of the country • Functioning of the healthcare market and the health insurance market, • Functioning of the pharmaceutical sector and the importance of innovation and the problems of incentives in both the supply and demand of healthcare and pharmaceutical products.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week Online Course. No Timetable requirements.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA	50%
						FA	50%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Mini-Thesis in Clinical Pharmacy 803/804		Module Code			PHC803/804	
Module Type	Core	NQF Level	9	Year Level	2/3	NQF Credit	120
Programme in which the module will be offered		M Clinical Pharmacy (3852)			Total Learning Time		1200 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of the research project students should be able to: Comprehend the principle steps in pharmaceutical research and development based on incomplete and often contradictory data and objectives.						

	• Evaluate the suitability of research methodologies for the purpose of undertaking research in clinical pharmacy. • Demonstrate expertise in formulating, conducting and reporting an independent and ethical research project. • Critically review and interpret the relevant literature. • Demonstrate the ability to formulate and employ effective and ethical tools in clinical pharmacy by completing a substantial body of research. • Demonstrate the ability to analyse and evaluate data, information and/or surveyed opinions gathered (depending on the nature of the research project) and discuss findings at a level appropriate to a Level 9 research project.						
Main Content	Applied Research Project / mini-thesis The general aims of the applied research project are to develop conceptual and academic rigour in applied research, and to develop participants' ability in the identification, analysis and presentation of independent applied research through appropriate inferences and recommendations in one of the following key subject areas within clinical pharmacy: • Paediatrics • HIV/AIDS • Oncology • Critical Care Participants will conduct research and report through a written mini-thesis.						
	Pre-requisite		None		Prohibited module Combination		None
	Co-requisite		None				
	Timetable Requirement per week No Timetable requirements.		0		Lectures p.w.		
			0		Practicals p.w.		
		0		Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)			Method of Student Assessment		FA	100%
Faculty	Natural Sciences			Home Department		School of Pharmacy	
Module Name	Clinical Pharmacy and Pharmaceutical Care 811			Module Code		PHC811	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	25
Programme in which the module will be offered		M Clinical Pharmacy (3852)			Total Learning Time		250 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Distinguish between the concepts of Clinical Pharmacy and Pharmaceutical Care. • Describe how Clinical Pharmacy and Pharmaceutical Care can contribute to improving quality of care. • Know the meaning of a profession, a practitioner, and a patient care practice. • Know the components of a professional practice: its philosophy of practice, patient care process, and practice management system.						

	- Describe the meaning of a therapeutic relationship, the ethical issues involved, and the rights and responsibilities of the patient and practitioner throughout the patient care process. - Read, interpret and evaluate general therapeutics literature. - Integrate and apply general therapeutics information to simulated patient cases. - Identify drug therapy problems for patients using a specific decision-making process. - Design a pharmacy care plan for individual patients, including goals of therapy, pharmacotherapy plan and a monitoring plan, in order to solve and prevent drug therapy problems. - Document a drug therapy problem and pharmacotherapy plan on patient charts and profiles. - Communicate, both verbally and in writing, the rationale for the identification and resolution of the drug therapy problem and the pharmacy care plan. - Acquire and apply patient-specific clinical and laboratory data to support therapeutic decisions.						
Main Content	- Definitions and descriptions of Clinical Pharmacy and Pharmaceutical Care. - Detailed information on the pharmaceutical care process, which includes patient assessment, medication history, medical history, identification of drug related needs and problems, implementation of the care plan, monitoring and follow-up. - Laboratory and diagnostic investigations.						
Pre-requisite	BPharm degree		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week No Timetable requirements.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%

Faculty	Natural Sciences		Home Department		School of Pharmacy		
Module Name	Evidence Based Medicine and Pharmacovigilance 812		Module Code		PHC812		
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	30
Programme in which the module will be offered		M Clinical Pharmacy (3852)			Total Learning Time	300 hours	
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: Evidence Based Medicine: - Know the concept of evidence-based medicine and use such in justifying therapeutic interventions. - Plan and carry out a search of the medical literature for the best external evidence. - Appraise critically medical literature for clinical decision making.						

	• Integrate individual clinical care with the best available clinical evidence from systematic research. • Know and apply the principles of biomedical statistics to assess therapeutic outcomes. Rational Medicine Use: • Define rational medicine use. • Assess the problem of rational medicine use. • Introduce systems and tool to improve rational medicine use. • Describe how indicators can be used to identify medicine use problems. • Perform a prescribing indicator study and identify medicine use problems. • Discuss the use of aggregate data including defined daily dose (DDD) to analyse the consumption of medicines. • Perform an ABC analysis and explain how it can be used to identify medicine use problems, reduce costs, and improve efficiency in the pharmaceutical supply system. • Discuss how VEN system for setting priorities will assist in medicine selection, purchasing, and inventory management. • Know the concept of MUE • Understand the process for implementing and performing a MUE. • Discuss the use of a MUE for improving pharmaceutical therapy. • Prepare criteria and thresholds for a MUE. Pharmacovigilance: • Demonstrate an understanding of, and critically evaluate, issues surrounding the risks and benefits of drug use in humans including the cause, manifestations and consequences of adverse drug effects (ADEs), the manner in which these are detected and monitored, and the related historic and legal frameworks. • Know and apply the principles for adverse drug reactions and reporting systems in the health service. • Assess the benefit, harm, effectiveness and risk of medicines. • To maximise therapeutic outcomes. • Detect and classify medicine-use problems and report findings timeously. • Provide recommendations to optimise patient health outcomes.
Main Content	Evidence Based Medicine: • Basic and Clinical Biostatistics • Review of study designs and therapeutic interventions • Meta-analysis Rational Medicine Use: • Methods to Investigate Medicine Use • Indicators for Health Care • Prescribing Indicators • Patient Care Indicators • Complementary Indicators • Methods to Investigate Medicine Use • Defined Daily Dose • VEN Analysis • VEN Applications • ABC Analysis • Applications of ABC Analysis

	• Steps in Performing ABC Analysis • Identifying the need to perform an MUE • Stepwise Approach to Performing a MUE Pharmacovigilance: • Relationship or Causality assessments • Classification of Factors affecting susceptibility to adverse drug reactions • Adverse drug reaction reporting systems				
Pre-requisite	BPharm degree	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week No Timetable requirements.		0	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Pharmacokinetics 814		Module Code			PHC814	
Module Type	Core	NQF Level	9	Year Level	1/2	NQF Credit	25
Programme in which the module will be offered		M Clinical Pharmacy (3852)				Total Learning Time	250 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to:						
	Identify key pharmacokinetics parameters necessary for therapeutic drug monitoring.						
	Identify clinically significant pharmacokinetic drug-drug interactions and design appropriate interventions.						
	Identify patient populations with clinically significant pharmacokinetic variability and design treatment interventions to manage such patients (e.g. neonates).						
	Adjust patient's drug dosing and therapy using relevant pharmacokinetic data and principles.						
Main Content	Perform therapeutic drug monitoring (TDM) in relevant patients.						
	Basic principles of pharmacokinetics						
	Pharmacokinetics of IV administration						
	Pharmacokinetics of oral administration						
	Biopharmaceutics						
Non-linear processes							
Pharmacokinetic Variation and Model-Independent relationships							
Absorption, distribution, metabolism and excretion of medicines in various patient populations (e.g. neonates, cystic fibrosis, etc.)							
Pre-requisite	None		Prohibited module			None	
Co-requisite	None		Combination				
Timetable Requirement per week			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table						
Block lectures and online delivery						
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment		CA	100%

Faculty	Natural Sciences		Home Department		School of Pharmacy		
Module Name	Adult Medicine (1) 815		Module Code		PHC815		
Module Type	Core	NQF Level	9	Year Level	1/2	NQF Credit	30
Programme in which the module will be offered		M Clinical Pharmacy (3852)			Total Learning Time	300 hours	
Duration	Year		Proposed Semester		Both Semesters		
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Apply disease orientated knowledge of the following areas: Cardiovascular Disorders, Respiratory Disorders, Renal Disorders, Neurologic Disorders, Psychiatric Disorders • Demonstrate competence in the pharmacotherapy of listed medical diseases/problems. • Review Standard Treatment Guidelines and pharmacotherapeutic options related to disorders in adults: • Assess patient-specific medical problems. • Evaluate patient-specific drug therapy and therapeutic problems. • Design a comprehensive drug therapy plan for patient-specific problems. • Justify recommendations based on patient-specific pharmacologic, pharmacokinetic, pharmacodynamic, ethical, legal, and evidence-based information. • Retrieve, interpret and integrate medical information • Educate patients						
Main Content	• Cardiovascular Disorders • Respiratory Disorders • Renal Disorders • Neurologic Disorders • Psychiatric Disorders						
	Content description: • Disease overview • Epidemiology • Etiology • Pathophysiology • Clinical presentation • Diagnosis • Laboratory and diagnostic investigations • Prognosis • Pharmacologic and Non-pharmacologic treatment • Pharmacoeconomic considerations • Evaluation of therapeutic outcomes						
Pre-requisite	None		Prohibited module		None		
Co-requisite	None		Combination				
Timetable Requirement per week			0	Lectures p.w.			
Block lectures and online delivery			0	Practicals p.w.			
No Timetable requirements.			0	Tutorials p.w.			

Other teaching modes that does not require time-table							
Block lectures and online delivery No Timetable requirements.							
Assessment Module Type	Continuous Assessment (CA)		Method of Student Assessment			CA	100%
Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Paediatrics 816		Module Code			PHC816	
Module Type	Core	NQF Level	9	Year Level	2	NQF Credit	30
Programme in which the module will be offered		M Clinical Pharmacy (3852)			Total Learning Time		300 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Apply disease-orientated knowledge of anatomy, physiology, pathophysiology, epidemiology, etiology, risk factors, signs and symptoms, natural course and prognosis, and laboratory and diagnostic test interpretation. • Demonstrate competence in the pharmacotherapy of listed medical diseases/problems. • Review Standard Treatment Guidelines and pharmacotherapeutic options related to paediatric disorders. • Monitor patients in the health care setting. • Assess patient-specific medical problems. • Evaluate patient-specific drug therapy and therapeutic problems. • Design a comprehensive drug therapy plan for patient-specific problems. • Justify recommendations based on patient-specific pharmacologic, pharmacokinetic, pharmacodynamic, pharmacogenomic, pharmacoeconomic, ethical, legal, and evidence-based information. • Retrieve, interpret and integrate medical information. • Educate patients.						
Main Content	• Gastrointestinal Disorders • Cardiovascular System • Central Nervous System • Infectious Diseases • Respiratory System						
	Content description: • Disease overview • Epidemiology • Etiology • Pathophysiology • Clinical Presentation • Diagnosis • Laboratory and diagnostic investigations • Prognosis • Pharmacologic and Non-pharmacologic treatment • Pharmacoeconomic considerations • Evaluation of therapeutic outcomes						
Pre-requisite	PHC811, PHC812		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week	0	Lectures p.w.		
Block lectures and online delivery	0	Practicals p.w.		
No Timetable requirements.	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Adult Medicine (2) 820		Module Code			PHC820	
Module Type	Core	NQF Level	9	Year Level	1/2	NQF Credit	85
Programme in which the module will be offered		M Clinical Pharmacy (3852)				Total Learning Time	850 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module student should be able to: • Apply disease-orientated knowledge of the following areas: ○ Gastrointestinal Disorders ○ Gynecologic and Obstetric Disorders ○ Urologic Disorders ○ Immunologic Disorders ○ Dermatologic Disorders ○ Infectious Diseases ○ Endocrinologic Disorders ○ Rheumatologic Disorders ○ Ophthalmic and Otolaryngological Disorders ○ Hematologic Disorders • Demonstrate competence in the pharmacotherapy of listed medical diseases/problems. • Review Standard Treatment Guidelines and pharmacotherapeutic options related to disorders in adults. • Assess patient-specific medical problems. • Evaluate patient-specific drug therapy and therapeutic problems. • Design a comprehensive drug therapy plan for patient-specific problems. • Justify recommendations based on patient-specific pharmacologic, pharmacokinetic, pharmacodynamic, ethical, legal, and evidence-based information. • Retrieve, interpret and integrate medical information • Educate patients						
Main Content	• Gastrointestinal Disorders • Gynecologic and Obstetric Disorders • Urologic Disorders • Immunologic Disorders • Dermatologic Disorders • Infectious Diseases Content description: • Disease overview • Epidemiology • Etiology • Pathophysiology • Clinical presentation						

	• Diagnosis • Laboratory and diagnostic investigations • Prognosis • Pharmacologic and Non-pharmacologic treatment • Pharmacoeconomic considerations • Evaluation of therapeutic outcomes				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week Block lectures and online delivery No Timetable requirements.		0	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment		CA	100%

Faculty	Natural Sciences			Home Department			School of Pharmacy		
Module Name	HIV 821			Module Code			PHC821		
Module Type	Elective	NQF Level	9	Year Level	2	NQF Credit		15	
Programme in which the module will be offered		M Clinical Pharmacy (3852)				Total Learning Time		150 hours	
Duration	Year		Proposed Semester			Both Semesters			
Main Learning Outcomes	On completion of this module student should be able to:								
	• Apply disease-orientated knowledge of anatomy, physiology, pathophysiology, epidemiology, etiology, risk factors, signs and symptoms, natural course and prognosis, and laboratory and diagnostic test interpretation. • Demonstrate competence in the pharmacotherapy of listed medical diseases/problems. • Review Standard Treatment Guidelines and pharmacotherapeutic options related to HIV/AIDS disorders. • Apply clinical principles in the setting of each disease state, patient population, and/or therapeutic category. • Monitor patients in the health care setting. • Know the general principles for the management of opportunistic infections. • Assess patient-specific medical problems. • Evaluate patient-specific drug therapy and therapeutic problems. • Design a comprehensive drug therapy plan for patient-specific problems. • Justify recommendations based on patient-specific pharmacologic, pharmacokinetic, pharmacodynamic, pharmacogenomic, pharmacoeconomic, ethical, legal, and evidence-based information. • Retrieve, interpret and integrate medical information. • Educate patients.								
Main Content	• Disease overview • Epidemiology • Etiology • Pathophysiology • Diagnosis and clinical presentation • Laboratory and diagnostic investigations								

	• Prognosis • Pharmacologic and Non-pharmacologic treatment • Pharmacoeconomic considerations • Evaluation of therapeutic outcomes			
Pre-requisite	PHC811, PHC812	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week Block lectures and online delivery.		0	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences			Home Department			School of Pharmacy	
Module Name	Oncology 822			Module Code			PHC822	
Module Type	Elective	NQF Level	9	Year Level	2	NQF Credit	15	
Programme in which the module will be offered		M Clinical Pharmacy (3852)				Total Learning Time	150 hours	
Duration	Year			Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module student should be able to:							
	• Apply disease-orientated knowledge of anatomy, physiology, pathophysiology, epidemiology, etiology, risk factors, signs and symptoms, natural course and prognosis, and laboratory and diagnostic test interpretation. • Demonstrate competence in the pharmacotherapy of listed medical diseases/problems. • Review Standard Treatment Guidelines and pharmacotherapeutic options related to HIV/AIDS disorders. • Monitor patients in the health care setting. • Assess patient-specific medical problems. • Evaluate patient-specific drug therapy and therapeutic problems. • Design a comprehensive drug therapy plan for patient-specific problems. • Justify recommendations based on patient-specific pharmacologic, pharmacokinetic, pharmacodynamic, pharmacogenomic, pharmacoeconomic, ethical, legal, and evidence-based information. • Retrieve, interpret and integrate medical information. • Educate patients.							
Main Content	• Cancer Treatment and Chemotherapy • Breast Cancer • Lung Cancer • Colorectal Cancer • Prostate Cancer • Lymphomas • Ovarian Cancer • Acute Leukemias • Multiple • Myelodysplastic Syndromes • Renal Cell Carcinoma • Melanoma							

	Content description • Disease overview • Epidemiology • Etiology • Pathophysiology • Clinical presentation • Diagnosis • Laboratory and diagnostic investigations • Prognosis • Pharmacologic and Non-pharmacologic treatment • Pharmacoeconomic considerations • Evaluation of therapeutic outcomes				
Pre-requisite	PHC811, PHC812	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week Block lectures and online delivery.		0	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment			CA

Faculty	Natural Sciences		Home Department			School of Pharmacy	
Module Name	Critical Care 823		Module Code			PHC823	
Module Type	Elective	NQF Level	9	Year Level	3	NQF Credit	15
Programme in which the module will be offered		M Clinical Pharmacy (3852)				Total Learning Time	150 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module student should be able to: • Apply disease-orientated knowledge of anatomy, physiology, pathophysiology, epidemiology, etiology, risk factors, signs and symptoms, natural course and prognosis, and laboratory and diagnostic test interpretation. • Demonstrate competence in the pharmacotherapy of listed medical diseases/problems. • Apply clinical principles in the setting of each disease state, patient population, and/or therapeutic category: • Monitor patients in the health care setting. • Assess patient-specific medical problems. • Evaluate patient-specific drug therapy and therapeutic problems. • Design a comprehensive drug therapy plan for patient-specific problems. • Justify recommendations based on patient-specific pharmacologic, pharmacokinetic, pharmacodynamic, pharmacogenomic, pharmacoeconomic, ethical, legal, and evidence-based information. • Retrieve, interpret and integrate medical information.						
Main Content	The main content will be for the following disorders. Organ-System Related • Pulmonary • Cardiovascular • Renal						

	• Neurology • Gastrointestinal • Hepatic • Dermatology • Immunology • Endocrine • Hematology • Psychiatry Specific Considerations • Infectious Diseases • Pharmacokinetics and Pharmacodynamics • Toxicological emergencies • Pediatric and Neonatal Considerations (optional) Support Care • Nutrition • Analgesia • Sedation • Delirium • Neuromuscular blocking agents • Venous thromboembolism prophylaxis • Stress ulcer prophylaxis • Bowel regimens Content description: • Disease overview • Epidemiology • Etiology • Pathophysiology • Diagnosis and clinical presentation • Laboratory and diagnostic investigations • Prognosis • Pharmacologic and Non-pharmacologic treatment • Pharmacoeconomic considerations • Evaluation of therapeutic outcomes			
Pre-requisite	PHC811, PHC812	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week Block lectures and online delivery.		0	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous Assessment (CA)	Method of Student Assessment	CA	100%

Faculty	Natural Sciences		Home Department	Physics and Astronomy	
Module Name	Quantum Mechanics 820		Module Code	PHY820	
Module Type	Elective	NQF Level	9	Year Level	1
Programme in which the module will be offered		MSc (Physical Science) (3081)		NQF Credit	30
				Total Learning Time	300 hours
Duration	Semester		Proposed Semester	First Semester	

Main Learning Outcomes	On completion of this module, students should be able to:			
	• Use second quantisation to perform nuclear structure calculations. • Solve relativistic quantum problems. • Calculate the scattering cross sections for simple systems.			
	Main Content			
	• Second quantisation • Dirac equation • Relativistic Quantum Mechanics • Scattering theory			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		4	Lectures p.w.	
		0	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences			Home Department			Physics and Astronomy		
Module Name	Nuclear Physics 830			Module Code			PHY830		
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	30		
Programme in which the module will be offered		MSc (Physical Science) (3081)				Total Learning Time	300 hours		
Duration	Semester		Proposed Semester			Second Semester			
Main Learning Outcomes	On completion of this module, students should be able to: • Calculate the excitation in nuclei using different nuclear models. • Use scattering codes to calculate the expected results in different types of experiments. • Calculate the influence of the standard model on nuclear physics.								
Main Content	• Nuclear models • Nuclear reactions • Interface between low energy nuclear physics and particle physics								
Pre-requisite	None		Prohibited module Combination			None			
Co-requisite	None								
Timetable Requirement per week			4	Lectures p.w.					
			0	Practicals p.w.					
			1	Tutorials p.w.					
Other teaching modes that does not require time-table									
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment			CA		50%	
						FA		50%	

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Solid State Physics 840		Module Code			PHY840	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	30
Programme in which the module will be offered		MSc (Physical Science) (3081)			Total Learning Time	300 hours	
Duration	Semester		Proposed Semester			Second Semester	

Main Learning Outcomes	On completion of this module, students should be able to: • Use Green functions to solve problems. • Calculate phonon properties of systems. • Solve aspects of simple nano-systems.			
Main Content	• Many particle Physics • Phonons and Superconductivity • Low temperature physics • Electronic Properties of materials • Nano physics			
Pre-requisite	None	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		4	Lectures p.w.	
		0	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Physics and Astronomy	
Module Name	Physics Education 850		Module Code			PHY850	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	30
Programme in which the module will be offered		MSc (Physical Science) (3081)				Total Learning Time	300 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain the effect of the curriculum on students learning’. • Describe the differences between research methods used in Physics education. • Describe the insights that the study of metacognition gives to teaching.						
Main Content	• Curriculum Design • Research Methodology in Physics education. • Metacognition						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			4	Lectures p.w.			
			0	Practicals p.w.			
			1	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Continuous and Final Assessment (CFA)		Method of Student Assessment		CA		50%
					FA		50%

Faculty	Natural Sciences		Home Department		Physics and Astronomy		
Module Name	Mathematical Methods of Physics 860		Module Code		PHY860		
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	30
Programme in which the module will be offered		MSc (Physical Science) (3081)			Total Learning Time		300 hours

Duration	Semester	Proposed Semester	Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to:			
	• Use Tensor analysis to describe elasticity and the covariance of systems. • Solve scattering problems using calculus of residues. • Use orthogonal functions in quantum and classical systems. • Solve differential equations using Green functions.			
Main Content	• Tensor analysis • Complex analysis • Sturm Liouville theory • Integral equations including Green functions.			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week		4	Lectures p.w.	
		0	Practicals p.w.	
		1	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Continuous and Final Assessment (CFA)	Method of Student Assessment	CA	50%
			FA	50%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Studies 807		Module Code			POP807	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MPhil (Population Studies) (3921)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Critically: • Calculate rates, proportions and ratios. • Draw a “Lexis’Diagram”. • Critically analyse mortality and any demographic event. • Read and construct any demographic event tables. • Know how to prepare and analyze data. • Know how to evaluate quality of secondary data. • Know how to respond to any demographic event. • Prepare policy statements on any demographic issue.						
Main Content	• Introduction to demography • Calculation of rates, ratios and proportions • Lexis’Diagram • Demographic analysis of mortality • Direct and indirect methods of analysis • Deep analysis of any demographic events • Transversal and longitudinal analysis • Advanced demographic analysis of mortality • Analysis of families and parity ratios • Analysis of fertility and fecundity • Overview of migrations issues						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						

Timetable Requirement per week	3	Lectures p.w.		
	0	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment	FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Studies 809		Module Code			POP809	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MPhil (Population Studies) (3921)				Total Learning Time	150 hours
Duration	Term		Proposed Semester			First Term	
Main Learning Outcomes	On completion of this module, students should be able to: Critically: • Appreciate the extent to which official statistics dominate the lives of citizens on a daily basis. • Describe the processes of data collection, processing, analysis, and dissemination by Statistics South Africa. • Learn some important calculations techniques used by Statistics South Africa that are not taught in other courses in the Department of Statistics.						
Main Content	• History of Official Statistics; • Why should governments gather statistics; • The future of statistics; • Census 1996 and Census 2001; • Vital statistics in South Africa; • Labor Force and General Household Surveys in South Africa; • Economic and Industrial Surveys; • Population projections; • Index numbers; • Development and poverty.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3		Lectures p.w.		
			0		Practicals p.w.		
			0		Tutorials p.w.		
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department		Statistics and Population Studies		
Module Name	Population Studies 811		Module Code		POP811		
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered	MPhil (Population Studies) (3921)				Total Learning Time	150 hours	

Duration	Term	Proposed Semester	First Term			
Main Learning Outcomes	On completion of this module, students should be able to: Critically: - Interpret and explain survey and census processes. - Develop questionnaires. - Conduct a demographic survey.					
Main Content	- Data collection processes - How to conduct a survey - Design of questionnaires - Practical experience					
Pre-requisite	None	Prohibited module Combination		None		
Co-requisite	None					
Timetable Requirement per week		3	Lectures p.w.			
		0	Practicals p.w.			
		0	Tutorials p.w.			
Other teaching modes that does not require time-table						
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment		FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Projections 812		Module Code			POP812	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MPhil (Population Studies) (3921)				Total Learning Time	150 hours
Duration	Term		Proposed Semester			Third Term	
Main Learning Outcomes	On completion of this module, students should be able to: • Select the appropriate projective methods for the estimation of future growth of population by age, urban/rural population, households, labour force and school enrolments. • Critically assess past demographic trends for their use as assumptions in the elaboration of projective outputs. • Relate the projective methods learned in class with those applied in the world reports produced by international organisations. • Use software to produce projective computations. • Produce a research report.						
Main Content	• Scope and data for population projections, projections by sex and age, projection of urban and rural population, projection of labour force, projection of households for local planning, projection of school level enrolments and projection by level of education, selected topics, computerised-assisted applications.						
Pre-requisite	POP807 or equivalent only.		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment	FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Studies 813		Module Code			POP813	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MPhil (Population Studies) (3921)				Total Learning Time	150 hours
Duration	Term		Proposed Semester			Third Term	
Main Learning Outcomes	On completion of this module, students should be able to: Examine the background of African historical demography and migration patterns						
Main Content	- Population in African history - Growth and Regulation of Prehistoric Populations - Demographic transition - Special African mortality patterns - Migration						
Pre-requisite	POP807 or equivalent only.		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			

Other teaching modes that does not require time-table				
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment	FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Studies 814		Module Code			POP814	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MPhil (Population Studies) (3921)				Total Learning Time	150 hours
Duration	Term		Proposed Semester			Fourth Term	
Main Learning Outcomes	On completion of this module, students should be able to: • Critically examine the linkages between demographic and social issues from an analytical perspective. • Up-dated examination of contemporary debates on the relationships between population and employment, education, elderly and social exclusion variables. • The student will be able to gain critical knowledge of theoretical perspectives on the social dimensions of demographic variables such as fertility, mortality and migration.						
Main Content	• Direct analysis of gender equity • Direct and indirect analysis of maternal and child mortality and morbidity • Direct analysis of contraception						

	• Critical analysis of data on reproductive health • Use of available data (surveys) • Use of Demographic and Health Surveys (DHS)			
Pre-requisite	POP807 or equivalent knowledge only	Prohibited module Combination		None
Co-requisite	None			
Timetable Requirement per week		3	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment	FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Studies 815		Module Code			POP815	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MPhil (Population Studies) (3921)				Total Learning Time	150 hours
Duration	Term		Proposed Semester			Fourth Term	
Main Learning Outcomes	On completion of this module, students should be able to: • Know how to critically analyse: gender issues; contraception questions; maternal and infant issues; reproductive health data. • Generate and structure draft policies on the above mentioned issues.						
Main Content	• Direct analysis of gender equity • Direct and indirect analysis of maternal and child mortality and morbidity • Direct analysis of contraception • Critical analysis of data on reproductive health • Use of available data (surveys) • Use of Demographic and Health Surveys (DHS)						
Pre-requisite	POP807 or equivalent knowledge only		Prohibited module Combination		None		
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment		FA		100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Population Studies 848		Module Code			POP848	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MPhil (Population Studies) (3921)				Total Learning Time	150 hours
Duration	Term		Proposed Semester			Third Term	

Main Learning Outcomes	On completion of this module, students should be able to: • Critically use mathematics to analyse demographic events from discrete to continuous forms. • Make use of different models in researching the relationships between population and socio-economic variables. • Investigate the adequacy of models of integrating population variables into development planning. • Use population models for prospective demographic analysis. • Produce a research report on the importance of population factors in development planning.				
Main Content	• Instantaneous mortality function. • Expression for expectancy of life. • Analytical expression for mortality functions. • Nuptiality function. • Fertility functions. • Selected applications. • Instantaneous crude birth and death rates. • Population dynamics functions (exponential, stable, Malthusian, stationary), • Population modelling and development planning. • Applications to sectoral development.				
Pre-requisite	POP807 and some mathematical background in differential and integral calculus or equivalent		Prohibited module Combination	None	
Co-requisite	None				
Timetable Requirement per week			3	Lectures p.w.	
			0	Practicals p.w.	
			0	Tutorials p.w.	
Other teaching modes that does not require time-table					
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment	FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 800		Module Code			STA800	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MSc (Statistical Science) (3838)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to:						
	• Critically design and execute various Data Mining techniques, from extraction of huge data repositories to modeling of the data. Techniques covered would include: • Decision tree analyses, clustering, visual data analyses, regression and classification. • Students would produce a scientific report of the findings.						

Main Content	• Data management and processing; statistical inference techniques explored, classification, association and clustering, post-processing of discovered structures.				
Pre-requisite	STA331; STA332 or equivalent	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment		FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 801/802		Module Code			STA801/802	
Module Type	Core	NQF Level	9	Year Level	1/2	NQF Credit	180
Programmes in which the module will be offered		MSc (Statistical Science) (3838), MSc (Computational Finance) (3093)				Total Learning Time	1800 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: • Make a worthy contribution to knowledge in the field of Statistics, Population Studies and Demography or Computational Finance and its interaction with another field or discipline, following upon a research question with the potential to make such a satisfactory contribution. • Carry out and report on this research in a format suitable for publication. • Concomitant outcomes will normally include personal skills development supporting the required research and writing activities under the guidance of the supervisor(s).						
Main Content	• The primary task is to develop the research proposal towards its envisaged goal(s), conducting the master's research and preparing the master's thesis under the guidance of the supervisor(s), before submitting it for assessment; • Research proposals may derive from any area of Statistics, Population Studies and Demography or Computational Finance in interaction with another field or discipline, and may include substantial historical and/or comparative studies.						
Pre-requisite	STA810 or equivalent		Prohibited module Combination			None	
Co-requisite	As required by department						
Timetable Requirement per week No Timetable requirements.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 803/804		Module Code			STA803/804	
Module Type	Core	NQF Level	9	Year Level	1/2	NQF Credit	120
Programmes in which the module will be offered		MSc (Statistical Science) (3838), MSc (Computational Finance) (3093), MPhil (Population Studies) (3921)				Total Learning Time	1200 hours
Duration	Year		Proposed Semester			Both Semesters	
Main Learning Outcomes	On completion of this module, students should be able to: Make a contribution knowledge in the field of: - Statistics, Population Studies and Demography or Computational Finance and its interaction with another field or discipline, following upon a research question with the potential to make such a satisfactory contribution. - Carry out and report on this research in a format suitable for publication. - Concomitant outcomes will normally include personal skills development supporting the required research and writing activities under the guidance of the supervisor(s).						
Main Content	- The primary task is to develop the research proposal towards its envisaged goal(s), conducting the research and preparing the mini-thesis under the guidance of the supervisor(s), before submitting it for assessment; - Research proposals may derive from any area of Statistics, Population Studies and Demography or Computational Finance in interaction with another field or discipline, and may include substantial historical and/or comparative studies.						
Pre-requisite	STA810 or equivalent		Prohibited module Combination			None	
Co-requisite	As required by department						
Timetable Requirement per week No Timetable requirements.			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 810		Module Code			STA810	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MSc (Statistical Science) (3838)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Integrate knowledge and understanding of practical project management, the planning and costing of a project, documentation and the presentation of the results.						

	• Develop the ability to solve business/decision making problems using appropriate qualitative and quantitative tools. • Develop the ability to effectively present and communicate relevant research results.				
Main Content	• Conceptual foundations of industry research. Time management and planning of a research project, meeting management, problem statement and objectives formulation, research design and measurement, data processing and analysis, report writing				
Pre-requisite	None	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		1	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment		FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 829		Module Code			STA829	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		MSc (Statistical Science) (3838), MPhil (Population Studies) (3921)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			Second Semester	
Main Learning Outcomes	On completion of this module, students should be able to: Critically analyse a multivariate data set using an appropriate statistical technique required by the research hypothesis and execute various techniques to check for the assumption of normality.						
Main Content	Data summaries and graphical techniques, analysis and covariance structures, inferences about multivariate and linear models, multivariate normality, discrimination and classification						
Pre-requisite	STA853 or equivalent and knowledge of SAS		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 830		Module Code			STA830	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programmes in which the module will be offered	MSc (Statistical Science) (3838), MPhil (Population Studies) (3921)					Total Learning Time	150 hours
Duration	Term		Proposed Semester			First Term	

Main Learning Outcomes	On completion of this module, students should be able to: • Apply advanced execution of principles in experimental design. • Critically apply skills in statistical analysis of data from experiments (chemistry, physics, industry, environment, biochemistry, medical science, pharmacy, education). • Examination of different techniques of experimental design such as completely randomised design, randomised complete block design, latin square design, split-plot design, means separation. • Exposure to practical examples drawn from different disciplines where such techniques are widely applied. • Use of Chi-square tests in experimental data. • Use of analysis of variance in assessing the robustness and fitness in regression analysis.				
Main Content	• Introduction and principles in experimental design • General considerations one one-way and two-way analysis of variance • The completely randomized design • The randomised complete block design • Mean separation methods • Latin square design • Split plot design • Incomplete block designs				
Pre-requisite	STA331, STA332 or equivalent	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment		FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 835		Module Code			STA835	
Module Type	Core/ Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programmes in which the module will be offered		MSc (Statistical Science) (3838), MSc (Medical Bioscience) (3819), MPhil (Population Studies) (3921)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Explain various terms used in Bio statistical analyses. • Design and execute various statistical techniques applicable to medical related data and critically evaluate and interpret the results.						
Main Content	• Study design in medical related research; exploratory medical research including risk ratio's, odds ratio's, mortality and morbidity rates; • Inferential medical research including non-parametric analyses, survival analysis and meta-analysis; regression analysis; and the analyses of a Bio statistical dataset using a statistical package						

Pre-requisite	STA331, STA332 or equivalent	Prohibited module Combination		None	
Co-requisite	None				
Timetable Requirement per week		3	Lectures p.w.		
		0	Practicals p.w.		
		0	Tutorials p.w.		
Other teaching modes that does not require time-table					
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment		FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 837		Module Code			STA837	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MSc (Statistical Science) (3838)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Critically apply knowledge on the development of statistical genetics. • Practical application and interpretation of linkage analysis, variance components methods and association analysis.						
Main Content	• Introduction to relevant aspects of human genetics; Mendelian inheritance, • Linkage analysis, • Association analysis of dichotomous and quantitative traits for case-control and family based study designs, • Software for human genetical analyses.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			3	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Industry Research Project 839		Module Code			STA839	
Module Type	Core	NQF Level	9	Year Level	1	NQF Credit	90
Programmes in which the module will be offered		MSc (Statistical Science) (3838), MSc (Computational Finance) (3093), MPhil (Population Studies) (3921)				Total Learning Time	900 hours
Duration	Semester		Proposed Semester			Second Semester	

Main Learning Outcomes	On completion of this module, students should be able to: • Make a contribution with limited scope to knowledge in the field of Statistics or Data Science, and its interaction with another field or discipline, following upon a research question with the potential to make such a satisfactory contribution. • Carry out and report on this research in a format suitable for publication. • Concomitant outcomes will normally include personal skills development supporting the required research and writing activities under the guidance of the supervisor(s).			
Main Content	• The primary task is to develop the research proposal towards its envisaged goal(s), conducting the research and preparing the project report under the guidance of the supervisor(s), before submitting it for assessment; • Research proposals may derive from any area of Statistics or Data Science in interaction with another field or discipline, and may include substantial historical and/or comparative studies.			
Pre-requisite	STA810 and passed three of the following modules: STA841, STA842, STA843 or COF827	Prohibited module Combination	None	
Co-requisite	As required by department			
Timetable Requirement per week No Timetable requirements.		3	Lectures p.w.	
		0	Practicals p.w.	
		0	Tutorials p.w.	
Other teaching modes that does not require time-table				
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment	FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Business Intelligence 841		Module Code			STA841	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MSc (Statistical Sciences) (3838)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module student should be able to: • Manage data, set up databases, datamarts and data warehouses. • Prepare relational databases. • Access and clean data including further querying and reporting around these data. • Statistically analyse databases. • Use data mining software tools.						
Main Content	• Business Intelligence techniques. • Ethical/professional data handling and communication. • Data management tools to prepare big data for statistical analysis and data mining. • Analytical tools for big data analysis.						

Pre-requisite	None	Prohibited module Combination	None		
Co-requisite	None				
Timetable Requirement per week	0	Lectures p.w.			
	0	Practicals p.w.			
	0	Tutorials p.w.			
Other teaching modes that does not require time-table					
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment	FA	100%	

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Contemporary Business Analysis 842		Module Code			STA842	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MSc (Statistical Sciences) (3838)				Total Learning Time	150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Mine text • Analyse web data • Analyse geospatial data • Analyse customer data.						
Main Content	• Marketing metrics for model assessment. • Recency, Frequency, Monetary (RFM) segmentation. • Customer lifetime value calculations. • Effectively communication skills for model deployment. • Ethical data handling.						
Pre-requisite	None		Prohibited module Combination			None	
Co-requisite	None						
Timetable Requirement per week			0	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)		Method of Student Assessment			FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Multicriteria Decision Making 843		Module Code			STA843	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MSc (Statistical Sciences) (3838)			Total Learning Time	150 hours	
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Use modeling and mathematical programming tools • Demonstrate an competency in the language of multi-criteria decision making						

	• Apply techniques for multi-criteria decision making			
Main Content	• Mathematical optimisation to formulate and solve decision making problems; • Programming techniques including linear, integer, mixed-integer and non-linear scenarios; • Model formulation and construction using multi-criteria decision making.			
Pre-requisite	None	Prohibited module Combination	None	
Co-requisite	None			
Timetable Requirement per week	0	Lectures p.w.		
	0	Practicals p.w.		
	0	Tutorials p.w.		
Other teaching modes that does not require time-table				
Assessment Module Type	Final Assessment (FA)	Method of Student Assessment	FA	100%

Faculty	Natural Sciences		Home Department			Statistics and Population Studies	
Module Name	Statistics 853		Module Code			STA853	
Module Type	Elective	NQF Level	9	Year Level	1	NQF Credit	15
Programme in which the module will be offered		MSc (Statistical Sciences) (3838)			Total Learning Time		150 hours
Duration	Semester		Proposed Semester			First Semester	
Main Learning Outcomes	On completion of this module, students should be able to: • Critically utilise matrix operations in a statistical context.						
Main Content	• Vector and Matrix theory, Linear transformations, Systems of Linear transformations						
Pre-requisite	STA331; STA332; MAT211; MAT221 or equivalent			Prohibited module Combination		None	
Co-requisite	None						
Timetable Requirement per week			1	Lectures p.w.			
			0	Practicals p.w.			
			0	Tutorials p.w.			
Other teaching modes that does not require time-table							
Assessment Module Type	Final Assessment (FA)			Method of Student Assessment		FA	100%

MODULES FROM OTHER FACULTIES

Refer to Faculty of Economic and Management Sciences Calendar

Undergraduate:

Economics	ECO151/ ECO152/ ECO231/ ECO232/ ECO235/ ECO242/ ECO311/ECO331/ ECO332/ ECO334/ ECO335/ ECO336
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Information Systems	IFS131/ IFS132/ IFS231/ IFS233/ IFS316/ IFS317
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Management	MAN131/ MAN132
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Postgraduate:

Financial Accounting	ACC781
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Information Systems	IFS735/ IFS736/ IFS740
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Management	MAN704/ MAN705/ MAN706/ MAN711/ MAN742/ MAN745/ MAN753/ / MAN756/ MAN762/ MAN780/ MAN787
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Refer to Faculty of Arts and Humanities Calendar

Undergraduate:

Introduction to Afrikaans	AFR002
English for Educational Development	EED117/ EED127
Introduction to Xhosa	XHO002

EXPLANATION OF SYMBOLS AND REMARKS ON ACADEMIC TRANSCRIPT

A	75-100%	Pass with Distinction
B	70-74%	Pass
C	60-69%	Pass
D	50-59%	Pass
E	45-49%	Fail
F	40-44%	Fail
G	39-0%	Fail
No Year mark		Absent from the examination: No results
SPG		Absent from the examination but with special permission to write the supplementary examination on medical or non-medical grounds.
SAG		Supplementary examination granted on academic grounds.
SUB		Failed to obtain the required sub minimum and have to repeat the course.
Ceased Programme		Ceased studying the programme.
DNQ		Did not qualify to write the examination
ABS		Absent from the examination
SDA		Senate Discretionary Assessment granted
External Credit Transfer		An external module completed at another institution deemed equivalent to be credited toward a qualification for which the student is registered.
Internal Credit Transfer		A module completed at this institution credited toward a qualification for which the student is registered.

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